

DC Circuit



Who is Jahid vai?

Engineer (EEE), BAERA

45th & 46th BCS Written candidate

Ex-EEE, BUET (2016)

Teaching Experience: More than 8 years

Jahid Hasan(EEE Engineer, BAERA // Ex-EEE,BUET// Whatsapp: 01829-415490)

DC Circuit

Topics:

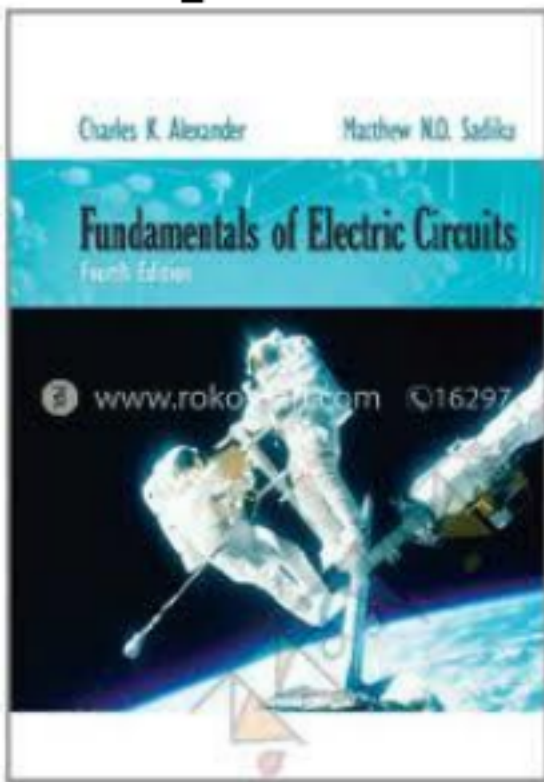
- DC & AC circuit
- Electronics & DLD
- Power system & protection
- Machine
- Communication
- Control system
- Signal & DSP
- Non-departmental

DC Circuit

Prepared by:
Engr. Muhammad Tanvirul Hasan
Assistant Engineer
Dhaka Electric Supply Company Ltd (DESCO)

60%

40%



Electrical Circuit---Alexander & Sadiku (4th Edition)			
Chapter	Example	Practice Problem	Exercise
2	2.6, 2.7, 2.10, 2.15	2.6, 2.7, 2.10, 2.15	2.18, 2.22, 2.25, 2.26, 2.32, 2.35, 2.38, 2.39, 2.44, 2.45(b), 2.47, 2.51, 2.55
3	3.2, 3.3, 3.4, 3.6, 3.7	3.2, 3.3, 3.4, 3.7, 3.13	3.5, 3.6, 3.8, 3.9, 3.10, 3.13, 3.15, 3.17, 3.18, 3.20, 3.21, 3.23, 3.44, 3.49, 3.50, 3.51, 3.60, 3.84, 3.87
4	4.4--4.13	4.4--4.13	4.13, 4.14, 4.24, 4.26, 4.27, 4.29, 4.30, 4.32, 4.37, 4.39, 4.40, 4.41, 4.44, 4.47, 4.48, 4.51, 4.53, 4.54, 4.55, 4.59, 4.60, 4.63, 4.64, 4.66, 4.67, 4.68, 4.72, 4.73
5	5.6--5.9	5.4, 5.5, 5.6, 5.8, 5.9, 5.10	5.8(b), 5.9(b), 5.10, 5.13, 5.14, 5.15, 5.19, 5.23, 5.27, 5.28, 5.30, 5.31, 5.33, 5.57, 5.58, 5.60, 5.61, 5.65, 5.66 (similar types), 5.72, 5.74
6	6.4, 6.5, 6.9, 6.10, 6.13, 6.14, 6.15 (only circuit diagram)	6.5, 6.9, 6.10, 6.15	6.46, 6.48, 6.78, 6.79, 6.80, 6.81, 6.82
7	7.1, 7.2, 7.3, 7.4, 7.5, 7.10, 7.11, 7.12, 7.20, 7.21	7.1, 7.2, 7.3, 7.4, 7.5, 7.10, 7.11, 7.12, 7.20, 7.21	7.40, 7.42, 7.44, 7.53, 7.54, 7.64

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Electrical Circuit---Alexander & Sadiku (4th Edition)

9	9.2, 9.7, 9.11, 9.12, 9.13, 9.14	9.2, 9.7, 9.13, 9.14	9.6, 9.22, 9.39, 9.43, 9.66, 9.80
10	10.1, 10.2, 10.3, 10.8, 10.9	10.1, 10.2, 10.3, 10.6, 10.8	10.3, 10.6, 10.7, 10.67
11	11.1--11.12, 11.14, 11.16	11.1--11.12, 11.14, 11.15, 11.16	11.7, 11.19, 11.23, 11.28, 11.30 (similar types), 11.32, 11.60, 11.61, 11.62, 11.63, 11.71, 11.72, 11.73, 11.74
12	12.2--12.10, 12.14, 12.15	12.2--12.10, 12.14, 12.15	12.8, 12.9, 12.10, 12.11, 12.13, 12.14, 12.15, 12.18, 12.19, 12.21, 12.25, 12.55, 12.57, 12.69, 12.71
14	14.7, 14.8, 14.9, 14.11	14.7, 14.8, 14.9	

DC Circuit

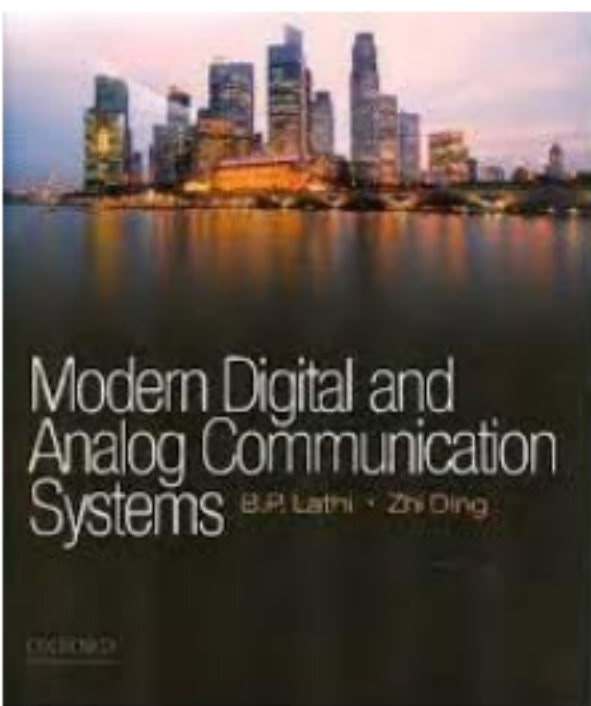
Topics to be covered for concept

Amplitude Modulation	DSB-WC	generation, detection, formulas, comparison
	DSB-SC	
	SSB-WC	
	SSB-SC	
	QAM	
	VSB	
Angle Modulation	FM	generation, detection, formulas
	PM	
Pulse Code Modulation	PCM	sampling, Nyquist Theorem, Shannon's Capacity Theorem, Quantization, Quantization Noise Error
Delta Modulation	DM	slope overload, granular noise
Line Coding		Unipolar NRZ, Polar NRZ, Unipolar RZ, Bipolar RZ Split/Manchester, Differential Manchester coding
Digital Modulation	BASK	generation , detection, bandwidth
	BFSK	
	BPSK	
	DPSK	
Multiplexing Techniques	TDM	Block diagram
	FDM	
	CDM	
	Joint TDM & FDM	
	WDM	

DC Circuit

B. L. Lathi (3rd Edition)

Chapter	Example	Problems
4	4.5, 4.8	4.2-1, 4.2-8, 4.2-9 4.3-1, 4.3-2, 4.3-3 4.5-2 4.6-1
5	5.1, 5.2, 5.3, 5.4, 5.5	5.1-3 5.2-1, 5.2-2, 5.2-3, 5.2-4 5.2-5, 5.2-6
6		6.2-2, 6.2-3, 6.2-4, 6.2-5



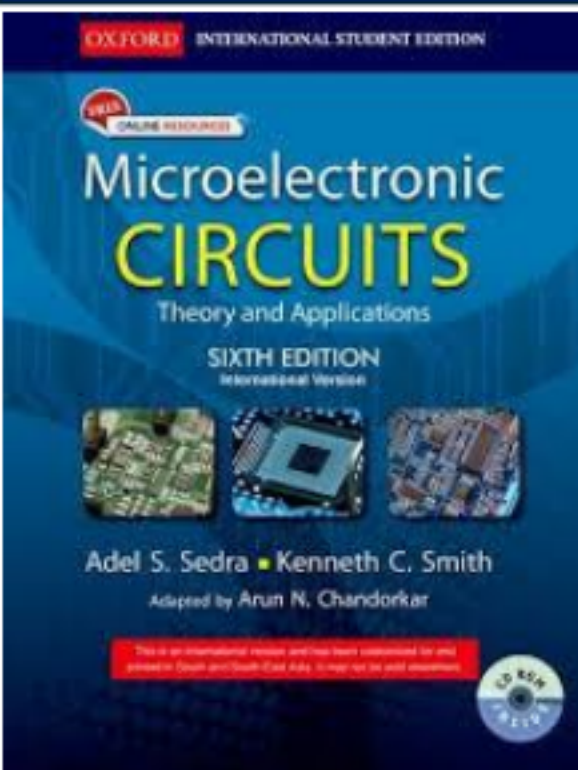
** Rony Parvej sheet on Communication

** Hand note for EEE job preparation by Md. Ismail Hossain

** Previous job Question (must)

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DC Circuit



Microelectronics Circuit (Sedra & Smith: 6th Edition)

Diode	Example	4.1, 4.2, 4.5, 4.6, 4.7, 4.8
	Exercise	4.4, 4.5, 4.11, 4.15, 4.16, 4.26, 4.6.3
	Problems	4.2, 4.3, 4.4, 4.9, 4.10, 4.11, 4.23, 4.85, 4.95

MOSFET	Example	5.1--5.8, 5.10
	Exercise	5.9, 5.14, 5.15
	Problem	5.49, 5.50, 5.51, 5.55

BJT	Example	6.2--6.10, 6.12
	Exercise	6.13, 6.14, 6.49
	Problems	6.28(b), (d), 6.29, 6.35

DC Circuit

Electronic Devices & Circuits (Boylested : 11th Edition)

*clipped
clamped*

Diode	Example	2.7, 2.9, 2.13, 2.17, 2.20, 2.22, 2.23, 2.24, 2.26, 2.27, 2.28, Fig-2.88, 2.100
	Exercise	5(b), 6, 7, 8, 10, 11, 31, 36, 42, 43, 44

BJT	Example	4.3, 4.4, 4.7--4.12, 4.14, 4.16, 4.18--4.23, 4.29--4.32
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BJT	Example	5.1, 5.2, 5.3, table-5.1 (first 3/4 nos.)
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DC Circuit

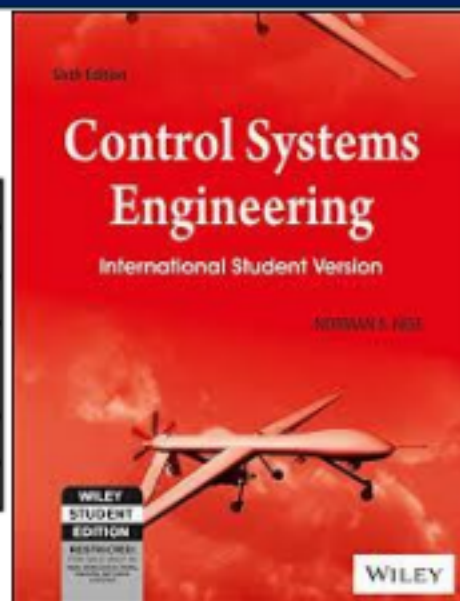
DLD (Digital Logic Design)

1. Design AND, OR, NOT logic gate using CMOS/NMOS/PMOS
2. Flip-Flop, Registers, Counter circuit (Block Diagram, Truth Table, Timing Diagram)
3. Implement AND, OR, NOT gate using universal NAND/NOR gate
4. Transmission gate
5. Boolean Algebra
6. K-map

DC Circuit

Control System NISE (6th Edition)

Chapter	Example	Skill-Assessment Exercise	Problem
2	2.3, 2.4, 2.5,	2.3, 2.5,	
4	4.2, 4.3, 4.5, 4.6, 4.7	4.2, 4.3, 4.4, 4.5	
5	5.1, 5.2, 5.3, 5.4, Fig 5.31 (signal flow diagram)	5.1	1, 2, 13
6	6.4, 6.9	6.3	
7	7.1, 7.2, 7.3, 7.4, 7.6	7.1, 7.2, 7.3	



Power Electronics

1. DC-DC converter : BUCK, BOOST, BUCK-BOOST- only circuit + input-output characteristic equation
2. single phase half wave controlled rectifier- SCR

DC Circuit

A TEXTBOOK OF ELECTRICAL TECHNOLOGY

IN S.I. UNITS

Volume II

AC & DC MACHINES

B.L. THERAJA
A.K. THERAJA

S. CHAND



Electrical Machine (B. L. Thereja: Vol-2)

Handwritten mark: a red 'X' with a checkmark inside.

Handwritten word: 'Ques' circled in red.

Chapter	Example
32	32.3, 32.5, 32.9, 32.11, 32.14, 32.16, 32.18, 32.21, 32.22, 32.23, 32.24, 32.27, 32.28, 32.29, 32.32, 32.33, 32.35, 32.38, 32.39, 32.42, 32.46, 32.49, 32.53, 32.55, 32.59, 32.61, 32.63, 32.72, 32.78, 32.90, 32.91, 32.92, 32.95
33	33.2, 33.3, 33.4, 33.5, 33.7, 33.8, 33.9, 33.10
34	34.3, 34.5, 34.8, 34.11, 34.14--34.19, 34.30, 34.31, 34.33, 34.34, 34.37, 34.38, 39(a), 34.40, 34.41
26	26.3--26.7, 26.8, 26.11(b), 26.12, 26.13, 26.15, 26.17, 26.20(a) 26.23, 26.24, 26.25, 26.26, 26.27, 26.30, 26.31
27	27.1, 27.2, 24.4--24.7
29	29.2--29.5, 29.6, 29.10, 29.11, 29.13(a)(d), 29.17, 29.21--29.25, 29.29, 29.30
30	30.8
37	37.14, 37.17, 37.18(a), 37.19, 37.22, 37.23, 37.24

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DC Circuit



Electrical Machinery (Chapman: 4th Edition)

Chapter	Example
2	2.1--2.9, fig-2.17, 2.18
5	5.1, 5.2, 5.5, 5.6, fig-5.13, 5.14
6	6.1, 6.2, 6.3, fig-6.9
7	7.1--7.4
Chapter	Exercise
2	2.1, 2.6, 2.9, 2.12, 2.18, 2.22
5	5.1, 5.7 (c)(d), 5.8, 5.9, 5.23(b)

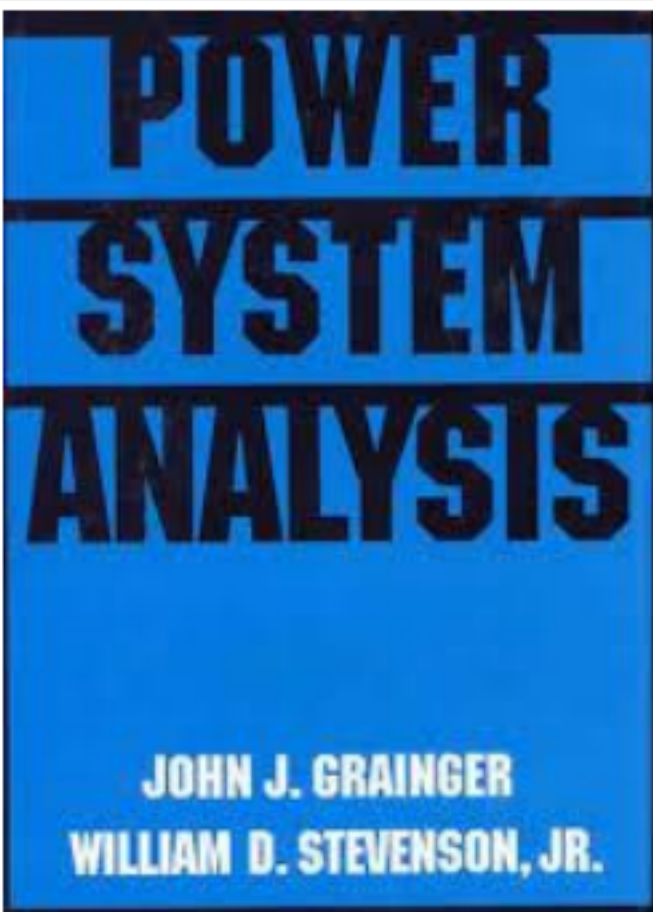
~~XFMR~~ Short cut

chapter	example
2	2.9, 2.10, 2.11, 2.13

Electric Machine (Charles I Hubert)

chapter	example
2	2.9, 2.10, 2.11, 2.13

DC Circuit



Power System

~~38~~ ~~Emp~~
Stevenson

Chapter	Example
1	1.4, 1.5, Fig-1.26,1.27
2	2.5, 2.6
3	3.8 Exercise-3.12, 3.13
6	6.1, 6.2
10	10.1, 10.3, 10.7
11	11.1, 11.2, 11.3, 11.8, 11.9, Fig-11.26, 11.27

DC Circuit

Revised Edition

PRINCIPLES OF POWER SYSTEM

V.K. MEHTA
ROHIT MEHTA

S. CHAND

V.K. Mehta

Chapter	Example
2	2.1, 2.2, 2.3, 2.5, 2.7, 2.9
3	3.1-3.16, 3.19
6	6.1-6.5, 6.9
8	8.17, 8.18, 8.19, 8.21, 8.22
10	10.1-10.4, 10.7, 10.17
17	17.2, 17.3, 17.7
18	18.2-18.9, 18.14-18.20
19	19.1, 19.2, 19.3
22	22.1-22.6

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Continuous and Discrete signal and system
(Soliman & Srinath, 2nd edition)

- Chapter-1
- Chapter-4 (only formula & math)

Com

- Optical Fiber
- Noise Figure
- Entropy of digital source
- Forouzan বই থেকে আনকমন ম্যাথ বেছে বেছে করুন

DC Circuit

Digital signal processing (DSP) topics:

- Sampling
- Quantization
- Find discrete signal from continuous
- Convolution (math) 
- LTI system (math)
- Finding step response (math)

DC Circuit



- টপিকভিত্তিক ১০০+ লাইভ ও রেকর্ডেড ক্লাস
- সম্পূর্ণ সিলেবাসের উপর ০৫টি পরীক্ষা
- নন-ডিপার্টমেন্টাল প্রিপারেশন
- প্রবলেম সলভিং এর জন্য **whatsapp** সাপোর্ট গ্রুপ



Group by Jahid Hasan

Jahid & Sakib's EEE Care

Public group · 2.6K members



$$\begin{array}{l} 10 \times 5 = 50 \\ 5 \times 2 = 10 \\ 20 \times 1 = 20 \end{array} \left. \vphantom{\begin{array}{l} 10 \times 5 = 50 \\ 5 \times 2 = 10 \\ 20 \times 1 = 20 \end{array}} \right\} 140$$

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DC Circuit



BCS



Pigest



Job solution



DC Circuit

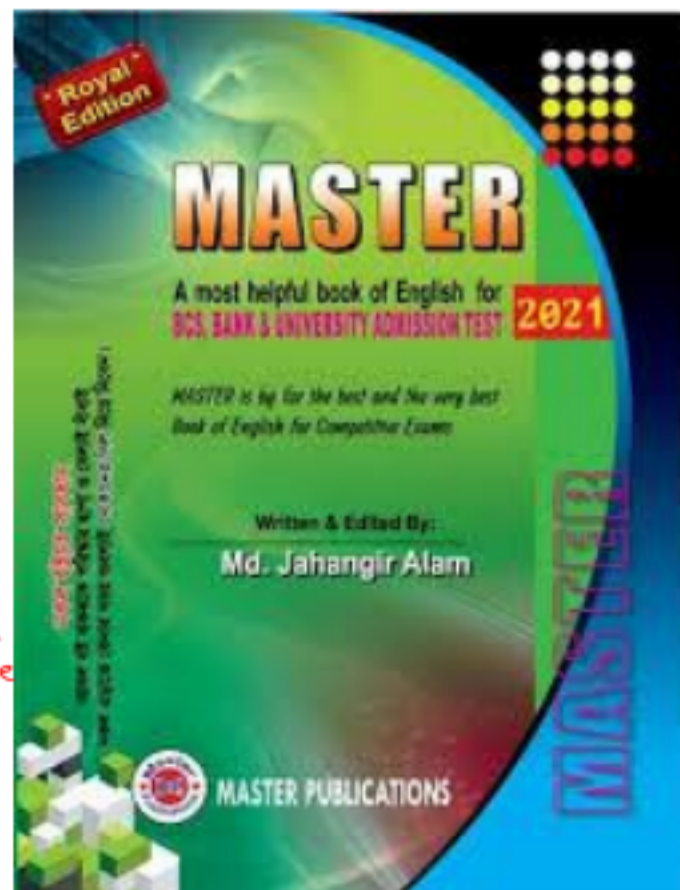


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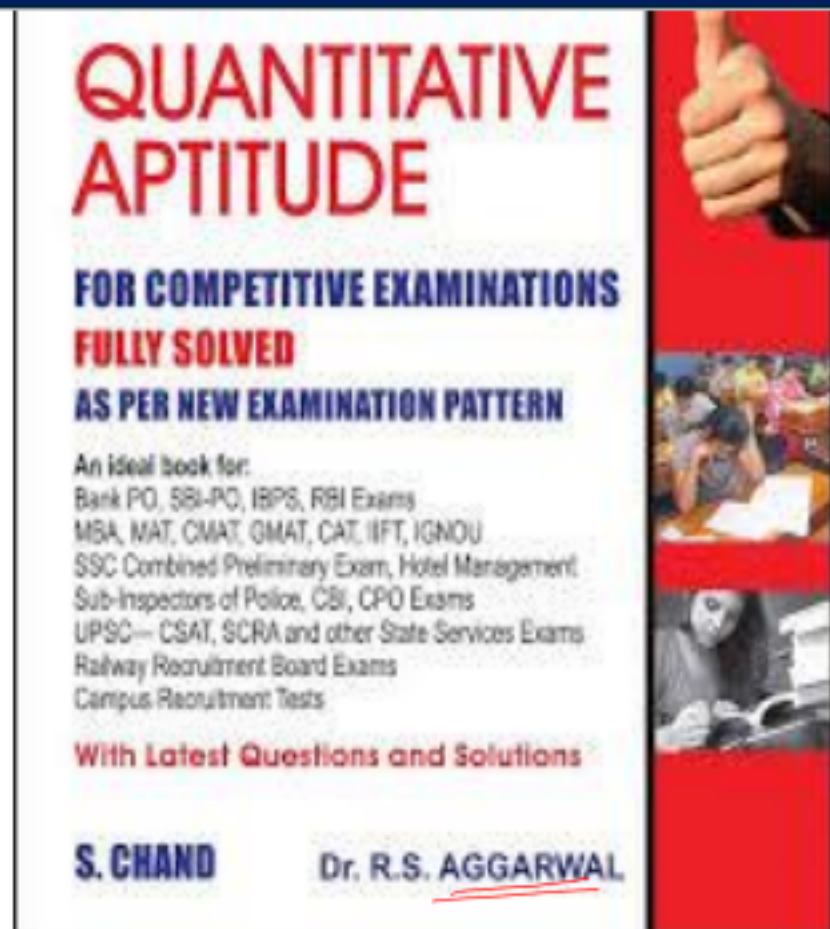
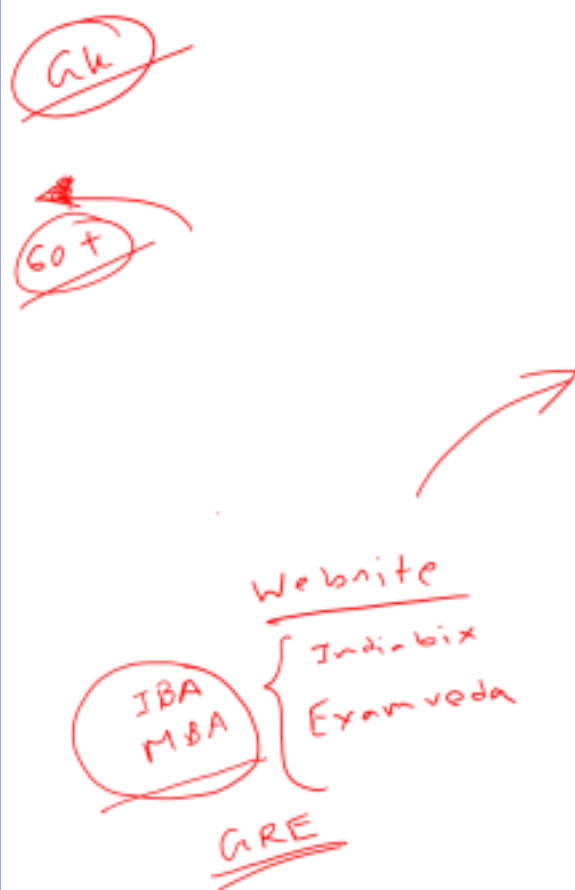
৯ম সংস্করণ

only Buet

Cliff's Trefl
T. J. Fitiakides → common mistake
Word smart



DC Circuit



DC Circuit

Websites & Apps

Power sector:

- Power cell
- PDB
- PGCB
- DPDC
- DESCO
- BREB
- BERC
- RNPP

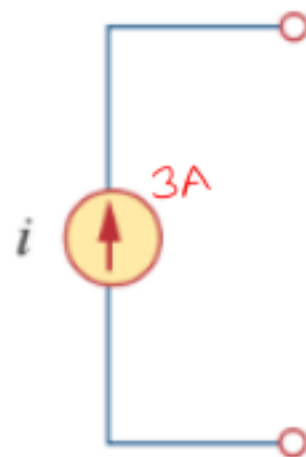
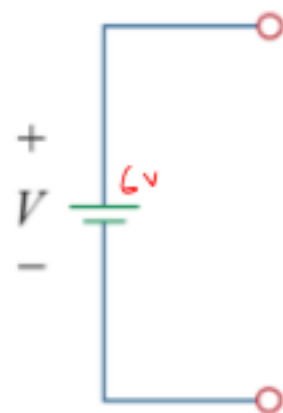
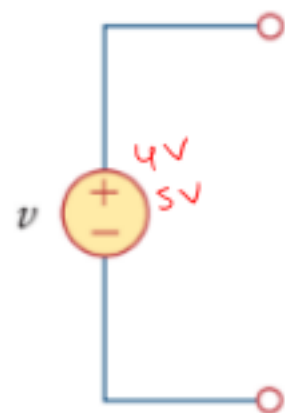


Recent issues
Buet Ck $\Rightarrow$ 15/20



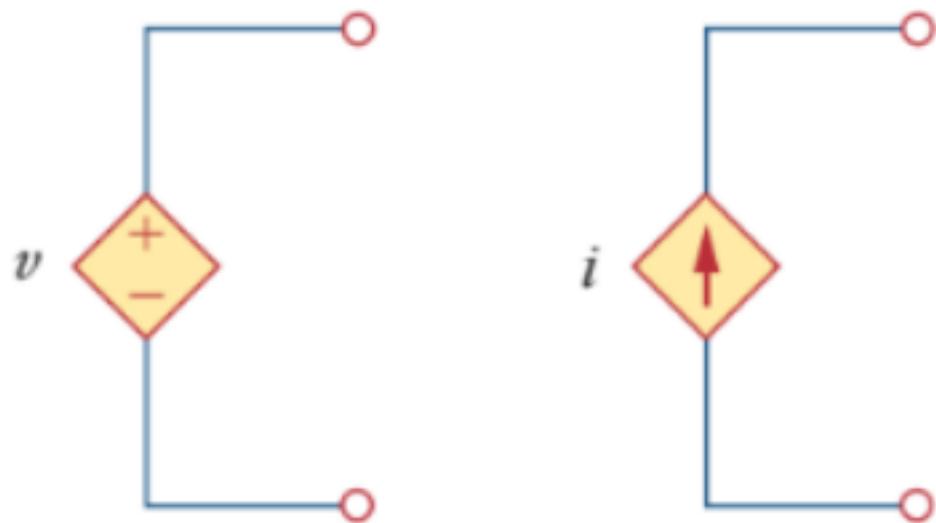
DC Circuit

Independent source:



DC Circuit

Dependent/controlled source:

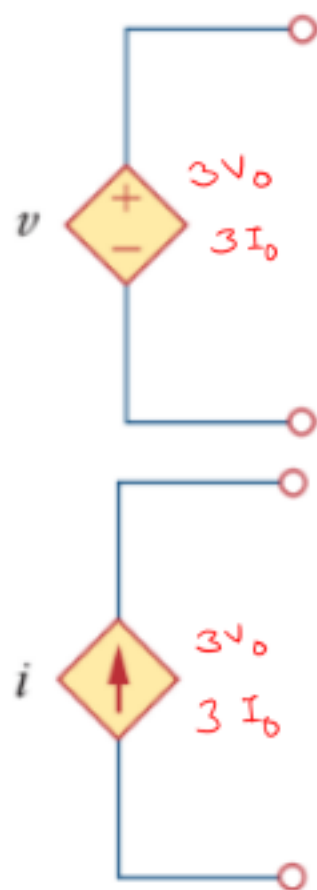


DC Circuit

Dependent/controlled source:

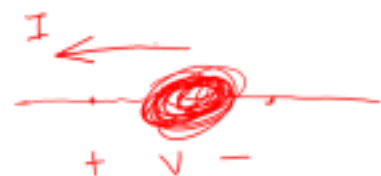
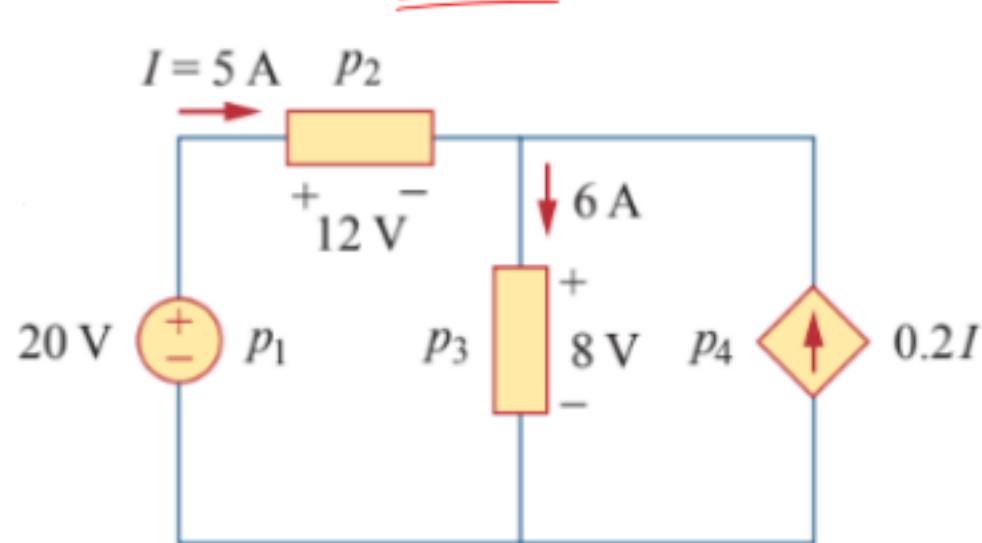
There are 4 possible types of dependent sources:

1. A voltage-controlled voltage source (VCVS).
2. A current-controlled voltage source (CCVS).
3. A voltage-controlled current source (VCCS).
4. A current-controlled current source (CCCS).



DC Circuit

Calculate the power supplied or absorbed by each element



$$P = VI$$

= $\oplus$ supply
= $\ominus$ absorbed

$$P_1 = 20 \times 5 = 100\text{ W} \rightarrow \text{supplied}$$

$$P_2 = 12 \times 5 = 60\text{ W} \rightarrow \text{absorbed}$$

$$P_3 = 8 \times 6 = 48\text{ W} \rightarrow \text{absorbed}$$

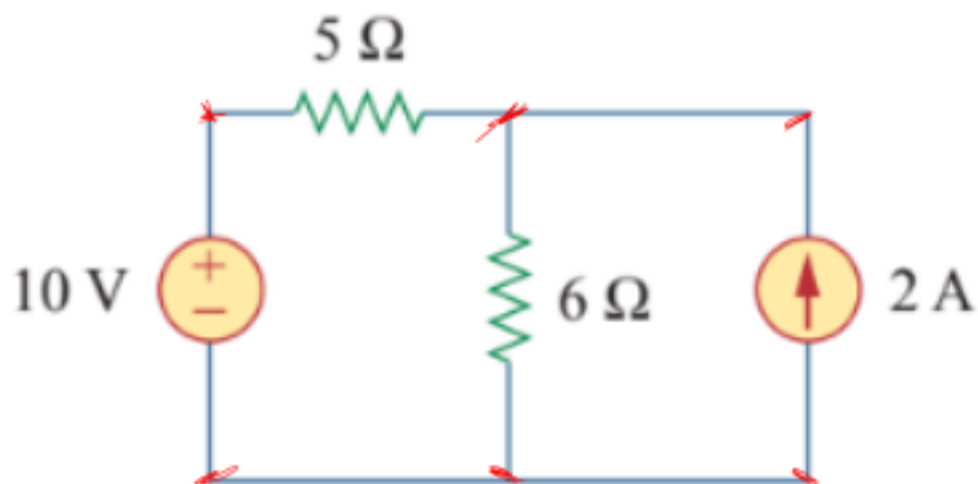
$$P_4 = 8 \times 0.2 \times 5 = 8\text{ W} \rightarrow \text{supplied}$$

$$\text{equal} \begin{cases} P_{\text{supplied}} = P_1 + P_4 = 108\text{ W} \\ P_{\text{absorbed}} = P_2 + P_3 = 108\text{ W} \end{cases}$$

DC Circuit

Branches:

A branch represents a single element such as a voltage source or a resistor.

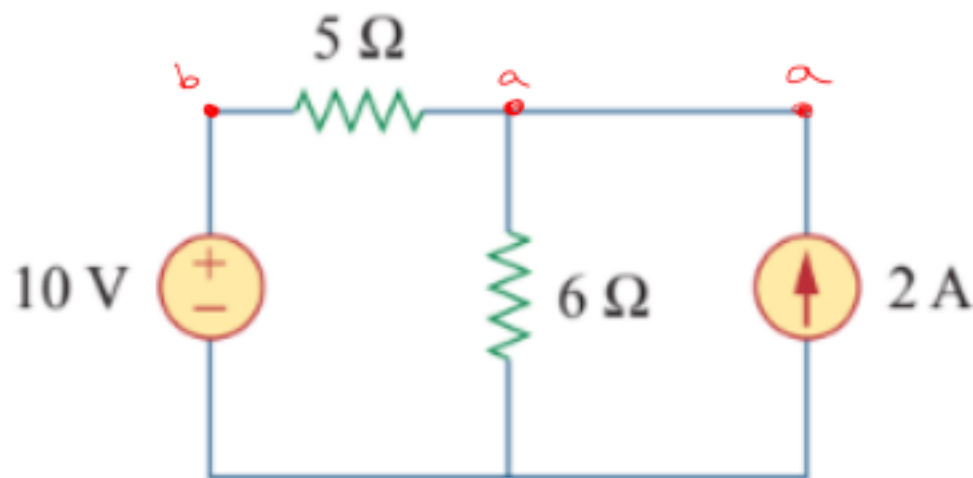
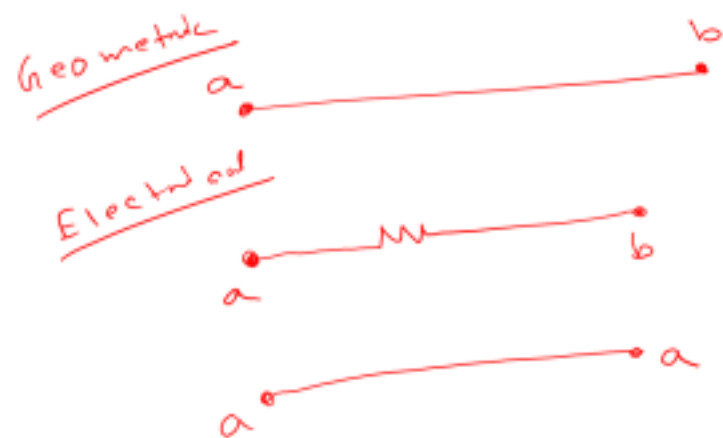


$$b = 4$$

DC Circuit

Node:

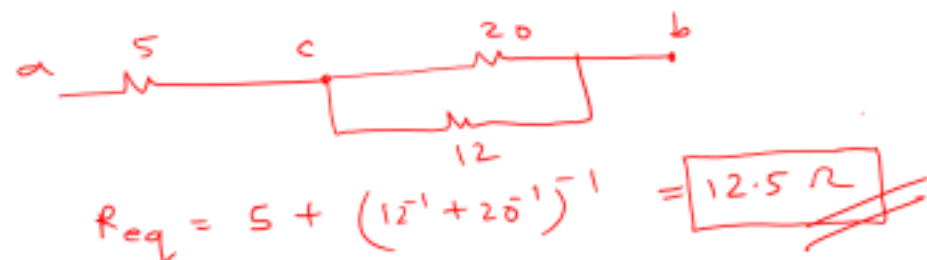
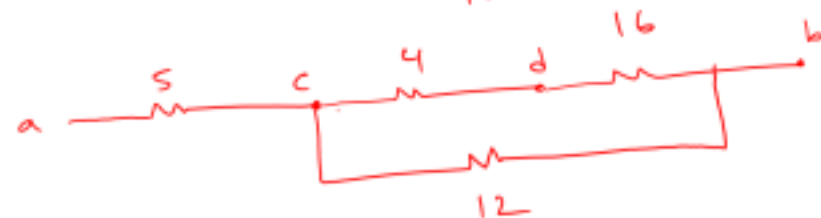
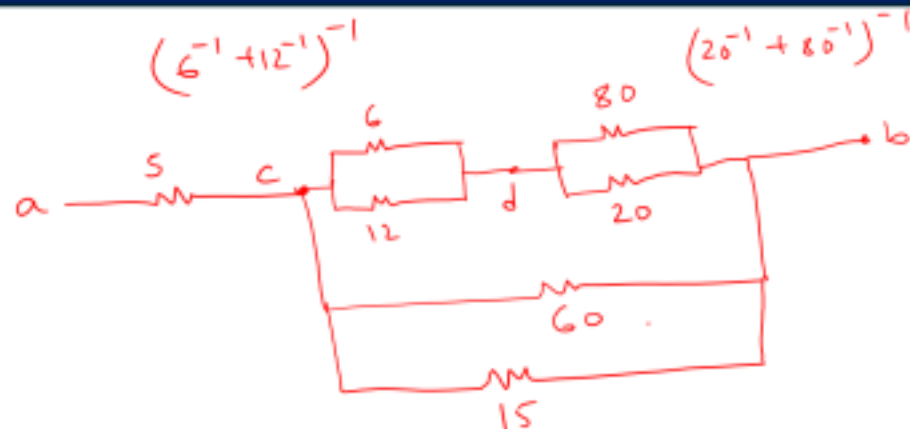
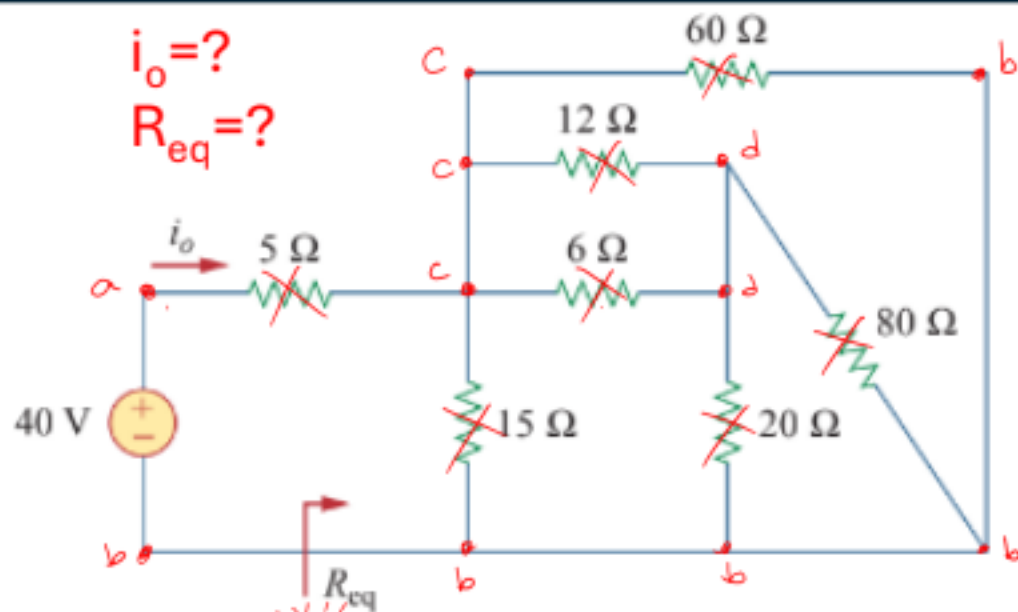
A node is the point of connection between two or more branches.



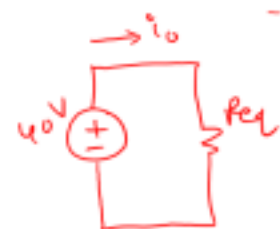
DC Circuit

$$i_o = ?$$

$$R_{eq} = ?$$



$$i_o = \frac{40}{R_{eq}} = \boxed{3.2 \text{ A}}$$

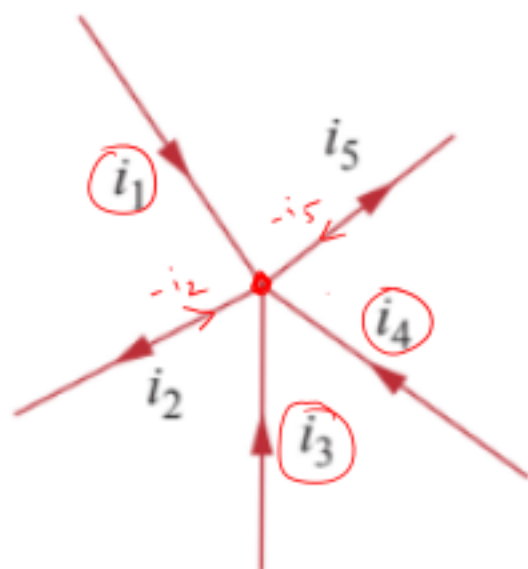


DC Circuit

Kirchhoff's current law (KCL)

The algebraic sum of currents entering a node (or a closed boundary) is zero.

$$\sum_{n=1}^N i_n = 0$$



$$i_1 + i_3 + i_4 = i_2 + i_5$$

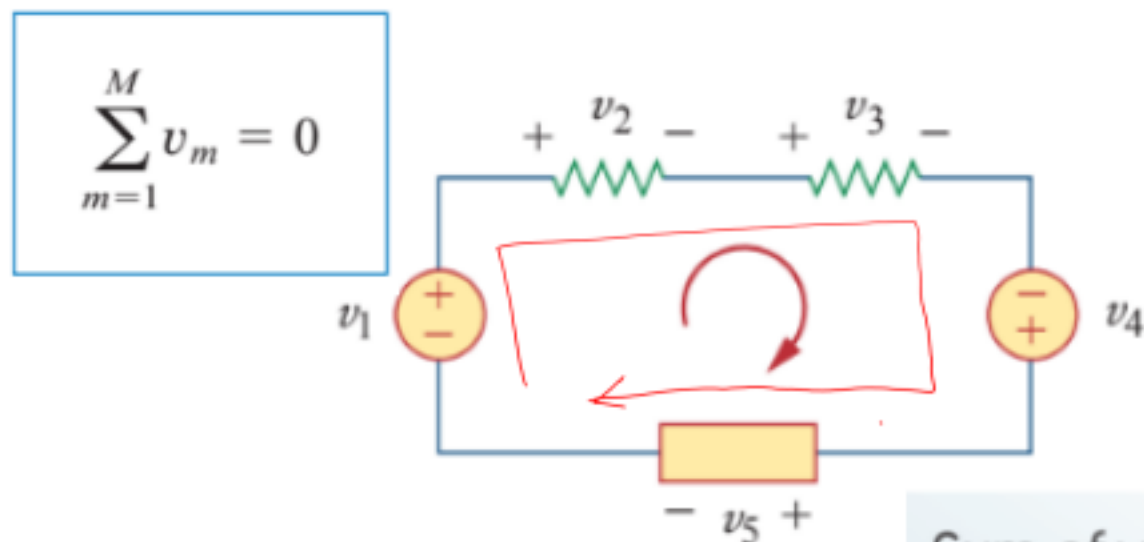
$$\Rightarrow i_1 + i_3 + i_4 + (-i_2) + (-i_5) = 0$$

$$\Rightarrow \boxed{\sum i = 0}$$

DC Circuit

Kirchhoff's voltage law (KVL)

The algebraic sum of all voltages around a closed path (or loop) is zero.



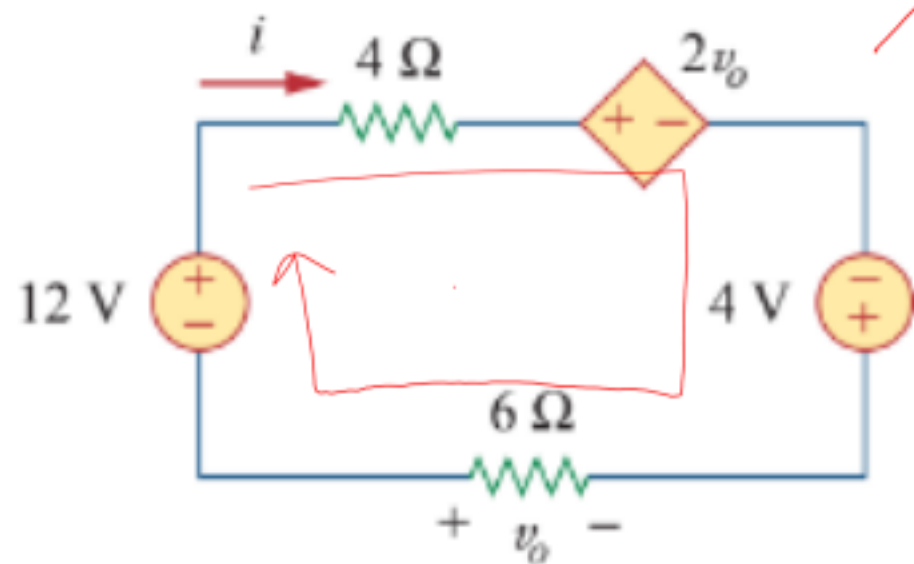
$$+v_2 + v_3 - v_4 + v_5 - v_1 = 0$$

$$\Rightarrow v_1 + v_4 = v_2 + v_3 + v_5$$

Sum of voltage drops = Sum of voltage rises

DC Circuit

Determine v_o and i in the circuit



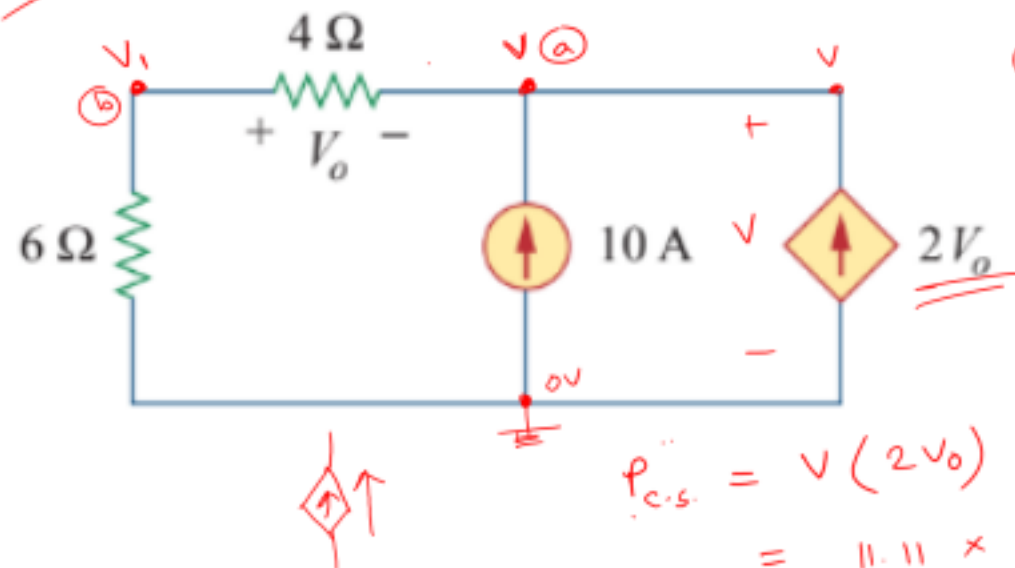
KVL $\Rightarrow$

$$4i + 2v_o - 4 + \boxed{-v_o} - 12 = 0$$
$$\Rightarrow 4i + 2(-6i) - 4 + 6i - 12 = 0$$
$$\Rightarrow 4i - 12i - 4 + 6i - 12 = 0$$
$$\Rightarrow 2i = -16 \Rightarrow \boxed{i = -8 \text{ A}}$$
$$v_o = -6(-8) = \boxed{48 \text{ V}}$$

DC Circuit

Find V_o in the circuit of Fig. 2.86 and the power dissipated by the controlled source.

Nodal



$$\textcircled{a} \Rightarrow 10 + 2V_o + \frac{V_1 - V}{4} = 0 \quad \begin{matrix} V_1 = ? \\ V = ? \end{matrix}$$

$$\textcircled{b} \Rightarrow \frac{V_1 - V}{4} + \frac{V_1}{6} = 0$$

$$V_1 - V = \textcircled{V_o}$$

$$\boxed{V_o = -4.44 \text{ V}}$$

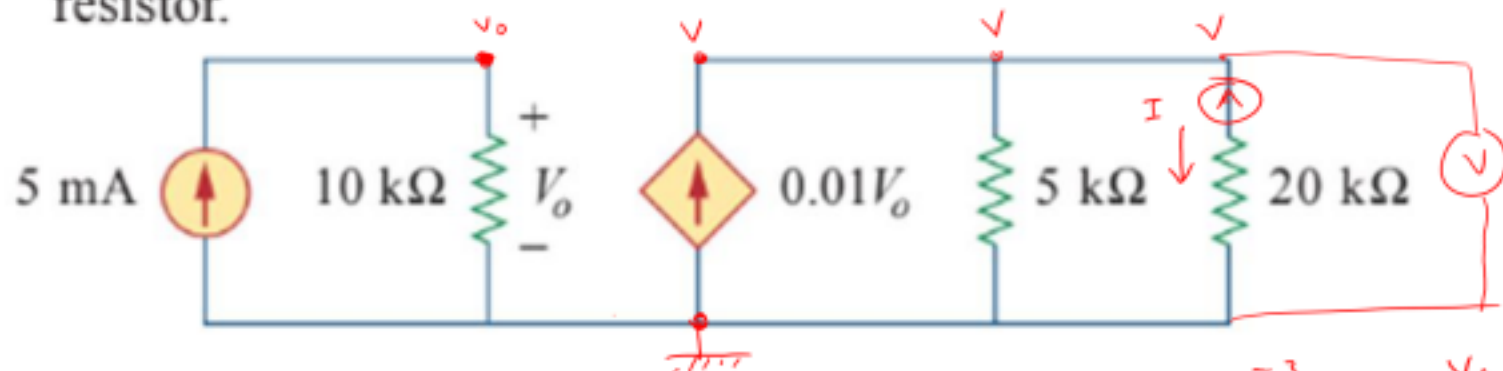
$$\boxed{V = 11.11 \text{ V}}$$

$$P_{c.s.} = V(2V_o)$$

$$= 11.11 \times 2 \times (-4.44) = \boxed{-99 \text{ W}} \rightarrow \text{absorbed}$$

DC Circuit

For the network in Fig. 2.89, find the current, voltage, and power associated with the 20-k Ω resistor.



$$P = I^2 R$$

$$= 0.1^2 \times 20 \times 10^3 = \boxed{200 \text{ W}}$$

$$0.01 V_o = \frac{V}{5 \text{ k}} + \frac{V}{20 \text{ k}}$$

$$\Rightarrow 0.01 V_o = \frac{V}{4 \text{ k}}$$

$$\Rightarrow V = 40 V_o$$

$$= 40 \times 50$$

$$= \boxed{2000 \text{ V}}$$

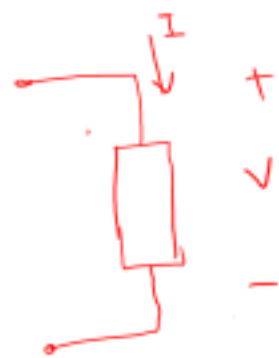
$$I = \frac{2 \times 10^3}{20 \times 10^3}$$

$$= \boxed{100 \text{ mA}}$$

$$5 \times 10^{-3} = \frac{V_o}{10 \text{ k}}$$

$$\Rightarrow \boxed{V_o = 50}$$

DC Circuit



$$P = VI = \begin{cases} +ve \rightarrow \text{absorbed} \\ -ve \rightarrow \text{supply} \end{cases}$$

$$\left. \begin{array}{l} \text{③} \rightarrow 3 \rightarrow \text{absorb} \rightarrow P_1 \\ \text{②} \rightarrow 2 \rightarrow \text{supply} \rightarrow P_2 \end{array} \right\} P_1 = P_2$$

$$P_1 + (-P_2) = 0$$

DC Circuit

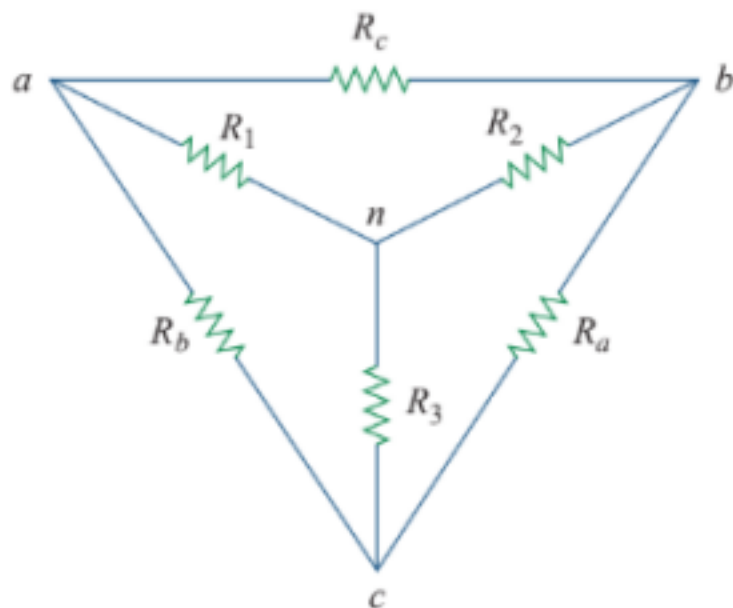
Power source can **absorb power** instead of supplying it, depending on the circuit conditions and the type of source.

When a battery is being **recharged**, the **current flows into it** in the opposite direction compared to when it is discharging, causing it to absorb power from the charger.



DC Circuit

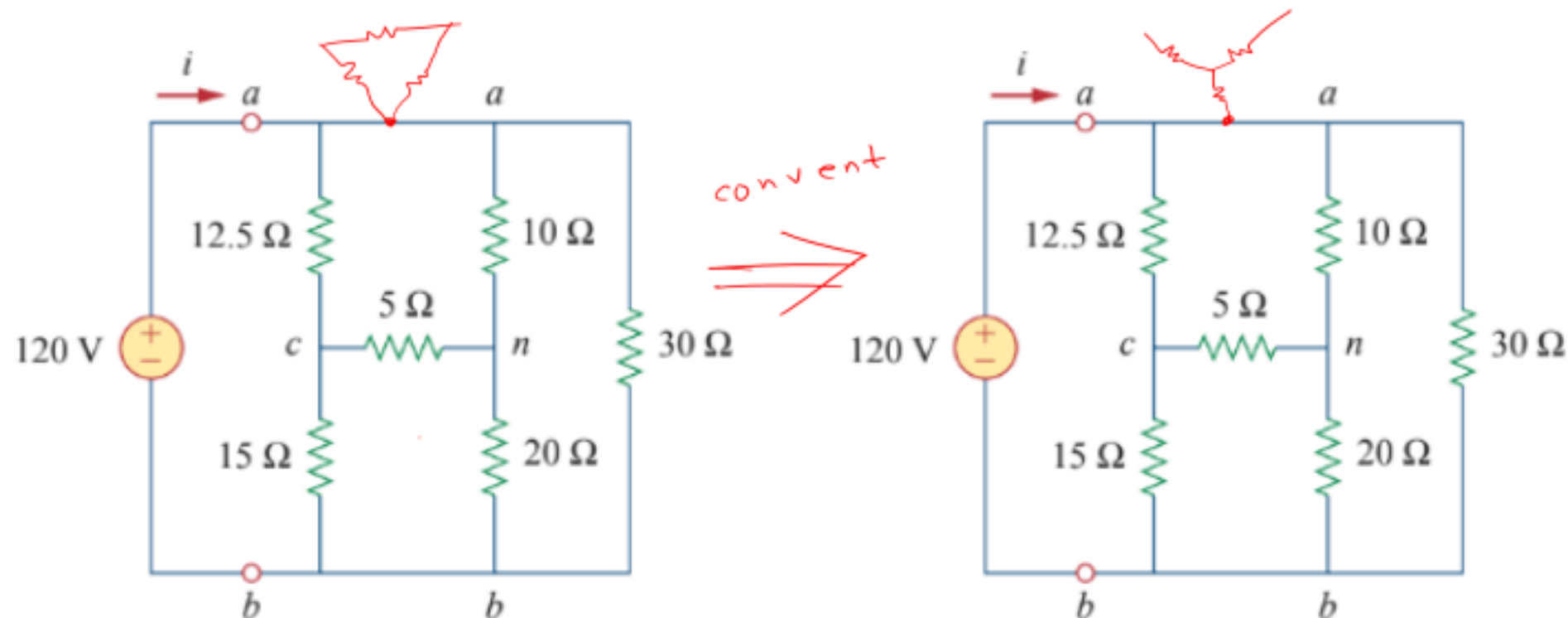
Wye-Delta Transformations



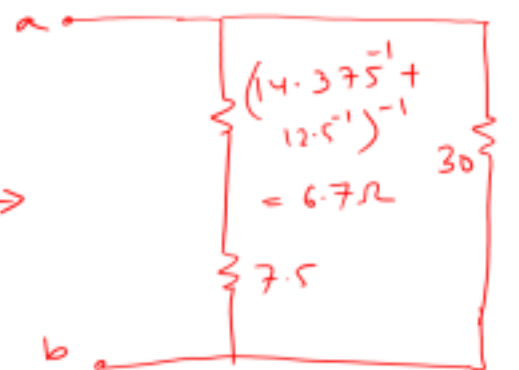
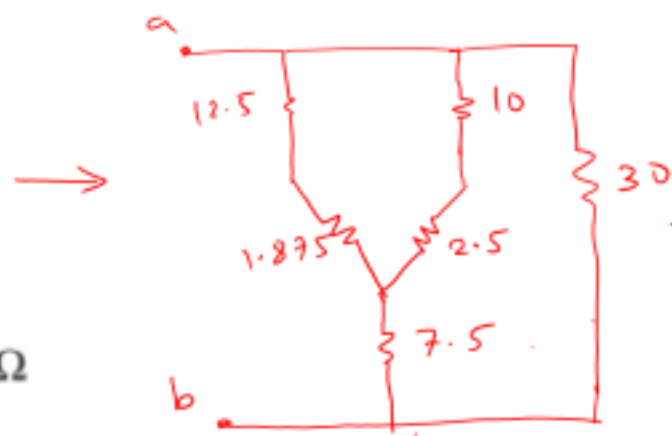
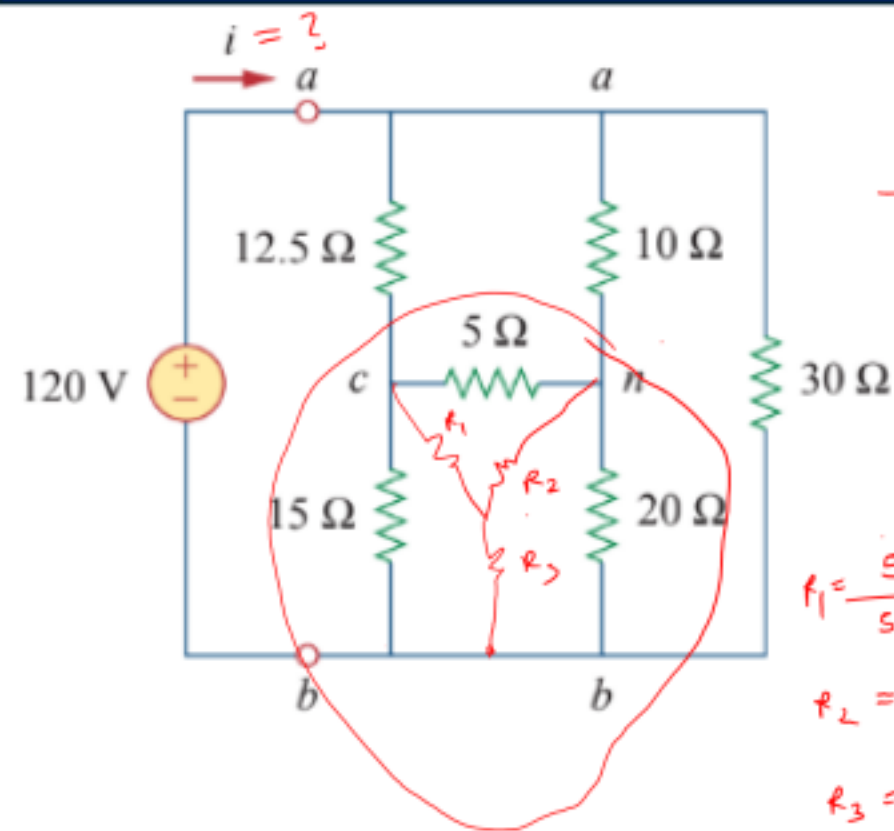
$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c}$$

$$R_a = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1}$$

DC Circuit



DC Circuit



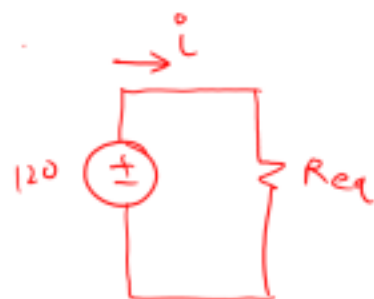
$$R_{eq} = [(6.7 + 7.5)^{-1} + 30^{-1}]^{-1}$$

$$= 9.63 \Omega$$

$$R_1 = \frac{5 \times 15}{5 + 15 + 20}$$

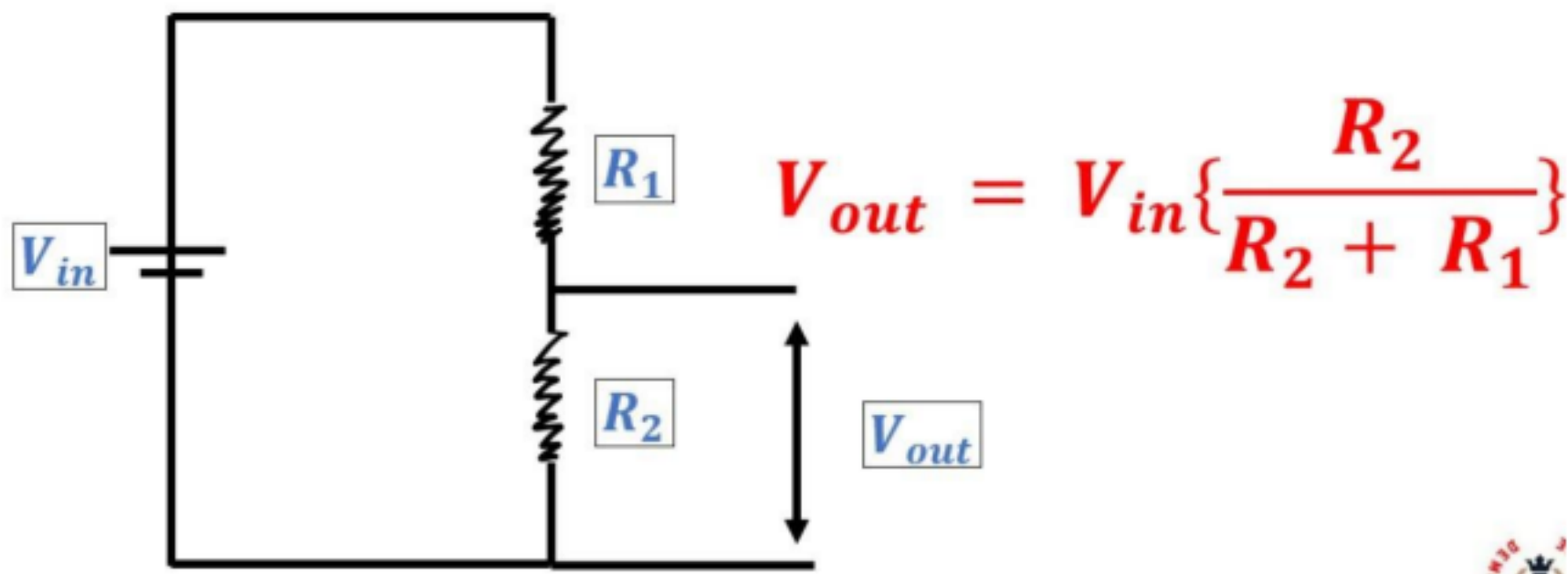
$$R_2 = \frac{5 \times 20}{40}$$

$$R_3 = \frac{15 \times 20}{40}$$



$$i = \frac{120}{9.63} = 12.45 \text{ A}$$

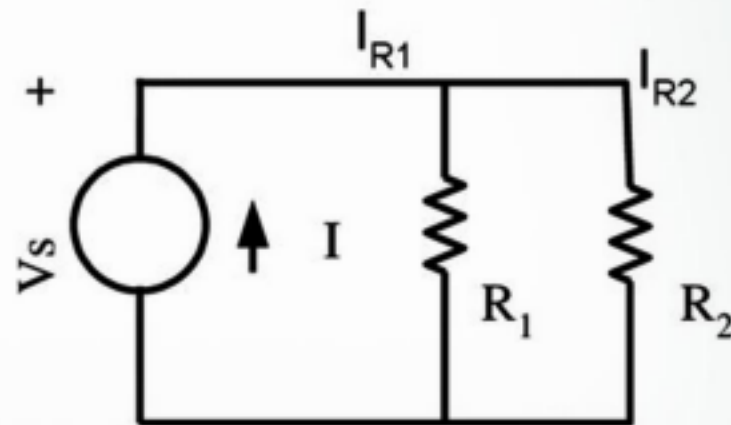
Voltage Divider Rule



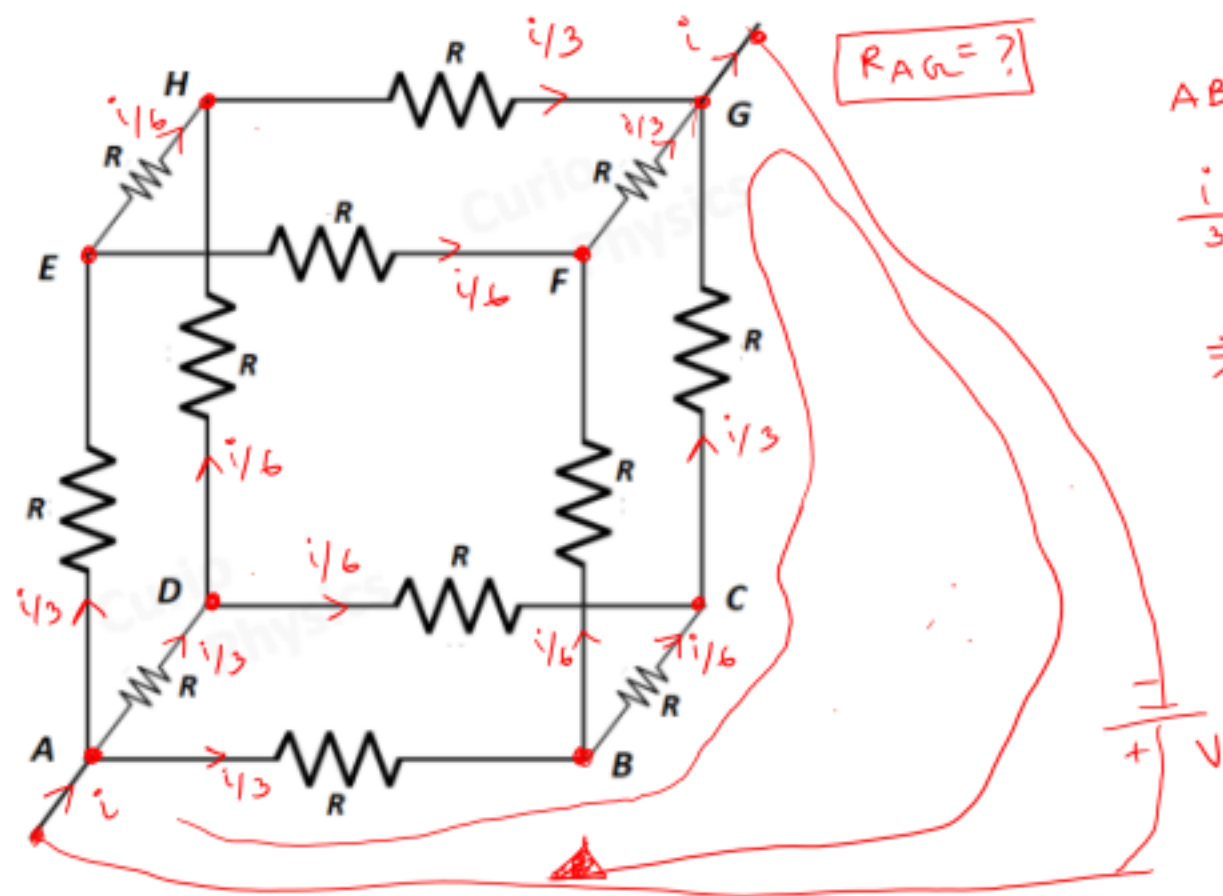
Current Divider

$$I_{R_1} = \frac{R_2}{R_1 + R_2} I$$

$$I_{R_2} = \frac{R_1}{R_1 + R_2} I$$



DC Circuit



ABCGA - Loop (KVL) $\Rightarrow$

$$\frac{i}{3}R + \frac{i}{6}R + \frac{i}{3}R = V$$

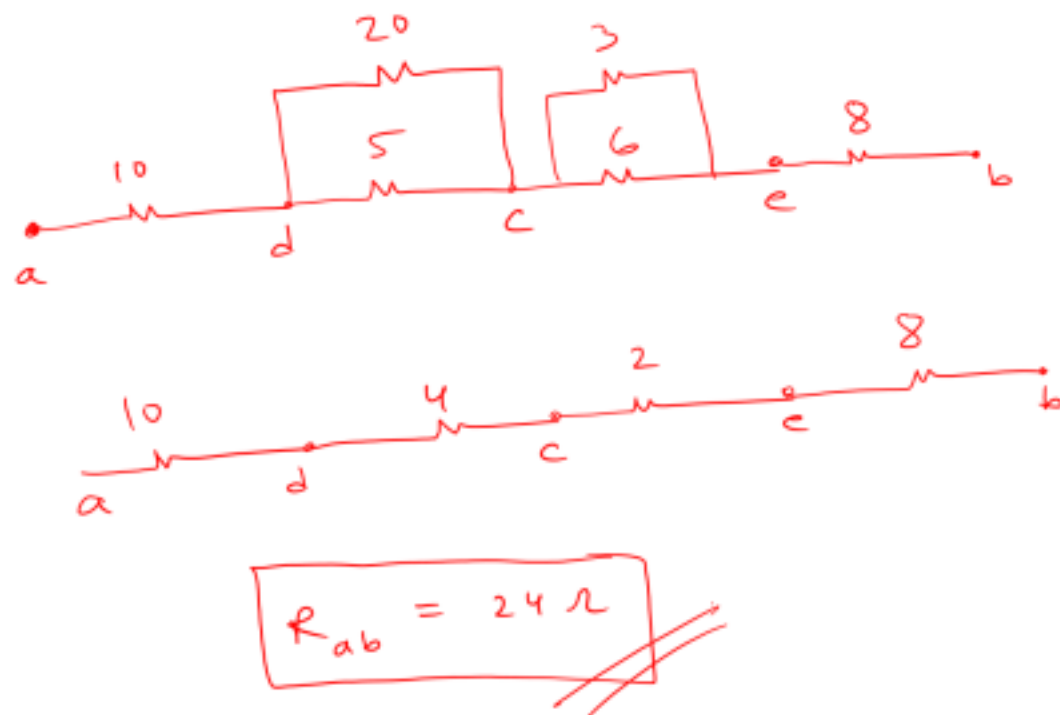
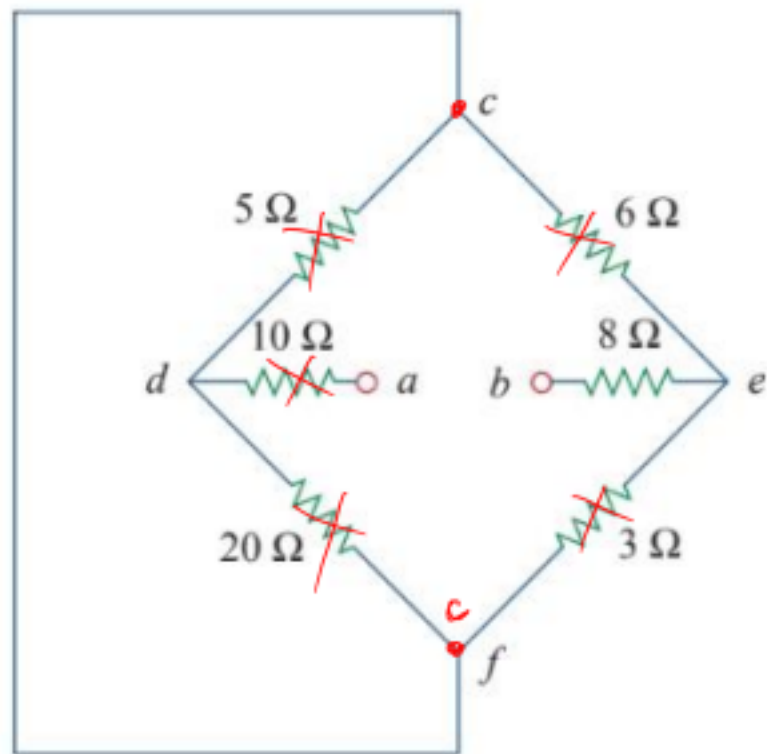
$$\Rightarrow iR \left(\frac{1}{3} + \frac{1}{6} + \frac{1}{3} \right) = V$$

$$\Rightarrow \left(\frac{V}{i} \right) = \frac{5}{6}R$$

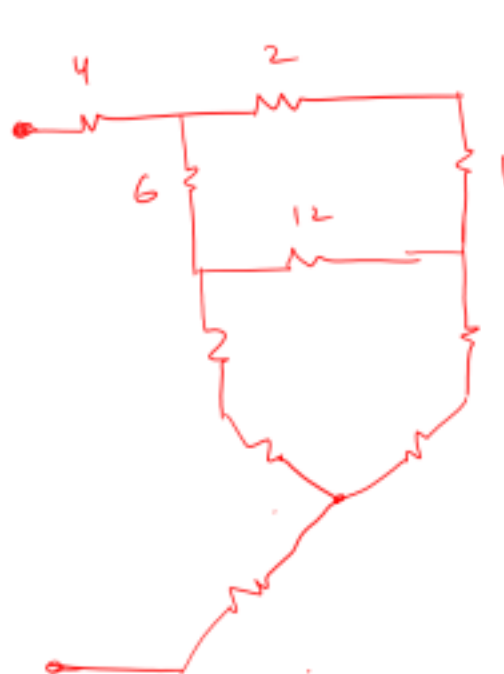
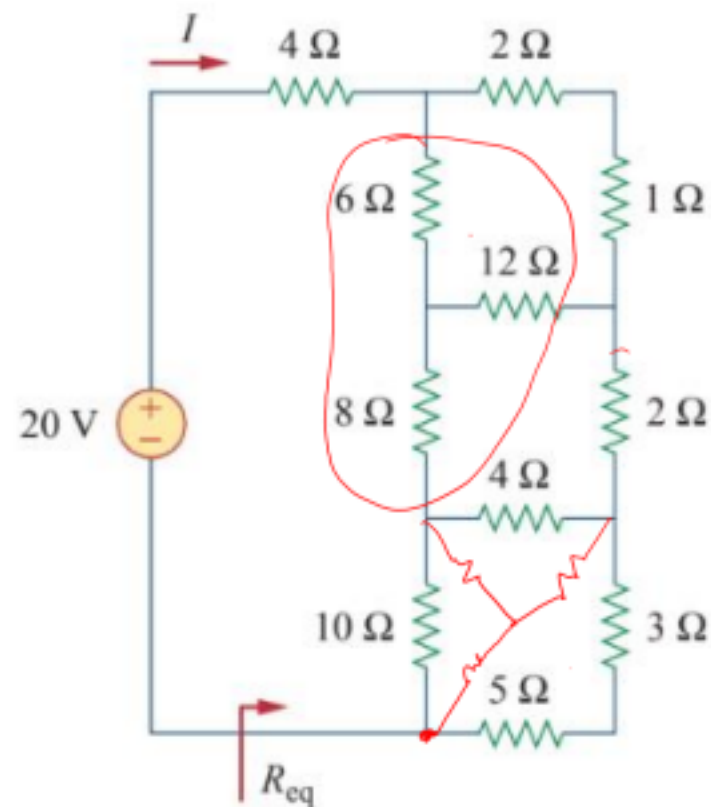
$$\Rightarrow \boxed{R_{AG} = \frac{5}{6}R}$$

DC Circuit

Find the equivalent resistance R_{ab}



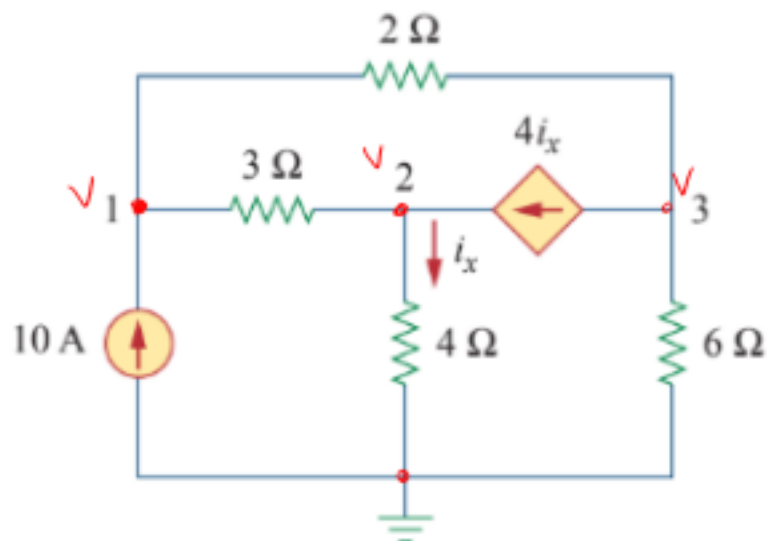
DC Circuit



Nodal Analysis

DC Circuit

Find the voltages at the three nonreference nodes in the circuit



Answer: $v_1 = 80 \text{ V}$, $v_2 = -64 \text{ V}$, $v_3 = 156 \text{ V}$.

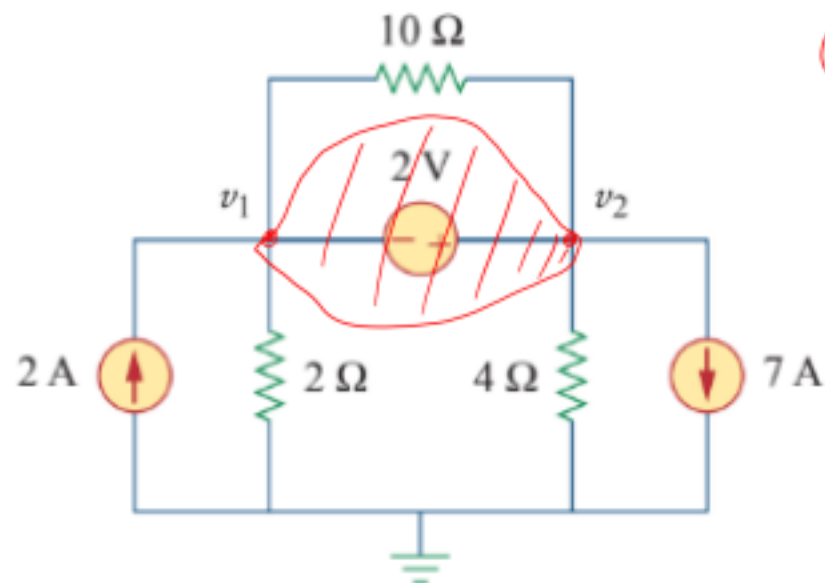
$$\textcircled{1} \Rightarrow 10 = \frac{v_1 - v_2}{3} + \frac{v_1 - v_3}{2} \quad \left| \quad i_x = \frac{v_2}{4} \right.$$

$$\textcircled{2} \Rightarrow \frac{v_2}{4} + \frac{v_2 - v_1}{3} = v_2$$

$$\textcircled{3} \Rightarrow v_2 + \frac{v_3}{6} + \frac{v_3 - v_1}{2} = 0$$

DC Circuit

Nodal Analysis with Voltage Sources (Supernode concept):



Find the node voltages

$$v_1 = -7.333 \text{ V}, v_2 = -5.333 \text{ V}.$$

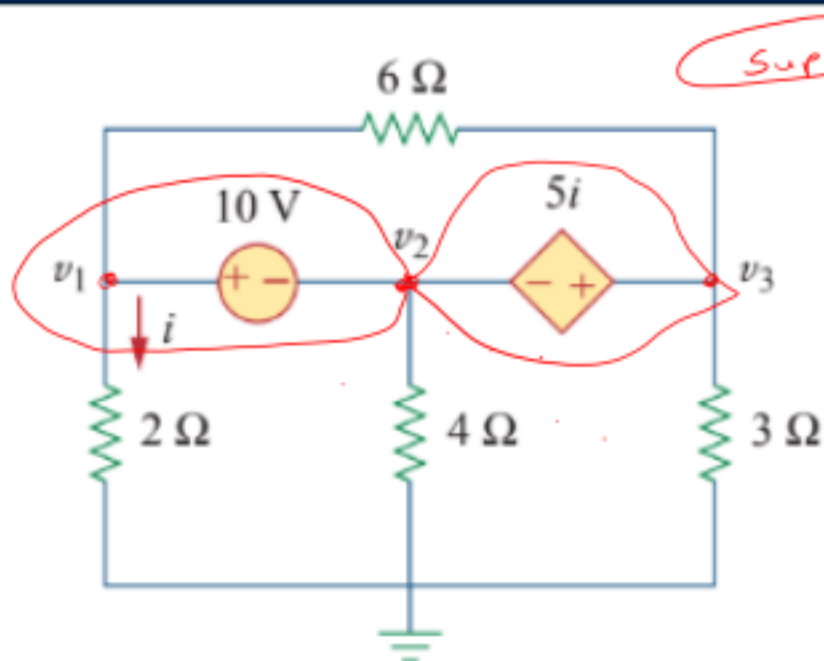
Supernode $\Rightarrow v_1 = v_2$

$$2 = \frac{v_1}{2} + \frac{v_2}{4} + 7$$



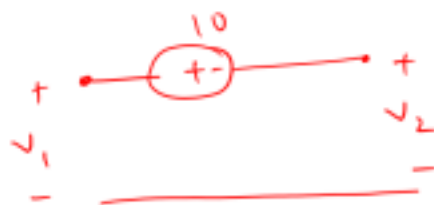
$$-v_1 - 2 + v_2 = 0$$

DC Circuit



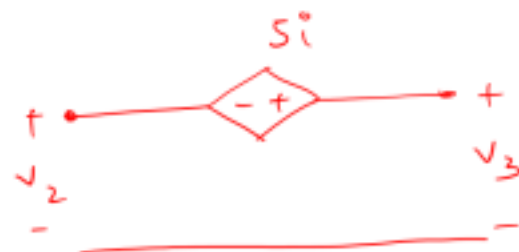
Supernode $\Rightarrow$
 $v_1 = v_2 = v_3$

$$\frac{v_1}{2} + \frac{v_2}{4} + \frac{v_3}{3} = 0$$



$$-v_1 + 10 + v_2 = 0$$

$$i = \frac{v_1}{2}$$

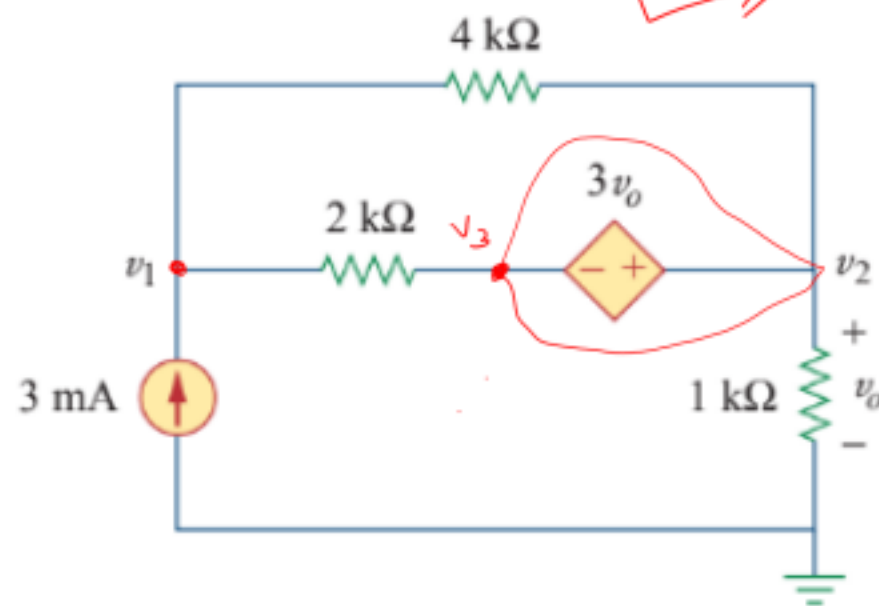


$$-v_2 - 5 \frac{v_1}{2} + v_3 = 0$$

Answer: $v_1 = 3.043 \text{ V}$, $v_2 = -6.956 \text{ V}$, $v_3 = 0.6522 \text{ V}$

DC Circuit

For the circuit in Fig. 3.70, find v_1 and v_2 using nodal analysis.



① $\Rightarrow$

$$3 \times 10^{-3} = \frac{v_1 - v_3}{2k} + \frac{v_1 - v_2}{4k}$$

Supernode $\Rightarrow v_2 = v_3$

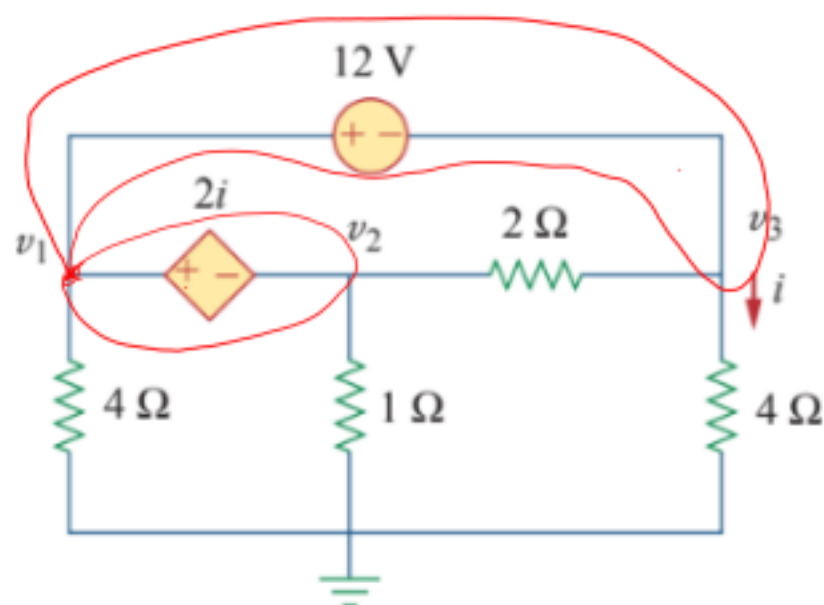
$$\frac{v_3 - v_1}{2k} + \frac{v_2}{1k} + \frac{v_2 - v_1}{4k} = 0$$



$$-v_3 - 3v_2 + v_2 = 0$$

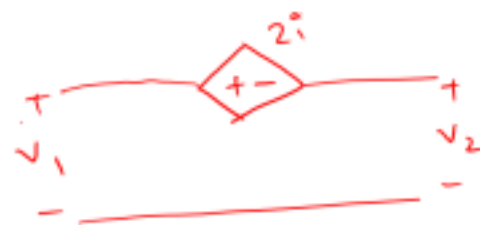
DC Circuit

For the circuit in Fig. 3.69, find v_1 , v_2 , and v_3 using nodal analysis.



$v_1 = v_2 = v_3$
supernode $\Rightarrow$

$$\frac{v_1}{4} + \frac{v_2}{1} + \frac{v_3}{4} = 0 \quad \left| \quad i = \frac{v_3}{4} \right.$$



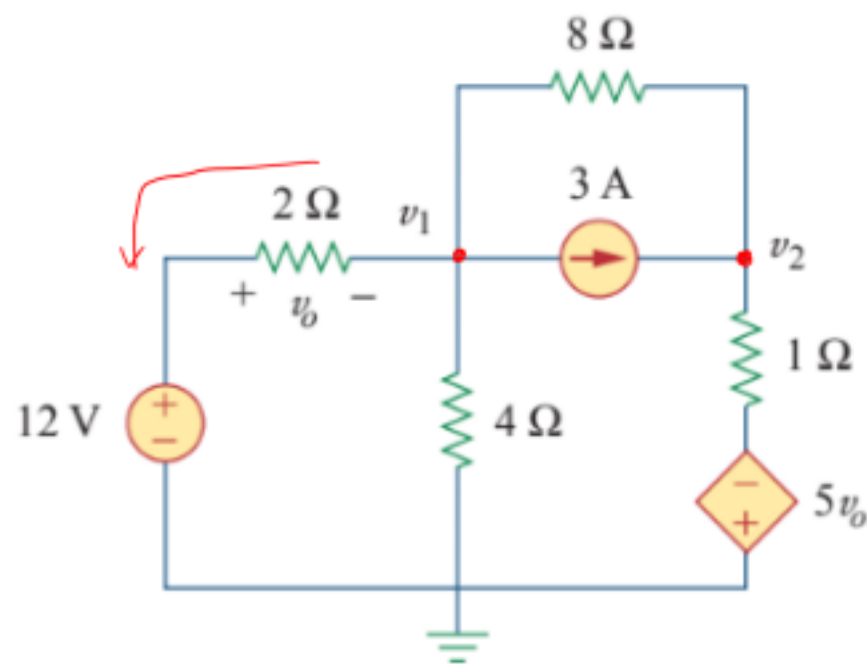
$$-v_1 + 2 \cdot \frac{v_3}{4} + v_2 = 0$$



$$-v_1 + 12 + v_3 = 0$$

DC Circuit

Determine V_1 and V_2 in the circuit

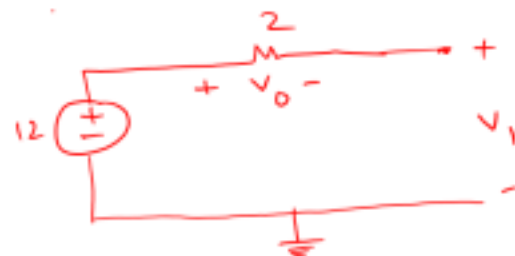


① $\Rightarrow$

$$3 + \frac{v_1}{4} + \frac{v_1 - 12}{2} + \frac{v_1 - v_2}{8} = 0$$

② $\Rightarrow$

$$3 = \frac{v_2 - v_1}{8} + \frac{v_2 - (-5v_o)}{1}$$



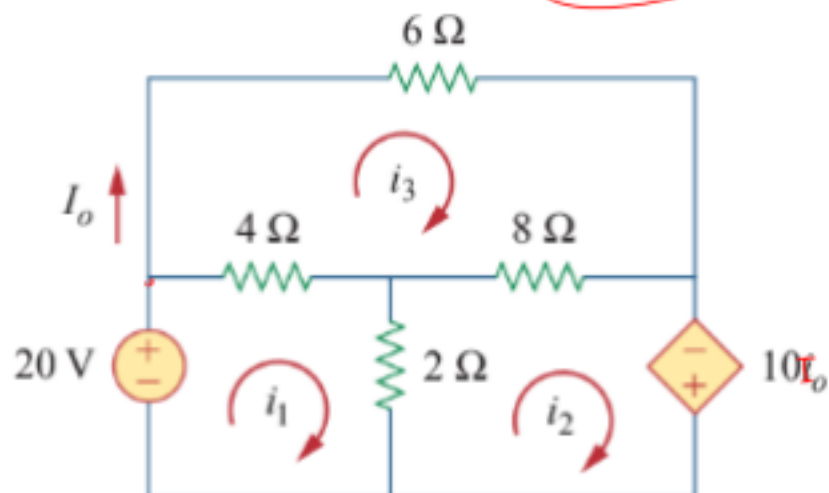
$$-12 + v_o + v_1 = 0$$

DC Circuit

Mesh Analysis:

Suggest

$$i_3 = I_o$$



Using mesh analysis, find I_o

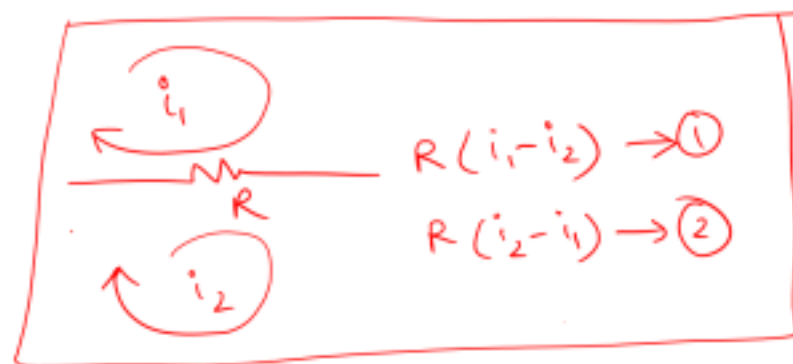
Answer: -5 A .

$$\textcircled{1} \Rightarrow 4(i_1 - i_3) + 2(i_1 - i_2) - 20 = 0$$

$$\textcircled{2} \Rightarrow 8(i_2 - i_3) - 10i_3 + 2(i_2 - i_1) = 0$$

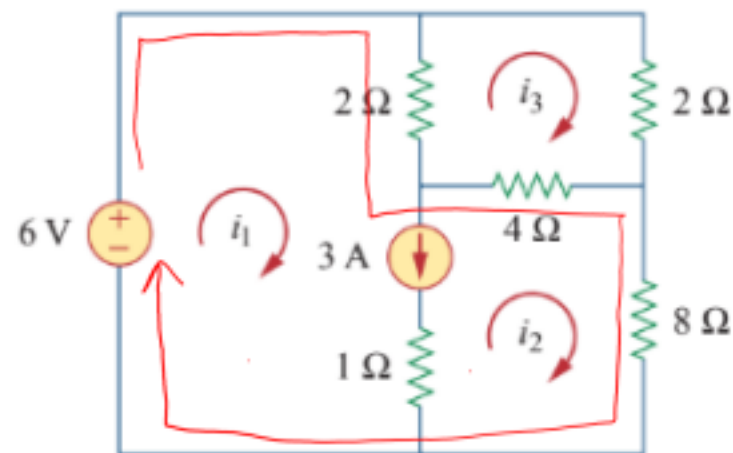
$$\textcircled{3} \Rightarrow 6i_3 + 8(i_3 - i_2) + 4(i_3 - i_1) = 0$$

$$I_o = ?$$



DC Circuit

Mesh Analysis with Current Sources (Supermesh concept):



Supermesh $\Rightarrow 2(i_1 - i_3) + 4(i_2 - i_3) + 8i_2 - 6 = 0$

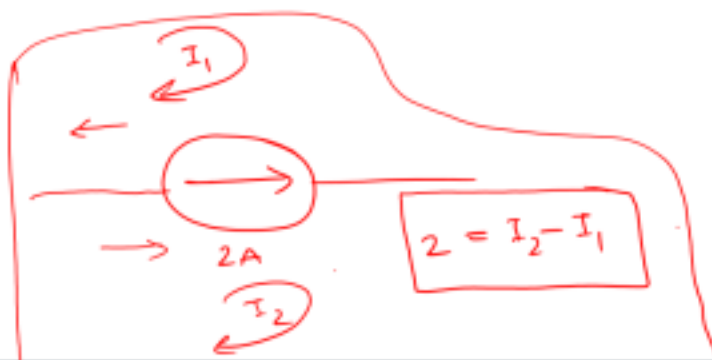
③ $\Rightarrow 2i_3 + 4(i_3 - i_2) + 2(i_3 - i_1) = 0$



$3 = i_1 - i_2$

Use mesh analysis to determine i_1 , i_2 , and i_3 in Fig.

Answer: $i_1 = 3.474 \text{ A}$, $i_2 = 0.4737 \text{ A}$, $i_3 = 1.1052 \text{ A}$



DC Circuit

Draw equivalent circuit for the following mesh equation.

$$25I_1 - 5I_2 - 15I_3 = 24$$

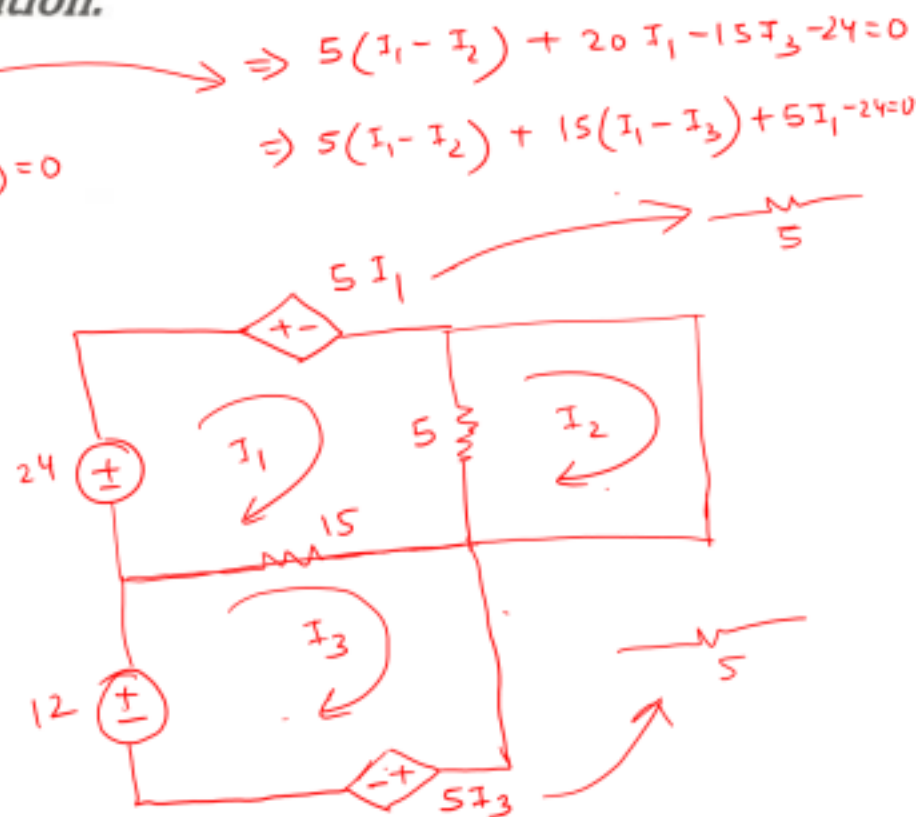
$$5I_1 - 5I_2 = 0 \Rightarrow 5(I_2 - I_1) = 0$$

$$15I_1 - 20I_3 = -12$$

$$\Rightarrow 15I_1 - 20I_3 + 12 = 0$$

$$\Rightarrow 15(I_1 - I_3) - 5I_3 + 12 = 0$$

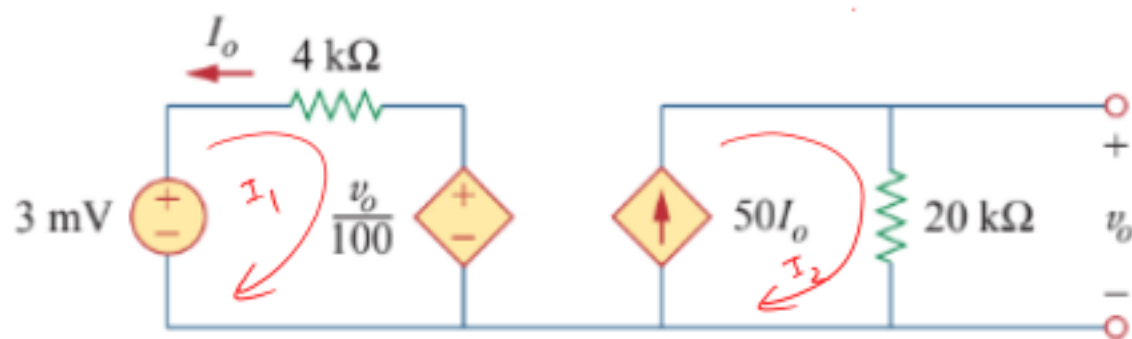
$$\Rightarrow 15(I_3 - I_1) + 5I_3 - 12 = 0$$



DC Circuit

Calculate v_o and I_o in the circuit

$$I_o = -I_1$$



$$4000 I_1 + \frac{v_o}{100} - 3 \times 10^{-3} = 0$$

$$I_2 = 50 I_o$$

$$v_o = 20 \times 10^3 (50 I_o)$$

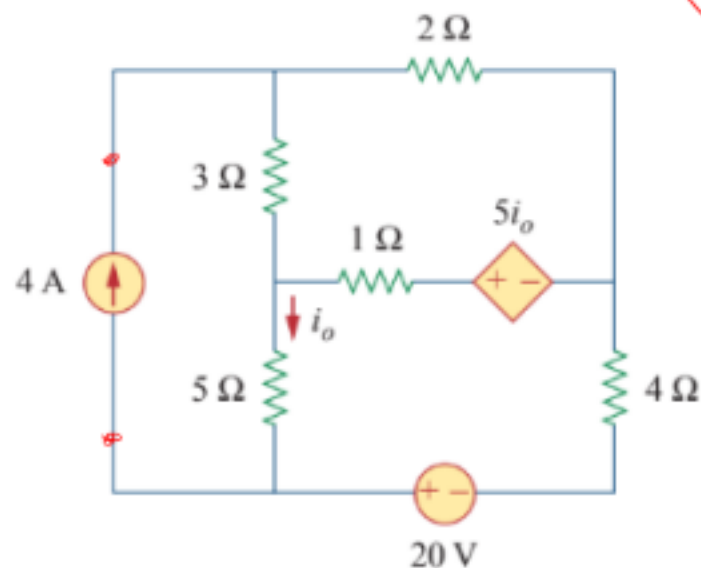
$$v_o = ?$$

$$I_o = ?$$

DC Circuit

Superposition:

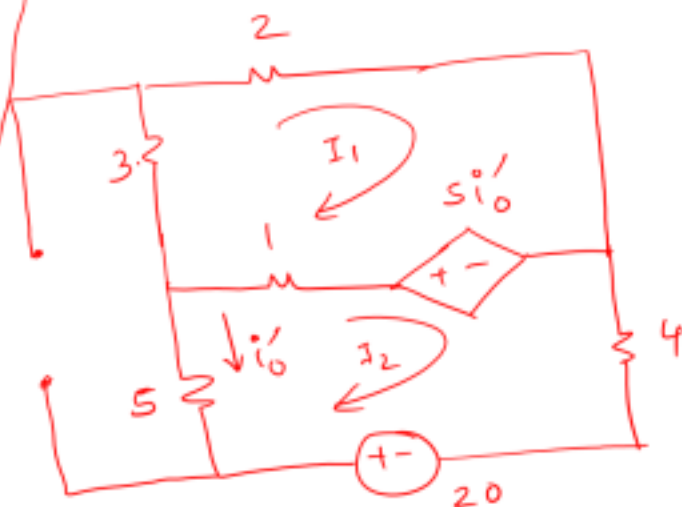
Find i_o in the circuit



-0.4706 A

1 $\rightarrow$ independent $\rightarrow$ ON
voltage $\rightarrow$ shunt
c.s. $\rightarrow$ open

$$i_o' = -I_2$$

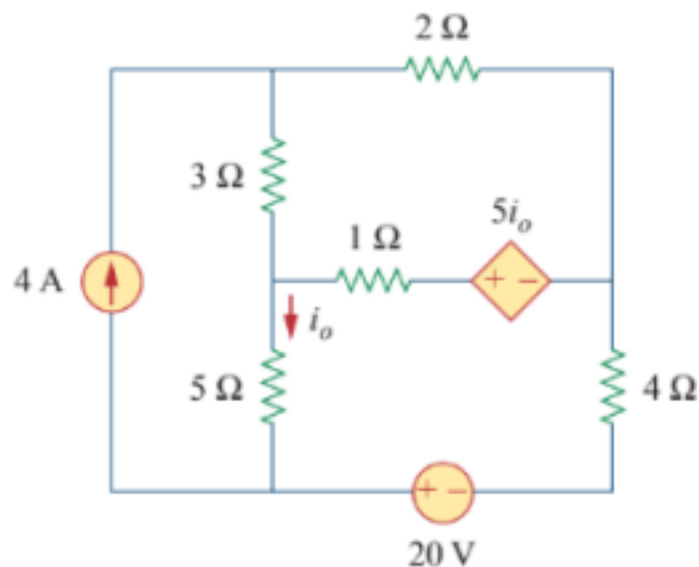


$$\begin{cases} \textcircled{1} \Rightarrow 2I_1 - 5i_o' + 1(I_1 - I_2) + 3I_1 = 0 \\ \textcircled{2} \Rightarrow 1(I_2 - I_1) + 5i_o' + 4I_2 - 20 + 5I_2 = 0 \end{cases} \quad \left. \vphantom{\begin{matrix} \textcircled{1} \\ \textcircled{2} \end{matrix}} \right\} i_o'$$

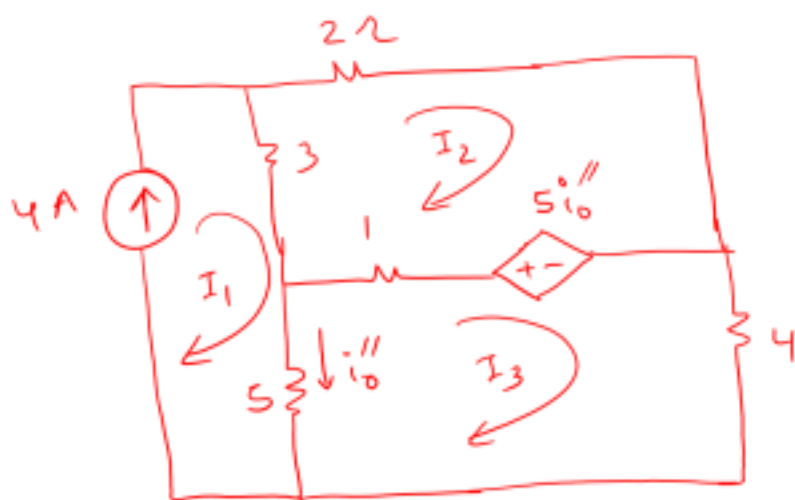
DC Circuit

Superposition:

Find i_o in the circuit



-0.4706 A



$$I_1 = 4$$

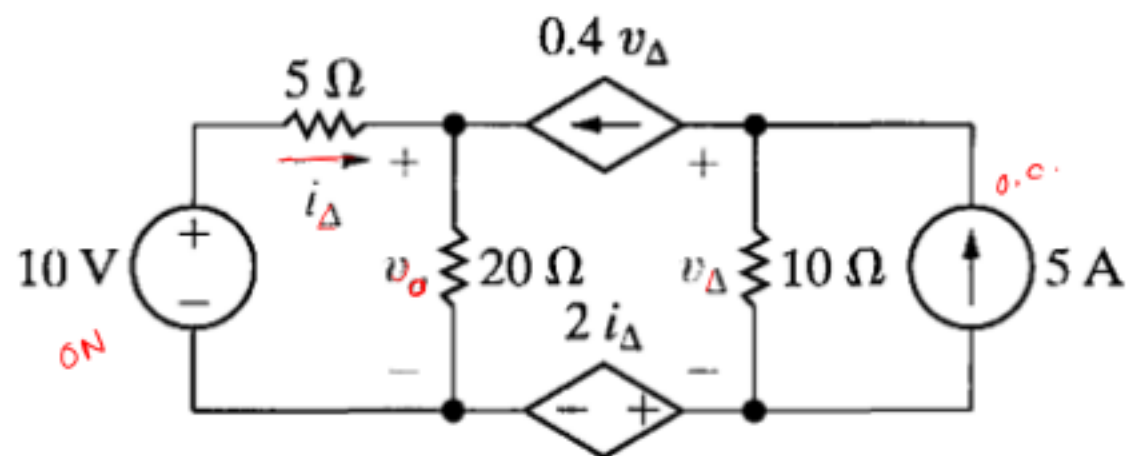
$$i_o'' = I_1 - I_3$$

$$\begin{aligned} \textcircled{2} \Rightarrow 2I_2 - 5i_o'' + 1(I_2 - I_3) + 3(I_2 - I_1) &= 0 \\ \textcircled{3} \Rightarrow 1(I_3 - I_2) + 5i_o'' + 4I_3 + 5(I_3 - I_1) &= 0 \end{aligned} \quad \left. \vphantom{\begin{aligned} \textcircled{2} \Rightarrow \\ \textcircled{3} \Rightarrow \end{aligned}} \right\} i_o''$$

$$i_o = i_o' + i_o'' = \boxed{?}$$

DC Circuit

Use the principle of superposition to find v_o

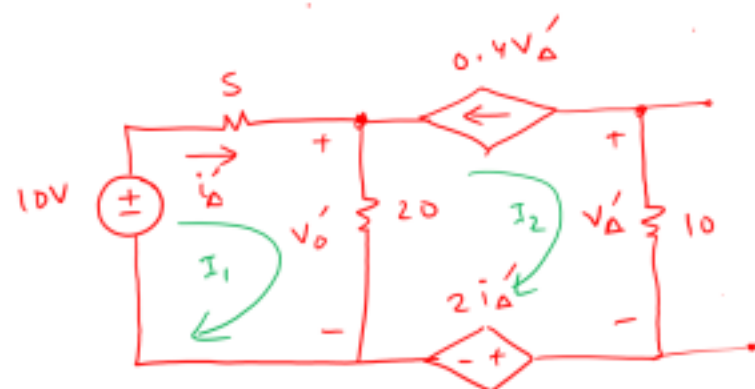


$$I_2 = -0.4v_\Delta'$$

$$v_\Delta' = 10 I_2 = 10(-0.4v_\Delta')$$

$$\Rightarrow v_\Delta' = -4v_\Delta'$$

$$\Rightarrow 5v_\Delta' = 0 \Rightarrow v_\Delta' = 0 \quad \boxed{I_2 = 0}$$



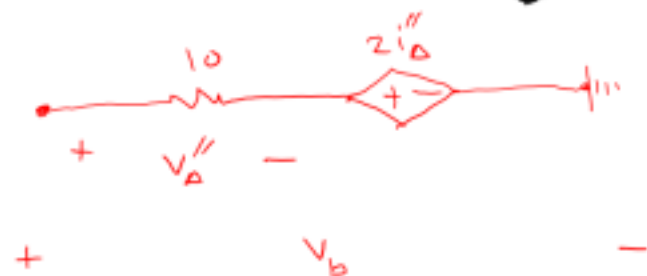
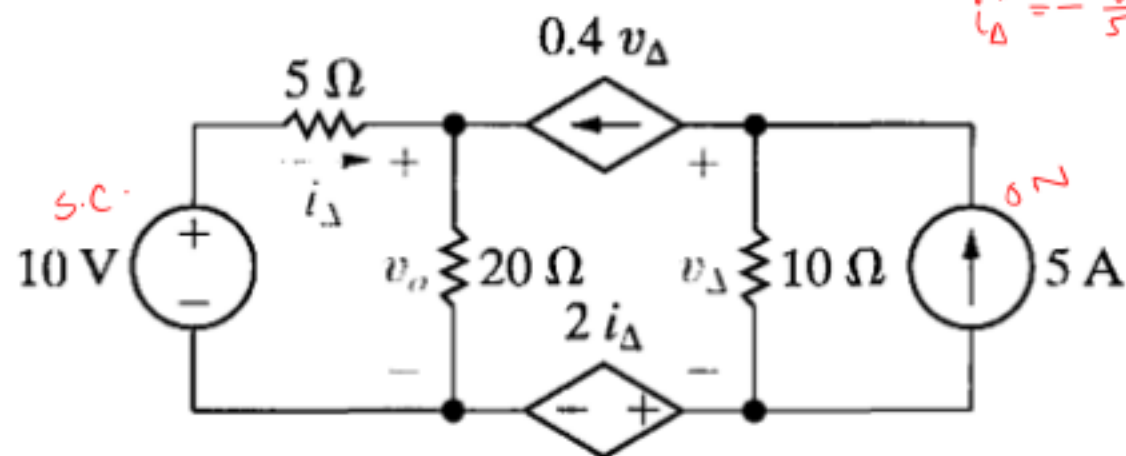
$$5i_\Delta' + 20i_\Delta' = 10 + 20I_2$$

$$\Rightarrow i_\Delta' = \frac{10}{25} = \frac{2}{5}$$

$$v_o' = 20i_\Delta' = \boxed{8V}$$

DC Circuit

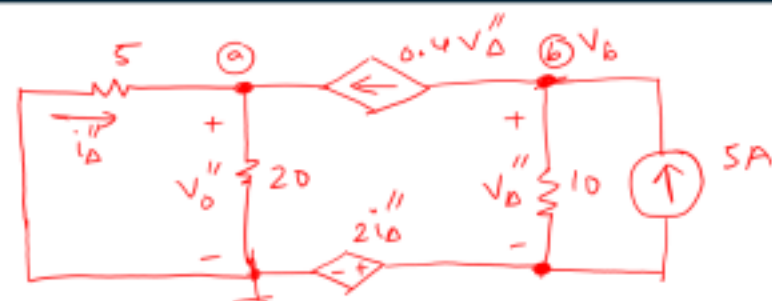
Use the principle of superposition to find v_o



$$v_{\Delta}'' + 2i_{\Delta}'' = v_b$$

$$v_b - 2i_{\Delta}'' = v_{\Delta}''$$

$$i_{\Delta}'' = -\frac{v_b''}{5}$$



$$\textcircled{a} \Rightarrow \frac{v_o''}{20} = i_{\Delta}'' + 0.4v_{\Delta}''$$

$$\textcircled{b} \Rightarrow 5 = 0.4v_{\Delta}'' + \frac{v_b - 2i_{\Delta}''}{10}$$

$$\Rightarrow 5 = 0.4v_{\Delta}'' + 0.1v_{\Delta}''$$

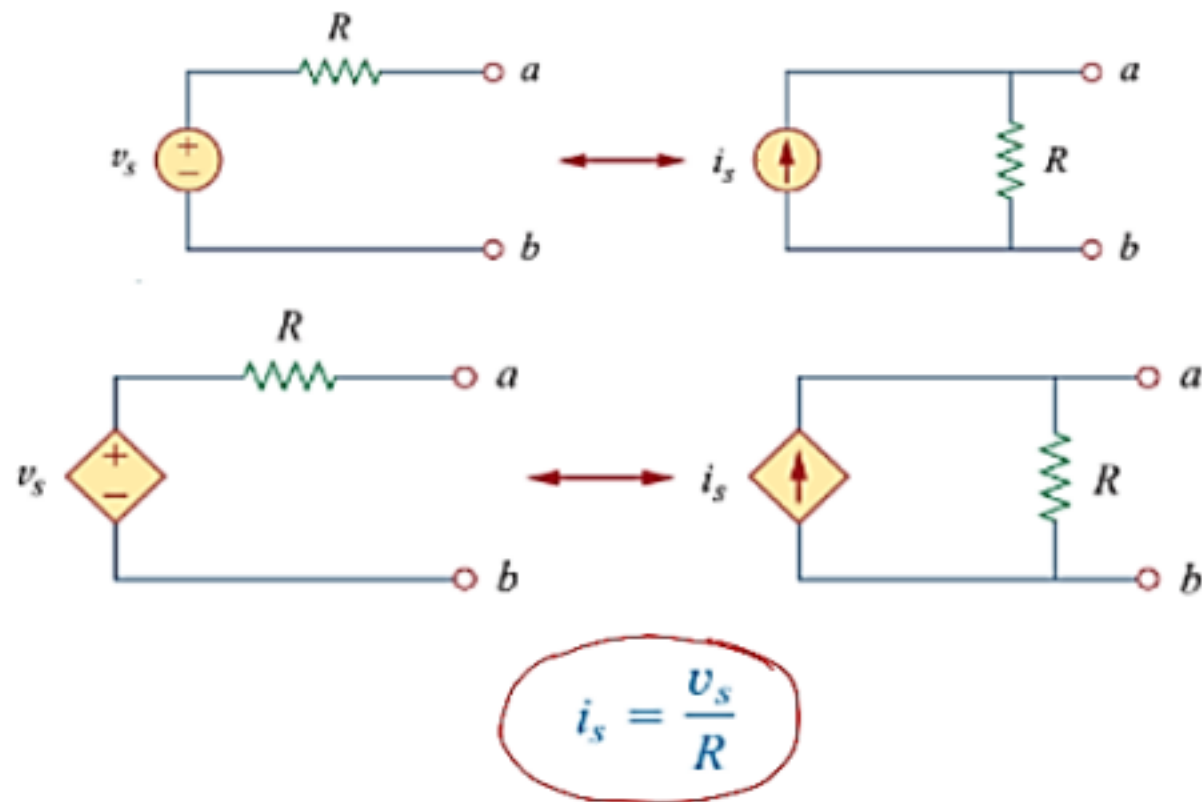
$$\Rightarrow v_{\Delta}'' = 10 \text{ V}$$

$$\textcircled{a} \Rightarrow \frac{v_o''}{20} + \frac{v_o''}{5} = 4 \quad \left\{ \begin{array}{l} v_o = v_o' + v_o'' \\ = 8 + 16 \\ = 24 \text{ V} \end{array} \right.$$

$$\Rightarrow v_o'' = 16 \text{ V}$$

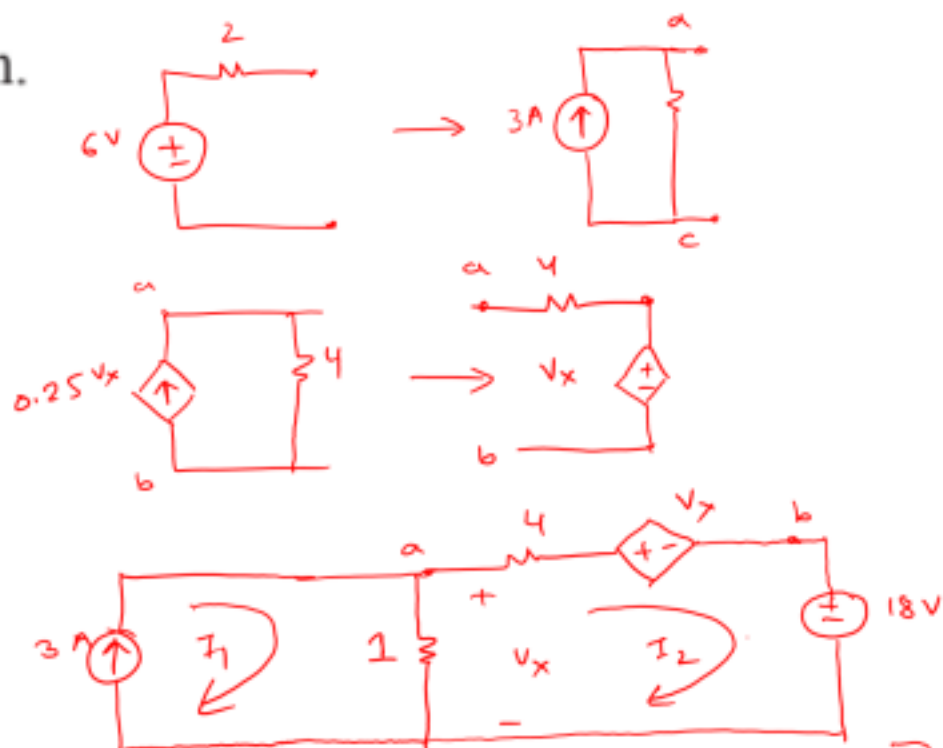
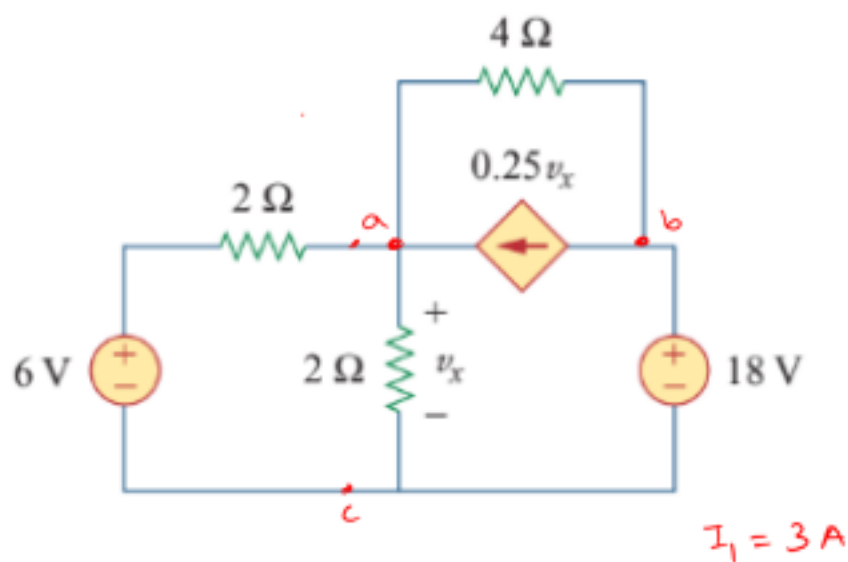
DC Circuit

Source Transformation:



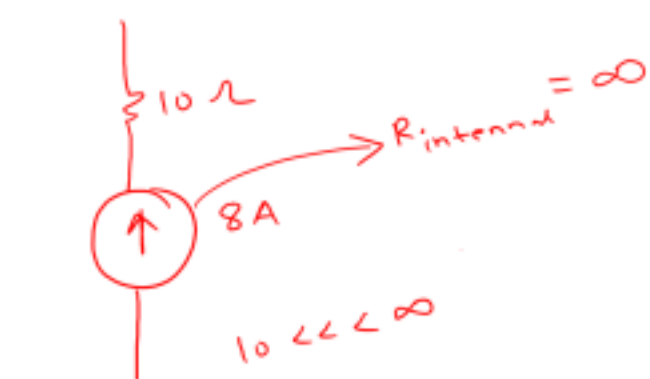
DC Circuit

Find v_x using source transformation.

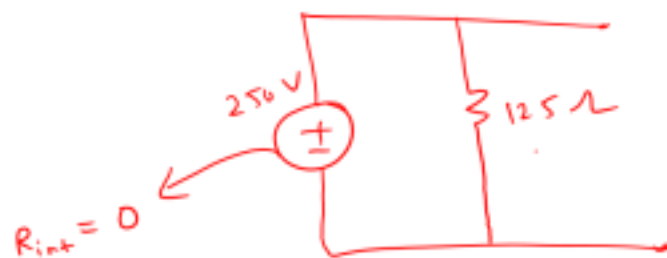


$$\left. \begin{aligned} 4I_2 + v_x + 18 + 1(I_2 - 3) &= 0 \\ v_x &= 1(3 - I_2) \end{aligned} \right\} \boxed{v_x = ?}$$

DC Circuit



(ignore 10 Ω)



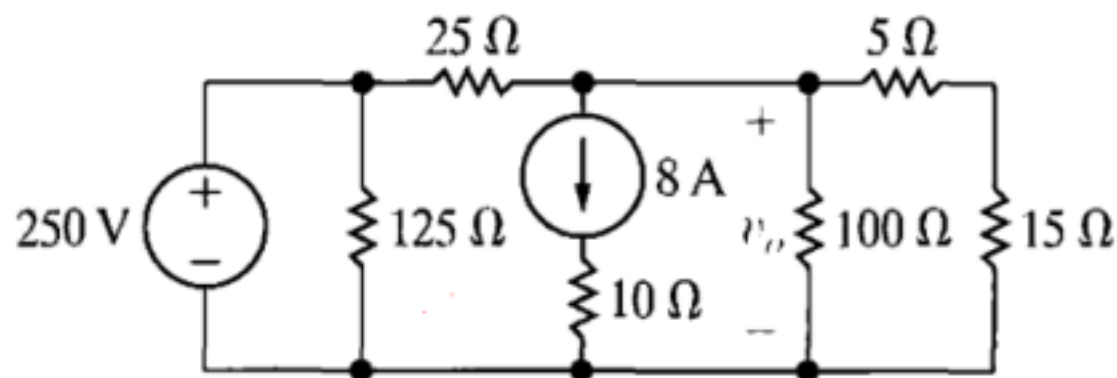
$$(0^{-1} + 125^{-1})^{-1} = 0$$



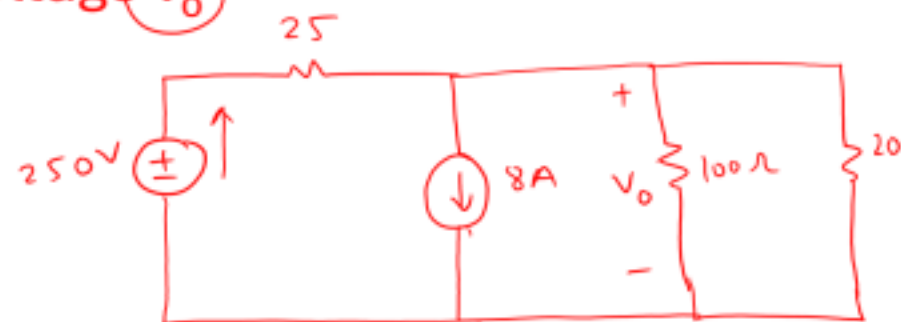
(ignore 125 Ω)

DC Circuit

a) Use source transformations to find the voltage v_o



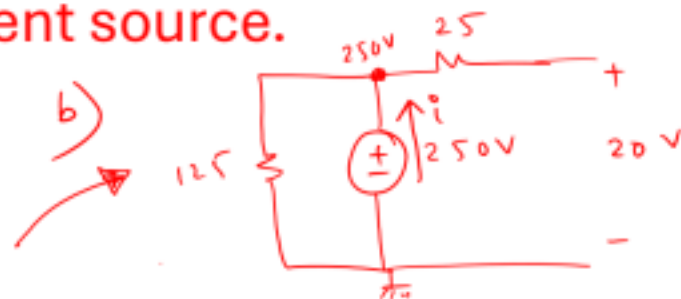
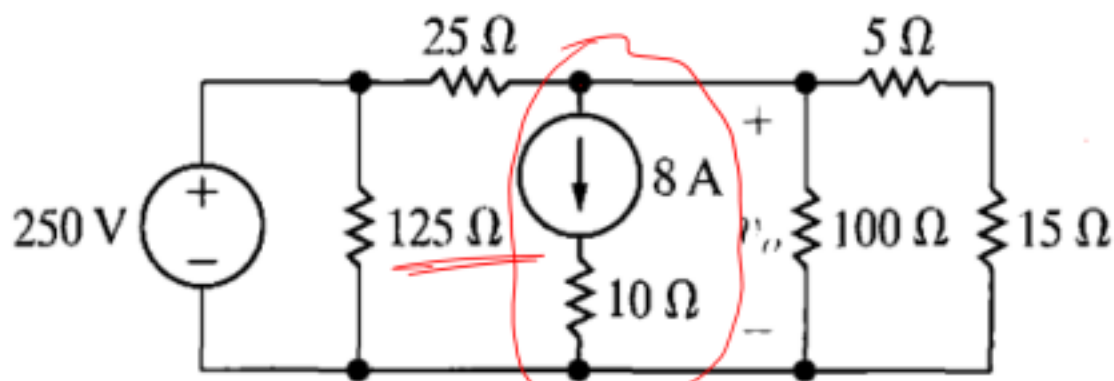
$$v_o = 10 \times 2 = \boxed{20 \text{ V}}$$



DC Circuit

b) Find the power developed by the 250 V voltage source.

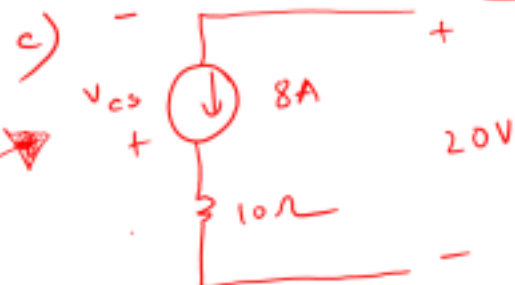
c) Find the power developed by the 8 A current source.



$$\frac{250 - 20}{25} + \frac{250}{125} = i$$

$$\Rightarrow i = 11.2 \text{ A}$$

$$P_{250V} = 250 \times 11.2 = 2800 \text{ W supply}$$

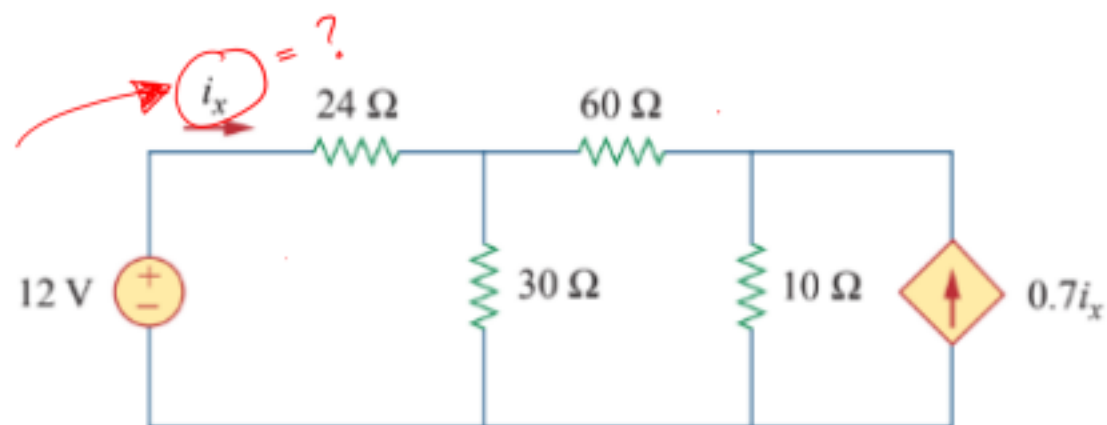


$$-20 - V_{cs} + 8 \times 10 = 0$$

$$\Rightarrow V_{cs} = 60 \text{ V}$$

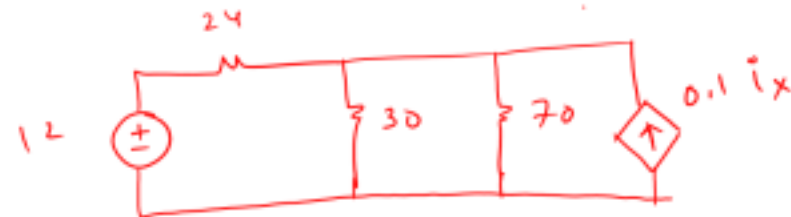
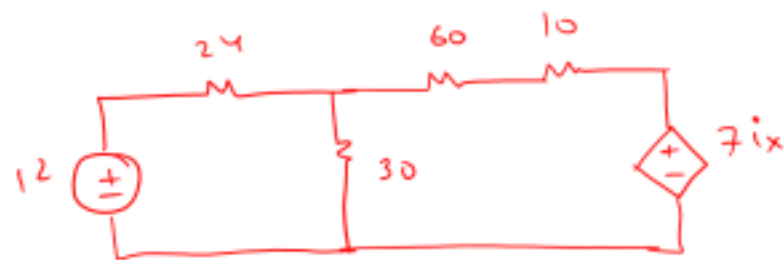
$$P_{8A} = (60) 8 = 480 \text{ W supply}$$

DC Circuit



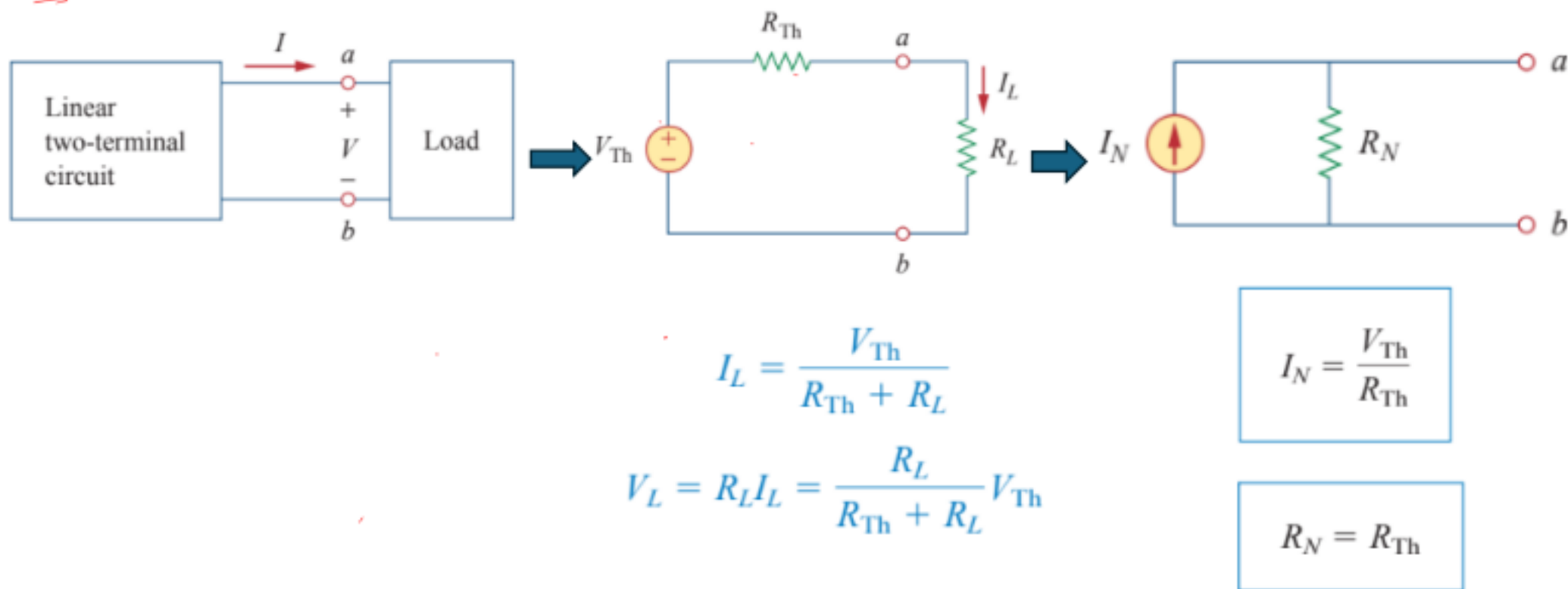
$$i_x = \frac{12 - 2.1 i_x}{45}$$

$$\Rightarrow i_x = ?$$



DC Circuit

Thevenin's & Norton's Theorem:

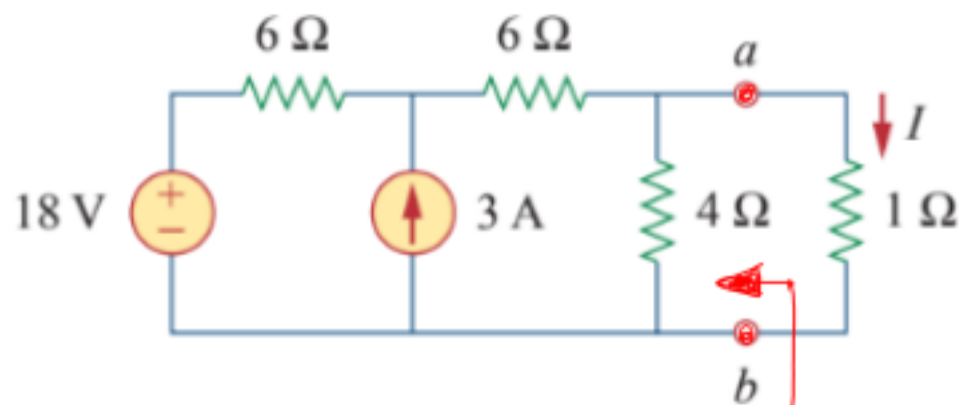


DC Circuit

	V_{Th}	R_{Th}
All dependent src	0	 $R_{Th} = \frac{1}{i}$
All independent src	 Nodal <u>Mesh</u> S.P. S.T.	v.s. $\rightarrow$ short c.s. $\rightarrow$ open  $R_{Th} = \frac{1}{i}$
Ind. + dep src		

DC Circuit

Using Thevenin's theorem, find the equivalent circuit to the left of the terminals in the circuit. Then find I .



$$V_{Th} = 9 \text{ V}, R_{Th} = 3 \Omega, I = 2.25 \text{ A.}$$



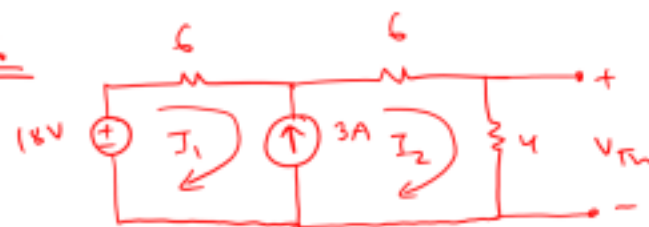
$$I = \frac{V_{Th}}{3 + 1}$$

R_{Th} :



$$R_{Th} = (4^{-1} + 12^{-1})^{-1} = 3 \Omega$$

V_{Th} :



$$I_2 - I_1 = 3$$

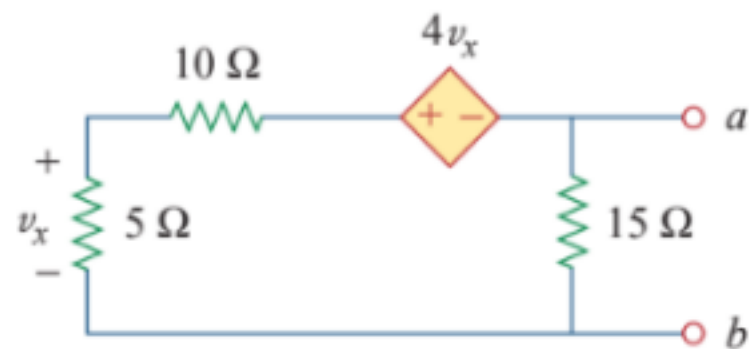
$$-18 + 6I_1 + 6I_2 + 4I_2 = 0$$

$$I_2 = ?$$

$$V_{Th} = 4I_2 = ?$$

DC Circuit

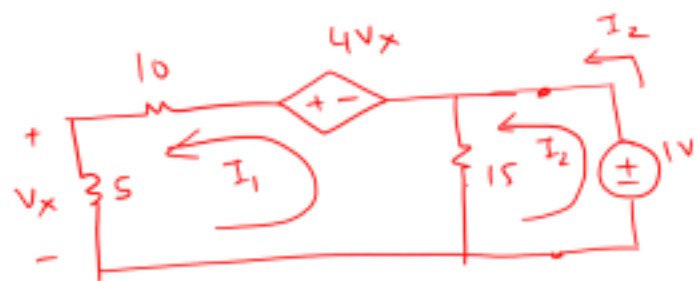
Obtain the Thevenin equivalent of the circuit



$V_{Th} = 0\text{ V}$, $R_{Th} = \boxed{-7.5\ \Omega}$

→ supply

$V_x = 5I_1$



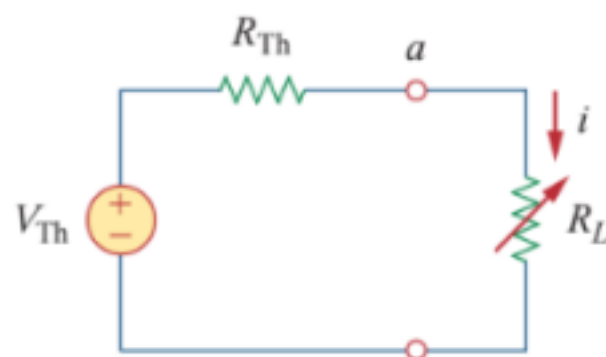
$I_2 = ?$

$$\begin{cases} -4v_x + 10I_1 + 5I_1 + 15(I_1 - I_2) = 0 \\ -1 + 15(I_2 - I_1) = 0 \end{cases}$$

$R_{Th} = \frac{1}{I_2} = \boxed{?}$

DC Circuit

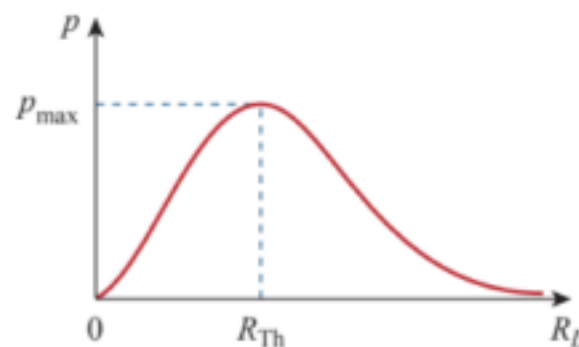
Maximum Power Transfer:



power delivered to the load, $p = i^2 R_L = \left(\frac{V_{Th}}{R_{Th} + R_L} \right)^2 R_L$

$$\begin{aligned} \frac{dp}{dR_L} &= V_{Th}^2 \left[\frac{(R_{Th} + R_L)^2 - 2R_L(R_{Th} + R_L)}{(R_{Th} + R_L)^4} \right] \\ &= V_{Th}^2 \left[\frac{(R_{Th} + R_L - 2R_L)}{(R_{Th} + R_L)^3} \right] = 0 \rightarrow \text{for } p_{\max} \end{aligned}$$

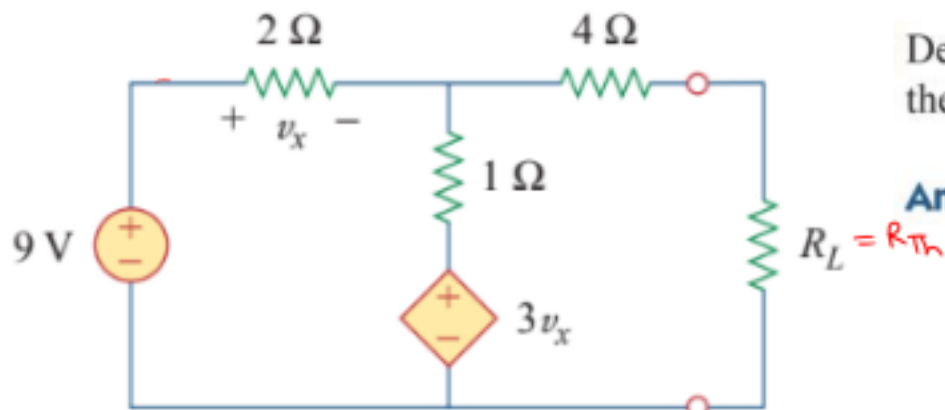
$$0 = (R_{Th} + R_L - 2R_L) = (R_{Th} - R_L)$$



$$R_L = R_{Th}$$

$$p_{\max} = \frac{V_{Th}^2}{4R_{Th}}$$

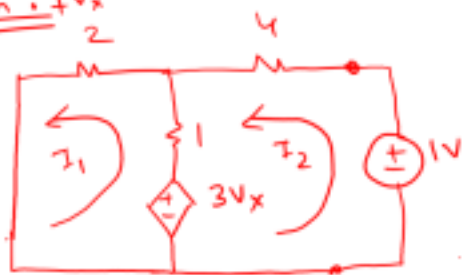
DC Circuit



Determine the value of R_L that will draw the maximum power from the rest of the circuit in Fig. 4.52. Calculate the maximum power.

Answer: 4.22 Ω , 2.901 W.

R_{Th} : $+v_x -$
2



$$-v_x = 2I_1$$

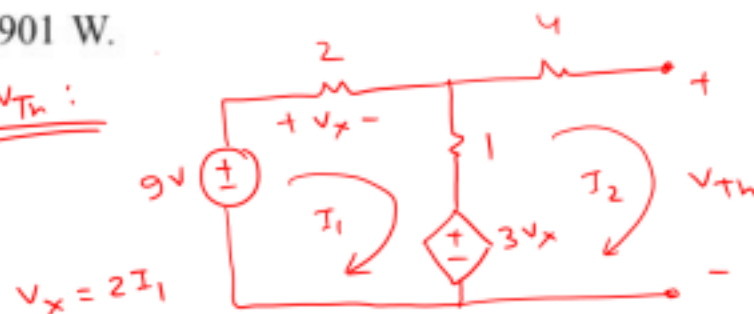
$$2I_1 - 3v_x + 1(I_1 - I_2) = 0$$

$$4I_2 + 1(I_2 - I_1) + 3v_x - 1 = 0$$

$$I_2 = ?$$

$$R_{Th} = \frac{1}{I_2} = \boxed{\infty} = \boxed{R_L}$$

V_{Th} :



$$v_x = 2I_1$$

$$-9 + v_x + I_1 + 3v_x = 0$$

$$\Rightarrow -9 + 2I_1 + I_1 + 6I_1 = 0$$

$$\Rightarrow I_1 = 1$$

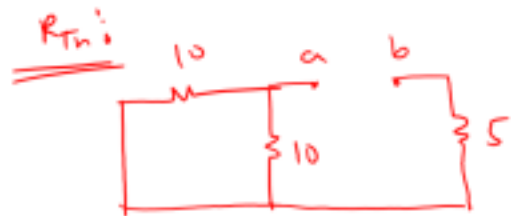
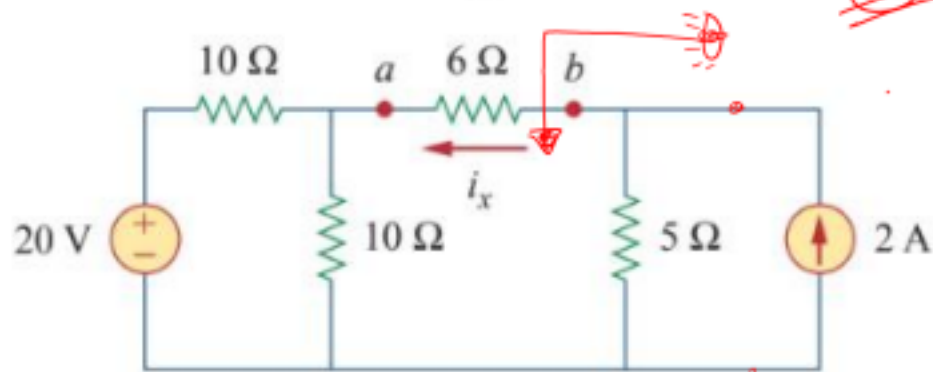
$$0 = V_{Th} - 3(2I_1) + 1(0 - 1)$$

$$\Rightarrow \boxed{V_{Th} = 7V}$$

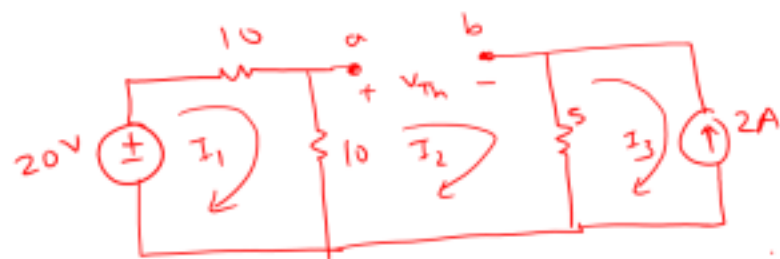
$$\left. \begin{array}{l} P_{max} = \frac{7^2}{4R_{Th}} \\ = \boxed{?} \end{array} \right\}$$

DC Circuit

Find the Thevenin equivalent looking into terminals a - b of the circuit in Fig. 4.110 and solve for i_x .



$$R_{Th} = (10^{-1} + 10^{-1})^{-1} + 5 = 10\Omega$$



$$I_2 = 0$$
$$I_3 = -2$$

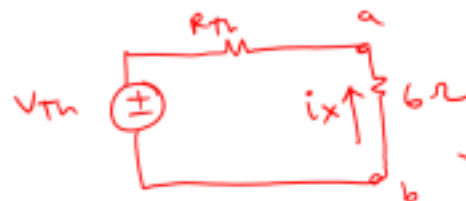
V_{Th} :

$$10I_1 + 10I_1 = 20$$

$$\Rightarrow I_1 = 1A$$

$$V_{Th} + 5(0+2) + 10(0-1) = 0$$

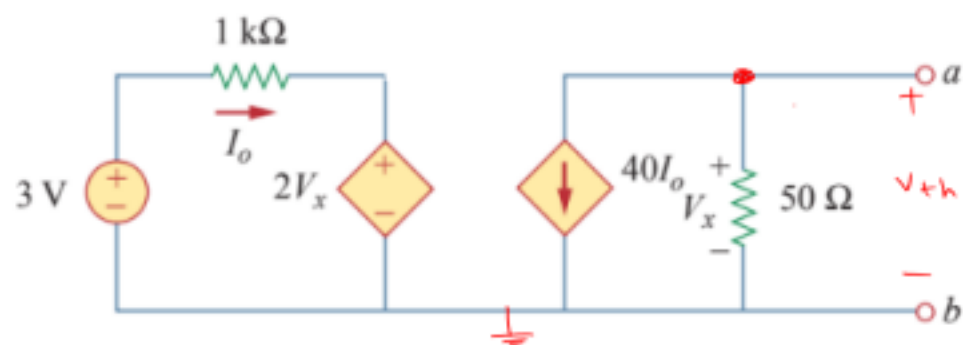
$$\Rightarrow V_{Th} = 0V$$



$i_x = 0A$

DC Circuit

Find the Thevenin equivalent between terminals $a-b$



$$\underline{V_{th}}$$

$$-3 + 1000 I_o + 2 V_{th} = 0$$

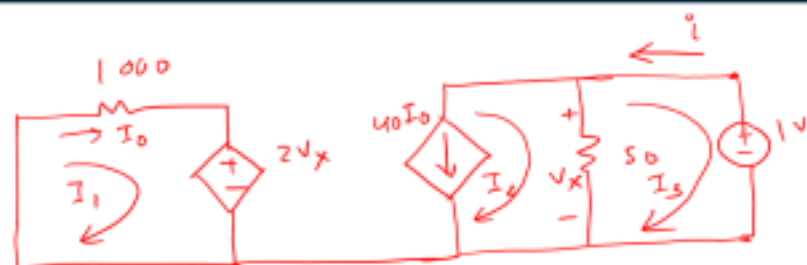
$$40 I_o + \frac{V_{th}}{50} = 0$$

$$V_{th} = ?$$

R_{th} :

$$I_o = I_1$$

$$I_3 = -i$$



$$1000 I_o + 2 V_x = 0$$

$$1 + 50 (I_3 - I_2) = 0$$

$$I_2 = -40 I_o$$

$$V_x = 50 (I_2 - I_3)$$

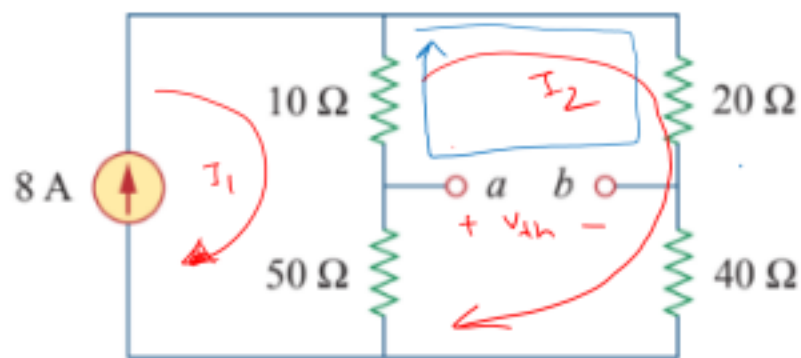
$$I_3 = ?$$

$$i = ?$$

$$R_{th} = \frac{1}{i}$$

DC Circuit

Determine the Thevenin and Norton equivalents at terminals $a-b$ of the circuit in Fig. 4.125.



$$I_1 = 8 \text{ A}$$

$$20 I_2 + 40 I_2 + 60 (I_2 - I_1) = 0$$

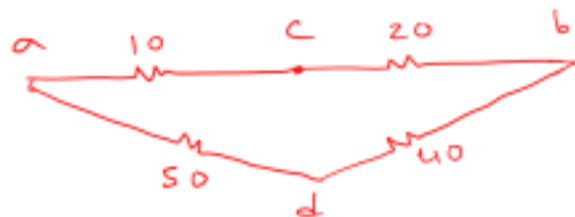
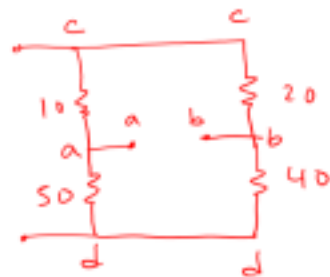
$$\Rightarrow 120 I_2 = 60 \times 8$$

$$I_2 = 4 \text{ A}$$

$$20 I_2 - V_{th} + 10 (I_2 - I_1) = 0$$

$$\Rightarrow V_{th} = 80 - 40 = 40 \text{ V}$$

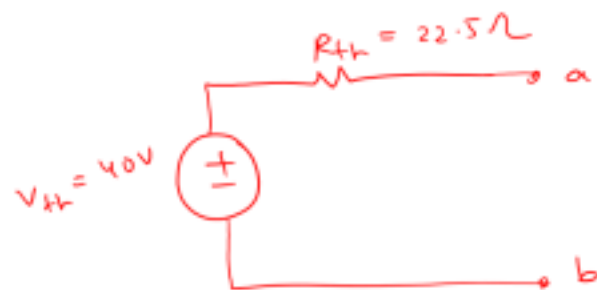
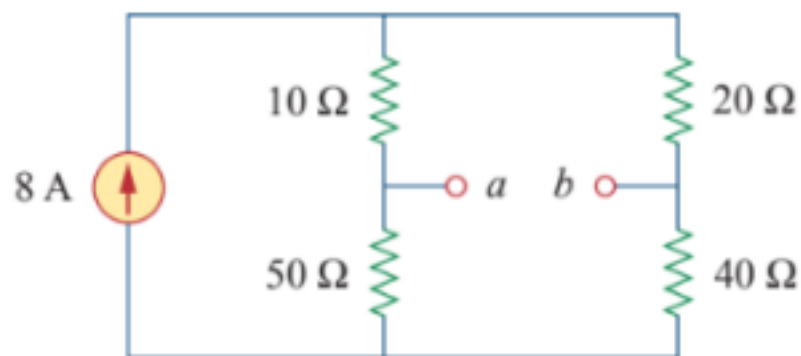
R_{th} :



$$R_{th} = (30' + 90')^{-1} = 22.5 \Omega$$

DC Circuit

Determine the Thevenin and Norton equivalents at terminals $a-b$ of the circuit in Fig. 4.125.

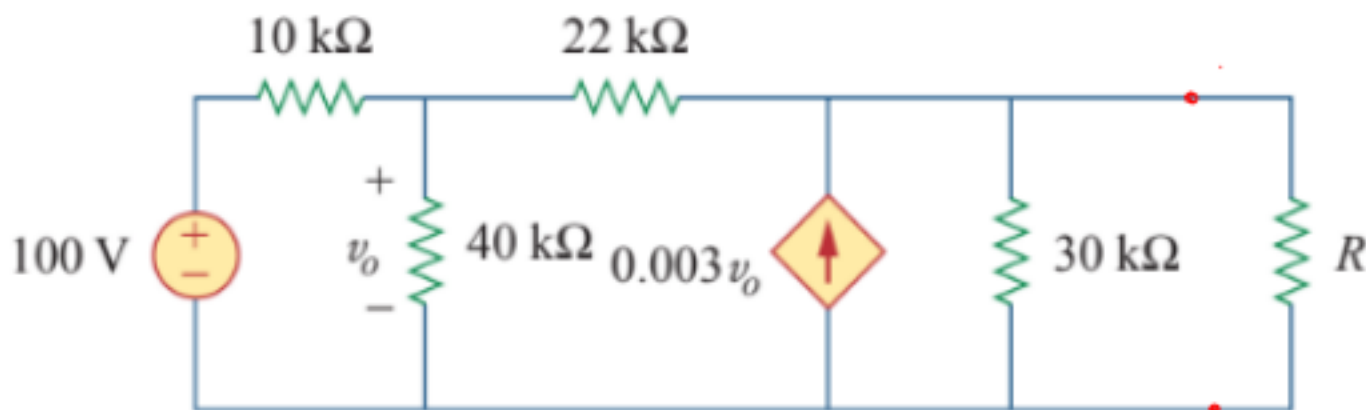


S.T.



DC Circuit

Find the maximum power transferred to resistor R

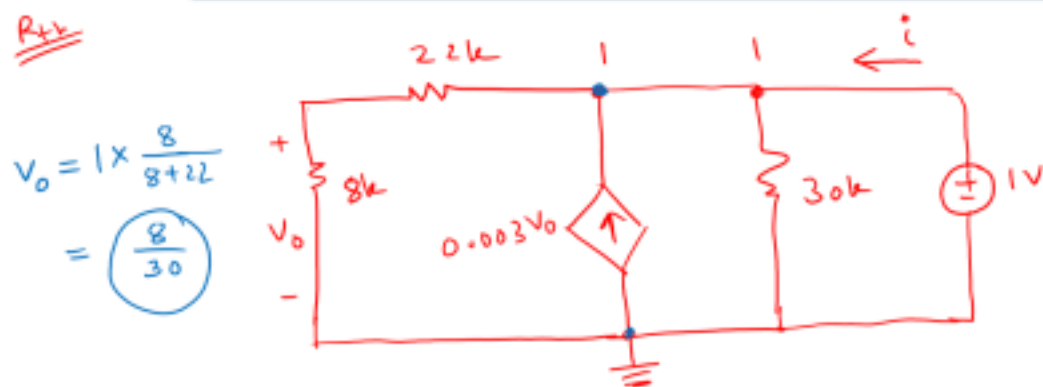


$$0.003v_o + i = \frac{1}{30k} + \frac{1}{30k}$$

$$\Rightarrow 0.003v_o + i = \frac{1}{15k}$$

$$\Rightarrow 0.003 \times \frac{8}{30} + i = \frac{1}{15k}$$

$$\Rightarrow i = -7.33 \times 10^{-4} \text{ A}$$

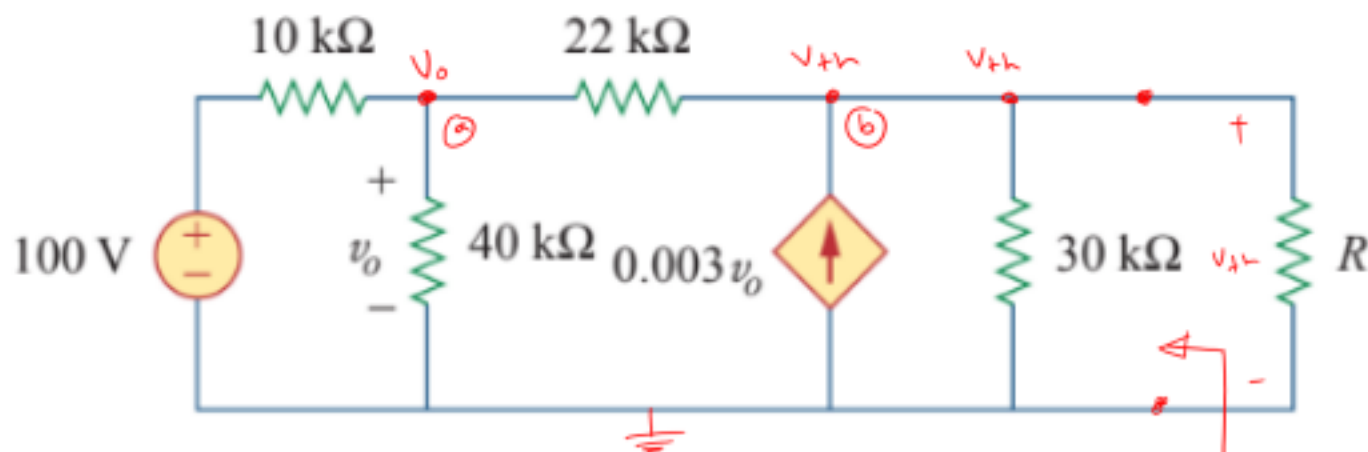


$$v_o = 1 \times \frac{8}{8+22}$$
$$= \left(\frac{8}{30} \right)$$

$$R_{th} = \frac{1}{i} = \boxed{-1363.7 \Omega}$$

DC Circuit

Find the maximum power transferred to resistor R

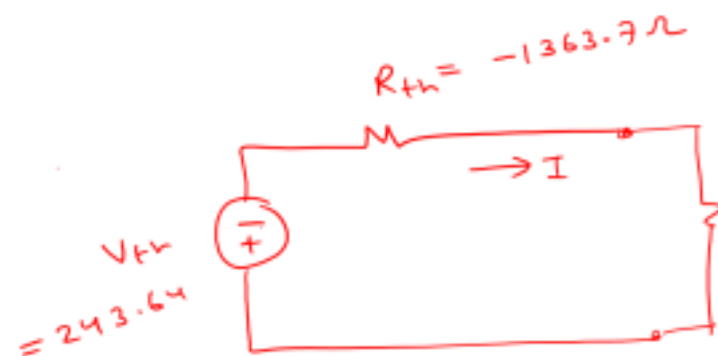


$$(a) \Rightarrow \frac{v_o - 100}{10k} + \frac{v_o}{40k} + \frac{v_o - v_{th}}{22k} = 0$$

$$(b) \Rightarrow 0.003 v_o = \frac{v_{th} - v_o}{22k} + \frac{v_{th}}{30k}$$

$$v_{th} = -243.64 \text{ V}$$

DC Circuit



$$R = |-1363.7| = 1363.7 \Omega$$

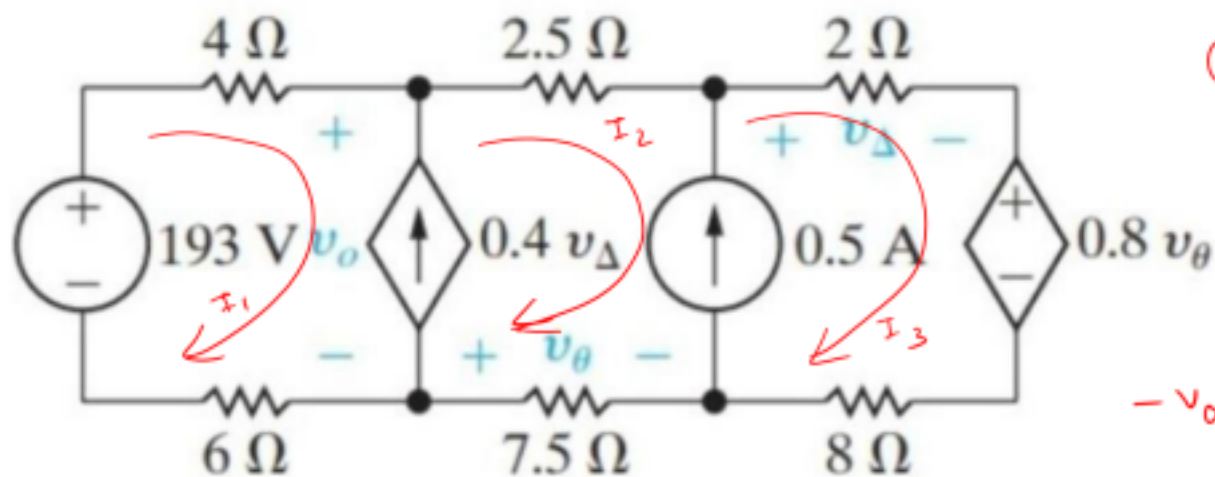
$$I = \frac{V_{th}}{R_{th} + R} = \frac{V_{th}}{0} = \infty$$

$$R = |R_{th}|$$

$$P_{max} = I^2 R \rightarrow \infty$$

DC Circuit

Find the voltage V_o



$$v_{\Delta} = 2I_3$$

$$v_{\theta} = -7.5I_2$$

$$I_2 - I_1 = 0.4v_{\Delta} = 0.8I_3$$

$$I_3 - I_2 = 0.5A$$

$$-193 + 4I_1 + v_o + 6I_1 = 0$$

$$\Rightarrow 10I_1 + v_o = 193$$

$$\Rightarrow 10I_1 + 4I_2 + 10I_3 = 193$$

$$-v_o + 10I_2 + 10I_3 + 0.8(-7.5I_2) = 0$$

$$\Rightarrow v_o = 4I_2 + 10I_3$$

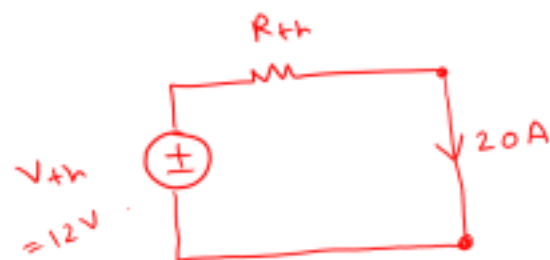
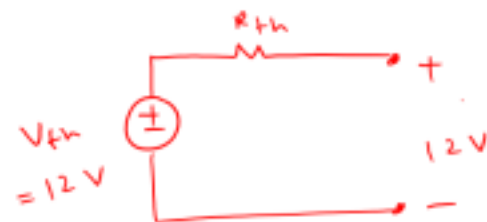
$$= \boxed{173V}$$

$$I_2 = 12$$

$$I_3 = 12.5$$

DC Circuit

A battery has a short-circuit current of 20 A and an open-circuit voltage of 12 V. If the battery is connected to an electric bulb of resistance $2\ \Omega$, calculate the power dissipated by the bulb.



$$R_{th} = \frac{12}{20} = 0.6$$



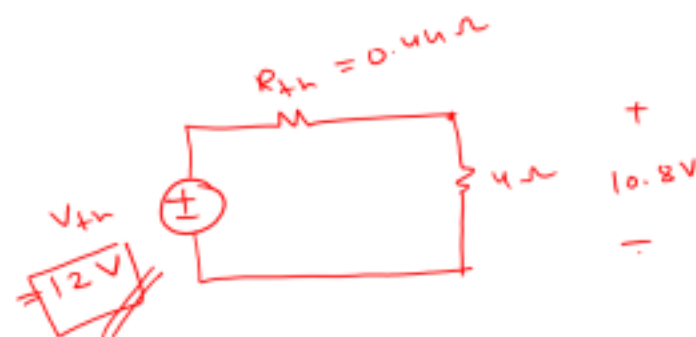
$$I = \frac{12}{0.6 + 2} = \frac{60}{13}$$

$$P = I^2 \times 2$$

$$= \boxed{42.6W}$$

DC Circuit

When connected to a 4-Ω resistor, a battery has a terminal voltage of 10.8 V but produces 12 V on an open circuit. Determine the Thevenin equivalent circuit for the battery.

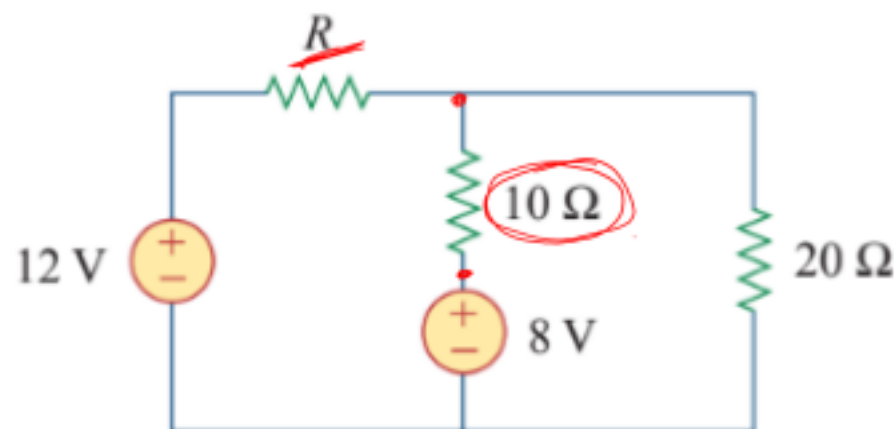


$$10.8 = \frac{4}{R_{th} + 4} \times 12$$

$$\Rightarrow \boxed{R_{th} = 0.44 \Omega}$$

DC Circuit

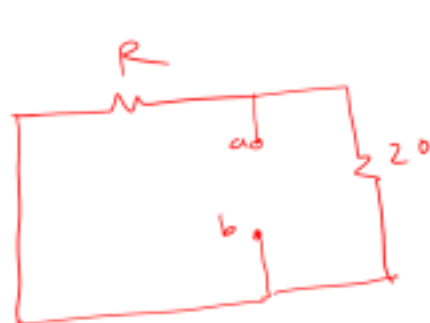
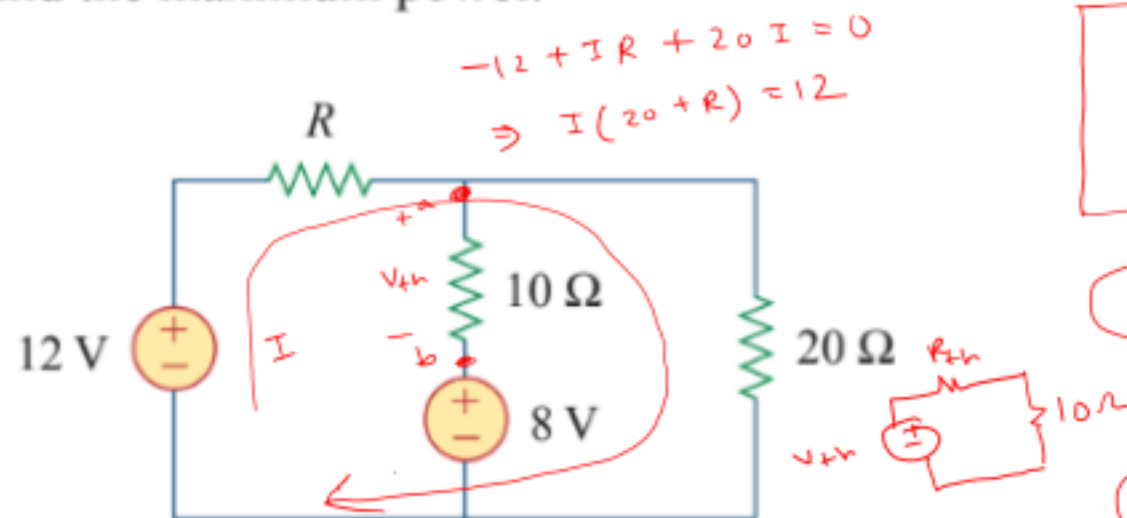
Compute the value of R that results in maximum power transfer to the $10\text{-}\Omega$ resistor in Fig. 4.134. Find the maximum power.



- The load is already specified and this is a "minimize losses" style problem.
- When a load is specified and internal losses can be adjusted, then the objective becomes, reduce R_{Th} as much as possible, which will result in maximum power transfer to the load.

DC Circuit

Compute the value of R that results in maximum power transfer to the $10\text{-}\Omega$ resistor in Fig. 4.134. Find the maximum power.



$$R_{th} = \frac{20R}{20+R}$$

Loss Minimized

$$R = 0$$

$$\left. \begin{array}{l} R_{th} = 0 \\ V_{th} = 4 \end{array} \right\} I = \frac{4}{10}$$
$$P_{max} = 4 \times \frac{4}{10} = 1.6 \text{ W}$$

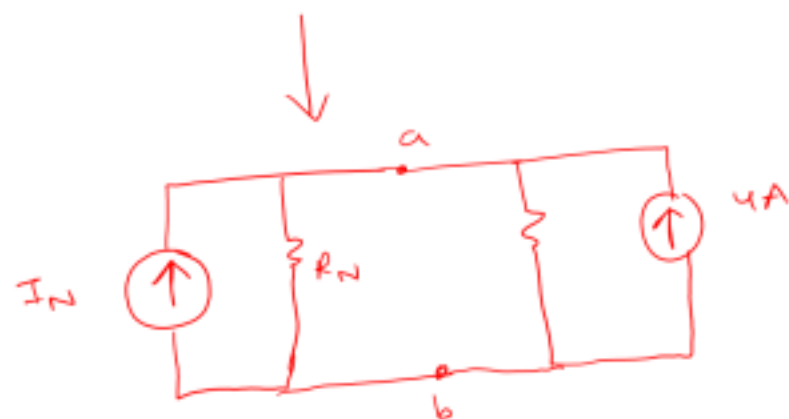
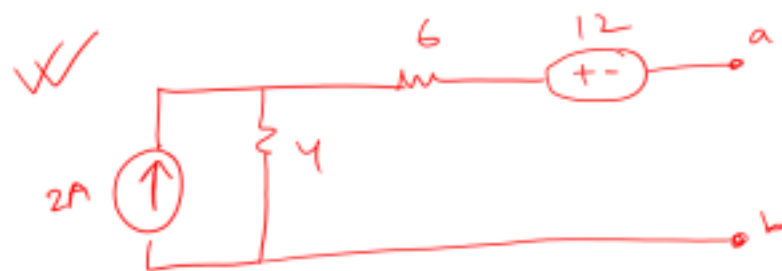
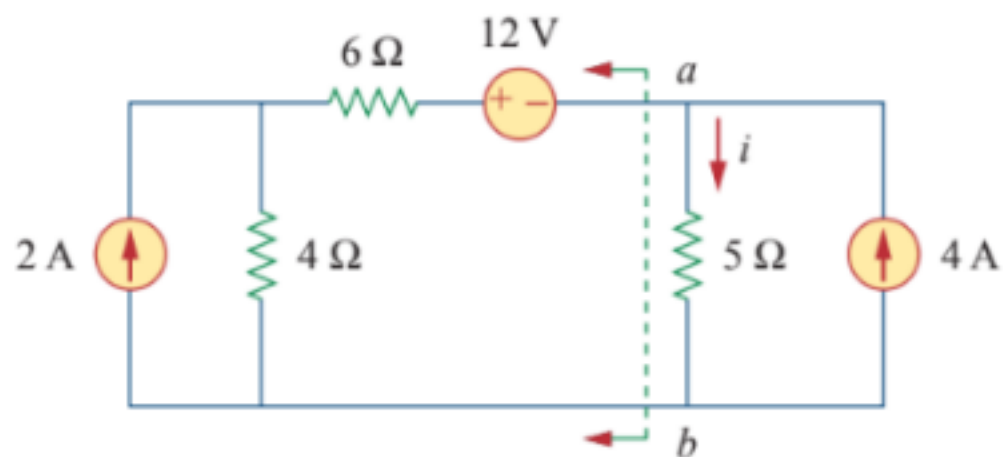
$$R = 20$$

$$\left. \begin{array}{l} R_{th} = 10 \\ V_{th} = -2 \end{array} \right\} P = \frac{(-2)^2}{4 \times 10} = 0.1 \text{ W}$$

$$R_{th} = R_L \Rightarrow \frac{20R}{20+R} = 10$$
$$\Rightarrow R = 20$$

DC Circuit

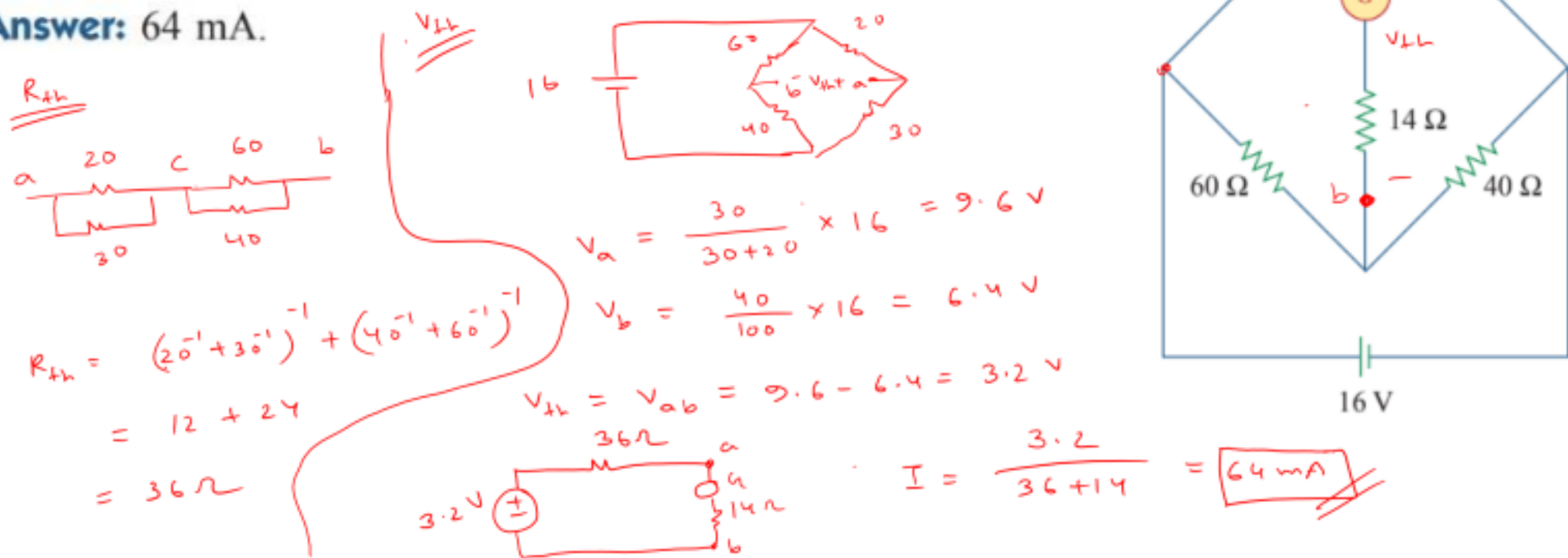
- 50 Obtain the Norton equivalent of the circuit in Fig. 4.116 to the left of terminals $a-b$. Use the result to find current i .



DC Circuit

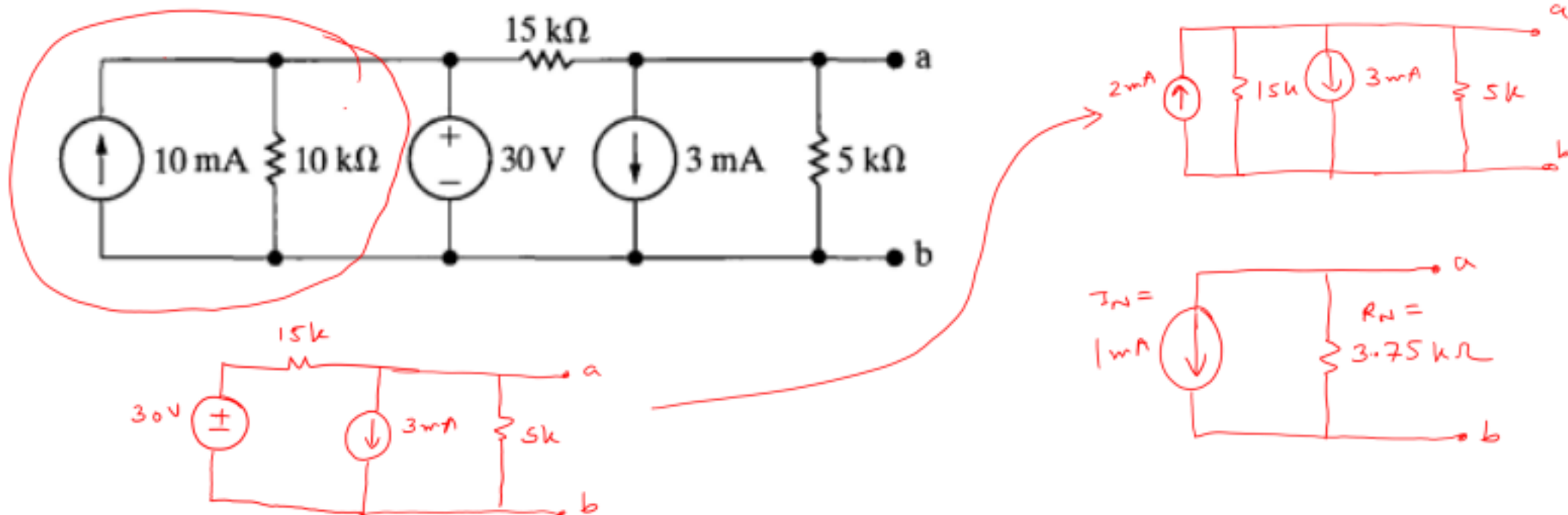
Obtain the current through the galvanometer, having a resistance of $14\ \Omega$, in the Wheatstone bridge shown in Fig. 4.66.

Answer: 64 mA.



DC Circuit

Find the Norton equivalent with respect to the terminals a,b in the circuit

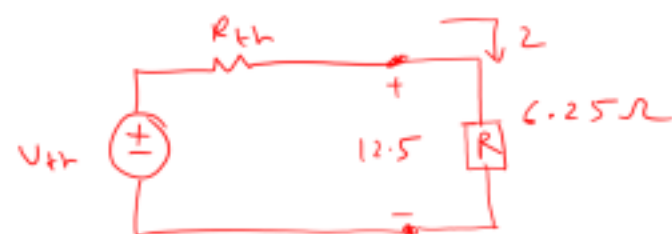
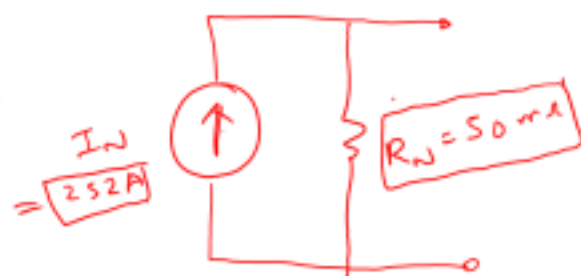


DC Circuit

An automobile battery, when connected to a car radio, provides 12.5 V to the radio. When connected to a set of headlights, it provides 11.7 V to the headlights. Assume the radio can be modeled as a 6.25Ω resistor and the headlights can be modeled as a 0.65Ω resistor. What are the Thévenin and Norton equivalents for the battery?

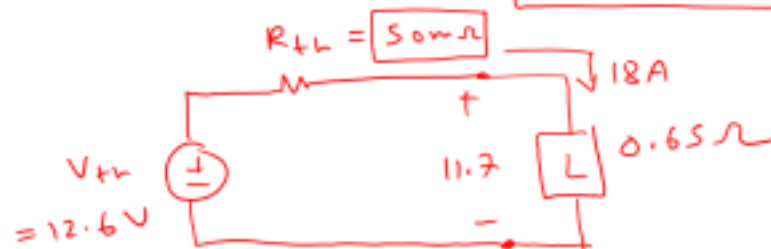
$$I_R = \frac{12.5}{6.25} = 2 \text{ A}$$

$$I_L = \frac{11.7}{0.65} = 18 \text{ A}$$



$$R_{th} = \frac{V_{th} - 12.5}{2} = \frac{V_{th} - 11.7}{18}$$

$$\Rightarrow V_{th} = 12.6 \text{ V}$$



Operational Amplifiers

DC Circuit

Active

- They require an external power supply to operate.
- They have a function of gain.
- They produce energy in the form of voltage and current.
- Ex- Semiconductor, diodes, transistors. *Op amp*

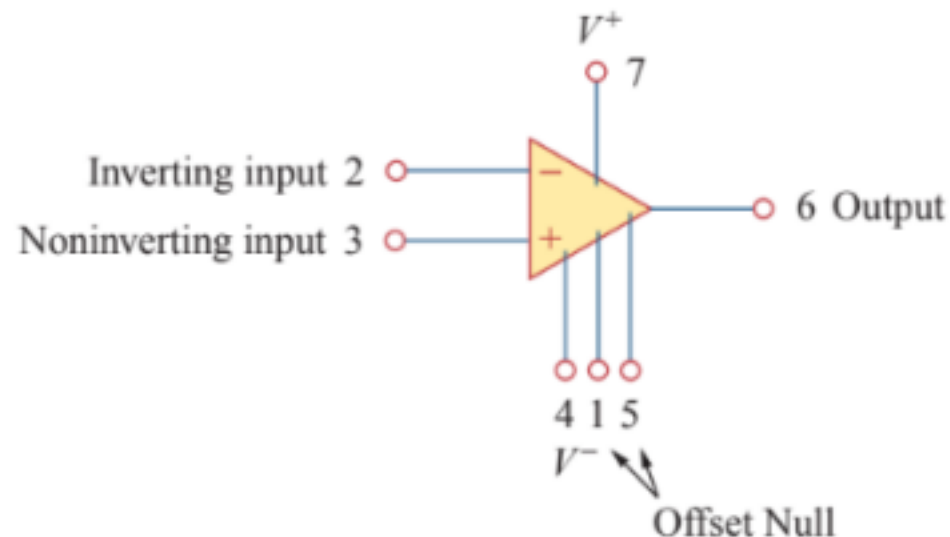
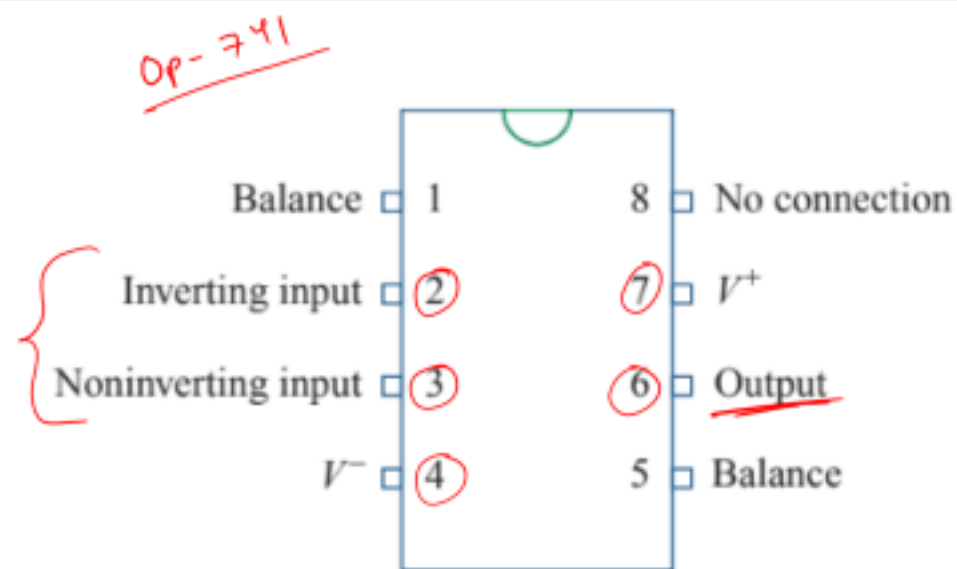
Passive

- They don't require an external power supply to operate.
- They have no function of gain.
- They store energy in the form of voltage or current.
- Ex- Resistor, Inductor, Capacitor.(R-L-C)

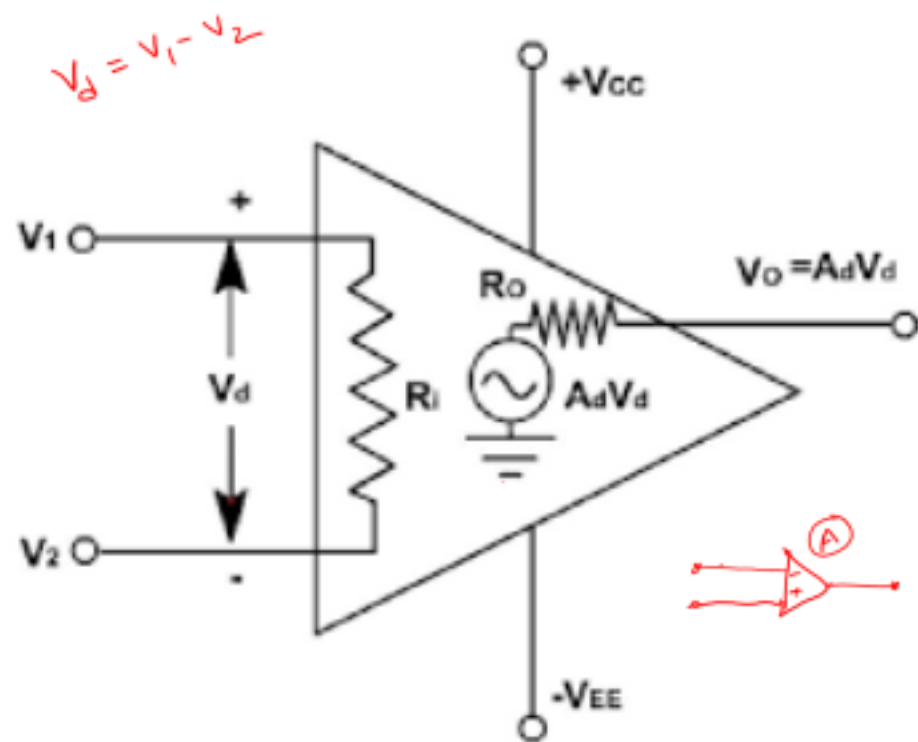
DC Circuit

- The op amp is an electronic unit that behaves like a voltage-controlled voltage source. It can also be used in making a voltage- or current-controlled current source.
- Op amp is an active circuit element designed to perform mathematical operations of addition, subtraction, multiplication, division, differentiation, and integration.

DC Circuit



DC Circuit

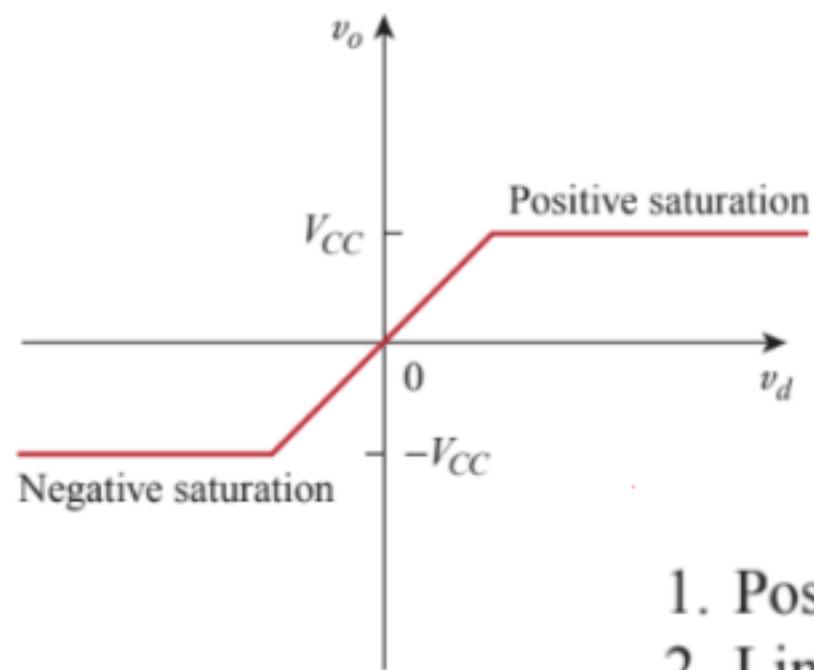


Ideal

Characteristics	Symbol	Values
Open loop voltage gain	A_{OL}	∞ <i>$10^5 \sim 10^8$</i>
Input impedance	R_{in}	∞
Output impedance	R_o	0
Offset voltage	V_{oo}	0
Bandwidth	B.W.	∞
C.M.R.R	ρ	∞
Slew rate	S	∞

A_d is the open-loop voltage gain because it is the gain without any external feedback from output to input.

DC Circuit



$$V_o = 16V$$

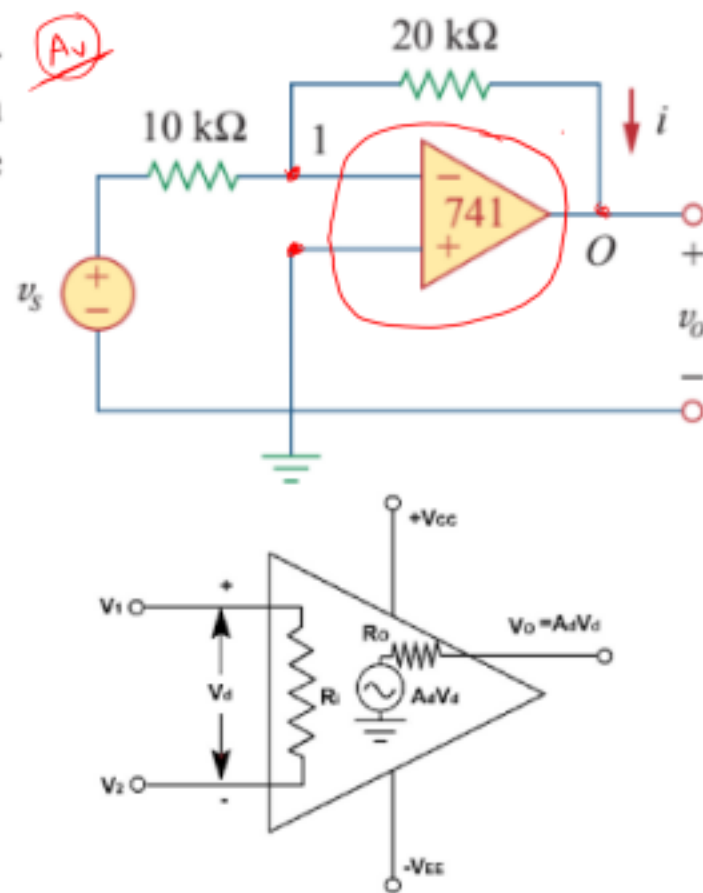
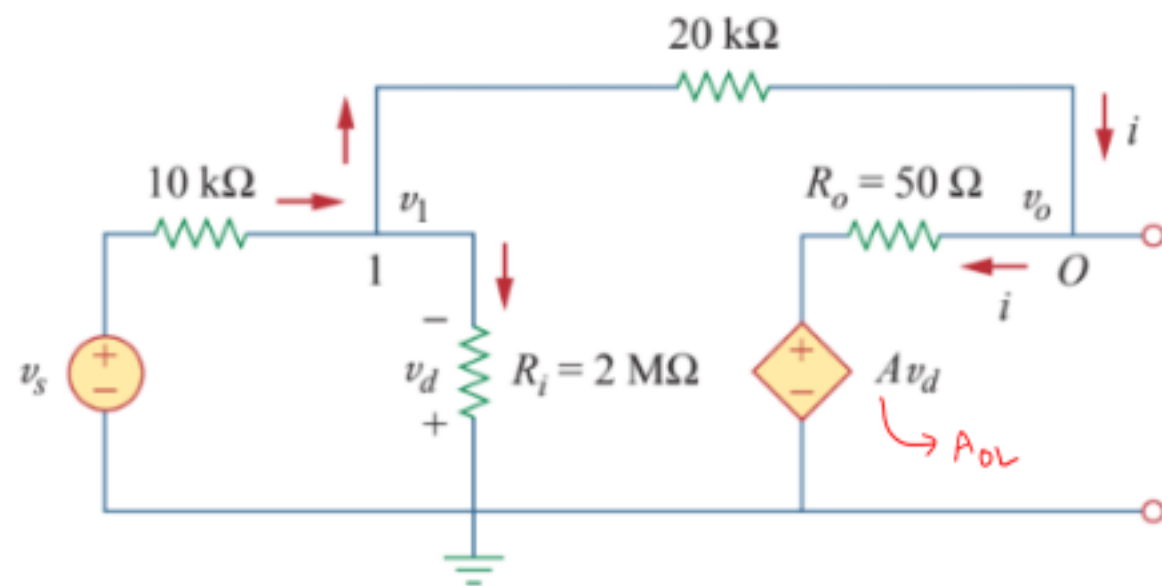
$$V_{sat} = +12V$$

Ans: $V_o = +12V$

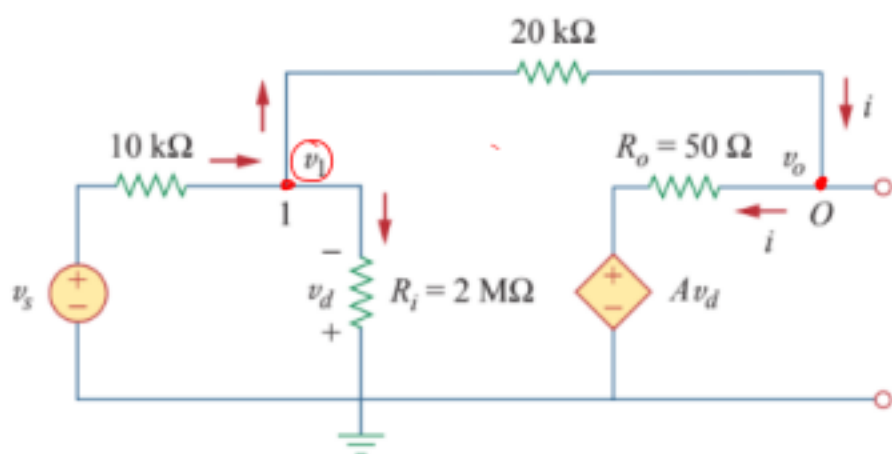
1. Positive saturation, $v_o = V_{CC}$.
2. Linear region, $-V_{CC} \leq v_o = Av_d \leq V_{CC}$.
3. Negative saturation, $v_o = -V_{CC}$.

DC Circuit

A 741 op amp has an open-loop voltage gain of 2×10^5 , input resistance of $2 \text{ M}\Omega$, and output resistance of 50Ω . The op amp is used in the circuit of Fig. 5.6(a). Find the closed-loop gain v_o/v_s . Determine current i when $v_s = 2 \text{ V}$.



DC Circuit



$$V_1 = -v_d$$

$$i = \frac{v_o + 2 \times 10^5 v_1}{50}$$

$$= \boxed{\checkmark}$$

$$\textcircled{1} \Rightarrow \frac{v_s - v_1}{10k} = \frac{v_1 - v_o}{20k} + \frac{v_1 - 0}{2M}$$

$$\textcircled{O} \Rightarrow \frac{v_1 - v_o}{20k} = \frac{v_o + 2 \times 10^5 v_1}{50}$$

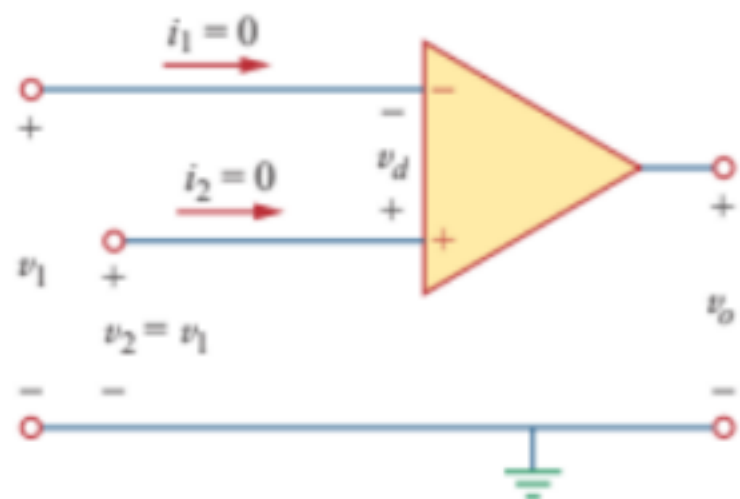
$$\Rightarrow \frac{v_1}{20k} - \frac{2 \times 10^5 v_1}{50} = \frac{v_o}{50} + \frac{v_o}{20k}$$

$$\Rightarrow v_1 = \cancel{\text{something}} v_o$$

$$\boxed{\frac{v_o}{v_s} = ?}$$

DC Circuit

Ideal Op Amp



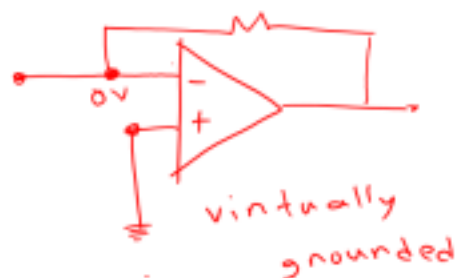
Assumptions

i) $i_1 = i_2 = 0$

ii) $v_d = 0$

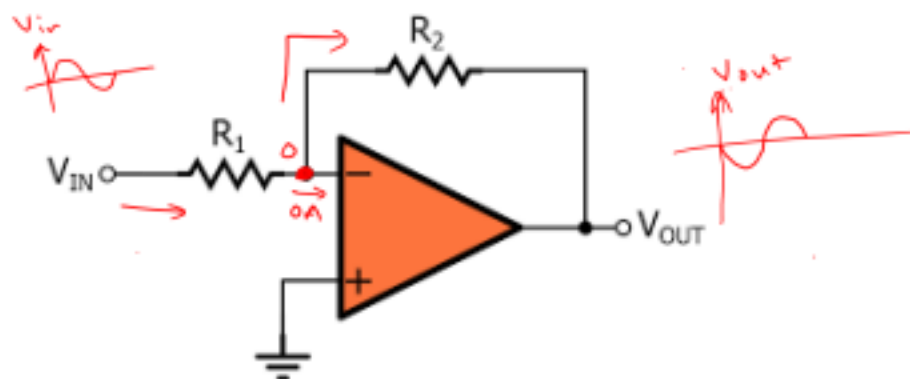
$\Rightarrow v_1 - v_2 = 0$

$\Rightarrow \boxed{v_1 = v_2} \rightarrow \text{virtually short}$



DC Circuit

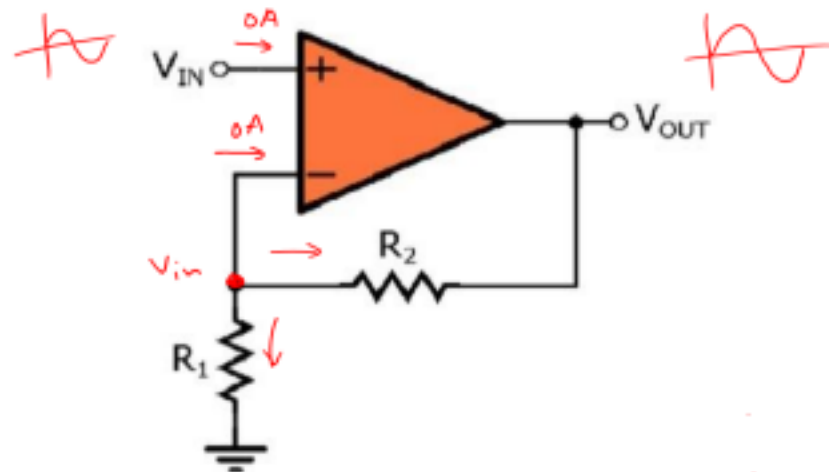
Inverting amplifier



$$\frac{V_{in} - 0}{R_1} = \frac{0 - V_{out}}{R_2}$$

$$\Rightarrow V_{out} = -\frac{R_2}{R_1} V_{in}$$

Non-inverting amplifier

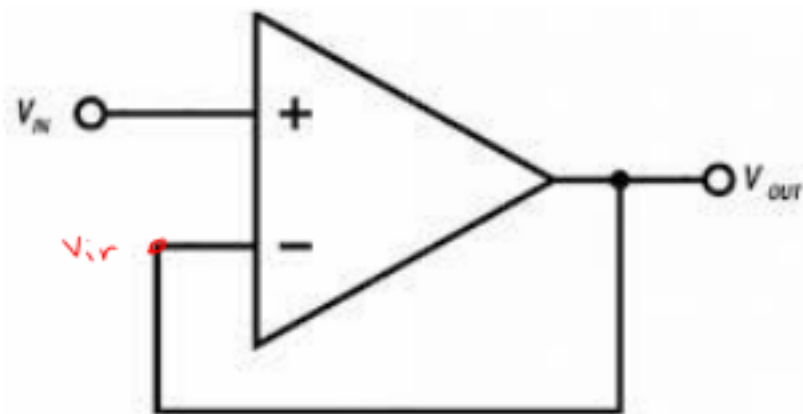


$$\frac{V_{in} - V_{out}}{R_2} + \frac{V_{in}}{R_1} = 0$$

$$\Rightarrow \frac{V_{out}}{R_2} = V_{in} \left(\frac{1}{R_2} + \frac{1}{R_1} \right) \Rightarrow V_{out} = V_{in} \left(1 + \frac{R_2}{R_1} \right)$$

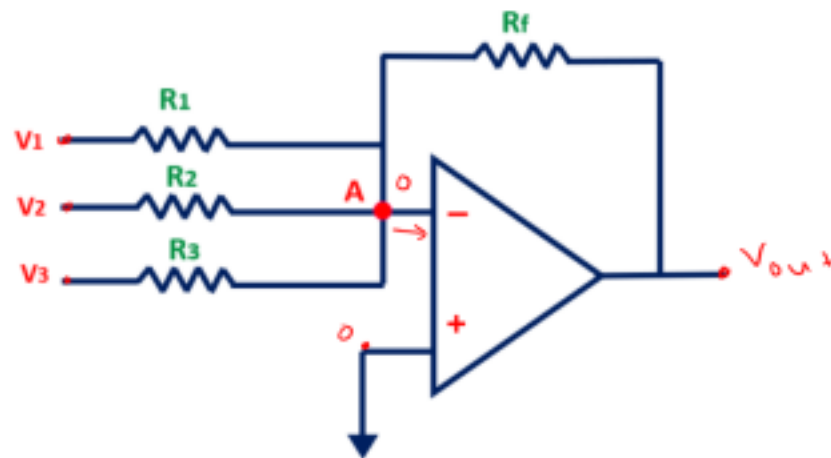
DC Circuit

voltage follower/buffer



$$V_{in} = V_{out}$$

Summing Amplifier

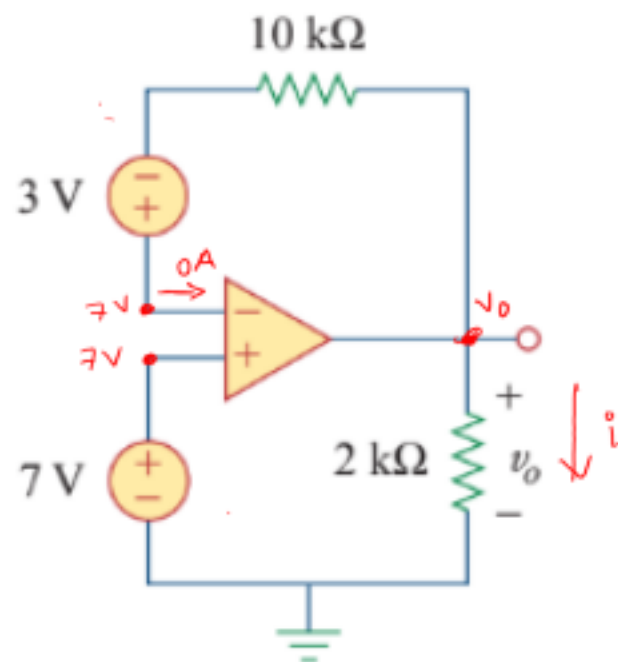


$$\frac{V_3}{R_3} + \frac{V_2}{R_2} + \frac{V_1}{R_1} = - \frac{V_{out}}{R_f}$$

$$\Rightarrow V_{out} = - \left(\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3 \right)$$

DC Circuit

Obtain v_o for the op amp circuit :



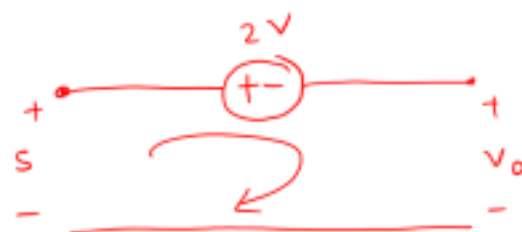
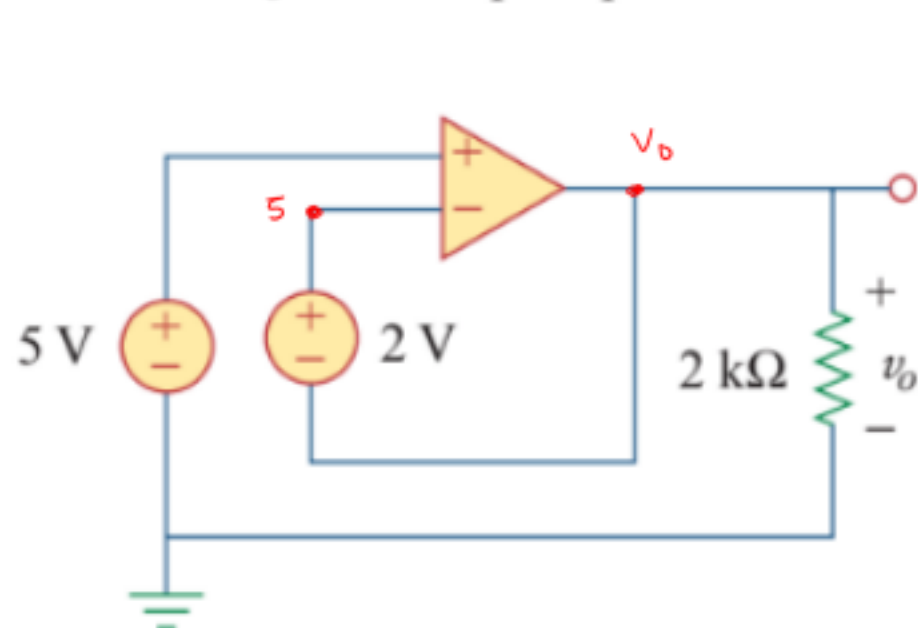
$$-v_o + 10i - 3 + 7 = 0$$

$$\Rightarrow \boxed{v_o = 4V}$$

$$i = \frac{v_o}{2k} = \frac{4}{2k} = \boxed{2mA}$$

DC Circuit

Obtain v_o for the op amp circuit:

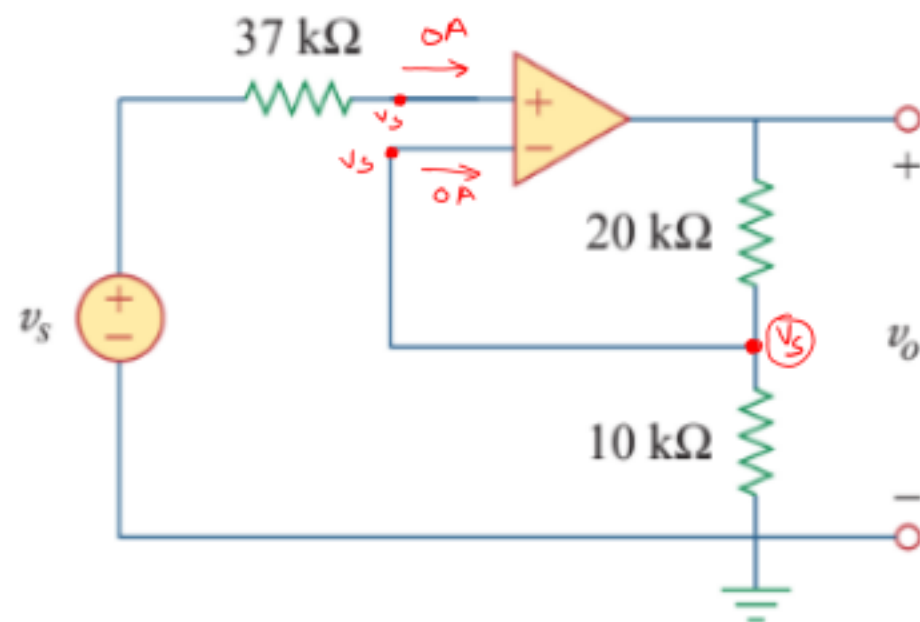


$$-5 + 2 + v_o = 0$$

$$\Rightarrow \boxed{v_o = 3\text{ V}}$$

DC Circuit

Find the gain v_o/v_s of the circuit



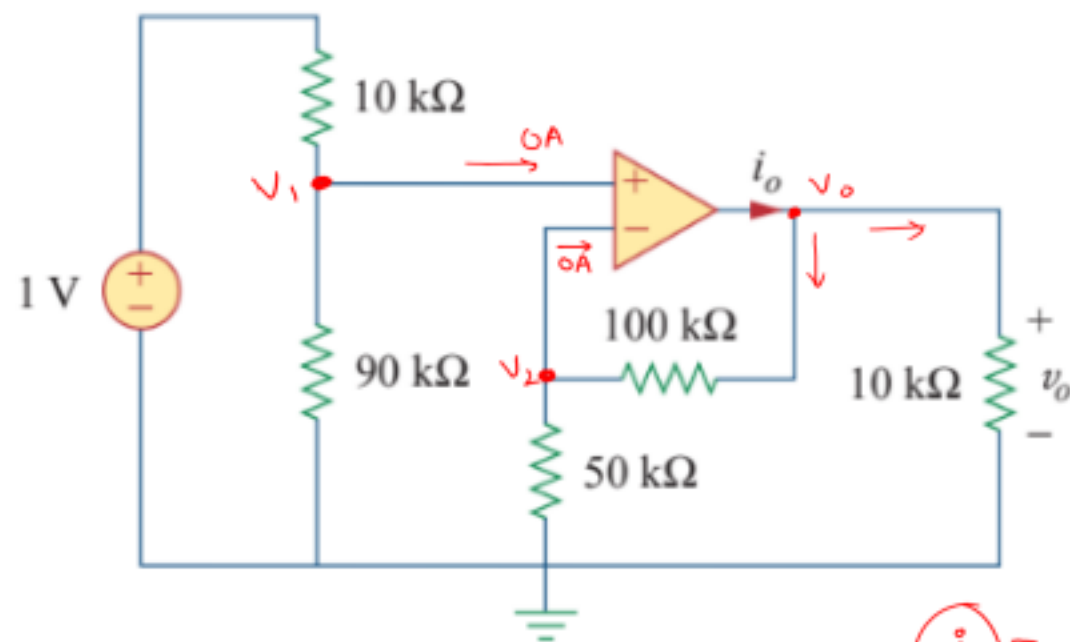
$$v_s = \frac{10}{10 + 20} v_o$$
$$= \frac{1}{3} v_o$$

$$\Rightarrow \boxed{\frac{v_o}{v_s} = 3}$$

DC Circuit

Find v_o and i_o in the circuit

$$v_1 = v_2$$



$$v_1 = \frac{90}{10 + 90} \times 1 = \frac{9}{10}$$

$$v_2 = \frac{50}{100 + 50} v_o = \frac{1}{3} v_o$$

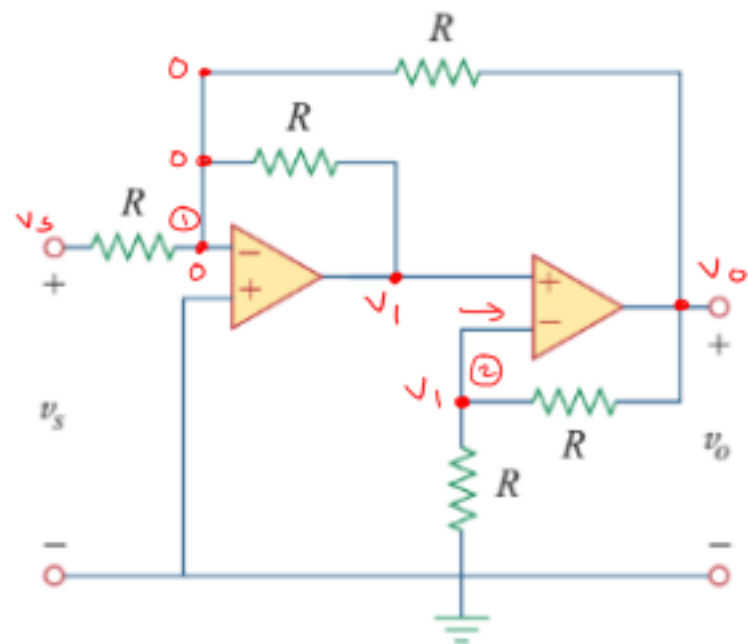
$$v_1 = v_2$$

$$\Rightarrow \frac{9}{10} = \frac{v_o}{3} \Rightarrow v_o = 2.7 \text{ V}$$

$$i_o = \frac{v_o}{10} + \frac{v_o}{150} = ?$$

DC Circuit

Q Determine the voltage transfer ratio v_o/v_s in the op amp circuit of Fig. 5.82, where $R = 10\text{ k}\Omega$.



$$\textcircled{1} \Rightarrow \frac{v_s}{R} = \frac{0 - v_1}{R} + \frac{0 - v_o}{R}$$

$$\Rightarrow v_s = -v_1 - v_o = -\frac{v_o}{2} - v_o$$

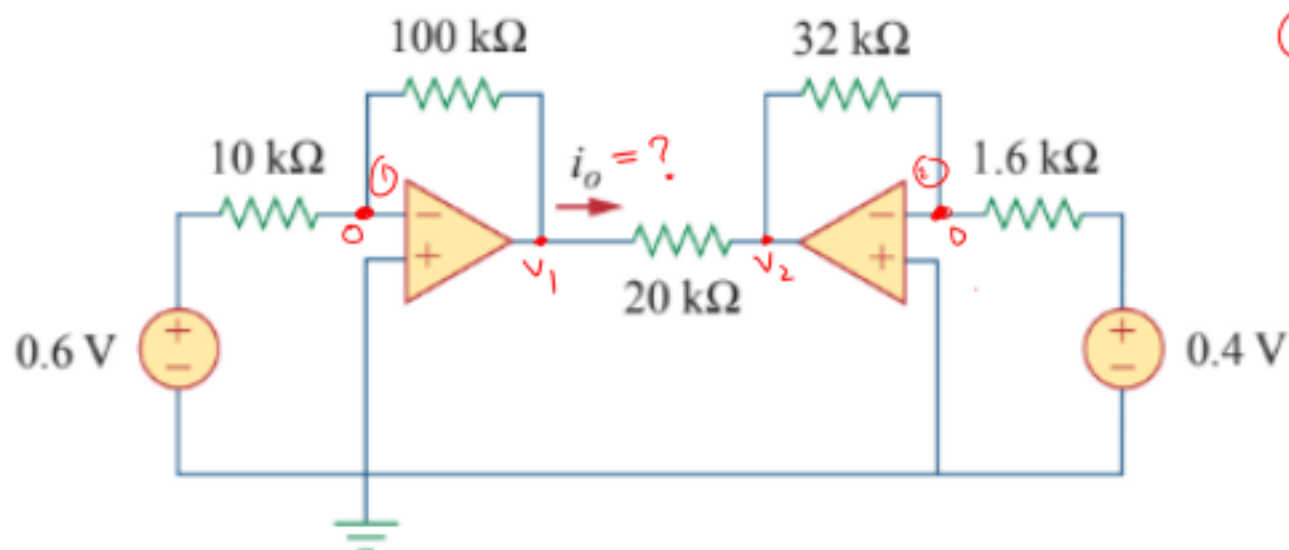
$$\textcircled{2} \Rightarrow \frac{v_1 - v_o}{R} + \frac{v_1}{R} = 0$$

$$\Rightarrow v_1 = v_o/2$$

$$\Rightarrow v_s = -\frac{3}{2}v_o$$

$$\Rightarrow \boxed{\frac{v_o}{v_s} = -\frac{2}{3}}$$

DC Circuit



$$\textcircled{1} \Rightarrow \frac{0.6}{10} = -\frac{v_1}{100}$$

$$\Rightarrow \boxed{v_1 = -6}$$

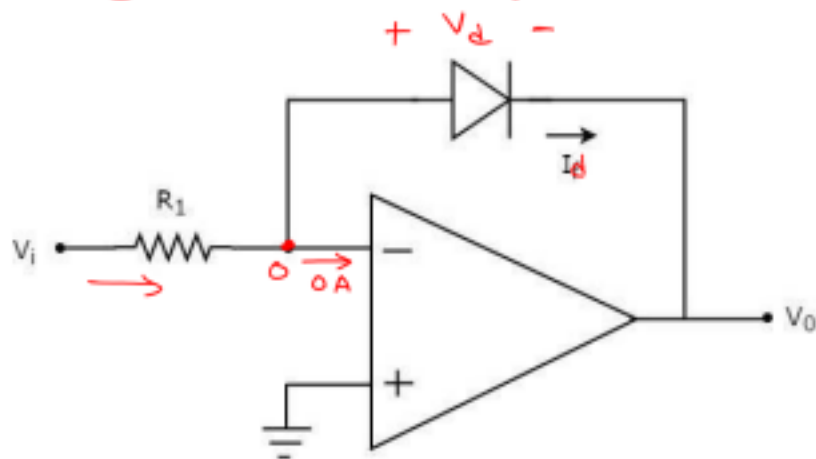
$$\textcircled{2} \Rightarrow \frac{0.4}{1.6} = -\frac{v_2}{32}$$

$$\Rightarrow \boxed{v_2 = -8}$$

$$i_o = \frac{v_1 - v_2}{20k} = \frac{-6 + 8}{20k} = \textcircled{?}$$

DC Circuit

Logarithmic amplifier:



$$\frac{V_i}{R_1} = I_D = I_S e^{\frac{V_d}{nV_T}}$$

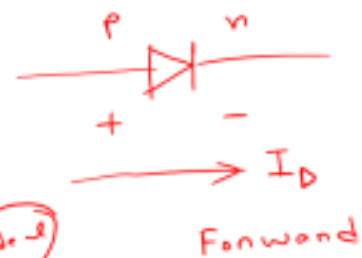
$$\Rightarrow \frac{V_i}{R} = I_S e^{\frac{-V_o}{nV_T}}$$

$$\Rightarrow \frac{V_i}{RI_S} = e^{\frac{-V_o}{nV_T}}$$

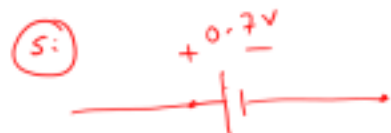
$$\Rightarrow \ln\left(\frac{V_i}{RI_S}\right) = -\frac{V_o}{nV_T}$$

$$\Rightarrow \boxed{V_o = -nV_T \ln\left(\frac{V_i}{RI_S}\right)}$$

$$\ln e^p = p \cdot \ln e = p$$



(i) Forward



(ii) 0.7V

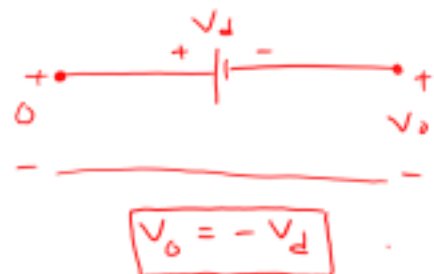
(iii) 0.3V

$$I_D = I_S e^{\frac{V_d}{nV_T}}$$

saturation current

$n = 1$

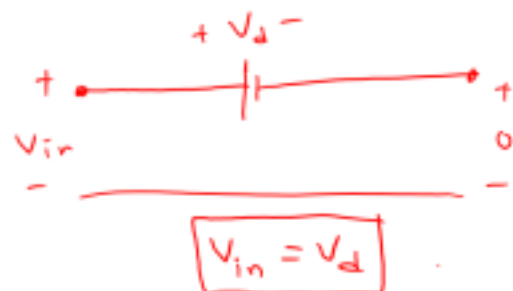
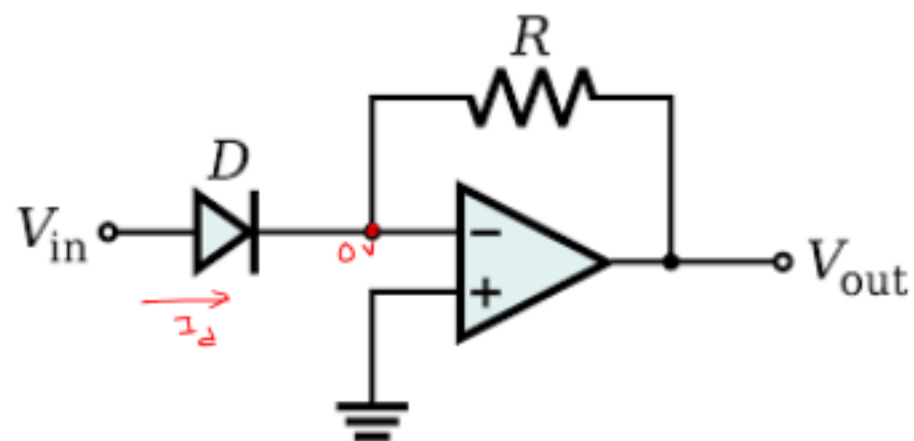
$V_T = 26 \text{ mV}$



$$\boxed{V_o = -V_d}$$

DC Circuit

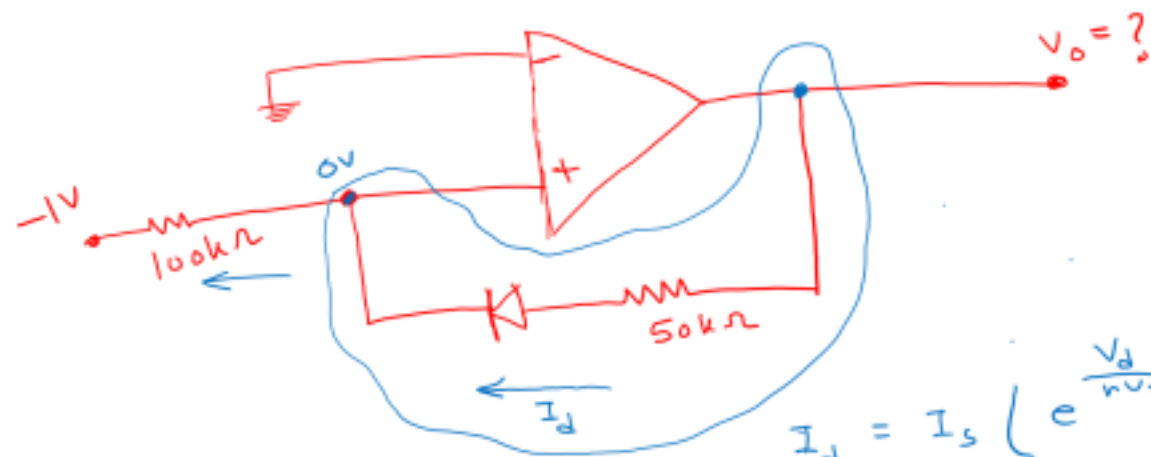
Exponential op amp



$$I_d = \frac{-V_{out}}{R} = I_s e^{\frac{V_d}{nV_T}}$$
$$\Rightarrow \boxed{V_{out} = -RI_s e^{\frac{V_{in}}{nV_T}}}$$

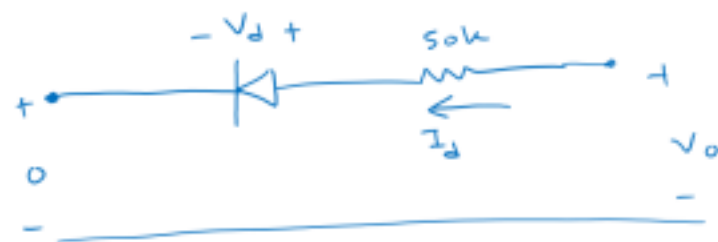
DC Circuit

$n=1$, $I_s = 10^{-15} \text{ A}$, $V_T = 26 \text{ mV}$



$$I_d = I_s \left(e^{\frac{V_d}{nV_T}} - 1 \right)$$
$$\Rightarrow \boxed{V_d = ?}$$

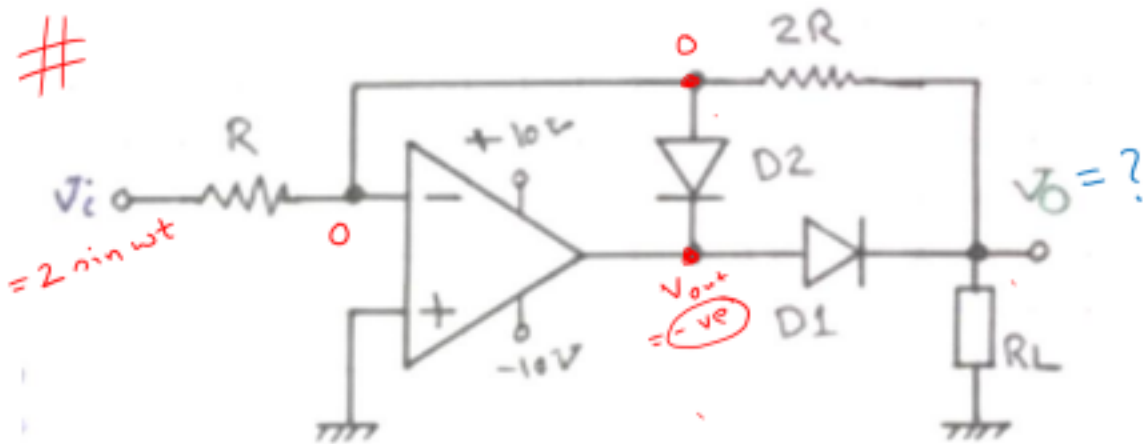
$$I_d = \frac{0 - (-1)}{100k} = \left(\frac{1}{100k} \right)$$



$$-V_o + 50k \times \frac{1}{100k} + V_d = 0$$

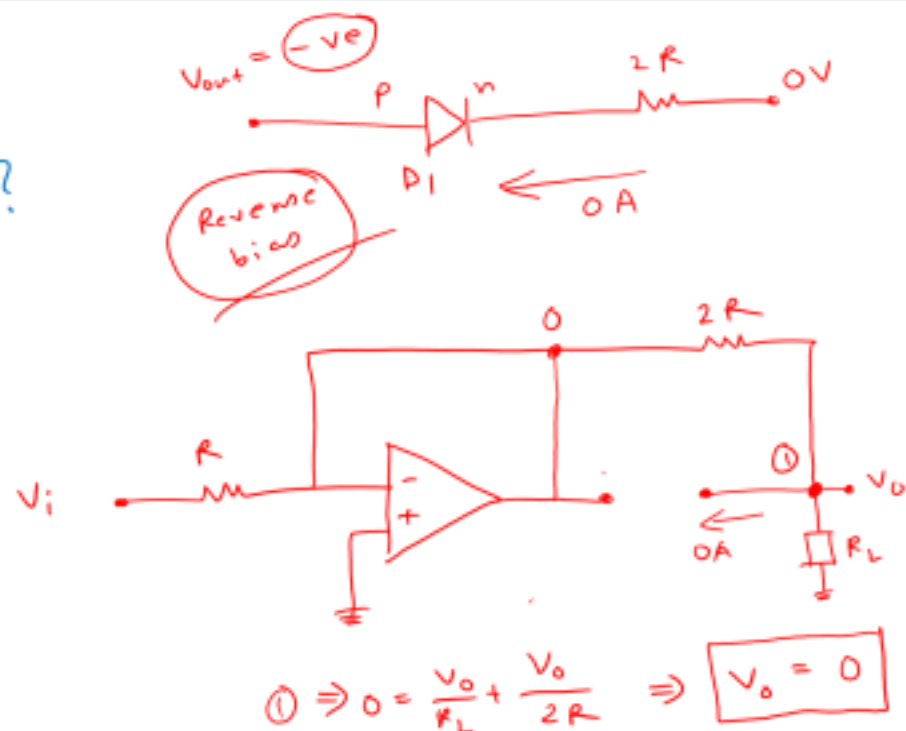
$$\Rightarrow V_o = \boxed{V_d} + 0.5$$
$$= \boxed{?}$$

DC Circuit

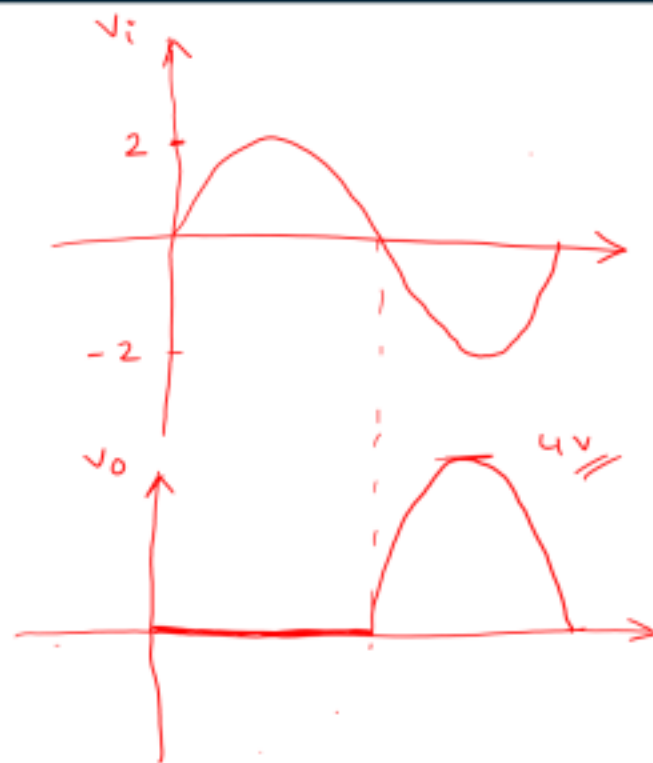
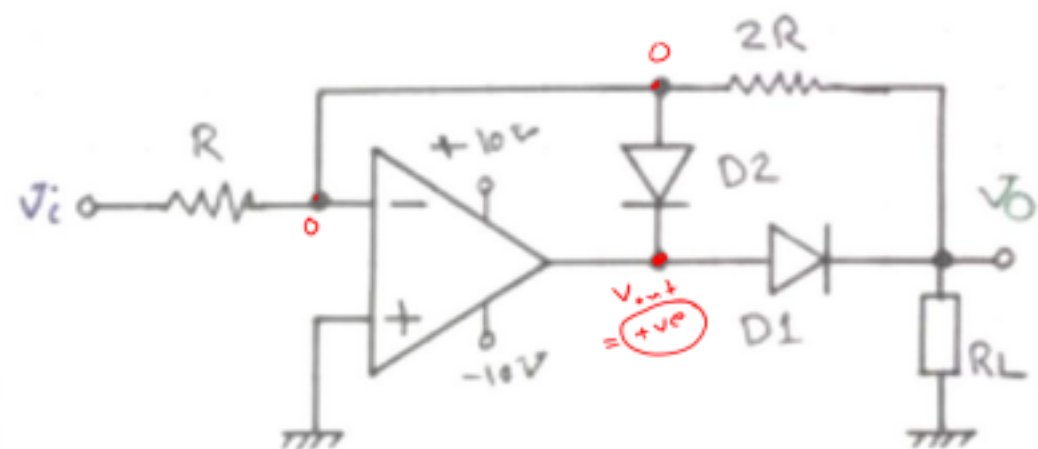


$V_i \geq 0 \Rightarrow D_2 \text{ ON}, D_1 \text{ off}$

$$V_o = 0$$



DC Circuit

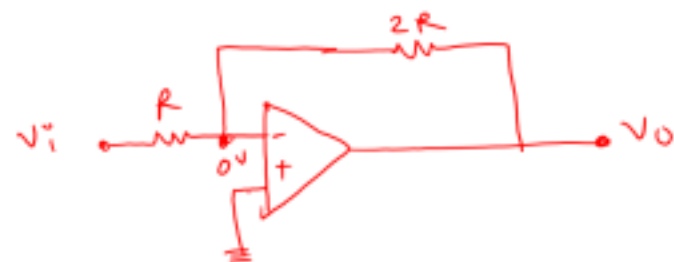


$V_i < 0$; D_2 off, D_1 ON

$$\frac{V_i}{R} = - \frac{V_o}{2R}$$

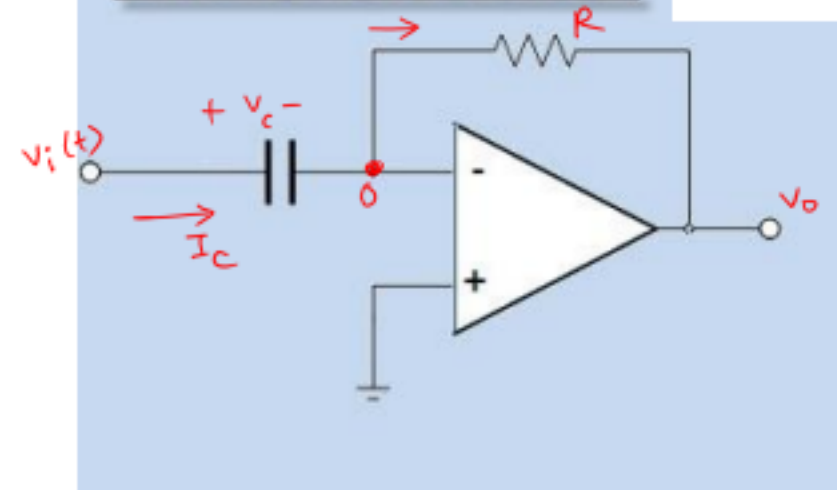
$$\Rightarrow V_o = -2V_i$$

$$V_i = -2V \Rightarrow \boxed{V_o = 4V}$$



DC Circuit

Differentiator

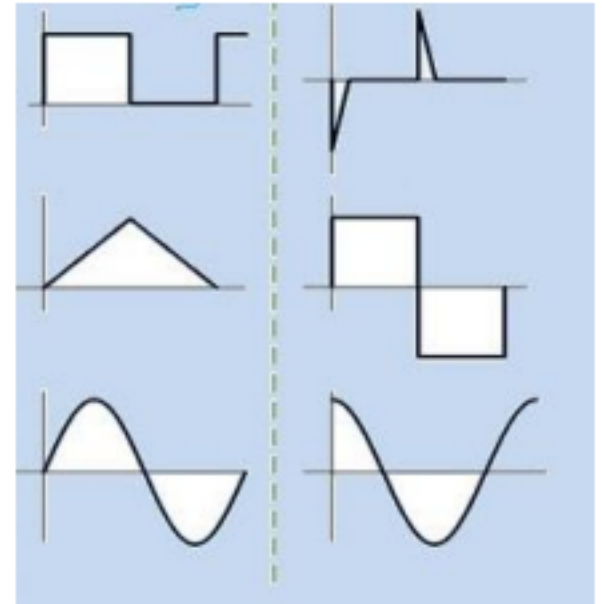


$$I_c = \frac{-v_o}{R}$$

$$\Rightarrow C \frac{dv_c}{dt} = \frac{-v_o}{R}$$

$$\Rightarrow v_o = -RC \frac{dv_c}{dt}$$

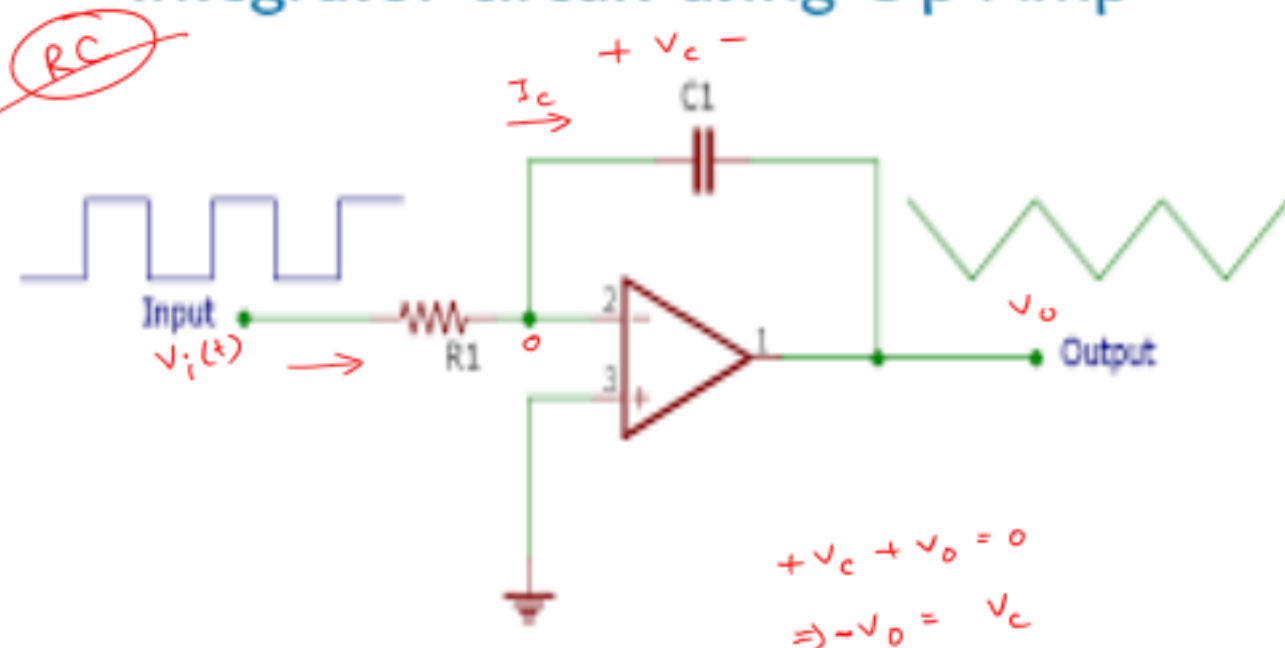
$$\Rightarrow v_o = -RC \frac{dv_i}{dt}$$



Signal

DC Circuit

Integrator circuit using Op-Amp



$$\frac{V_i}{R} = I_c$$

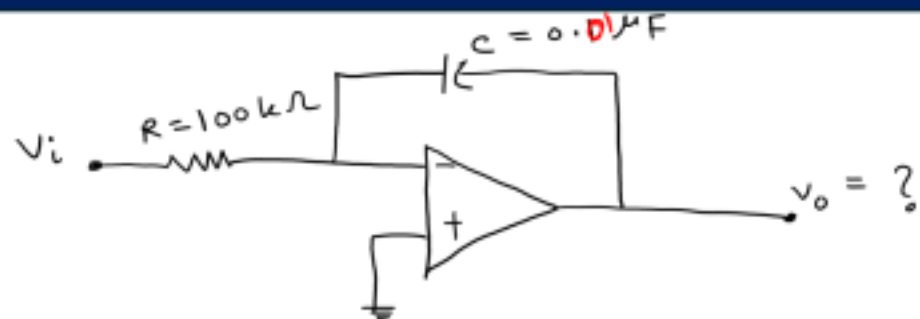
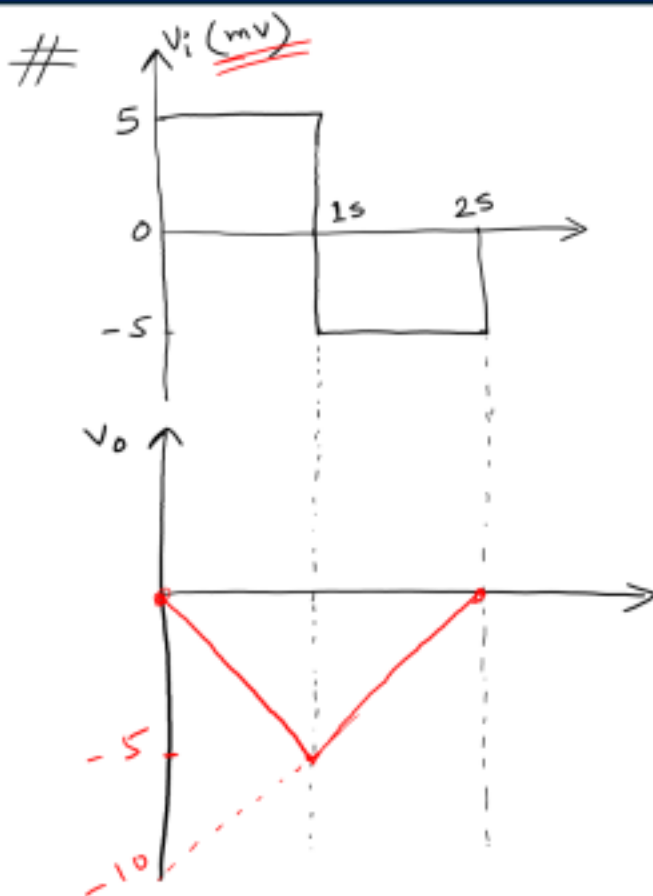
$$\Rightarrow \frac{V_i}{R} = C \frac{dV_c}{dt}$$

$$\Rightarrow \frac{V_i}{R} = -C \frac{dV_o}{dt}$$

$$\Rightarrow -\frac{1}{RC} \int V_i dt = \int dV_o$$

$$\Rightarrow \boxed{V_o = -\frac{1}{RC} \int V_i(t) dt}$$

DC Circuit



$$\text{(1st)} \quad V_o = -\frac{1}{RC} \int_0^t V_i dt = -\frac{5 \times 10^{-3}}{100 \times 10^3 \times 0.01 \times 10^{-6}} \int_0^1 dt$$

$$= (-5t) = -5[t]_0^1 = -5(1-0) = -5 \text{ mV}$$

$$\text{(2nd)} \quad V_o = 5 \int_1^t dt + (-5)$$

$$= 5[t]_1^t - 5 = 5(t-1) - 5 = 5t - 10$$

$$\text{(2s)} \Rightarrow V_o = 0$$

DC Circuit

Design problem

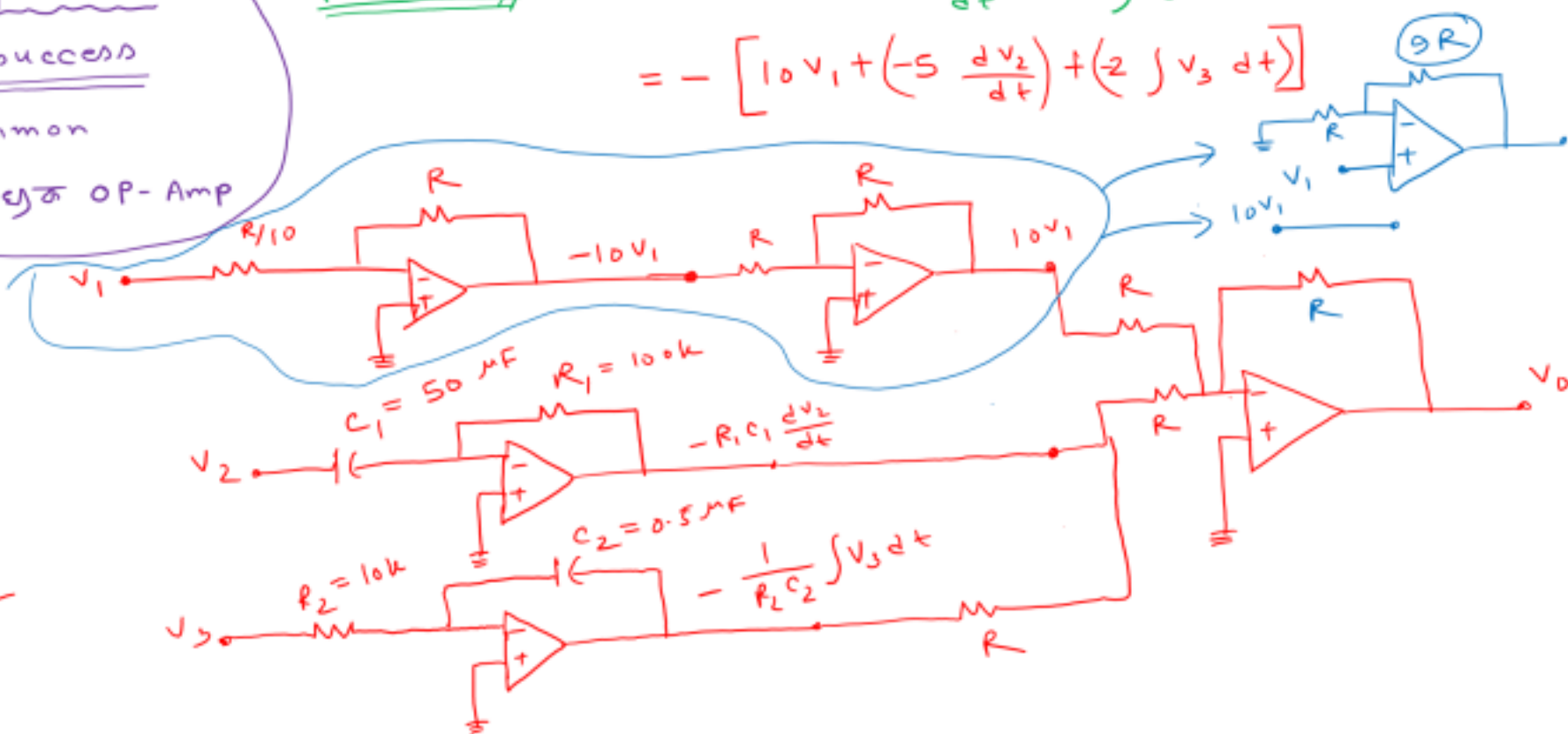
key to success

⊖ common

Less steps OP-Amp

PDB-2015 $V_o = -10V_1 + 5 \frac{dV_2}{dt} + 2 \int V_3 dt$

$$= - \left[10V_1 + \left(-5 \frac{dV_2}{dt} \right) + \left(2 \int V_3 dt \right) \right]$$



$$R_1 C_1 = 5$$

$$\frac{1}{R_2 C_2} = 2$$

DC Circuit

Slew Rate

Maximum rate of output voltage change per unit time without distortion

- To achieve the best performance, the slew rate should be maximized, allowing for the greatest undistorted output voltage swing.

$$S = \left. \frac{dV_o}{dt} \right|_{\text{maximum}} \text{ Volts}/\mu\text{S}$$

$$V_o = V_m \sin(\omega t)$$

$$\frac{dV_o}{dt} = \omega V_m \boxed{\cos(\omega t)}$$

$$\left| \frac{dV_o}{dt} \right|_{\text{max}} = \omega V_m = 2\pi f V_m = \text{S.R.}$$

$$\Rightarrow \boxed{f_{\text{max}} = \frac{\text{S.R.}}{2\pi V_m}}$$

DC Circuit

How to Measure Slew Rate?

The slew rate is measured by applying a step signal to the input stage of the op-amp and measuring the rate of change occurs at the output from 10% to 90% of the output signal's amplitude. Generally, the applied step signal is large and it is about 1 V.

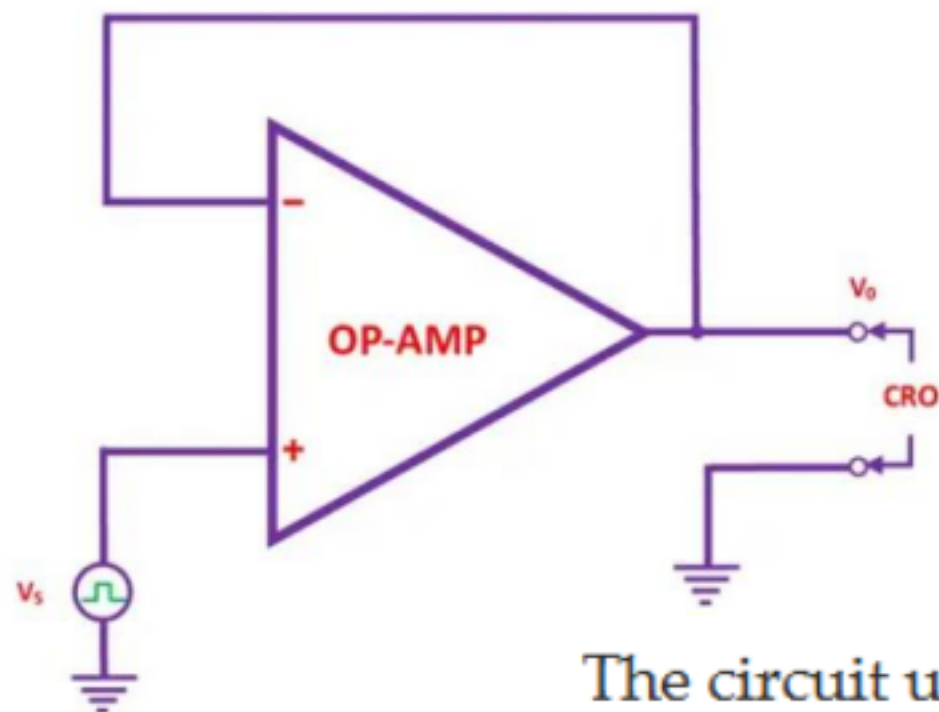
The slew rate is measured from the output voltage waveform as:

$$\text{Slew Rate}(S) = \frac{dV_0}{dt} = \frac{V_{0(90\%)} - V_{0(10\%)}}{t_{(90\%)} - t_{(10\%)}}$$

The slew rate can be measured by using an oscilloscope and a function generator.

DC Circuit

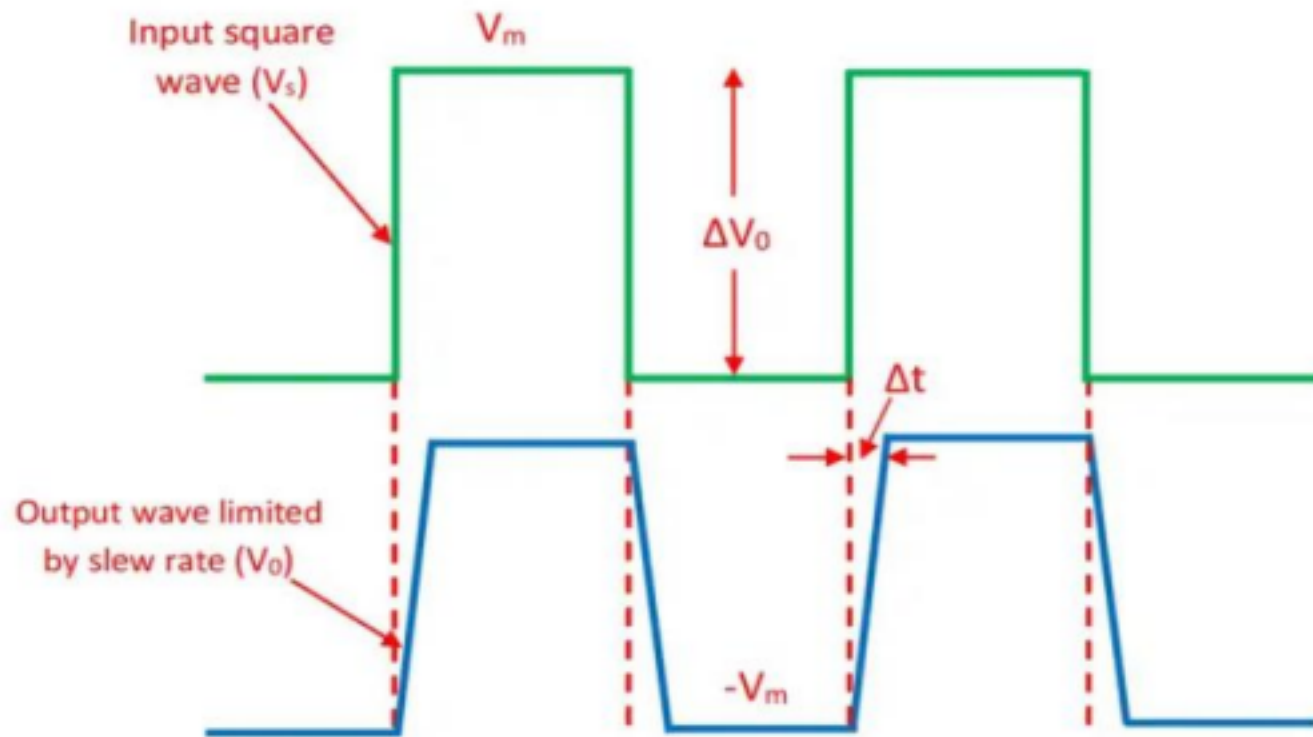
Ideally, the slew rate should be infinite. However, practical devices like the IC 741 op-amp have a slew rate of only about $0.5 \text{ V}/\mu\text{s}$, limiting its use in high-frequency applications.



The circuit used for slew rate measurement

DC Circuit

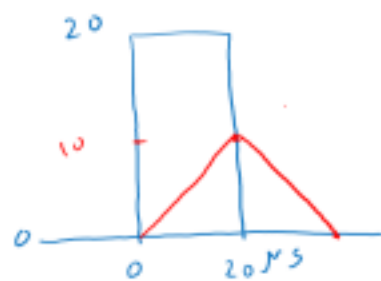
The input and slew limited output voltage waveform are shown in the figure below.



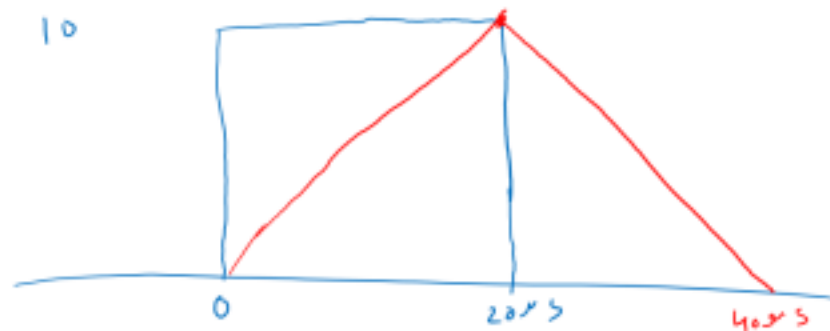
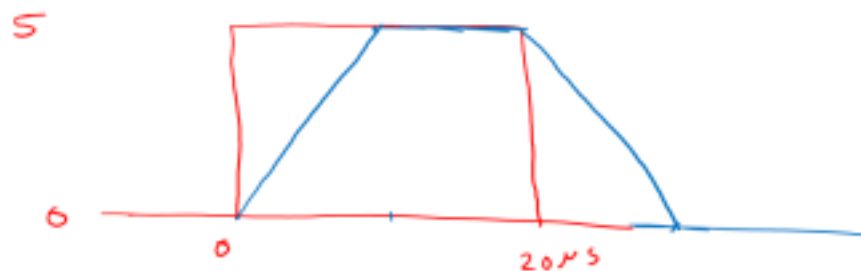
DC Circuit

Effect of s.r. on pulse

$$S.R. = 0.5 \text{ V}/\mu\text{s}$$



$$\frac{20}{0.5} = 40 \mu\text{s}$$

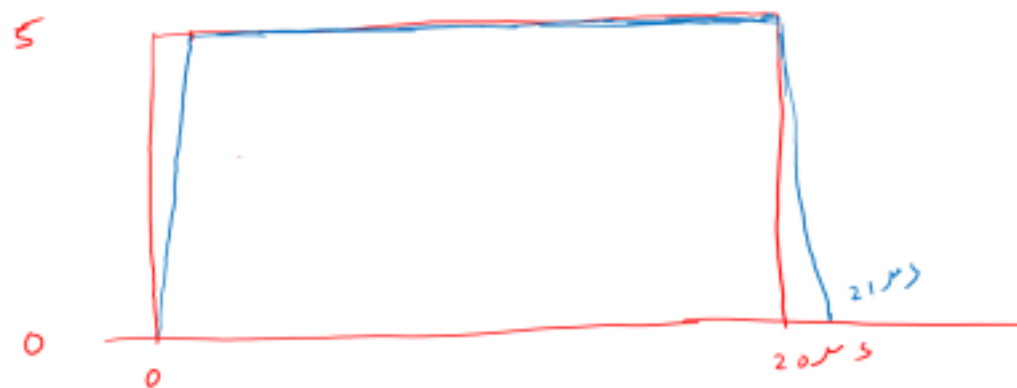


$$\frac{5}{0.5} = 10 \mu\text{s}$$

$$S.R. = 0.5 \text{ V}/\mu\text{s}$$

$$\frac{10}{0.5} = 20 \mu\text{s}$$

DC Circuit



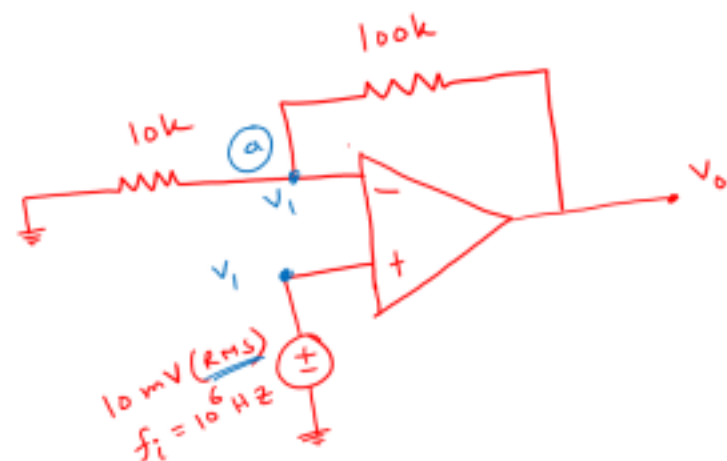
$$S.R. = 5V/\mu s$$

$$\frac{5}{5} = 1 \mu s$$

S.R. $\rightarrow$ important

DC Circuit

#



$$S.R. = 0.5V/\mu s$$

Determine whether V_o will be distorted?

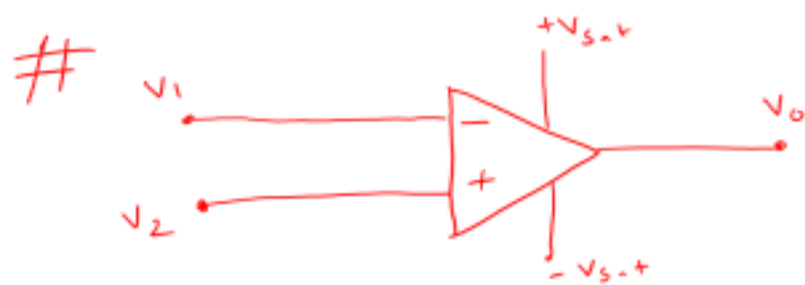
$$\begin{aligned} \textcircled{a} \Rightarrow \frac{V_i}{10k} + \frac{V_i - V_o}{100k} &= 0 \\ \Rightarrow \frac{V_o}{100k} &= V_i \left(\frac{1}{10k} + \frac{1}{100k} \right) \\ \Rightarrow V_o &= 11 V_i \\ &= 11 \times 10 \times 10^{-3} \times \sqrt{2} \\ &= 0.11 \times \sqrt{2} \end{aligned}$$

$$\begin{aligned} f_{max} &= \frac{S.R.}{2\pi V_m} \\ &= \frac{0.5 \times 10^6}{2\pi \times 0.11 \times \sqrt{2}} \\ &= 0.52 \times 10^6 \text{ Hz} \end{aligned}$$

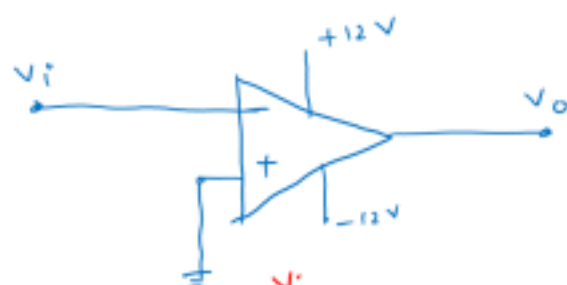
$$f_i > f_{max}$$

distorted signal

DC Circuit

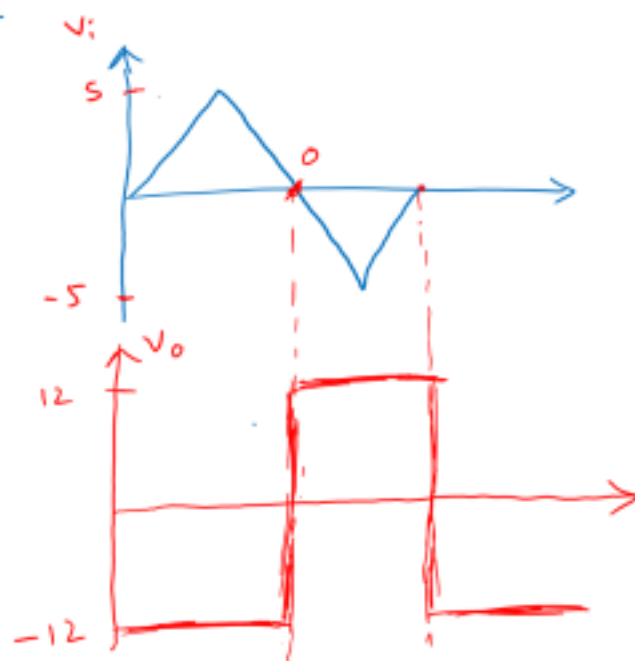


$$\begin{cases} V_1 > V_2 \Rightarrow V_0 = -V_{sat} \\ V_1 < V_2 \Rightarrow V_0 = +V_{sat} \end{cases}$$



$$V_i > 0 \Rightarrow V_0 = -12$$

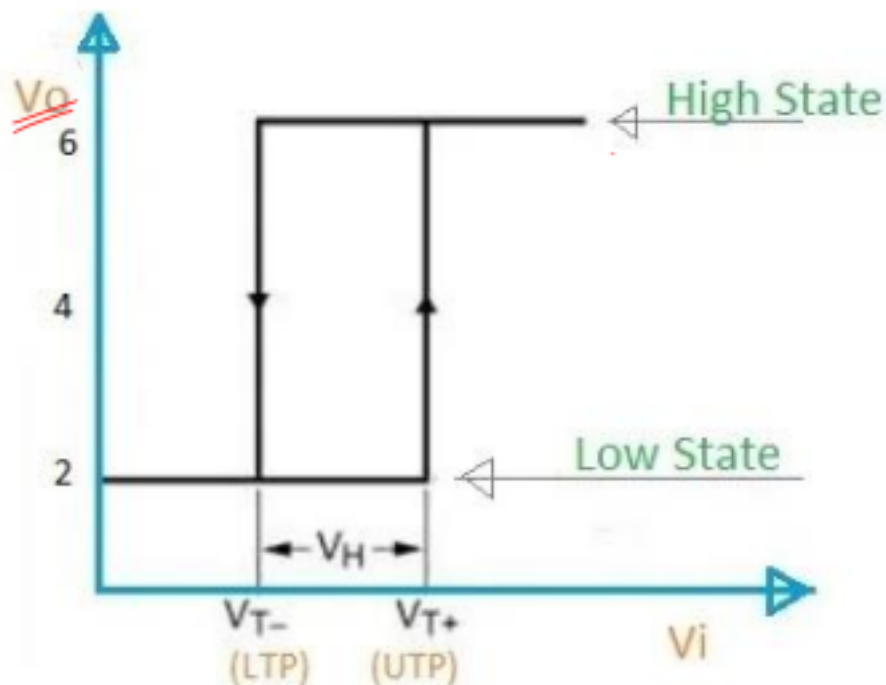
$$V_i < 0 \Rightarrow V_0 = +12$$



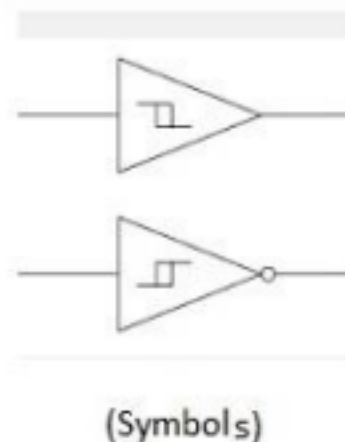
DC Circuit

→ The value of V_i triggers the schmitt trigger circuit and V_o will jump from "LOW" to "HIGH" state. It is called **UTP** or **Upper Trip Point**.

While value of V_i which causes jump from "HIGH" to "LOW" state is referred as **LTP** or **Lower Trip Point**.



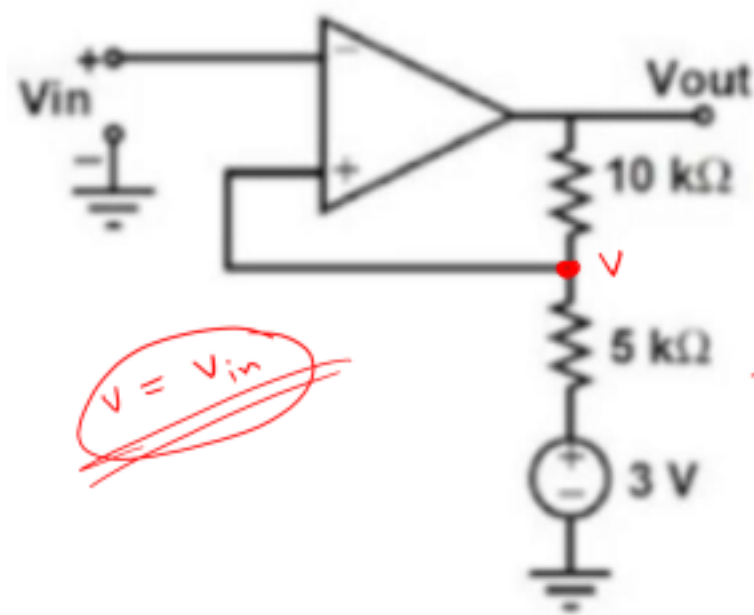
$V_H \rightarrow$ Hysteresis Range
(Hysteresis Curve)



Schmitt trigger

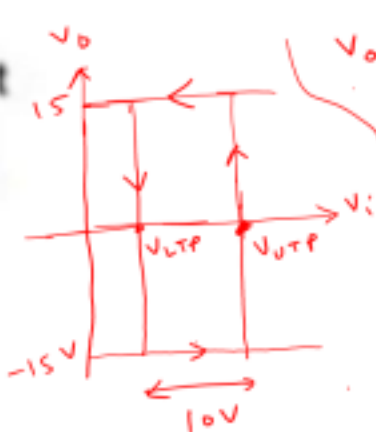
DC Circuit

For the operational amplifier circuit shown, the output saturation voltages are $\pm 15\text{ V}$. Find the upper and lower threshold voltages for the circuit



$$V = V_{in}$$

$$V_{sat} = \pm 15\text{ V}$$



$$\frac{V - V_{out}}{10} + \frac{V - 3}{5} = 0$$

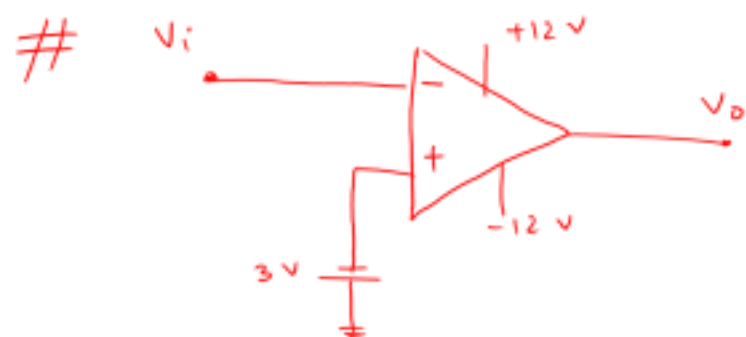
$$\Rightarrow \frac{V}{10} + \frac{V}{5} = \frac{V_{out}}{10} + \frac{3}{5}$$

$$\Rightarrow V_{in} = \frac{V_{out}}{3} + 2$$

$$V_{out} = +15\text{ V} \Rightarrow V_{UTP} = \frac{15}{3} + 2 = 7\text{ V}$$

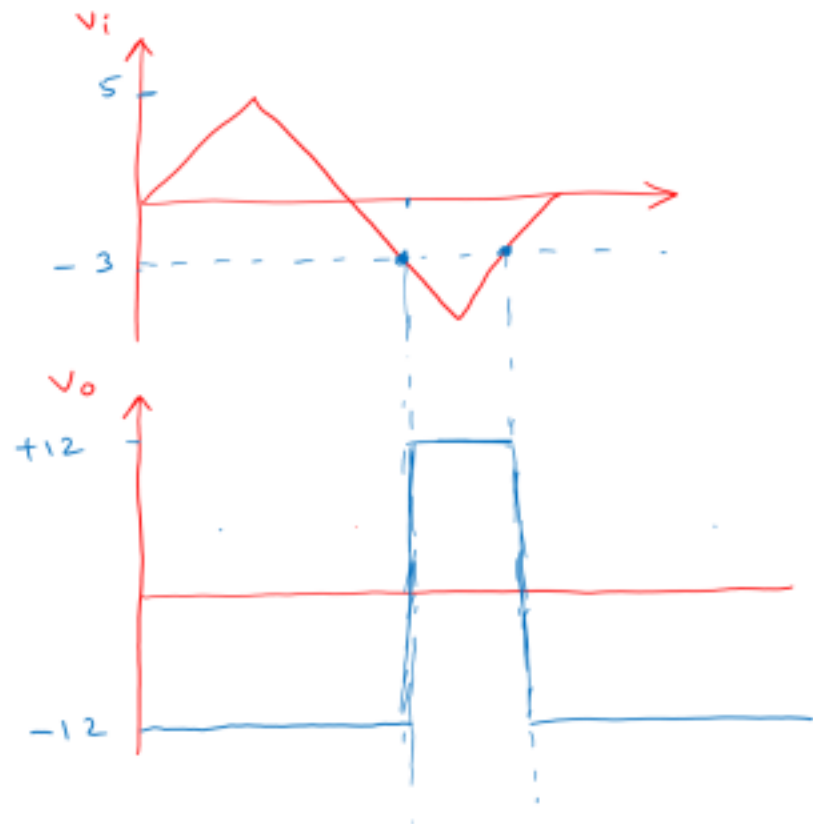
$$V_{out} = -15\text{ V} \Rightarrow V_{LTP} = \frac{-15}{3} + 2 = -3\text{ V}$$

DC Circuit

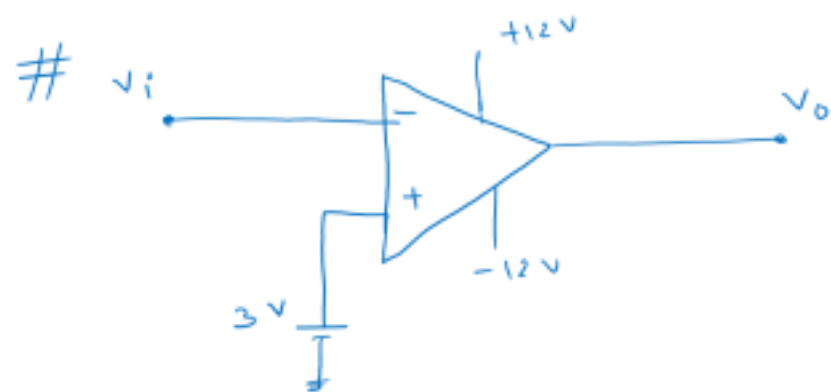


$$V_i > -3 \Rightarrow V_o = -12 \text{ V}$$

$$V_i < -3 \Rightarrow V_o = +12 \text{ V}$$

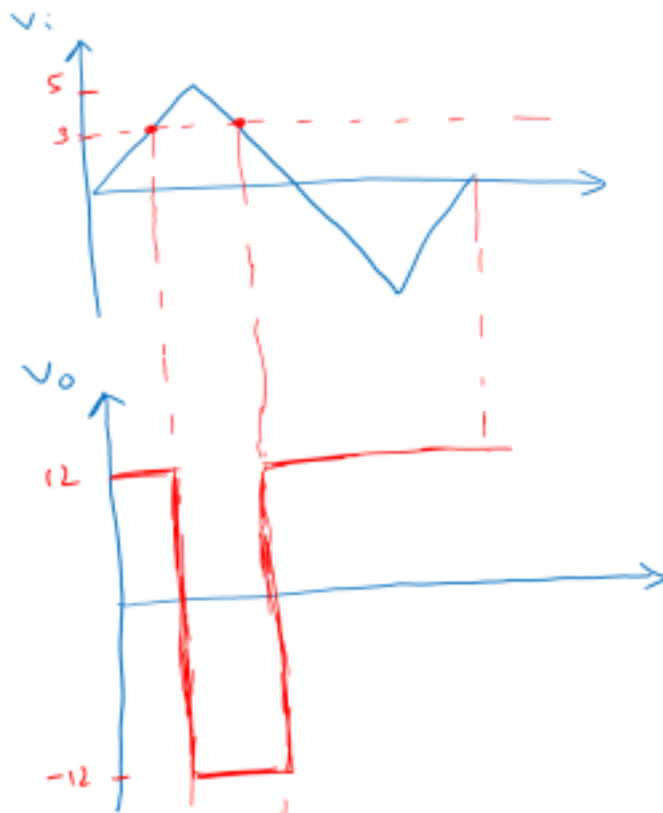


DC Circuit



$$v_i > 3 \Rightarrow v_o = -12 \text{ V}$$

$$v_i < 3 \Rightarrow v_o = +12 \text{ V}$$

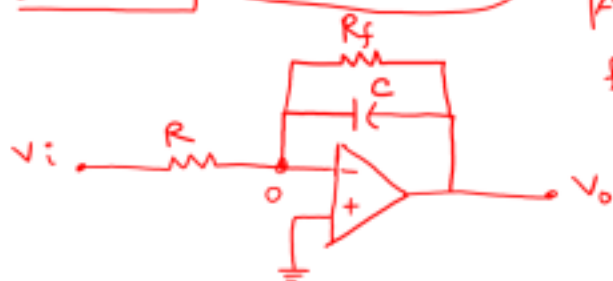


DC Circuit

DNCC'20] Active filter

$$|A_v| = ?$$

$$f_c = ?$$



$$\frac{V_i}{R} + \frac{V_o}{R_f} + \frac{V_o}{1/j\omega C} = 0$$

$$\Rightarrow V_o \left(\frac{1}{R_f} + j\omega C \right) = - \frac{V_i}{R}$$

$$\Rightarrow \frac{V_o}{V_i} = A_v = H(\omega) = - \frac{R_f}{R} \cdot \frac{1}{1 + j\omega C R_f}$$

$$|A_v| = \frac{R_f}{R} \cdot \frac{1}{\sqrt{1 + (\omega C R_f)^2}} = H(\omega)$$

$$H(0) = \left(\frac{R_f}{R} \right)$$

$$H(\infty) = 0$$



$$\text{at cut-off } |H(\omega_c)| = \frac{1}{\sqrt{2}} \cdot \left(\frac{R_f}{R} \right) = \frac{R_f/R}{\sqrt{1 + (\omega_c C R_f)^2}}$$

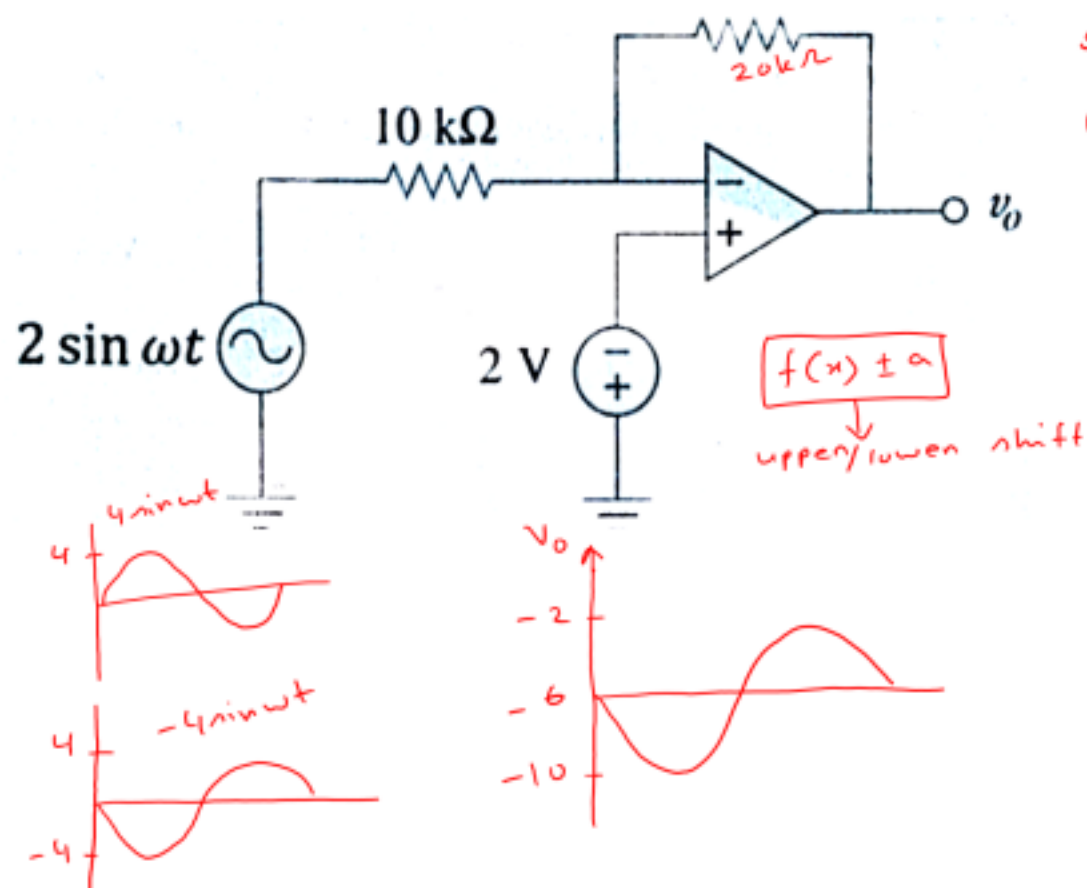
$$\Rightarrow 2 = 1 + \omega_c^2 C^2 R_f^2$$

$$\Rightarrow \omega_c^2 C^2 R_f^2 = 1$$

$$\Rightarrow \omega_c = \frac{1}{R_f C}$$

$$\Rightarrow f_c = \frac{1}{2\pi R_f C}$$

DC Circuit



Superposition $\Rightarrow$

AC active

$$\frac{2 \sin \omega t}{10k} = \frac{-v_{o1}}{20k}$$

$$\Rightarrow v_{o1} = -4 \sin \omega t$$

-2V active

$$\frac{0 - (-2)}{10k} = \frac{-2 - v_{o2}}{20k}$$

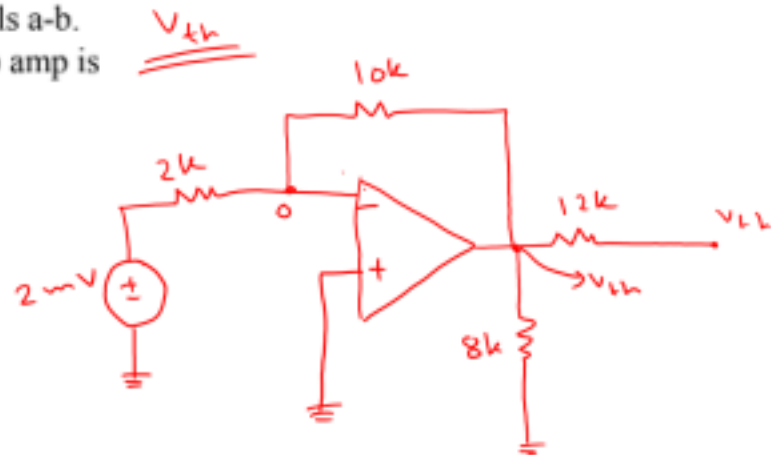
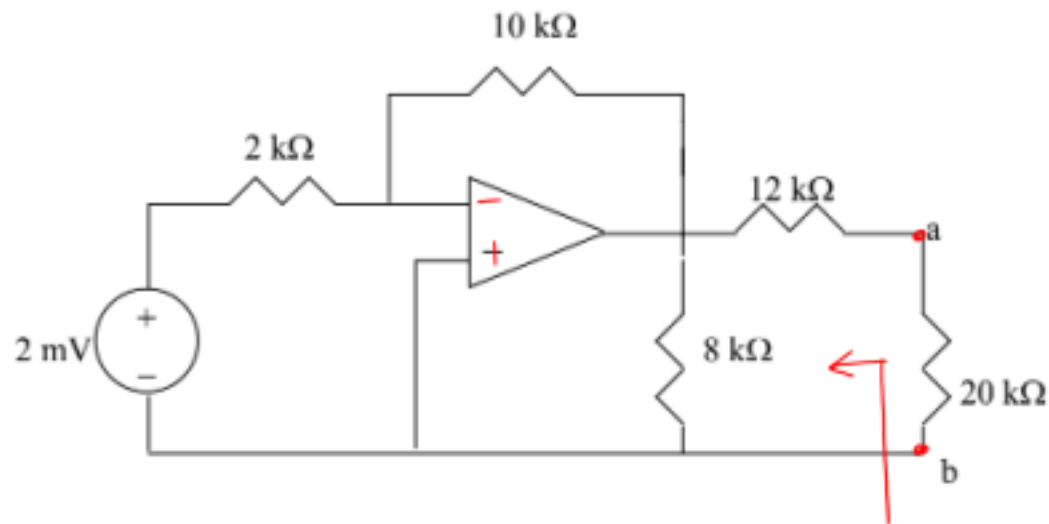
$$\Rightarrow 4 = -2 - v_{o2}$$

$$\Rightarrow v_{o2} = -6$$

$$v_o = v_{o1} + v_{o2} = -4 \sin \omega t - 6$$

DC Circuit

For the circuit in Fig. 5.57, find the Thevenin equivalent to the left of terminals a-b. Then calculate the power absorbed by the 20-k Ω resistor. Assume that the op amp is ideal.

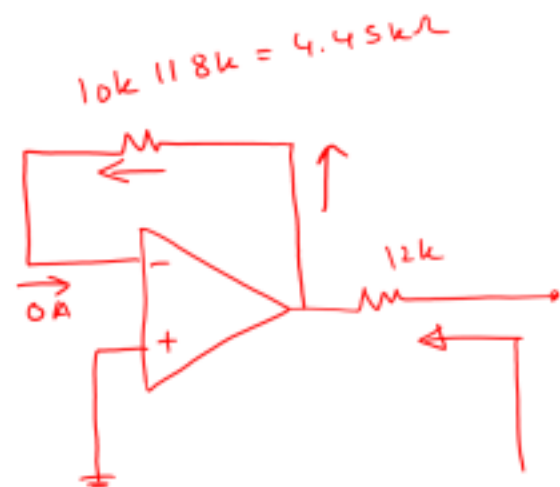
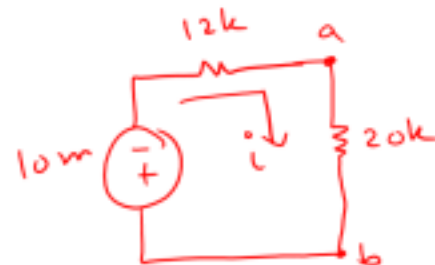
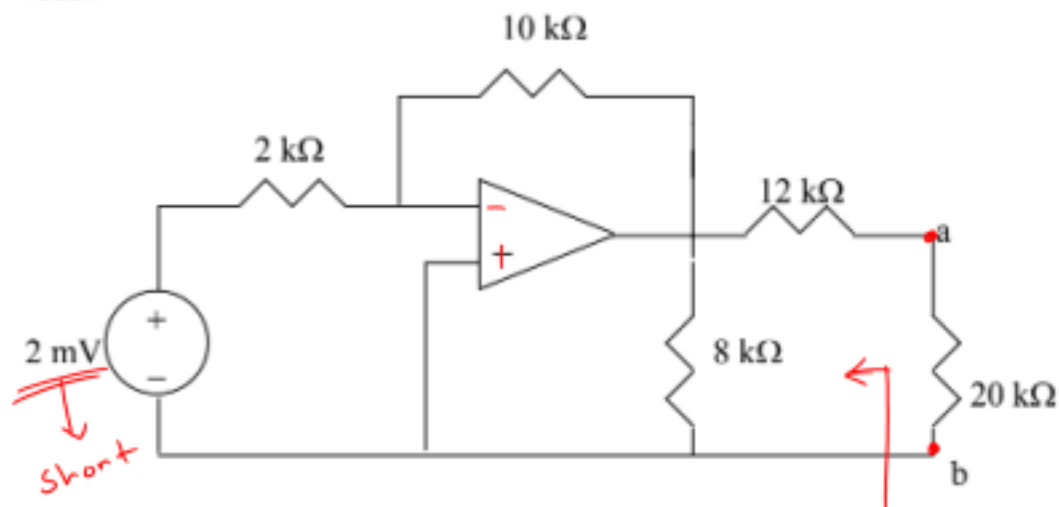


$$\frac{2\text{m} - 0}{2\text{k}} = \frac{0 - V_{th}}{10\text{k}}$$

$$\Rightarrow \boxed{V_{th} = -10\text{mV}}$$

DC Circuit

For the circuit in Fig. 5.57, find the Thevenin equivalent to the left of terminals a-b. Then calculate the power absorbed by the 20-k Ω resistor. Assume that the op amp is ideal.



$$R_{th} = 12 \text{ k}\Omega$$

$$i = \frac{-10 \text{ mV}}{12 \text{ k}\Omega + 20 \text{ k}\Omega}$$

$$P_{20} = i^2 \times 20 \text{ k}\Omega = \boxed{?}$$

DC Circuit

Common-Mode Rejection Ratio, CMRR

The accepted definition of CMRR for an op-amp is the open-loop voltage gain (A_{ol}) divided by the common-mode gain (A_{cm}).

$$\text{CMRR} = \frac{A_{ol}}{A_{cm}}$$

CMRR is commonly expressed in decibels as follows:

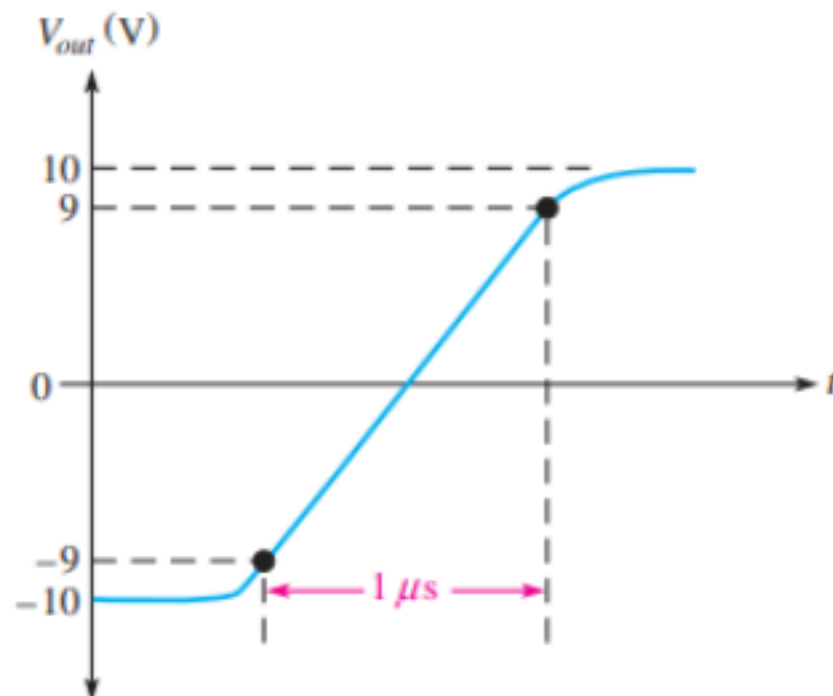
$$\text{CMRR} = 20 \log \left(\frac{A_{ol}}{A_{cm}} \right)$$

DC Circuit

The output voltage of a certain op-amp appears as shown in Figure below in response to a step input. Determine the slew rate

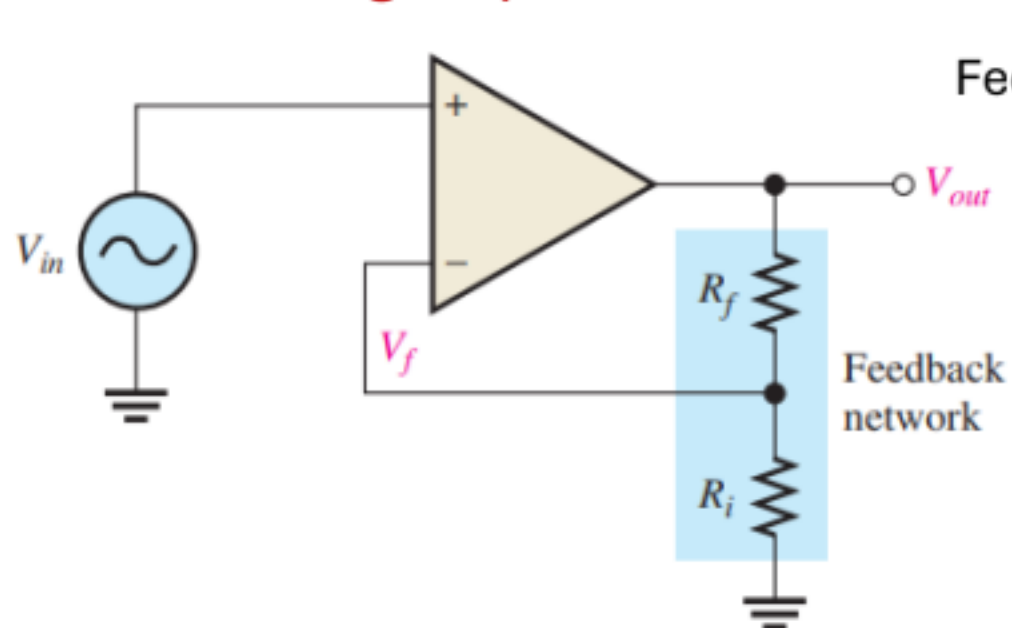
The output goes from the lower to the upper limit in $1\ \mu\text{s}$. Since this response is not ideal, the limits are taken at the 90% points, as indicated. Thus, the upper limit is $+9\ \text{V}$ and the lower limit is $-9\ \text{V}$. The slew rate is

$$\text{Slew rate} = \frac{\Delta V_{out}}{\Delta t} = \frac{+9\ \text{V} - (-9\ \text{V})}{1\ \mu\text{s}} = 18\ \text{V}/\mu\text{s}$$



DC Circuit

Noninverting Amplifier



Feedback voltage $V_f = \left(\frac{R_i}{R_i + R_f} \right) V_{out} = BV_{out}$

$$V_{diff} = V_{in} - V_f$$

This input differential voltage is forced to be very small as a result of the negative feedback and the high open-loop gain, A_{ol} . Therefore, a close approximation is $V_{in} \cong V_f$

DC Circuit

By substitution,

$$V_{in} \cong BV_{out}$$

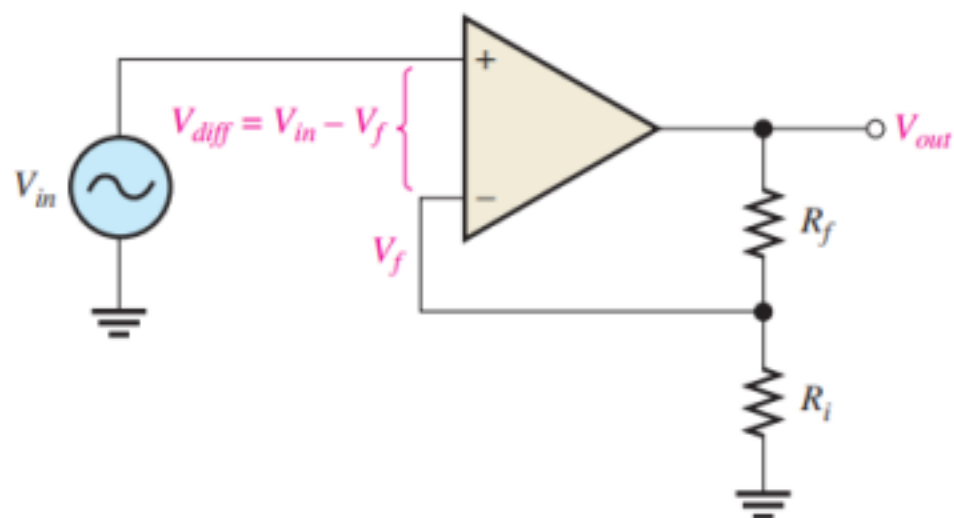
Rearranging,

$$\frac{V_{out}}{V_{in}} \cong \frac{1}{B}$$

The ratio of the output voltage to the input voltage is the closed-loop gain. This result shows that the closed-loop gain for the noninverting amplifier, $A_{cl(NI)}$, is approximately

$$A_{cl(NI)} = \frac{V_{out}}{V_{in}} \cong \frac{1}{B}$$

DC Circuit



$$V_{in} \cong BV_{out} \cong \left(\frac{R_i}{R_i + R_f} \right) V_{out}$$

$$\frac{V_{out}}{V_{in}} = \left(\frac{R_i + R_f}{R_i} \right)$$

which can be expressed as follows:

$$A_{cl(NI)} = 1 + \frac{R_f}{R_i}$$

DC Circuit

Resistances of a Noninverting Amplifier

$$R_{in(NI)} = (1 + A_{ol}B)R_{in}$$


Internal input resistance of op-amp itself (without feedback)

$$R_{out(NI)} = \frac{R_{out}}{1 + A_{ol}B}$$


internal output resistance

DC Circuit

- (a) Determine the input and output resistances of the amplifier in Figure 27. op-amp data sheet gives $R_{in} = 2 \text{ M}\Omega$, $R_{out} = 75 \text{ }\Omega$, and $A_{ol} = 200,000$.
- (b) Find the closed-loop voltage gain.

(a) The attenuation, B , of the feedback circuit is

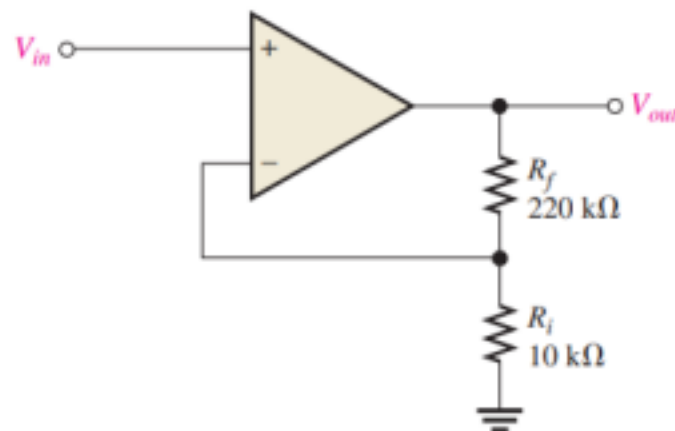
$$B = \frac{R_i}{R_i + R_f} = \frac{10 \text{ k}\Omega}{230 \text{ k}\Omega} = 0.0435$$

$$R_{in(NI)} = (1 + A_{ol}B)R_{in} = [1 + (200,000)(0.0435)](2 \text{ M}\Omega) = \mathbf{17.4 \text{ G}\Omega}$$

$$R_{out(NI)} = \frac{R_{out}}{1 + A_{ol}B} = \frac{75 \text{ }\Omega}{1 + 8700} = \mathbf{8.6 \text{ m}\Omega}$$

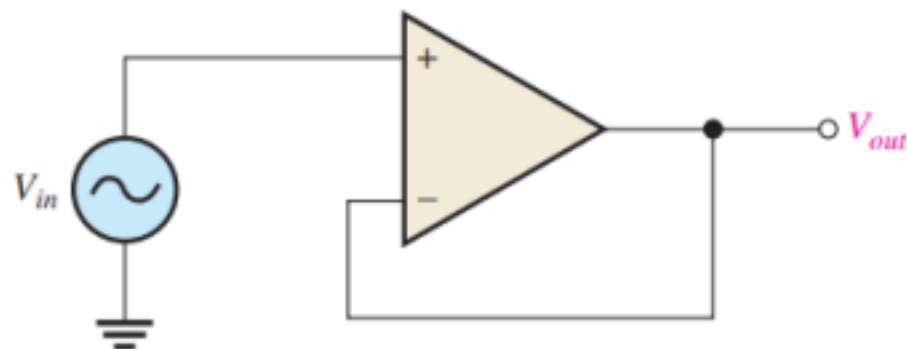
Notice that $R_{in(NI)}$ and $R_{out(NI)}$ are close to ideal values.

(b) $A_{cl(NI)} = \frac{1}{B} = \frac{1}{0.0435} = \mathbf{23.0}$



DC Circuit

Resistances of a Voltage-Follower



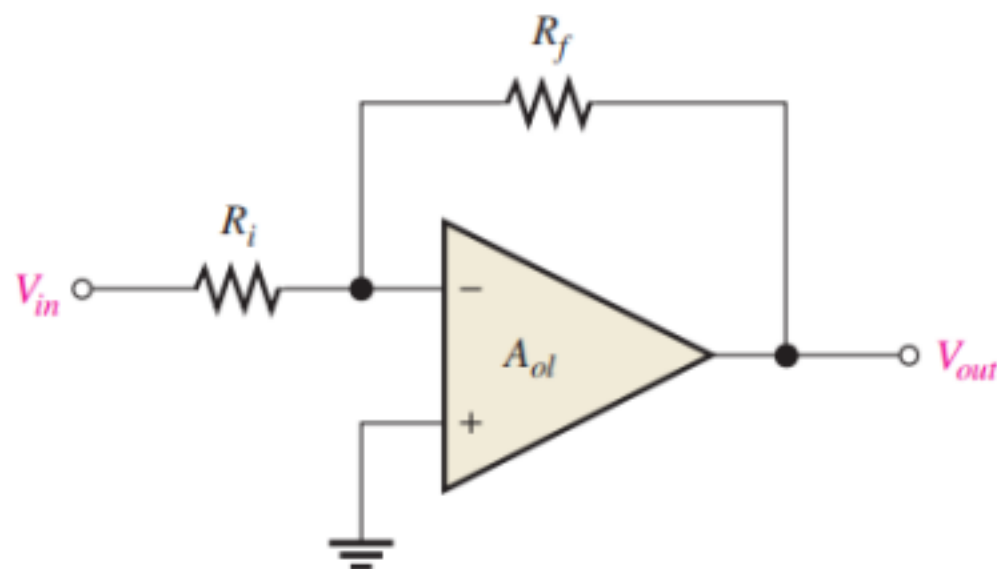
$$R_{in(VF)} = (1 + A_{ol})R_{in}$$

$$R_{out(VF)} = \frac{R_{out}}{1 + A_{ol}}$$

Since the voltage-follower is a special case of the noninverting configuration, the same resistance formulas are used, but with $B = 1$.

DC Circuit

Resistances of an Inverting Amplifier

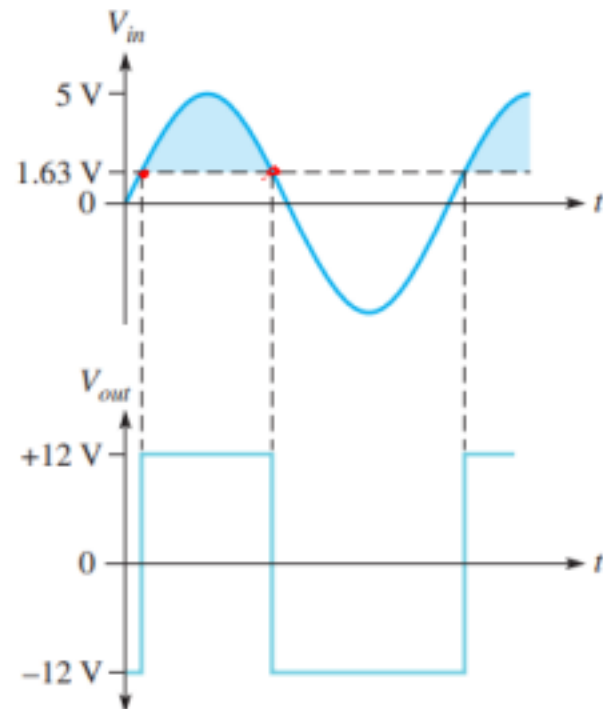
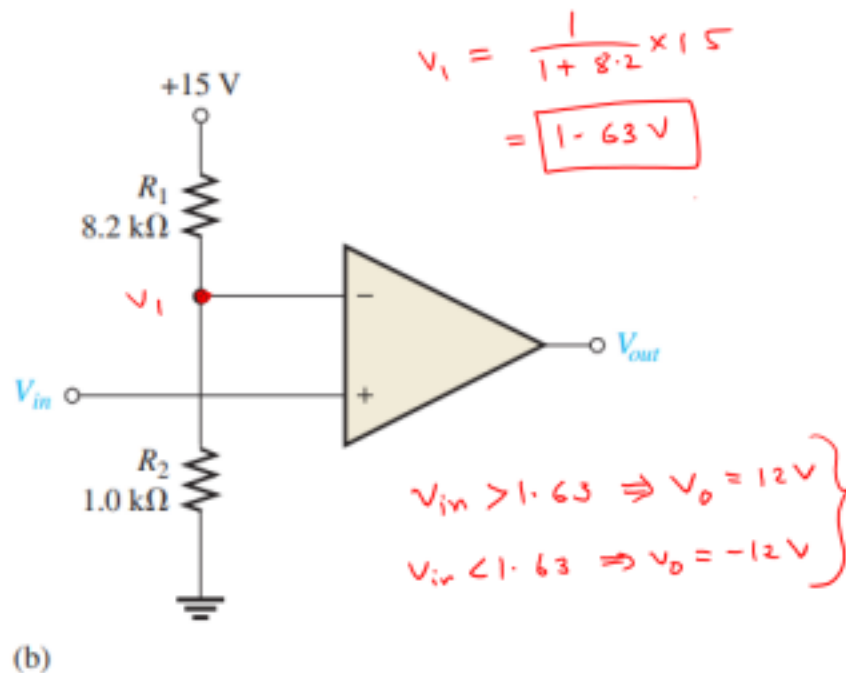


$$R_{in(I)} \cong R_i$$

$$R_{out(I)} = \frac{R_{out}}{1 + A_{ol}B}$$

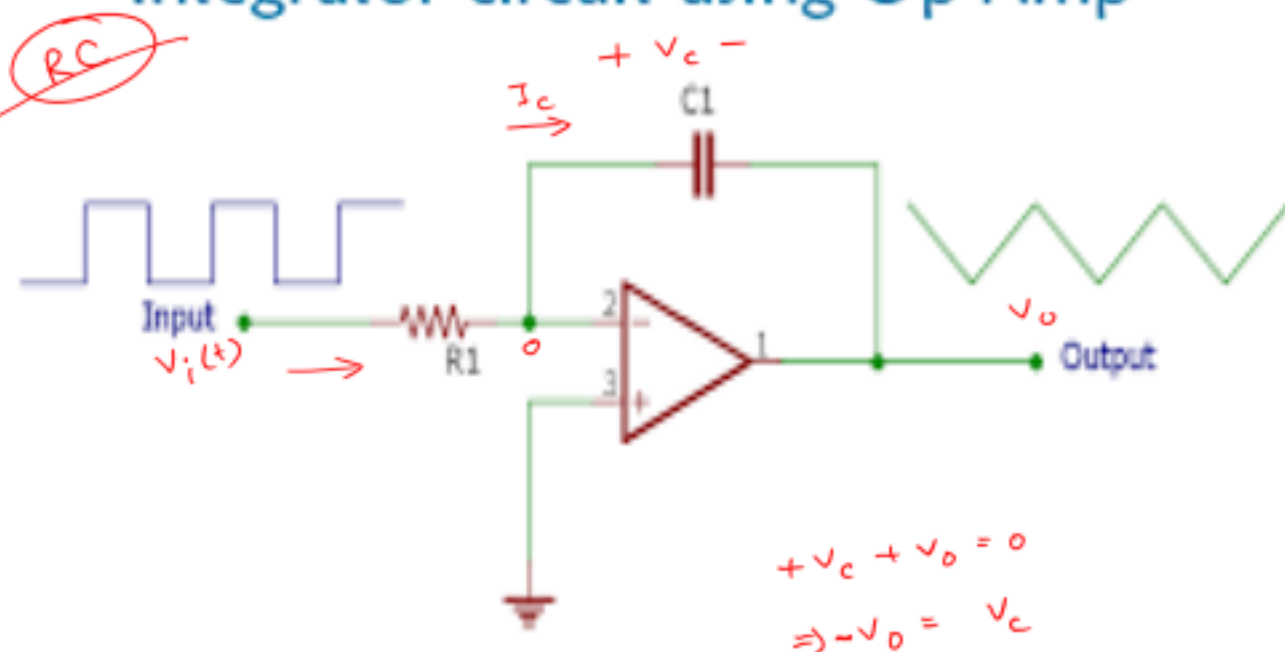
DC Circuit

The input signal in Figure 3(a) is applied to the comparator circuit in part (b). Draw the output showing its proper relationship to the input signal. Assume that the maximum output levels of the op-amp are ± 12 V.



DC Circuit

Integrator circuit using Op-Amp



$$\frac{V_i}{R} = I_c$$

$$\Rightarrow \frac{V_i}{R} = C \frac{dV_c}{dt}$$

$$\Rightarrow \frac{V_i}{R} = -C \frac{dV_o}{dt}$$

$$\Rightarrow -\frac{1}{RC} \int V_i dt = \int dV_o$$

$$\Rightarrow \boxed{V_o = -\frac{1}{RC} \int V_i(t) dt}$$

DC Circuit

Floyd

Rate of Change of the Output The rate at which the capacitor charges, and therefore the slope of the output ramp, is set by the ratio I_C/C , as you have seen. Since $I_C = V_{in}/R_i$, the rate of change or slope of the integrator's output voltage is $\Delta V_{out}/\Delta t$.

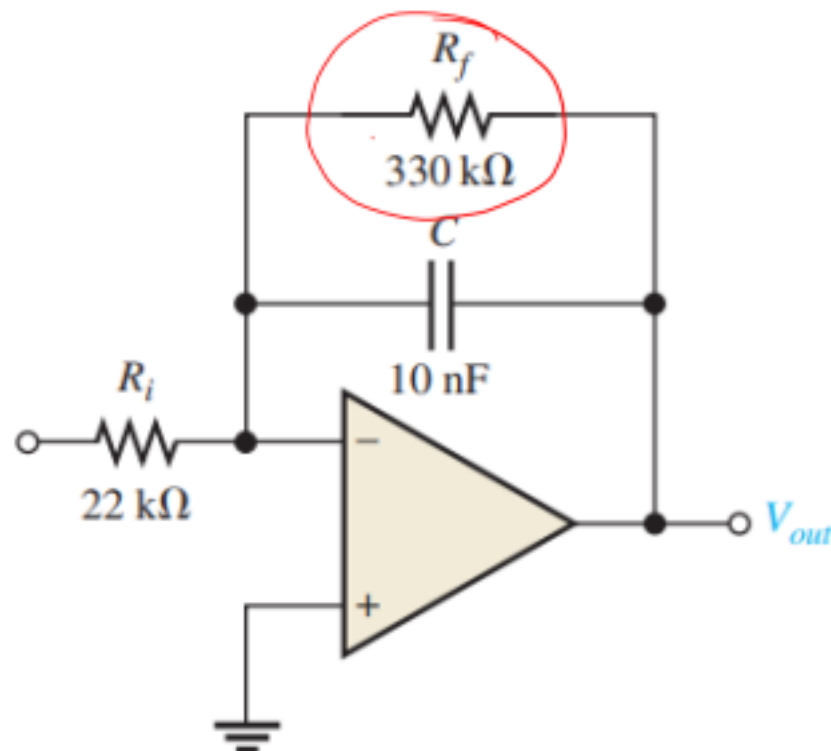
$$\frac{\Delta V_{out}}{\Delta t} = -\frac{V_{in}}{R_i C}$$

DC Circuit

A practical integrator using a feedback resistor to prevent the op-amp from saturating due to small imbalances in the capacitor charging and discharging current.

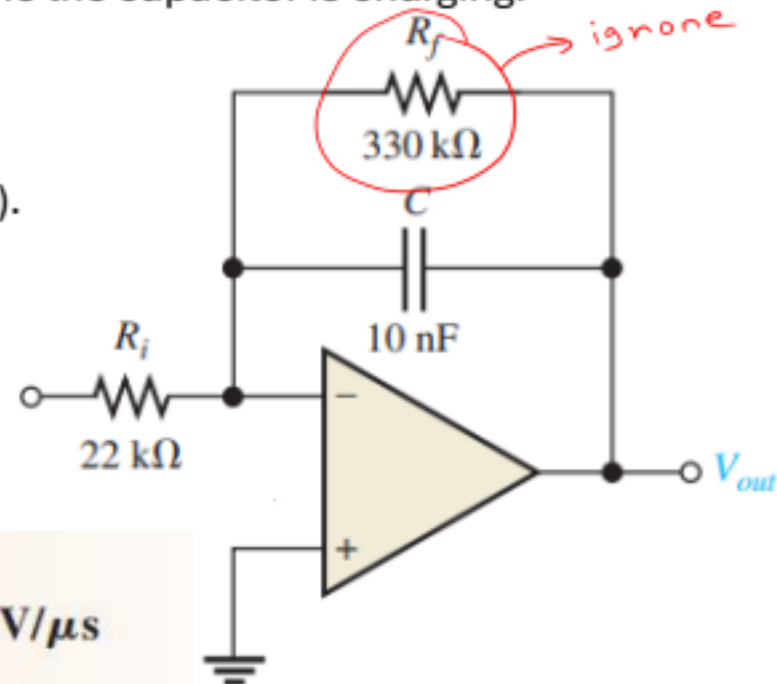
The feedback resistor has only a small effect on the circuit and is selected to have about 7% of the charging current, so its effect on the general circuit operation is negligible.

However, the feedback resistor will force the output to be centered about the zero line after several cycles.



DC Circuit

- (a) Calculate the rate of change of the output voltage during the time the capacitor is charging.
- (b) Calculate the total change in voltage.
- (c) Describe the output and draw the waveform (centered about 0 V).

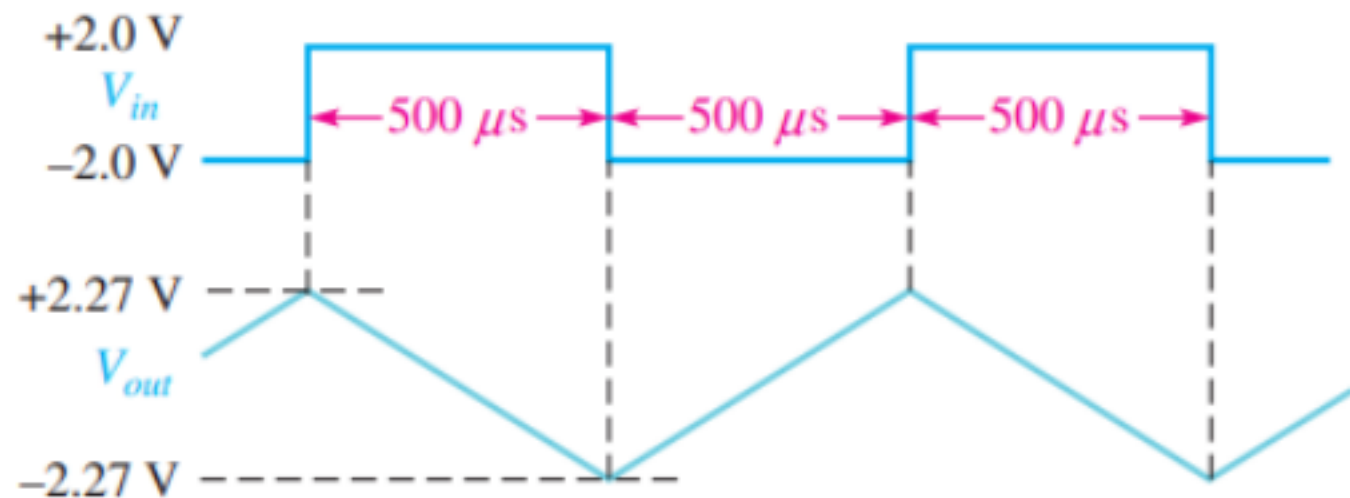


$$(a) \frac{\Delta V_{out}}{\Delta t} = -\frac{V_{in}}{R_i C} = -\frac{2.0 \text{ V}}{(22 \text{ k}\Omega)(10 \text{ nF})} = -9.09 \text{ kV/s} = -9.09 \text{ mV}/\mu\text{s}$$

(b) In $500 \mu\text{s}$, the output will change by $(-9.09 \text{ mV}/\mu\text{s})(500 \mu\text{s}) = 4.54 \text{ V}$, producing a negative-going ramp.

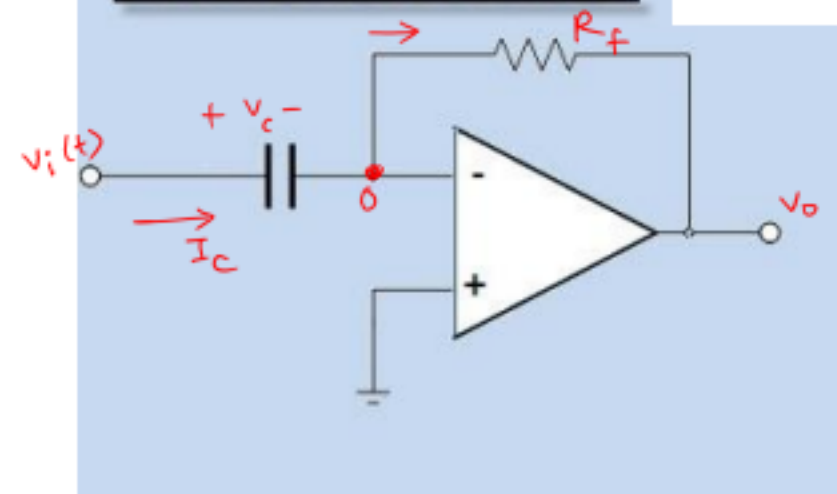
DC Circuit

- (c) When the capacitor discharges, the rate is the same except it produces a positive-going ramp. Because the output is centered about 0 V (after a few cycles), the output will vary from a minimum of $-\frac{1}{2}(4.54 \text{ V}) = -2.27 \text{ V}$ to a maximum of $\frac{1}{2}(4.54 \text{ V}) = +2.27 \text{ V}$. The waveform is shown in Figure



DC Circuit

Differentiator



$$I_c = \frac{-v_o}{R_f}$$

$$\Rightarrow C \frac{dv_c}{dt} = \frac{-v_o}{R_f}$$

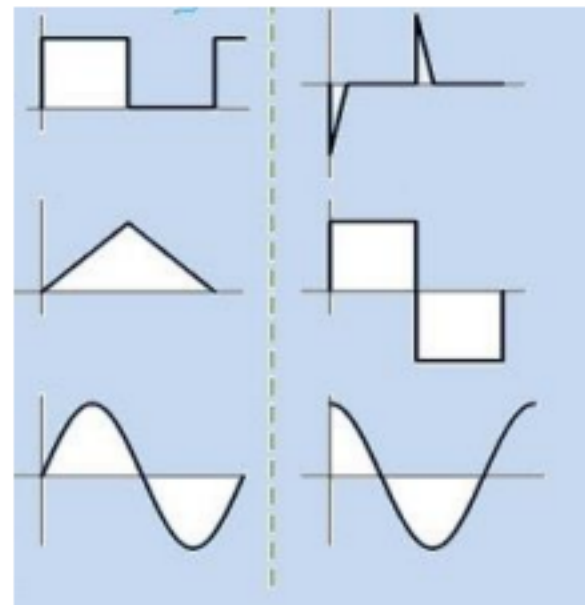
$$\Rightarrow v_o = -R_f C \frac{dv_c}{dt}$$

$$\Rightarrow v_o = -R_f C \frac{dv_i}{dt}$$

$$v_i = v_c$$



$$V_{out} = -\left(\frac{V_c}{t}\right) R_f C$$

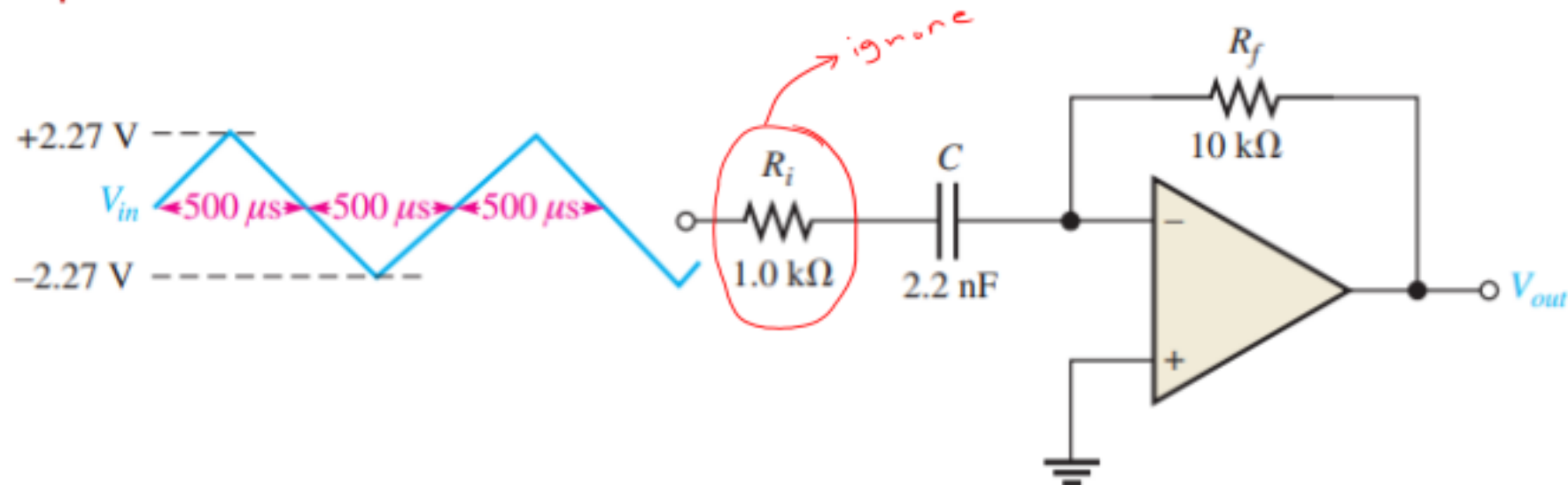


Signal

DC Circuit

A practical differentiator circuit with a series resistor on the input to limit high frequency gain (reducing noise and ringing). This resistor has a negligible effect on the general circuit operation.

Draw the output waveform.



DC Circuit

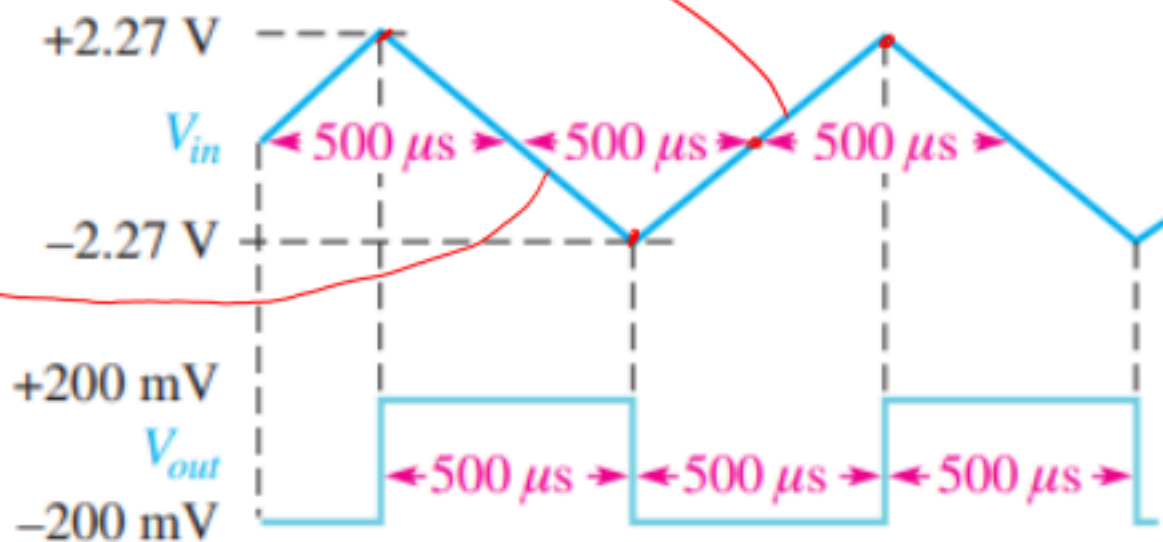
$$V_o = -R_f C \left(\frac{V_c}{t} \right)$$

$$= -10 \times 10^3 \times 2.2 \times 10^{-9} \left(\frac{2.27 - (-2.27)}{500 \times 10^{-6}} \right)$$

$$= -0.2 \text{ V}$$

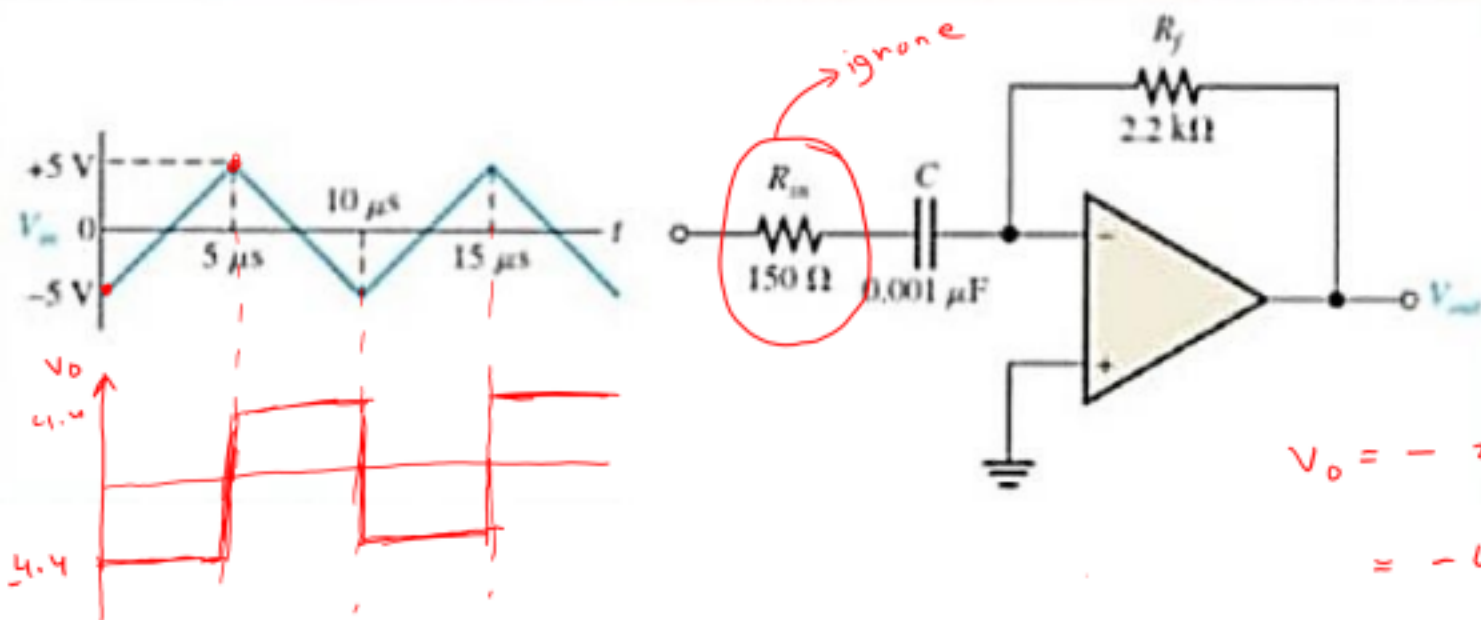
$$V_o = -R_f C \left(\frac{-2.27 - 2.27}{500 \times 10^{-6}} \right)$$

$$= 0.2 \text{ V}$$



DC Circuit

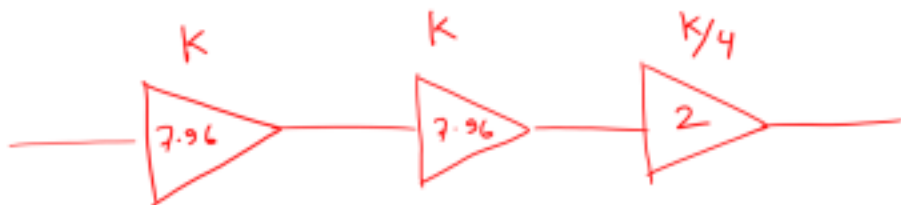
Determine the output voltage of the practical op-amp differentiator in Figure 13-42 for the triangular-wave input shown. The input resistor can be ignored as it is small compared to R_f .



$$V_o = -2.2 \times 10^3 \times 0.001 \times 10^{-6} \times \frac{5 - (-5)}{5 \times 10^{-6}} \\ = -4.4 \text{ V}$$

DC Circuit

In a certain electronic device, a three-stage amplifier is desired, whose overall voltage gain is 42 dB. The individual voltage gains of the first two stages are to be equal, while the gain of the third is to be one-fourth of each of the first two. Calculate the voltage gain of each.



$$20 \log \left(\frac{K^3}{4} \right) = 42$$

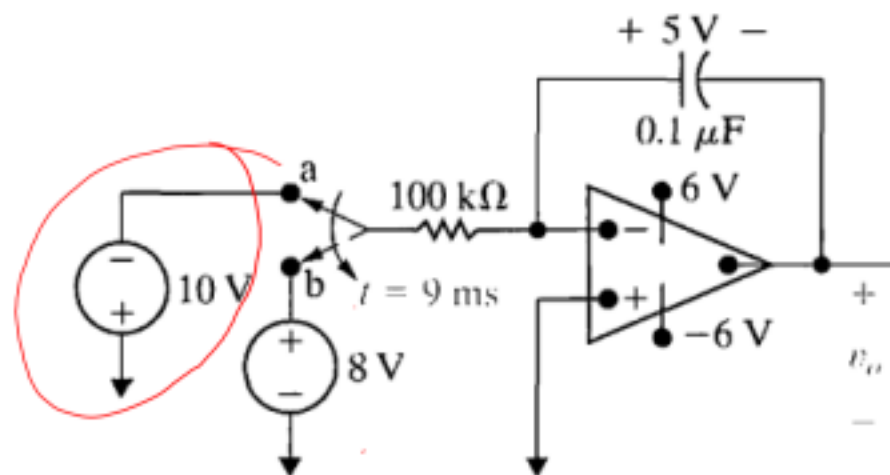
$$\Rightarrow \frac{K^3}{4} = 10^{2.1}$$

$$\Rightarrow K = \sqrt[3]{4 \times 10^{2.1}} = \boxed{7.96}$$

DC Circuit

At the instant the switch makes contact with terminal a in the circuit shown in Fig. 7.44, the voltage on the $0.1 \mu\text{F}$ capacitor is 5 V . The switch remains at terminal a for 9 ms and then moves instantaneously to terminal b. How many milliseconds after making contact with terminal b does the operational amplifier saturate?

$$t < 0 \Rightarrow 0 + 5 + v_o = 0 \Rightarrow v_o = -5$$



$$v_o = - \frac{(-10)}{100 \times 10^3 \times 0.1 \times 10^{-6}} \int_0^t dt - 5$$

$$= 1000t - 5$$

$$\text{at } t = 9 \text{ ms} \Rightarrow v_o = 1000 \times 9 \times 10^{-3} - 5 = \boxed{4 \text{ V}}$$

$$v_o = 4 - \frac{8}{10^{-2}} \int_{9 \times 10^{-3}}^t dt$$

$$\Rightarrow -6 = 4 - 800(t - 9 \times 10^{-3})$$

$$\Rightarrow \boxed{t = 21.5 \text{ ms}}$$

$$21.5 - 9 = \boxed{12.5 \text{ ms}}$$

Capacitors & Inductors

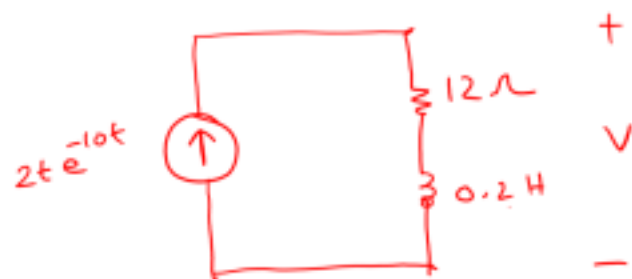
DC Circuit

Relation	Resistor (R)	Capacitor (C)	Inductor (L)
$v-i$:	$v = iR$	$v = \frac{1}{C} \int_{t_0}^t i dt + v(t_0)$	$v = L \frac{di}{dt}$
$i-v$:	$i = v/R$	$i = C \frac{dv}{dt}$	$i = \frac{1}{L} \int_{t_0}^t v dt + i(t_0)$
p or w :	$p = i^2 R = \frac{v^2}{R}$	$w = \frac{1}{2} C v^2$	$w = \frac{1}{2} L i^2$
Series:	$R_{eq} = R_1 + R_2$	$C_{eq} = \frac{C_1 C_2}{C_1 + C_2}$	$L_{eq} = L_1 + L_2$
Parallel:	$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$	$C_{eq} = C_1 + C_2$	$L_{eq} = \frac{L_1 L_2}{L_1 + L_2}$
At dc:	Same	Open circuit	Short circuit

DC Circuit

An electric motor can be modeled as a series combination of a $12\text{-}\Omega$ resistor and 200-mH inductor. If a current $i(t) = 2te^{-10t}$ A flows through the series combination, find the voltage across the combination.

$$\begin{aligned} V &= 24te^{-10t} + 0.2 \frac{d}{dt} (2te^{-10t}) \\ &= 24te^{-10t} + 0.4 (-10te^{-10t} + e^{-10t}) \\ &= (20te^{-10t} + 0.4e^{-10t}) \text{ V} \end{aligned}$$



DC Circuit

Design an analog computer circuit to solve the differential equation:

$$\frac{d^2v_o}{dt^2} + 2\frac{dv_o}{dt} + v_o = 10 \sin 4t, \quad t > 0$$

subject to $v_o(0) = -4$, $v_o'(0) = 1$, where the prime refers to the time derivative.

We first solve for the second derivative as

$$\frac{d^2v_o}{dt^2} = 10 \sin 4t - 2\frac{dv_o}{dt} - v_o$$

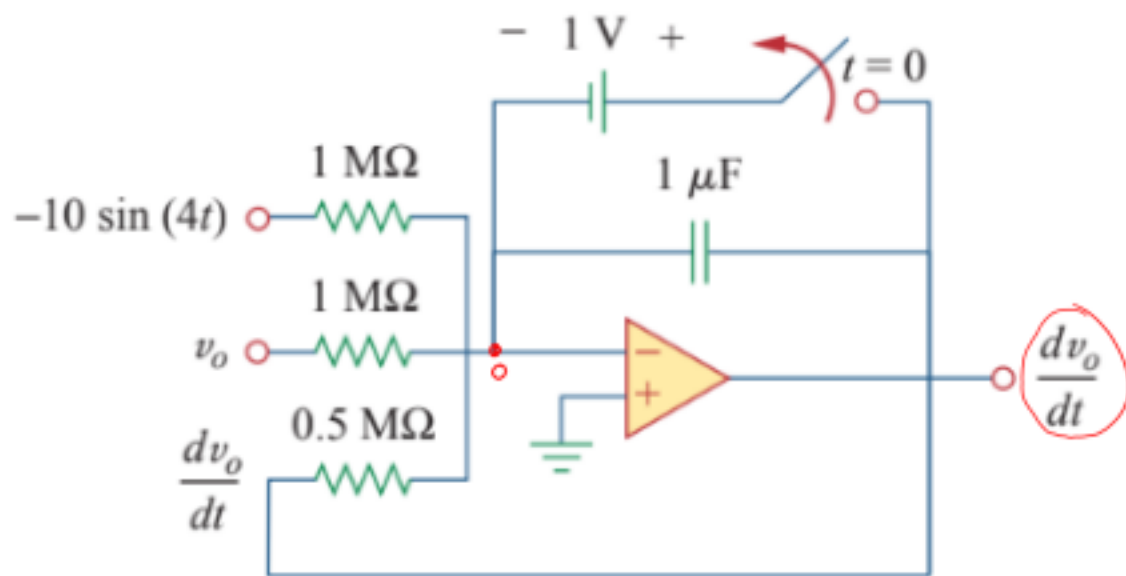
DC Circuit

$$\frac{dv_o}{dt} = - \int_0^t \left(-10 \sin 4t + 2 \frac{dv_o}{dt} + v_o \right) dt + v'_o(0)$$

$$\begin{aligned} 0 - 1 + v'_o(0) &= 0 \\ \Rightarrow v'_o(0) &= 1 \\ \Rightarrow v'_o(0) &= 1 \end{aligned}$$

where $v'_o(0) = 1$.

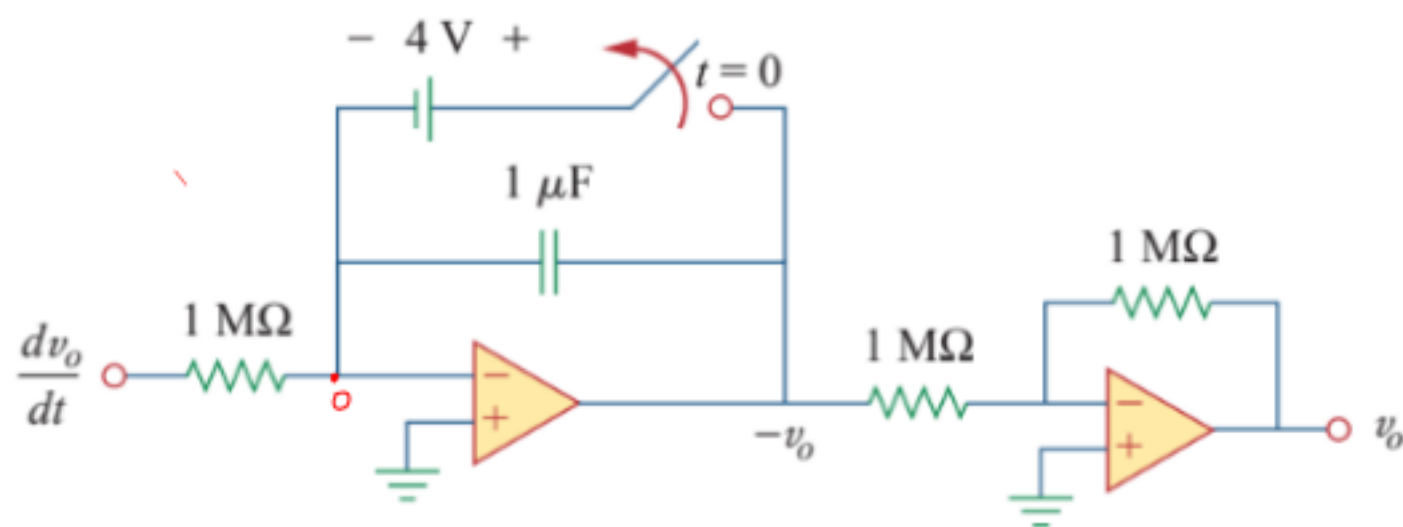
$$\frac{1}{0.5 \times 1} = 2$$



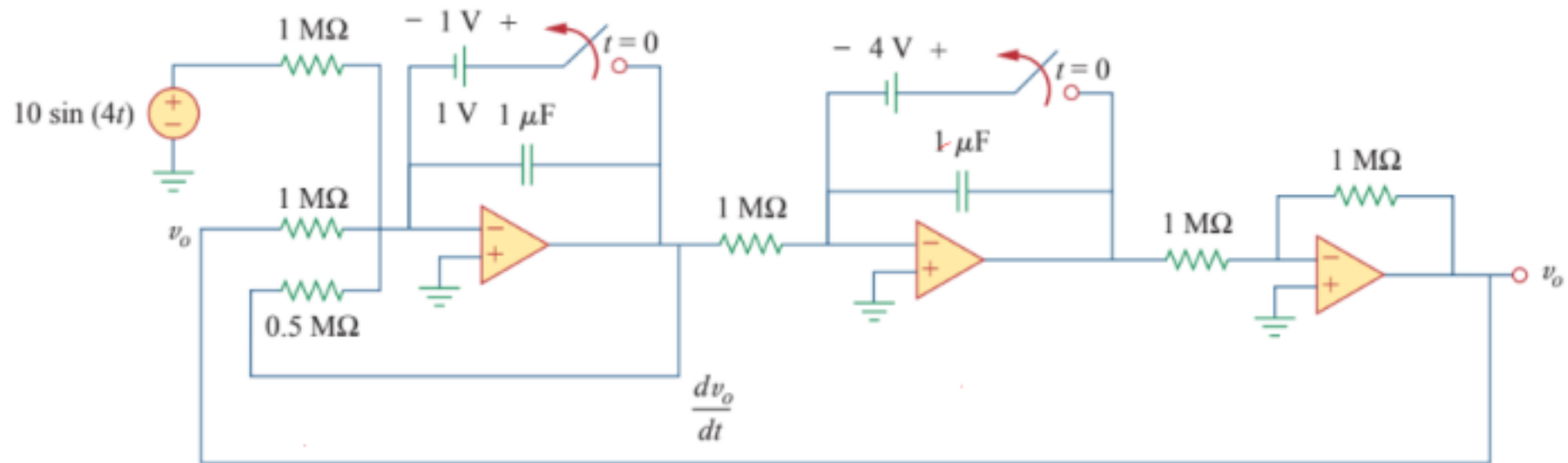
DC Circuit

$$-v_o = -\left(\int_0^t \frac{dv_o}{dt} dt + v_o(0)\right)$$

$$\begin{aligned} 0 - 4 - v_o &= 0 \\ \Rightarrow v_o(0) &= -4 \end{aligned}$$



DC Circuit



First-Order Circuit

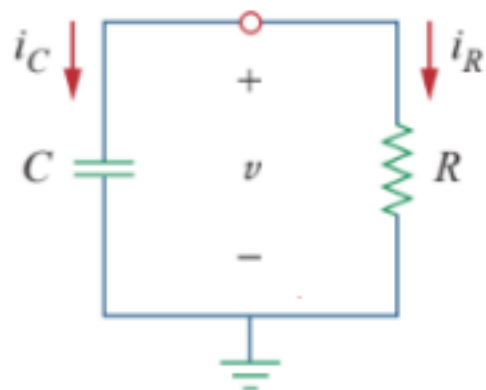
DC Circuit

A first-order circuit is characterized by a first-order differential equation.

Two types of first-order circuits : RC and RL

Two ways to excite the circuits:

- 1) In source-free circuits, we assume that energy is initially stored in the capacitive or inductive element. The energy causes current to flow in the circuit and is gradually dissipated in the resistors. Although source-free circuits are by definition free of independent sources, they may have dependent sources.
- 2) The second way of exciting first-order circuits is by independent sources.



DC Circuit

Source-Free RC Circuit

at time $t = 0$, the initial voltage

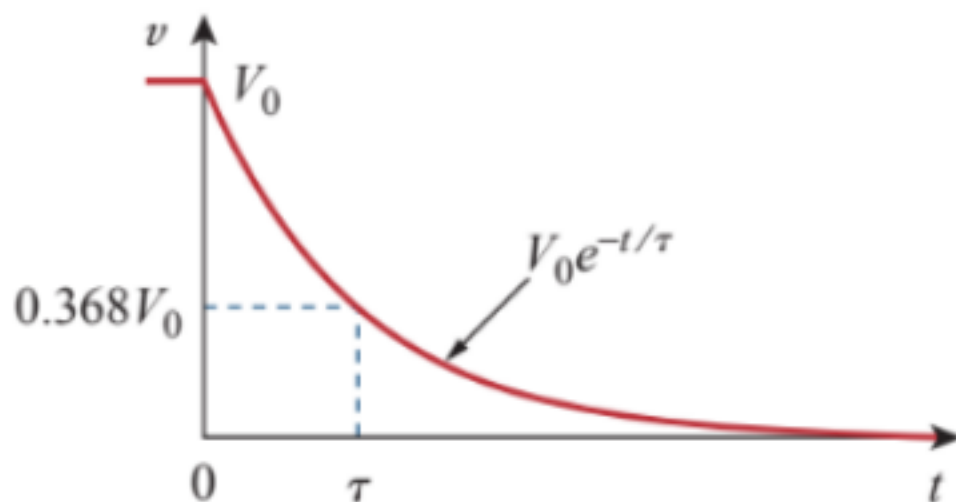
$$v(0) = V_0$$

$$v(t) = V_0 e^{-t/RC}$$



Natural Response

(depends on the nature of the circuit alone,
with no external sources)

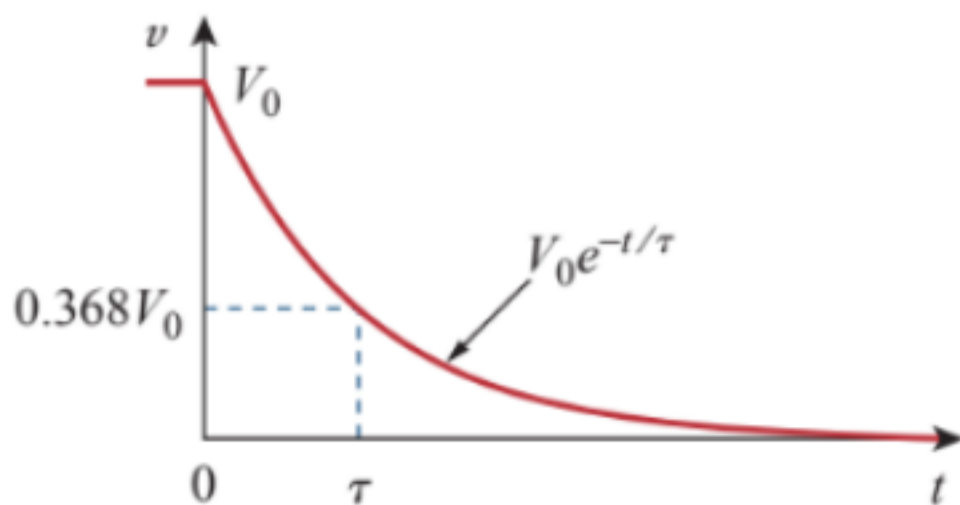


DC Circuit

The **time constant** of a circuit is the time required for the response to decay to a factor of $1/e$ or 36.8 percent of its initial value.

$$\tau = RC$$

$$v(t) = V_0 e^{-t/\tau}$$



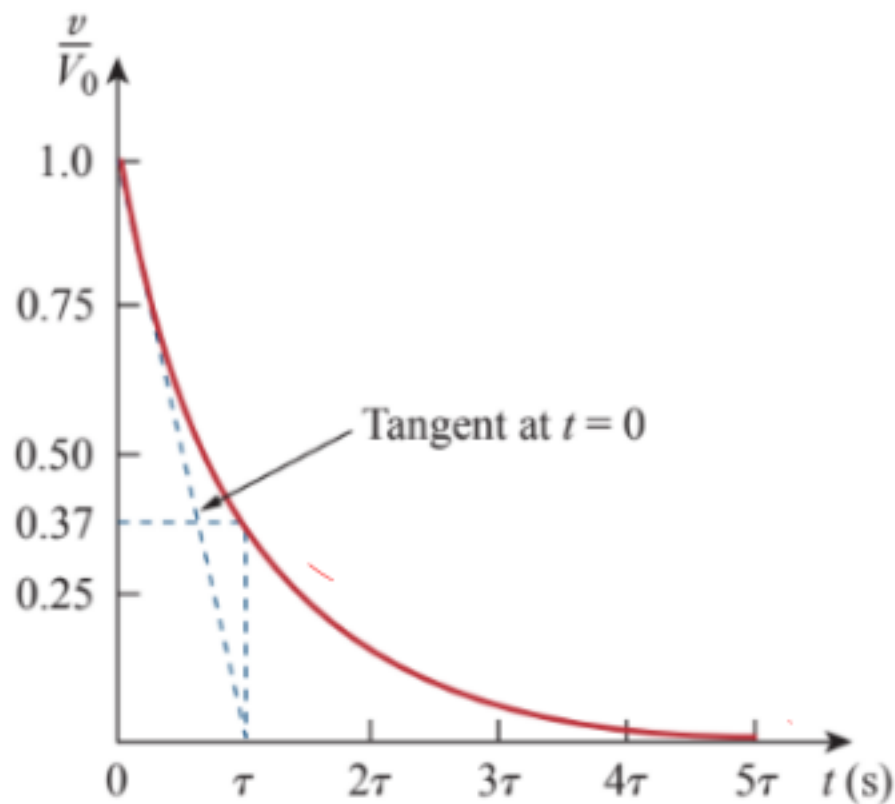
$$e^{-1} = 0.368$$

DC Circuit

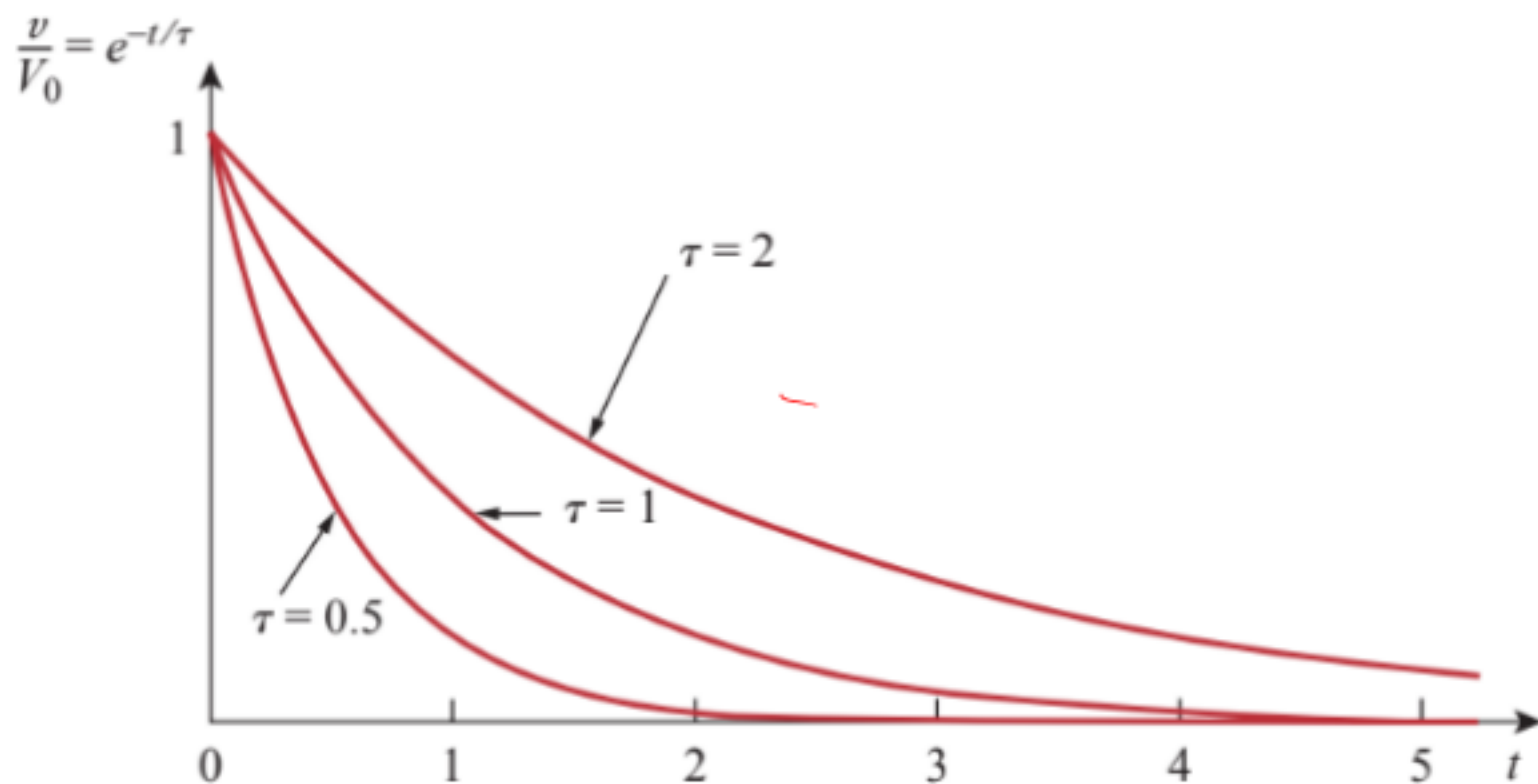
Values of $v(t)/V_0 = e^{-t/\tau}$.

t	$v(t)/V_0$
τ	0.36788
2τ	0.13534
3τ	0.04979
4τ	0.01832
5τ	0.00674

$$\tau = RC$$



DC Circuit

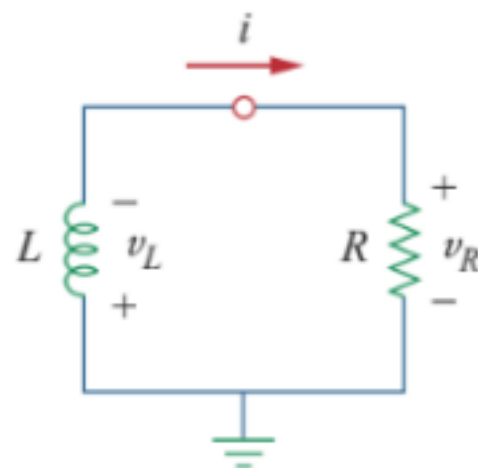
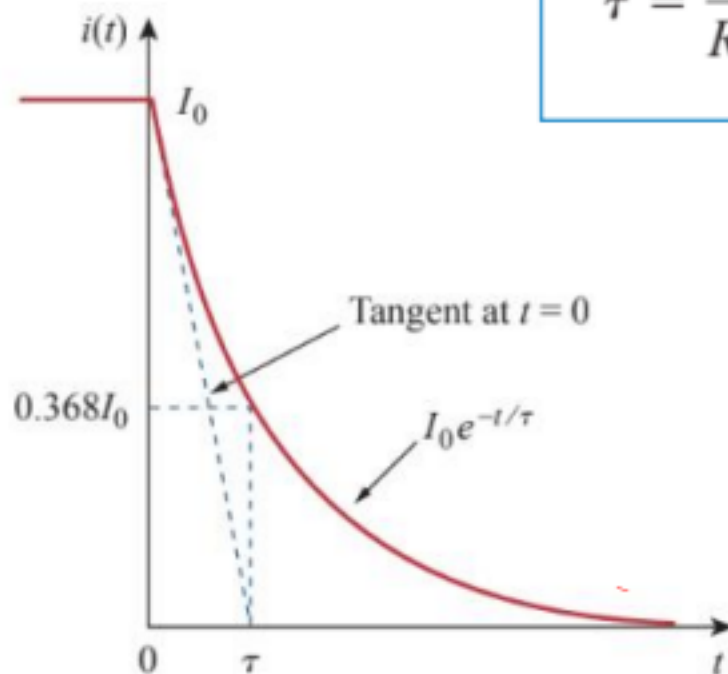


DC Circuit

Source-Free RL Circuit:

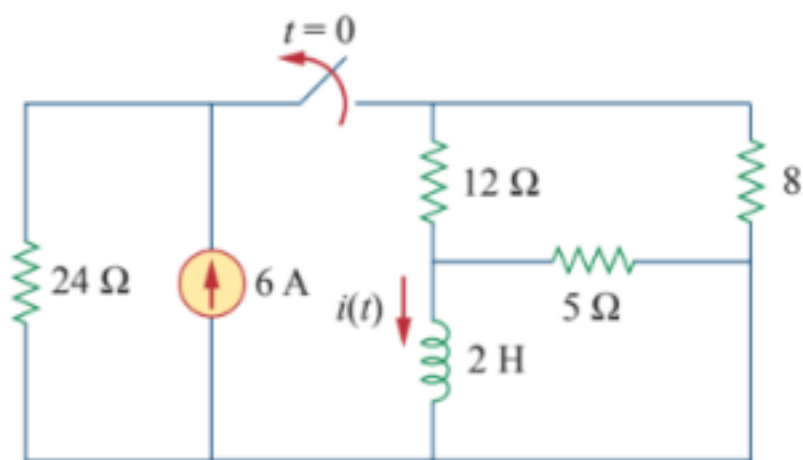
$$i(t) = I_0 e^{-t/\tau}$$

$$\tau = \frac{L}{R}$$



DC Circuit

find $i(t)$ for $t > 0$.



$$i(t) = i(0) e^{-t/\tau}$$

$$= \boxed{2 e^{-2t} \text{ A} ; t > 0}$$



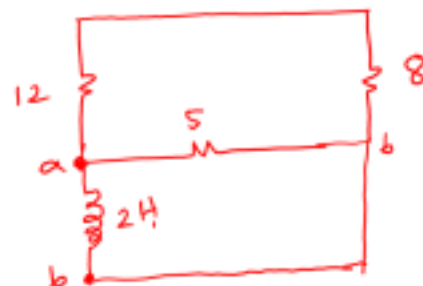
$$(8^{-1} + 24^{-1})^{-1} = 6 \Omega$$



$$i(0^-) = i(0)$$

$$= \frac{6}{18} \times 6 = \boxed{2 \text{ A}}$$

$t > 0$,

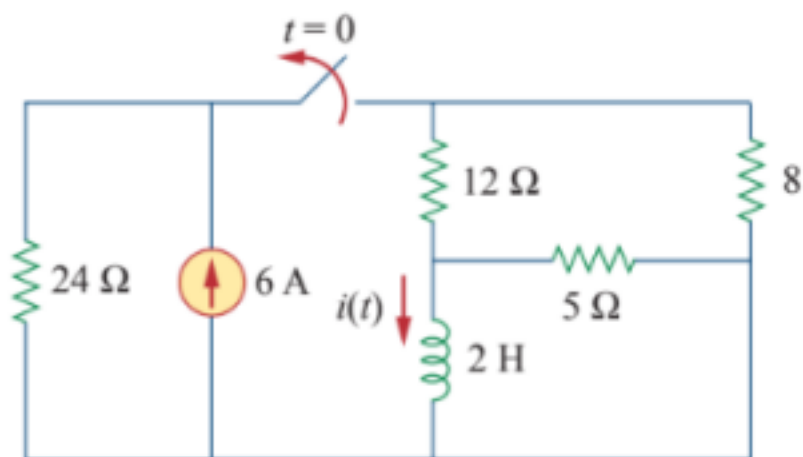


$$(20^{-1} + 8^{-1})^{-1} = 4 \Omega$$

$$\tau = \frac{L}{R} = \frac{2}{4} = \frac{1}{2}$$

DC Circuit

find $i(t)$ for $t > 0$.



DC Circuit

Singularity Functions:

Singularity functions are functions that either are discontinuous or have discontinuous derivatives.

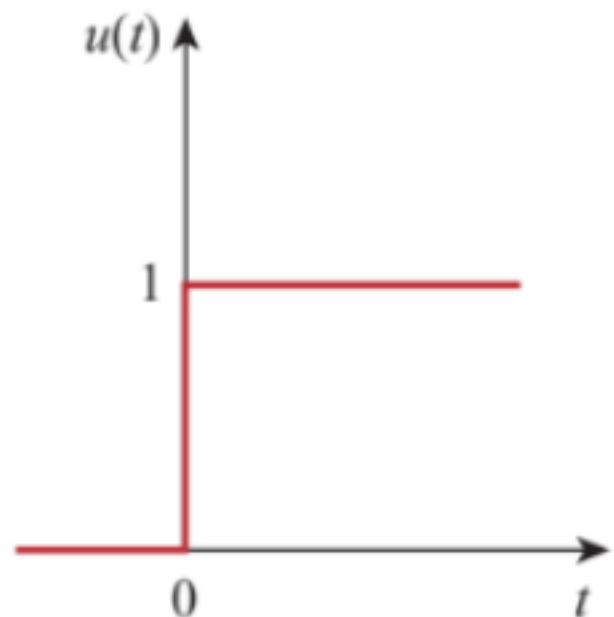
- 
- unit step
 - unit impulse
 - unit ramp

DC Circuit

The unit step function $u(t)$ is 0 for negative values of t and 1 for positive values of t .

$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t > 0 \end{cases}$$

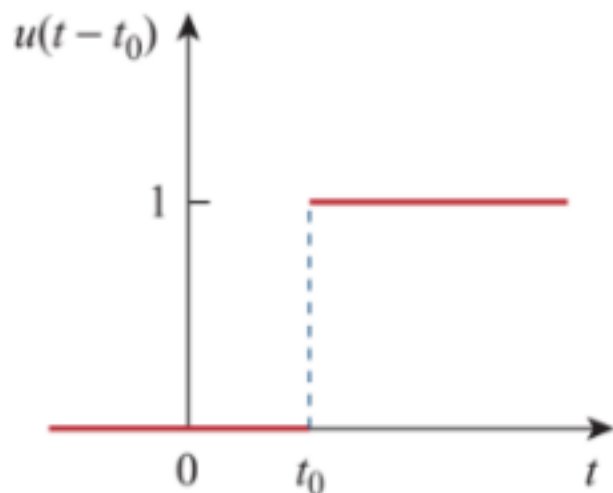
$$u(-t) = \begin{cases} 1, & t < 0 \\ 0, & t > 0 \end{cases}$$



DC Circuit

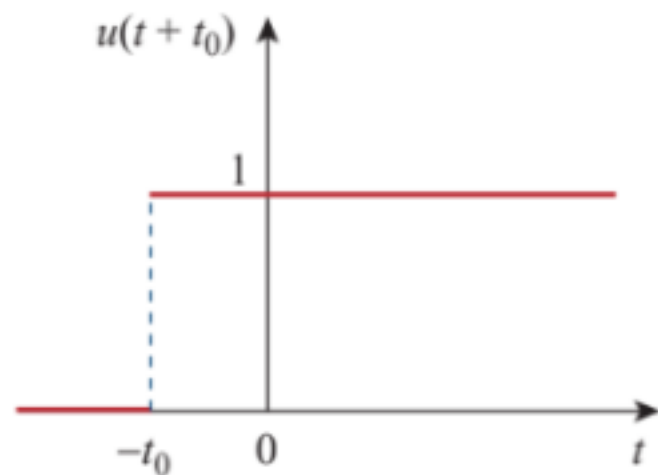
$$u(t - t_0) = \begin{cases} 0, & t < t_0 \\ 1, & t > t_0 \end{cases}$$

right shift
 $f(x-a)$

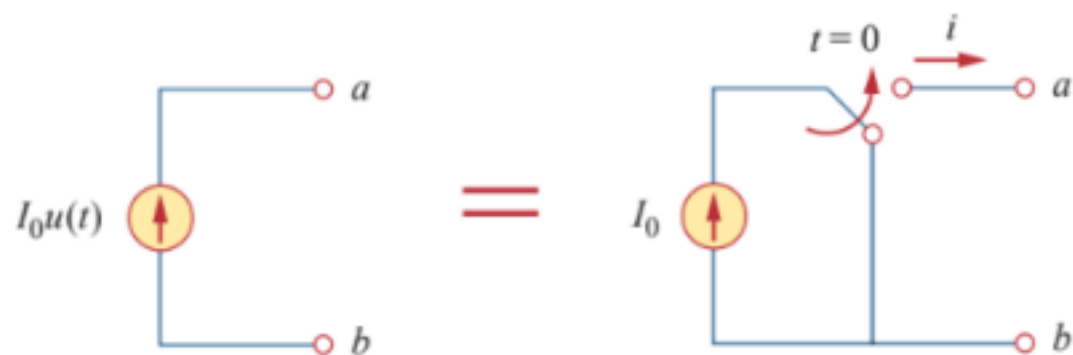
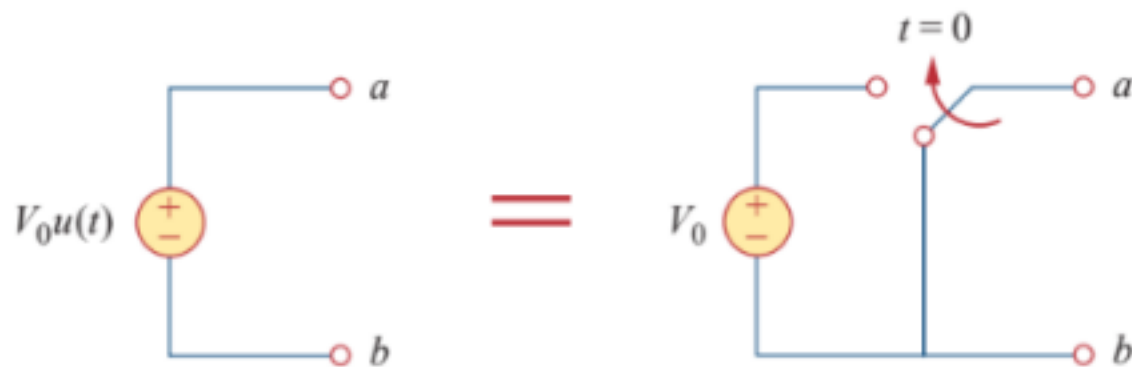


$$u(t + t_0) = \begin{cases} 0, & t < -t_0 \\ 1, & t > -t_0 \end{cases}$$

left shift
 $f(x+a)$



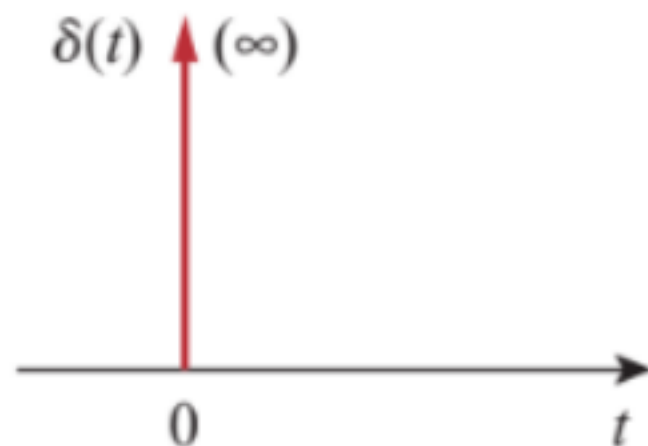
DC Circuit



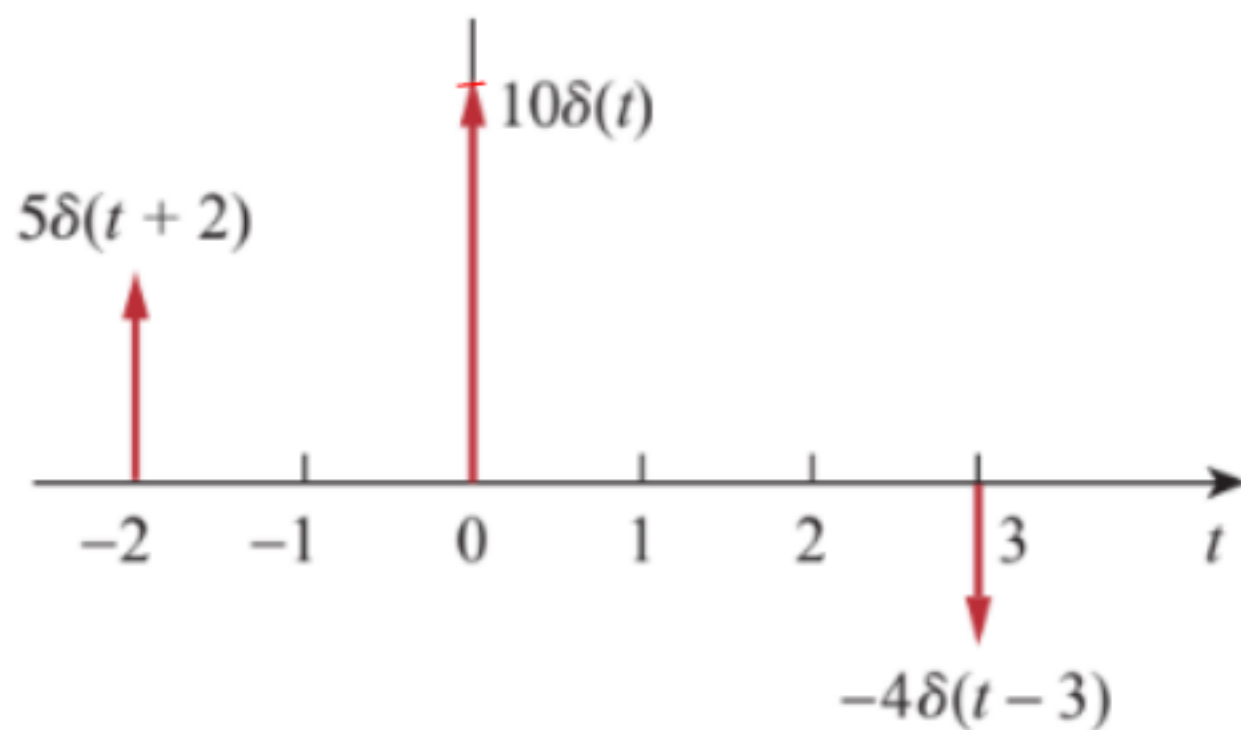
DC Circuit

The **unit impulse function** $\delta(t)$ is zero everywhere except at $t = 0$, where it is undefined.

$$\delta(t) = \frac{d}{dt}u(t) = \begin{cases} 0, & t < 0 \\ \text{Undefined}, & t = 0 \\ 0, & t > 0 \end{cases}$$



DC Circuit

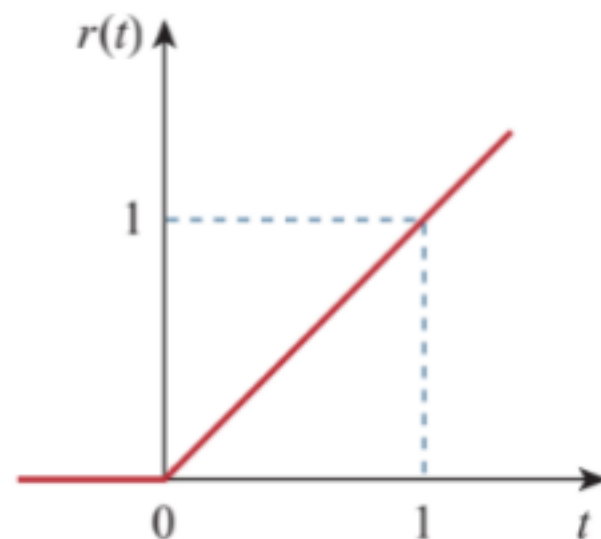


DC Circuit

The **unit ramp function** is zero for negative values of t and has a unit slope for positive values of t .

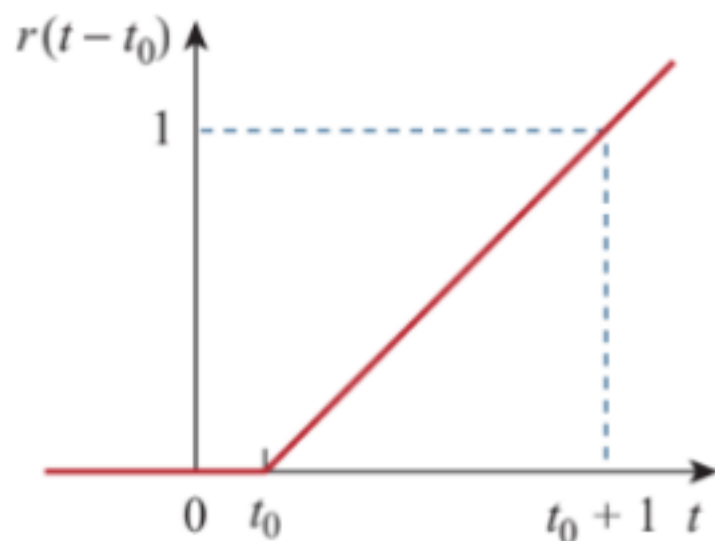
$$r(t) = \int_{-\infty}^t u(t) dt = tu(t)$$

$$r(t) = \begin{cases} 0, & t \leq 0 \\ t, & t \geq 0 \end{cases}$$

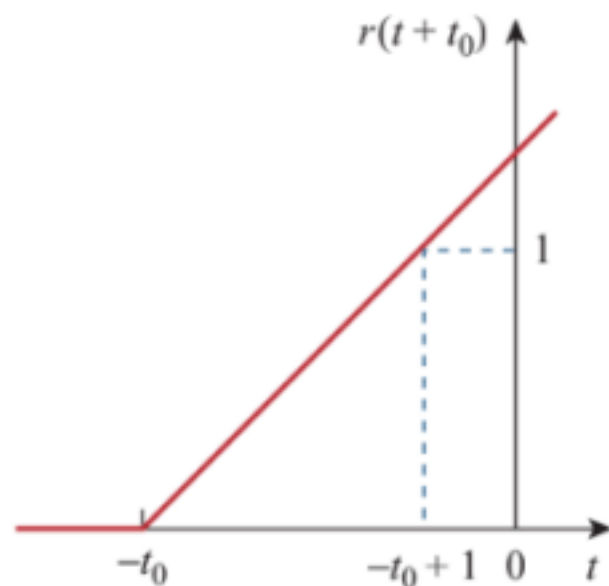


DC Circuit

$$r(t - t_0) = \begin{cases} 0, & t \leq t_0 \\ t - t_0, & t \geq t_0 \end{cases}$$



$$r(t + t_0) = \begin{cases} 0, & t \leq -t_0 \\ t + t_0, & t \geq -t_0 \end{cases}$$



DC Circuit

$$\delta(t) = \frac{du(t)}{dt}, \quad u(t) = \frac{dr(t)}{dt}$$

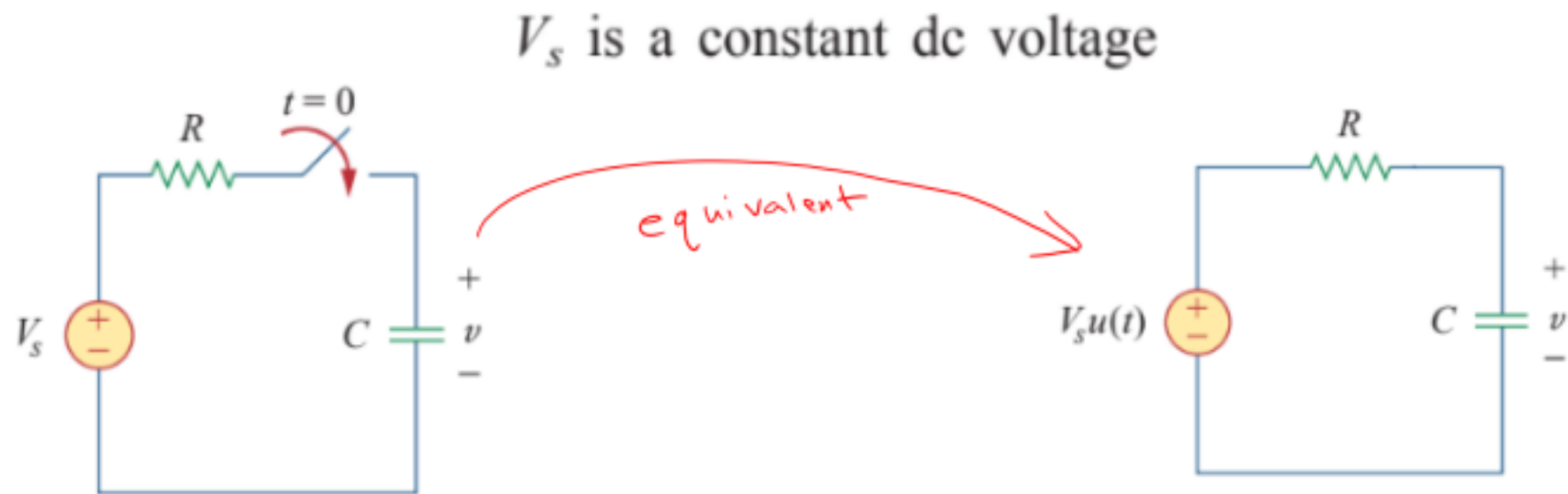
by integration as

$$u(t) = \int_{-\infty}^t \delta(t) dt, \quad r(t) = \int_{-\infty}^t u(t) dt$$

DC Circuit

Step Response of an RC Circuit:

The **step response** of a circuit is its behavior when the excitation is the step function, which may be a voltage or a current source.



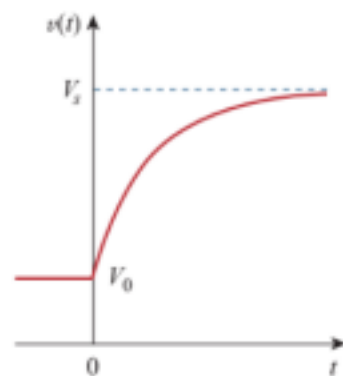
DC Circuit

We assume an initial voltage V_0 on the capacitor
voltage of a capacitor cannot change instantaneously,

$$v(0^-) = v(0^+) = V_0 = v(0)$$

voltage across the capacitor just **before switching** and **immediately after switching**

$$v(t) = \begin{cases} V_0, & t < 0 \\ V_s + (V_0 - V_s)e^{-t/\tau}, & t > 0 \end{cases}$$



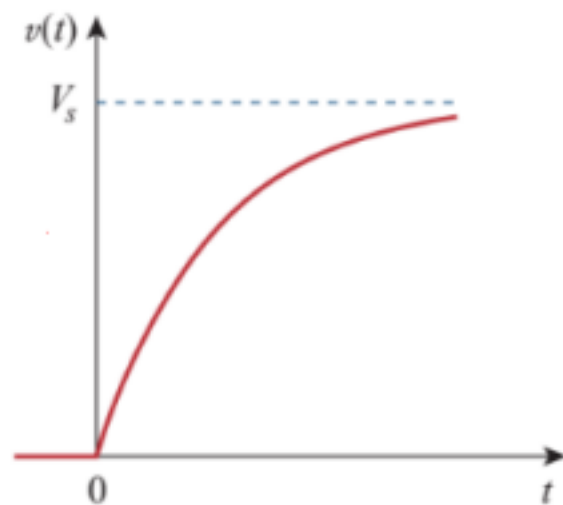
DC Circuit

If we assume that the capacitor is uncharged initially, $V_0 = 0$ in Eq. (7.46) so that

$$v(t) = \begin{cases} 0, & t < 0 \\ V_s(1 - e^{-t/\tau}), & t > 0 \end{cases}$$

which can be written alternatively as

$$v(t) = V_s(1 - e^{-t/\tau})u(t)$$



DC Circuit

$$v(t) = V_s + (V_0 - V_s)e^{-t/\tau}, \quad t > 0$$

$$v = v_n + v_f$$

$$v_n = V_0 e^{-t/\tau}$$

$$v_f = V_s(1 - e^{-t/\tau})$$

Complete response = natural response + forced response
stored energy independent source

forced response because it is produced by the circuit when an external “force” is applied.

DC Circuit

$$v(t) = V_s + (V_0 - V_s)e^{-t/\tau}, \quad t > 0$$

$$v = v_t + v_{ss}$$

Complete response = transient response + steady-state response
temporary part permanent part

$$v_t = (V_0 - V_s)e^{-t/\tau}$$

$$v_{ss} = V_s$$

- Transient response is circuit's temporary response that will die out with time.
- The steady-state response is the behavior of the circuit a long time after an external excitation is applied.

DC Circuit

সুপার

$$v(t) = v(\infty) + [v(0) - v(\infty)]e^{-t/\tau}$$

1. The initial capacitor voltage $v(0)$.
2. The final capacitor voltage $v(\infty)$. $\rightarrow$ steady-state value.
3. The time constant τ .

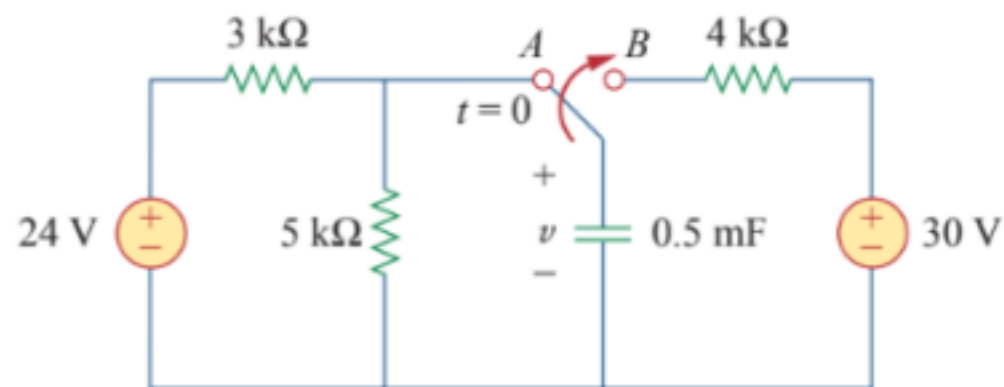
if the switch changes position at time $t = t_0$

✓✓

$$v(t) = v(\infty) + [v(t_0) - v(\infty)]e^{-(t-t_0)/\tau}$$

DC Circuit

The switch has been in position A for a long time. At $t = 0$, the switch moves to B. Determine $v(t)$ for $t > 0$ and calculate its value at $t = 1\text{s}$ and 4s .



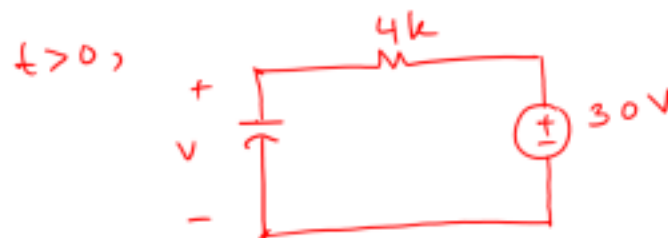
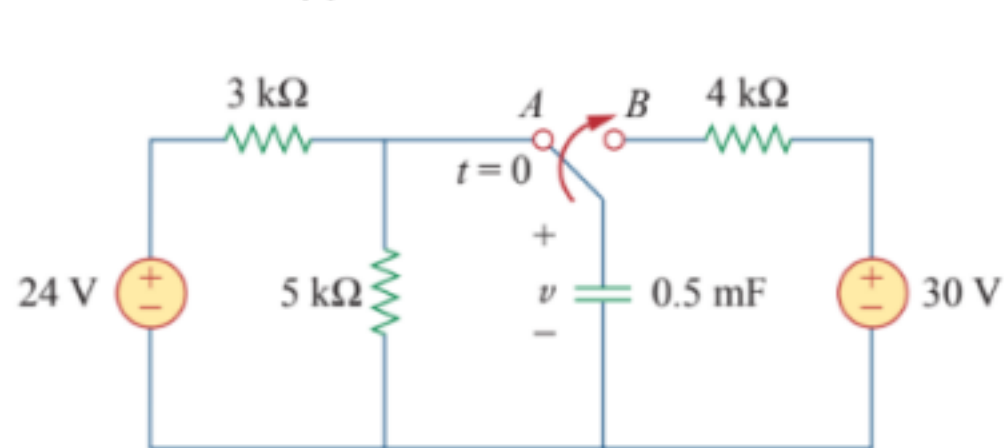
$t < 0$,



$$v(0^-) = v(0) = v(0^+) = \frac{5}{8} \times 24 = \boxed{15\text{V}}$$

DC Circuit

The switch has been in position A for a long time. At $t = 0$, the switch moves to B. Determine $v(t)$ for $t > 0$ and calculate its value at $t = 1\text{s}$ and 4s .



$$\begin{aligned}\tau &= RC \\ &= 4 \times 10^3 \times 0.5 \times 10^{-3} \\ &= 2\end{aligned}$$

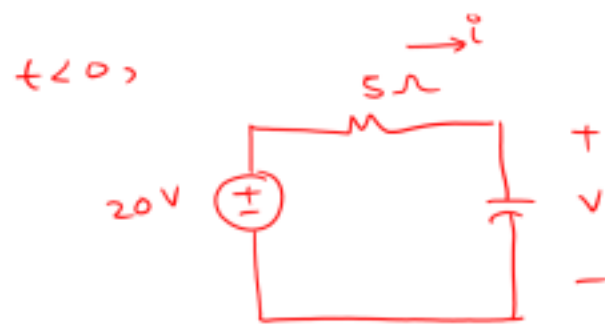
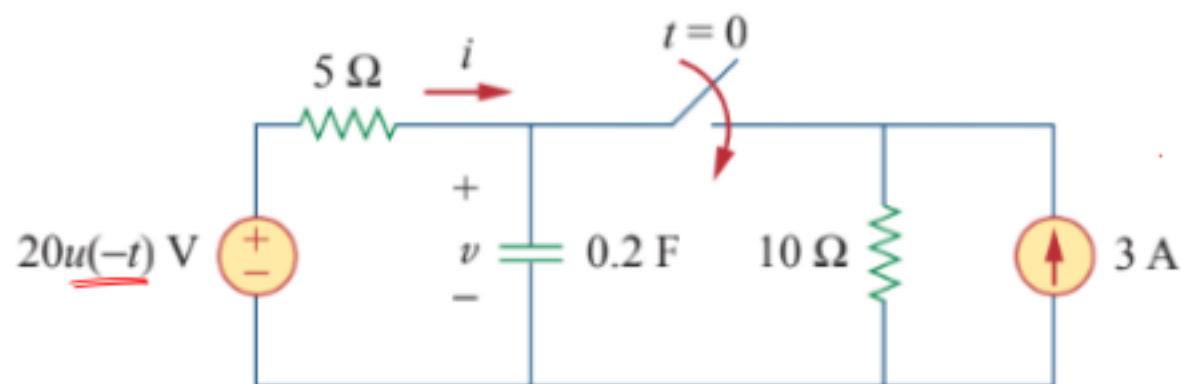
$$\begin{aligned}v(\infty) &= 30\text{V} \\ v(t) &= 30 + (15 - 30)e^{-0.5t} = (30 - 15e^{-0.5t})\text{V}\end{aligned}$$

$$t = 1\text{s}, \quad v(1) = 30 - 15e^{-0.5} = \boxed{20.9\text{V}}$$

$$t = 4\text{s}, \quad v(4) = 30 - 15e^{-2} = \boxed{27.96\text{V}}$$

DC Circuit

Find $i(t)$ and $v(t)$ for all time.



$$v(0^-) = v(0) = 20\text{ V}$$

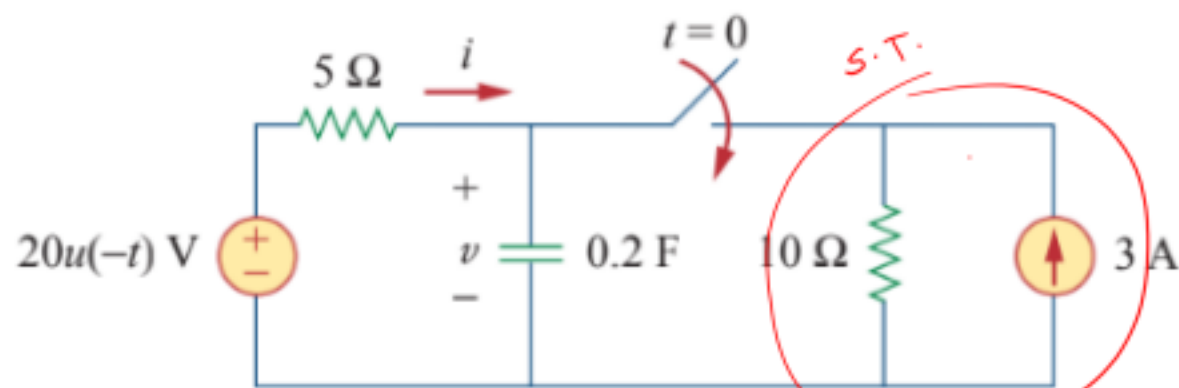
$$i(0^-) = i(0) = 0\text{ A}$$

Answer:
$$i(t) = \begin{cases} 0, & t < 0 \\ -2(1 + e^{-1.5t})\text{ A}, & t > 0 \end{cases}$$

$$v = \begin{cases} 20\text{ V}, & t < 0 \\ 10(1 + e^{-1.5t})\text{ V}, & t > 0 \end{cases}$$

DC Circuit

Find $i(t)$ and $v(t)$ for all time.



$$R_{Th} = (5^{-1} + 10^{-1})^{-1} = \frac{10}{3}$$

$$\tau = \frac{10}{3} \times 0.2 = \frac{2}{3}$$

Answer: $i(t) = \begin{cases} 0, & t < 0 \\ -2(1 + e^{-1.5t}) \text{ A}, & t > 0 \end{cases}$

$$v = \begin{cases} 20 \text{ V}, & t < 0 \\ 10(1 + e^{-1.5t}) \text{ V}, & t > 0 \end{cases}$$

$t > 0$,



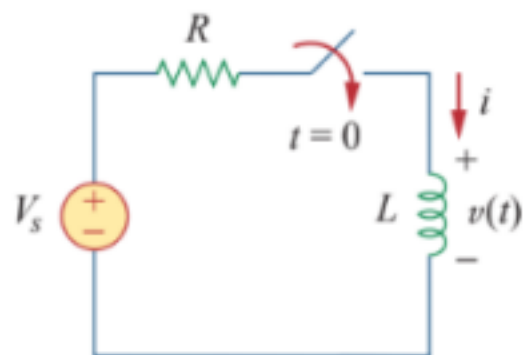
$$v(\infty) = \frac{5}{15} \times 30 = 10 \text{ V}$$

$$v(t) = 10 + [20 - 10] e^{-1.5t} = 10(1 + e^{-1.5t}) \text{ V} ; t > 0$$

$$i(t) = -\frac{v(t)}{5} = -2(1 + e^{-1.5t}) \text{ A} ; t > 0$$

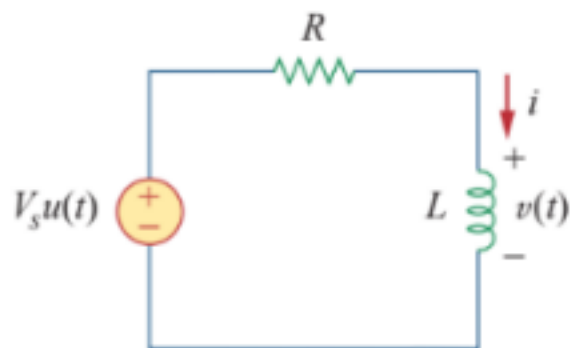
DC Circuit

Step Response of RL Circuit:



$$i = i_t + i_{ss}$$

$$i_t = Ae^{-t/\tau}, \quad \tau = \frac{L}{R}$$



$$i_{ss} = \frac{V_s}{R}$$

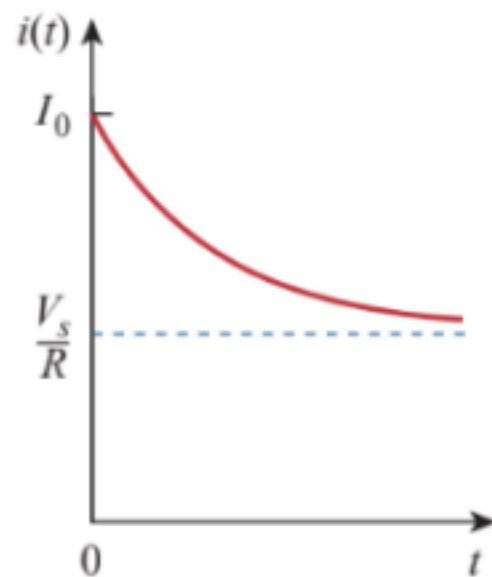
DC Circuit

Since the current through the inductor cannot change instantaneously,

$$i(0^+) = i(0^-) = I_0$$

$$i(t) = i(\infty) + [i(0) - i(\infty)]e^{-t/\tau}$$

1. The initial inductor current $i(0)$ at $t = 0$.
2. The final inductor current $i(\infty)$.
3. The time constant τ .



DC Circuit

if the switching takes place at time $t = t_0$ instead of $t = 0$,

$$i(t) = i(\infty) + [i(t_0) - i(\infty)]e^{-(t-t_0)/\tau}$$

If $I_0 = 0$, then

$$i(t) = \begin{cases} 0, & t < 0 \\ \frac{V_s}{R}(1 - e^{-t/\tau}), & t > 0 \end{cases}$$

or

$$i(t) = \frac{V_s}{R}(1 - e^{-t/\tau})u(t)$$

DC Circuit

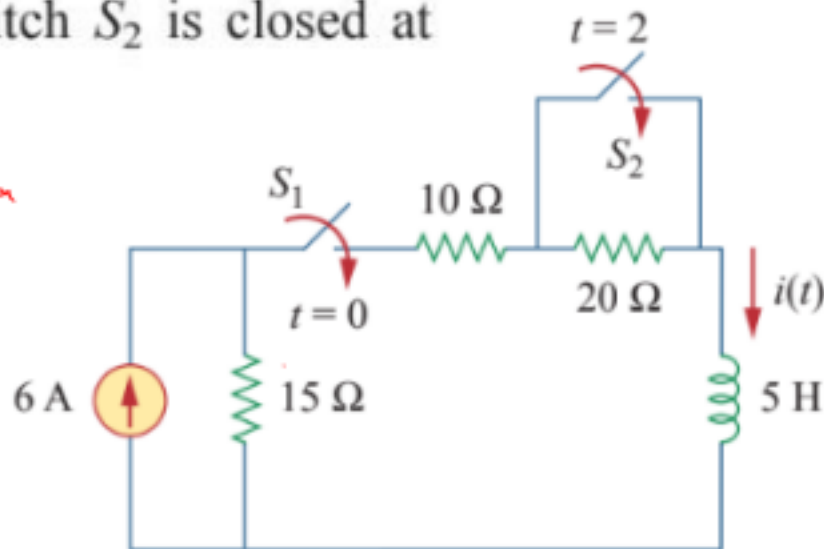
Switch S_1 in Fig. 7.54 is closed at $t = 0$, and switch S_2 is closed at $t = 2$ s. Calculate $i(t)$ for all t . Find $i(1)$ and $i(3)$.

$i(0^-) = i(0) = 0$
 $0 < t < 2$

$R_{in} = 45 \Omega$
 $\tau = \frac{5}{45} = \frac{1}{9}$

$i(\infty) = \frac{15}{15+30} \times 6 = 2 \text{ A}$
 $i(t) = 2 + (0-2) e^{-9t}$
 $= 2 [1 - e^{-9t}] \text{ A}$

$i(2) = i(2^-) = 2 \text{ A}$



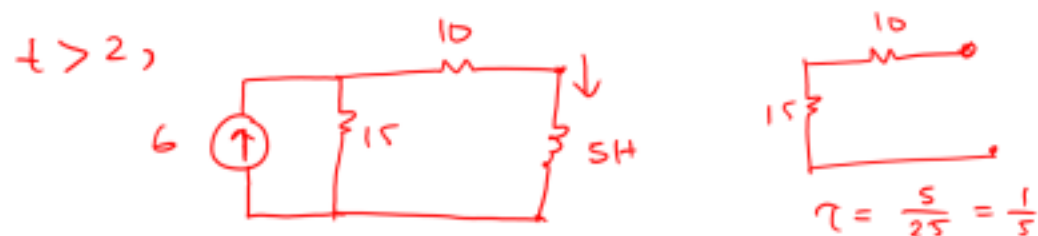
Answer:

$$i(t) = \begin{cases} 0, & t < 0 \\ 2(1 - e^{-9t}), & 0 < t < 2 \\ 3.6 - 1.6e^{-5(t-2)}, & t > 2 \end{cases}$$

$$i(1) = 1.9997 \text{ A}, i(3) = 3.589 \text{ A}.$$

DC Circuit

Switch S_1 in Fig. 7.54 is closed at $t = 0$, and switch S_2 is closed at $t = 2$ s. Calculate $i(t)$ for all t . Find $i(1)$ and $i(3)$.

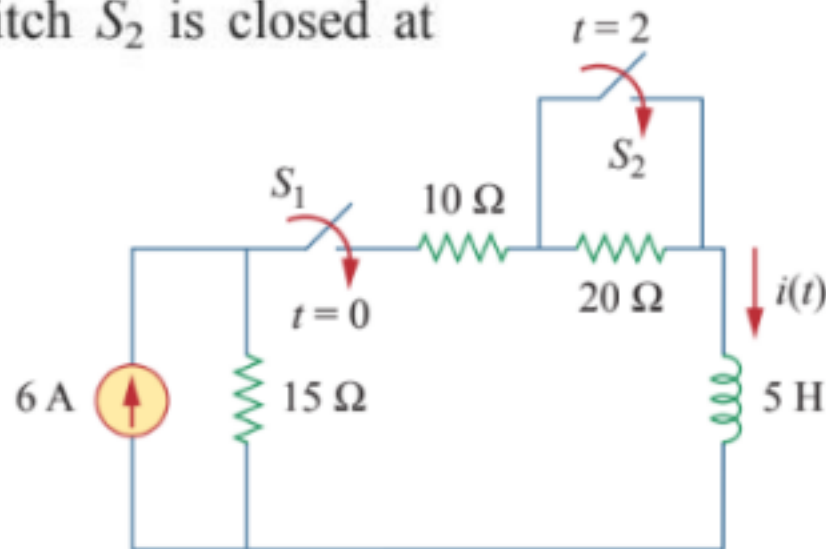


$$i(\infty) = \frac{15}{25} \times 6 = 3.6 \text{ A}$$

$$i(t) = i(\infty) + [i(2) - i(\infty)] e^{-(t-2)/\tau}$$

$$= 3.6 + [2 - 3.6] e^{-5(t-2)}$$

$$= \boxed{3.6 - 1.6 e^{-5(t-2)} ; t > 2}$$



Answer:

$$i(t) = \begin{cases} 0, & t < 0 \\ 2(1 - e^{-9t}), & 0 < t < 2 \\ 3.6 - 1.6e^{-5(t-2)}, & t > 2 \end{cases}$$

$$i(1) = 1.9997 \text{ A}, i(3) = 3.589 \text{ A}.$$

DC Circuit

First-Order Op Amp Circuits:

For the op amp circuit in Fig. 7.56, find v_o for $t > 0$ if $v(0) = 4$ V. Assume that $R_f = 50$ k Ω , $R_1 = 10$ k Ω , and $C = 10$ μ F.

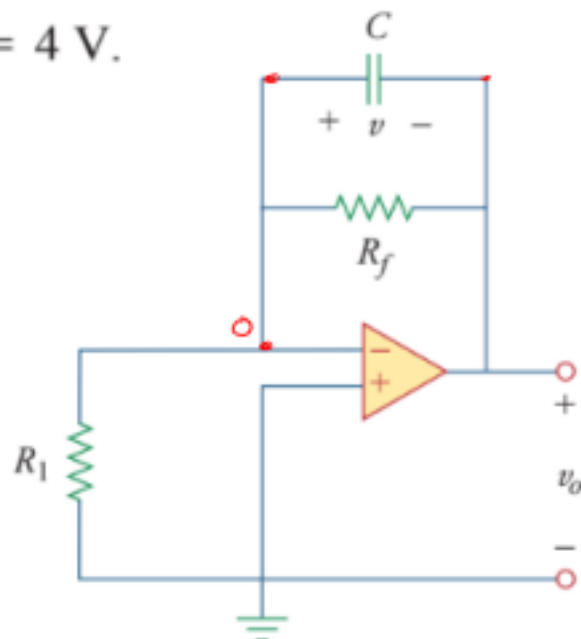
$$v(0^-) = v(0^+) = v(0) = 4 \text{ V}$$

$$+v + v_o = 0$$

$$\Rightarrow v_o(0) = -v(0) = -4 \text{ V}$$

$$v_o(\infty) = 0 \text{ V} \rightarrow \text{source-free circuit}$$

$$v_o(t) = v_o(\infty) + [v_o(0) - v_o(\infty)] e^{-t/\tau}$$
$$= \boxed{-4 e^{-2t} \text{ V}}; t > 0$$
$$\left. \begin{aligned} \tau &= R_f C \\ &= 50 \times 10^3 \times 10 \times 10^{-6} \\ &= \frac{1}{2} \end{aligned} \right\}$$



Answer: $-4e^{-2t}$ V, $t > 0$

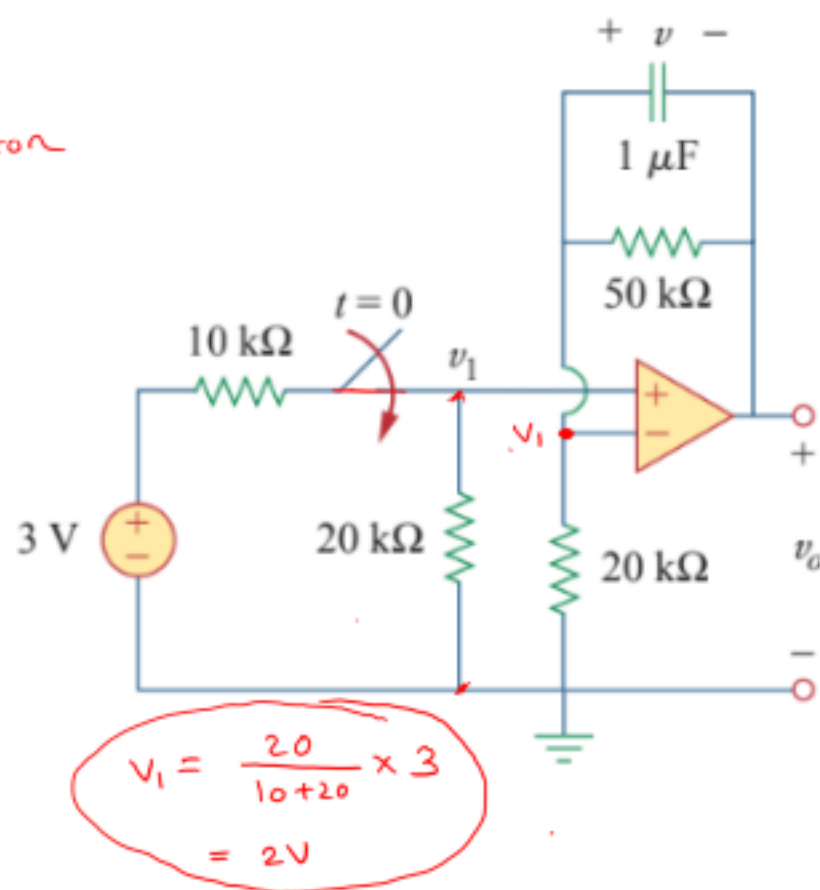
DC Circuit

Determine $v(t)$ and $v_o(t)$ in the circuit

$t < 0$, $v(0) = 0 \rightarrow$ no voltage across capacitor

$t > 0$,

$$\begin{aligned} \Rightarrow -v_1 + v + v_o &= 0 & \left| \begin{aligned} \frac{v_1}{20} + \frac{v_1 - v_o}{50} &= 0 \\ \Rightarrow \frac{2}{20} + \frac{2 - v_o}{50} &= 0 \\ \Rightarrow v_o(\infty) &= 7V \end{aligned} \right. \\ \Rightarrow -2 + v + 7 &= 0 \\ \Rightarrow v(\infty) &= -5V \end{aligned}$$



$$\begin{aligned} v_1 &= \frac{20}{10+20} \times 3 \\ &= 2V \end{aligned}$$

DC Circuit

Determine $v(t)$ and $v_o(t)$ in the circuit

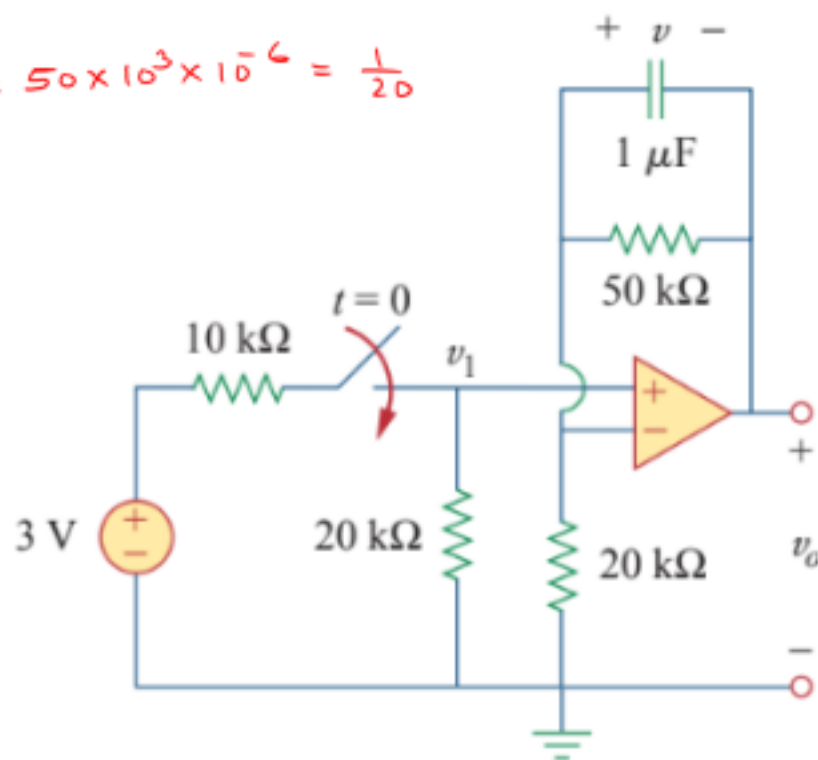
$$v(t) = -5 + [0 - (-5)] e^{-t/20}$$
$$= -5 + 5 e^{-t/20} = \boxed{5(e^{-t/20} - 1)}; t > 0$$

$$-v_1 + v + v_o = 0$$

$$\Rightarrow -2 - 5 + 5 e^{-t/20} = -v_o(t)$$

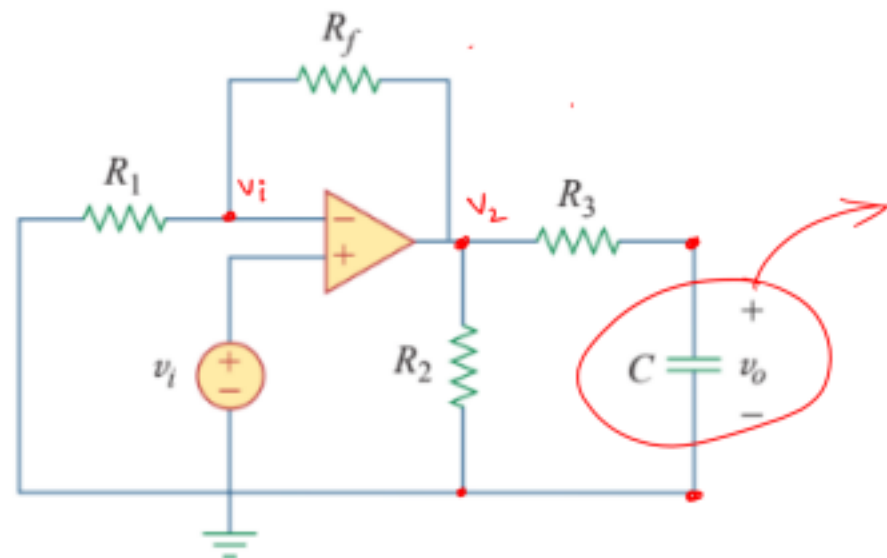
$$\Rightarrow v_o(t) = \boxed{7 - 5 e^{-t/20}}; t > 0$$

$$\tau = 50 \times 10^3 \times 10^{-6} = \frac{1}{20}$$



DC Circuit

Obtain the step response $v_o(t)$ for the circuit in Fig. 7.62. Let $v_i = 3u(t)$ V, $R_1 = 20$ k Ω , $R_f = 40$ k Ω , $R_2 = R_3 = 10$ k Ω , $C = 2$ μ F.



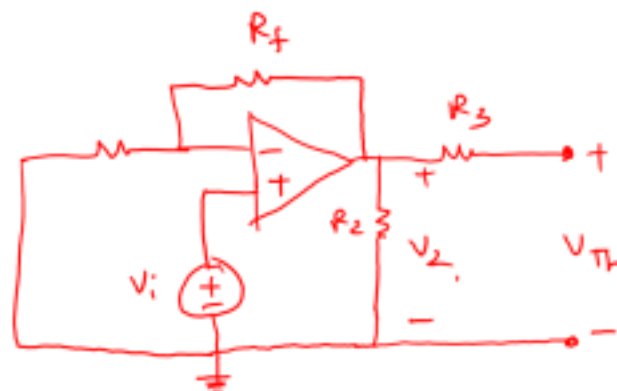
Answer: $9(1 - e^{-50t})u(t)$ V.

$$\frac{v_i}{R_1} + \frac{v_i - v_2}{R_f} = 0$$

$$\Rightarrow v_2 = \left(1 + \frac{R_f}{R_1}\right) v_i$$

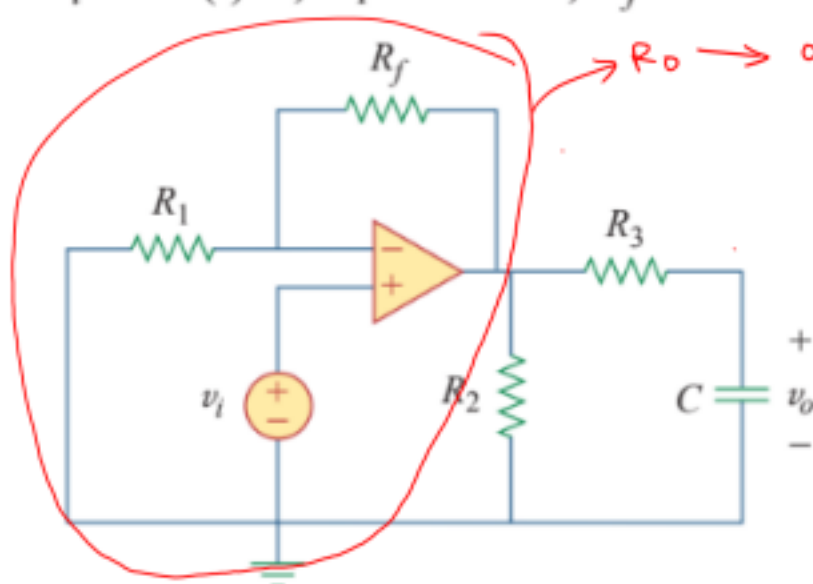
$$\Rightarrow v_{Th} = 3 v_i$$

$$\Rightarrow v_{Th} = 9 u(t)$$

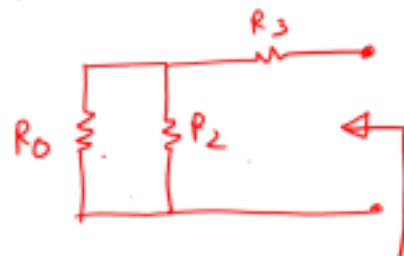


DC Circuit

Obtain the step response $v_o(t)$ for the circuit in Fig. 7.62. Let $v_i = 3u(t)$ V, $R_1 = 20$ k Ω , $R_f = 40$ k Ω , $R_2 = R_3 = 10$ k Ω , $C = 2$ μ F.



$R_o \rightarrow$ output resistance of op amp = 0Ω



$$R_{Th} = R_3 = 10 \text{ k}\Omega$$

$$\tau = 10 \times 10^3 \times 2 \times 10^{-6} = \frac{1}{50}$$

$$t < 0, \quad v_o(0) = 0$$

$$t > 0, \quad v_o(\infty) = 9$$



$$v_o(t) = 9 + (0 - 9)e^{-50t}$$
$$= \boxed{9(1 - e^{-50t})u(t)} \text{ V}$$

Answer: $9(1 - e^{-50t})u(t)$ V.

DC Circuit

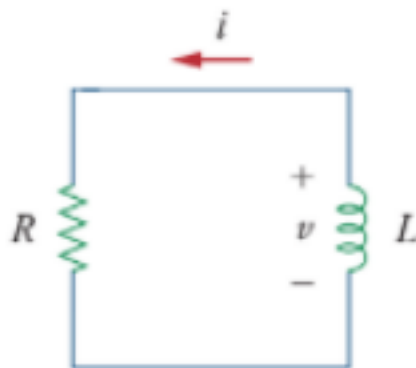
$$v(t) = 20e^{-10^3 t} \text{ V}, \quad t > 0$$

$$i(t) = 4e^{-10^3 t} \text{ mA}, \quad t > 0$$

(a) Find R , L , and τ .

(b) Calculate the energy dissipated in the resistance for $0 < t < 0.5 \text{ ms}$.

$$P = \frac{dw}{dt}$$



$$a) \quad \tau = \frac{1}{10^3} = \boxed{1 \text{ ms}}$$

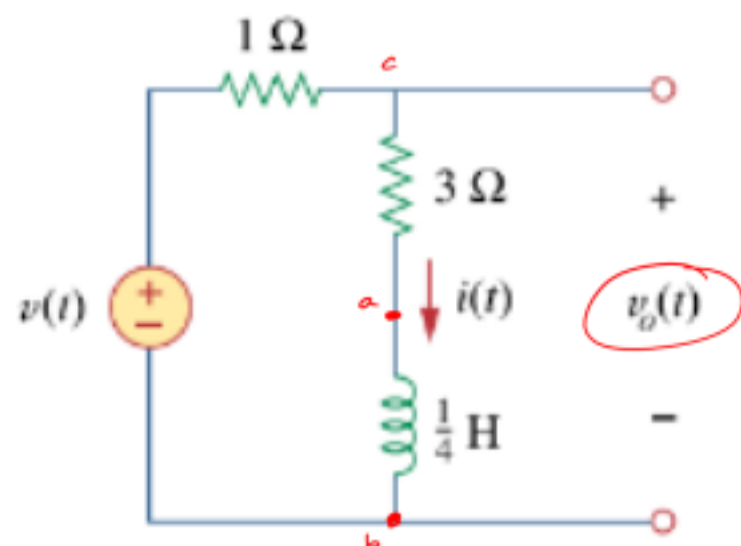
$$R = \frac{20}{4 \times 10^{-3}} = \boxed{5 \text{ k}\Omega}$$

$$\tau = \frac{L}{R} \Rightarrow L = 10^{-3} \times 5 \times 10^3 = \boxed{5 \text{ H}}$$

$$\begin{aligned} b) \quad W &= \int P \, dt = \int_0^{0.5 \text{ ms}} 80 \cdot e^{-2 \times 10^3 t} \, dt \quad \text{mJ} \\ &= - \frac{80}{2 \times 10^3} e^{-2 \times 10^3 t} \bigg|_0^{0.5 \times 10^{-3}} \quad \text{mJ} \\ &= \boxed{??} \end{aligned}$$

DC Circuit

Find $v_o(t)$ if $i(0) = 2$ A and $v(t) = 0$.

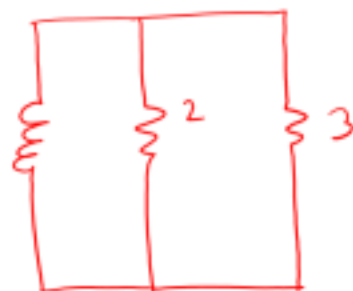
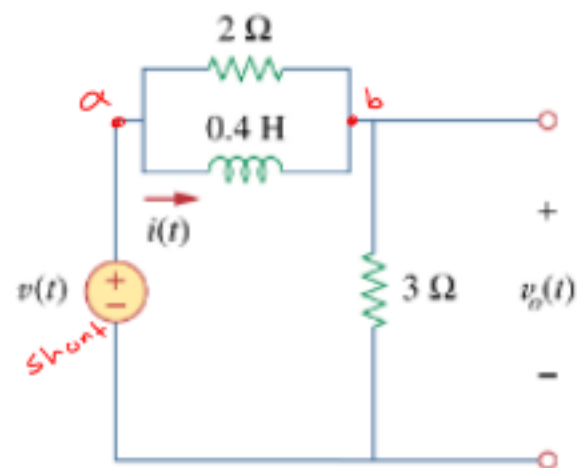


$$i(t) = i(0) e^{-t/\tau} = 2 e^{-16t}$$
$$\tau = \frac{L}{R} = \frac{1/4}{3+1} = \frac{1}{16}$$

$$v_o(t) = 3 i(t) + L \frac{di}{dt}$$
$$= 6 e^{-16t} + \frac{1}{4} \cdot 2 e^{-16t} (-16)$$
$$= \boxed{-2 e^{-16t}} \quad ; \quad t > 0$$

DC Circuit

determine $v_o(t)$ when $i(0) = 1$ A and $v(t) = 0$.



$$R_{eq} = 2 \parallel 3 = \frac{6}{5}$$

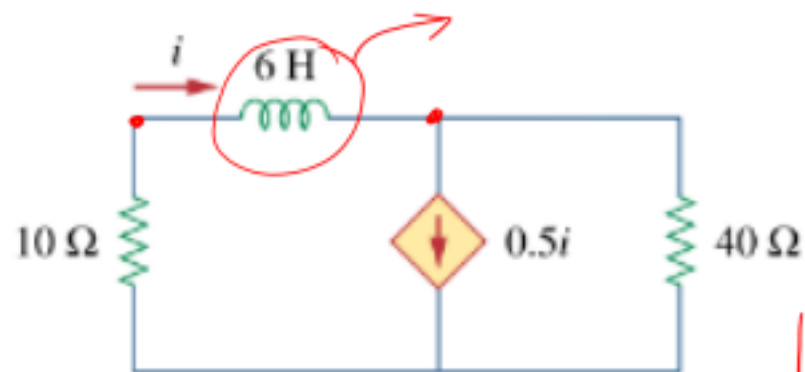
$$\tau = \frac{2}{5} \times \frac{5}{6} = \frac{1}{3}$$

$$i(t) = 1 e^{-3t}$$

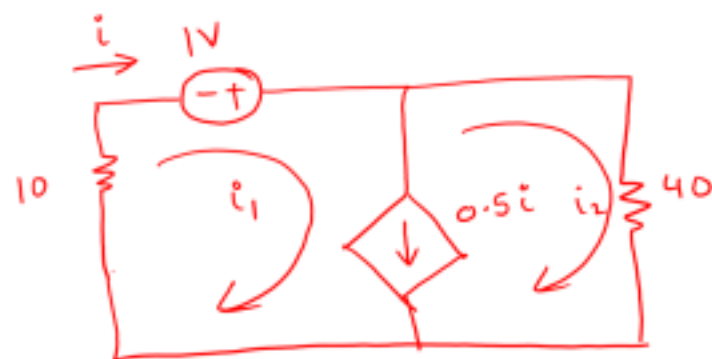
$$\begin{aligned} v_o(t) &= -L \frac{di}{dt} = -0.4 (-3) e^{-3t} \\ &= \boxed{1.2 e^{-3t}} \end{aligned}$$

DC Circuit

find $i(t)$ for $t > 0$ if $i(0) = 2$ A.



$$i(t) = 2 e^{-5t} \text{ A}$$



$$i_1 - i_2 = 0.5i_1$$
$$\Rightarrow 0.5i_1 = i_2$$

$$\tau = \frac{L}{R}$$
$$= \frac{6}{30} = \frac{1}{5}$$

$$-1 + 40i_2 + 10i_1 = 0$$

$$\Rightarrow 30i_1 = 1$$

$$\Rightarrow i_1 = \frac{1}{30}$$

$$R_{th} = \frac{1}{i_1} = 30\ \Omega$$

DC Circuit

$$v = 120e^{-50t} \text{ V}$$

and

$$i = 30e^{-50t} \text{ A}, t > 0$$

- (a) Find L and R .
- (b) Determine the time constant.
- (c) Calculate the initial energy in the inductor.
- (d) What fraction of the initial energy is dissipated in 10 ms?



$$a) \quad \tau = \frac{1}{50} \Rightarrow \frac{L}{R} = \frac{1}{50} \Rightarrow \boxed{L = 80 \text{ mH}}$$

$$R = \boxed{4 \Omega}$$

$$b) \quad \tau = \boxed{20 \text{ ms}}$$

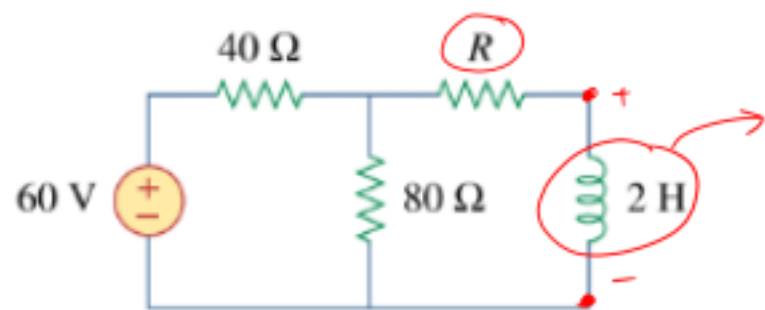
$$c) \quad W = \frac{1}{2} L i(0)^2 = \frac{1}{2} \times 80 \times 10^{-3} \times 30^2$$

$$d) \quad W_{10\text{ms}} = \frac{1}{2} \times 80 \times 10^{-3} (30 e^{-50 \times 10 \times 10^{-3}})^2 = \boxed{36 \text{ J}}$$
$$= ?$$

$$\frac{36 - ?}{36} = \boxed{\checkmark}$$

DC Circuit

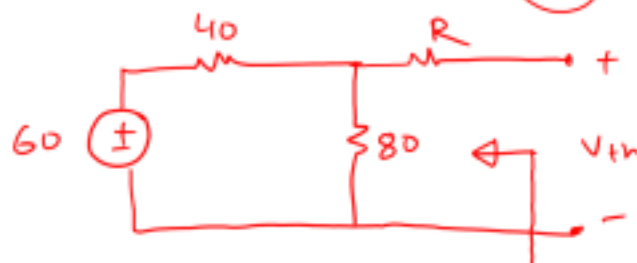
find the value of R for which the steady-state energy stored in the inductor will be 1 J .



$$W = \frac{1}{2} L \left(\frac{40}{\frac{80}{3} + R} \right)^2$$

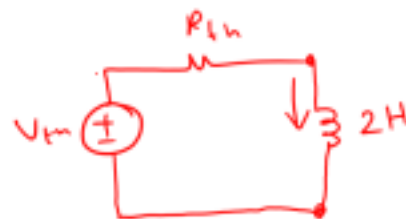
$$\Rightarrow 1 = \frac{1}{2} \times 2 \left(\frac{40}{\frac{80}{3} + R} \right)^2$$

$$\Rightarrow \boxed{R = 13.33\Omega}$$



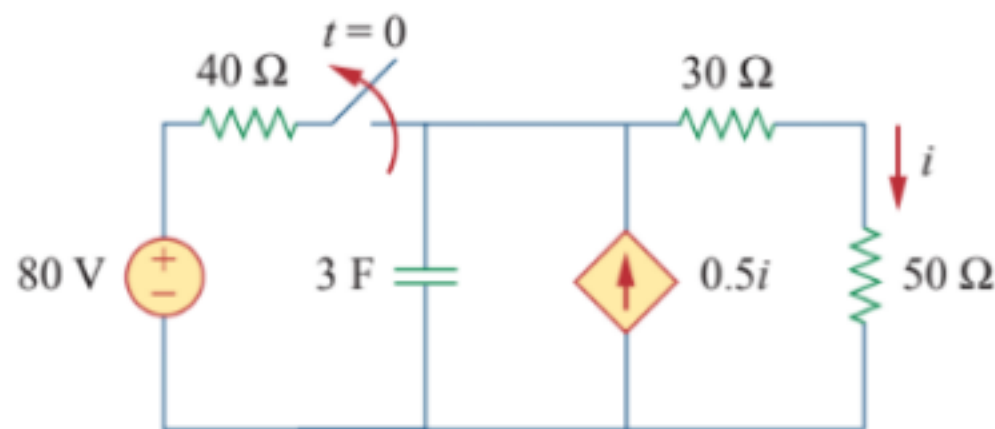
$$\begin{aligned} R_{th} &= (40 \parallel 80) + R \\ &= \frac{80}{3} + R \end{aligned}$$

$$\begin{aligned} V_{th} &= \frac{80}{80+40} \times 60 \\ &= \boxed{40\text{V}} \end{aligned}$$



DC Circuit

Find $i(t)$ for $t < 0$ and $t > 0$.



$t < 0$,

$$0.5i = \frac{v_1}{80} + \frac{v_1 - 80}{40} \quad \left| \quad \frac{v_1 - 80}{40} = -0.5i \right.$$

$$\Rightarrow 0 = \frac{v_1}{80} + \frac{v_1 - 80}{20}$$

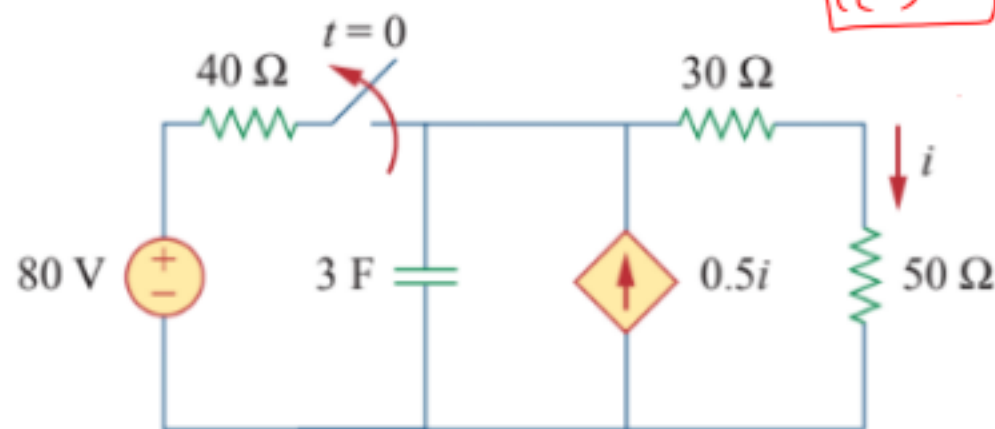
$$\Rightarrow 4 = \frac{v_1}{80} + \frac{v_1}{20}$$

$$\Rightarrow v_1 = 64 \text{ V}$$

$$i(0^-) = i(0) = \boxed{0.8 \text{ A}}$$

DC Circuit

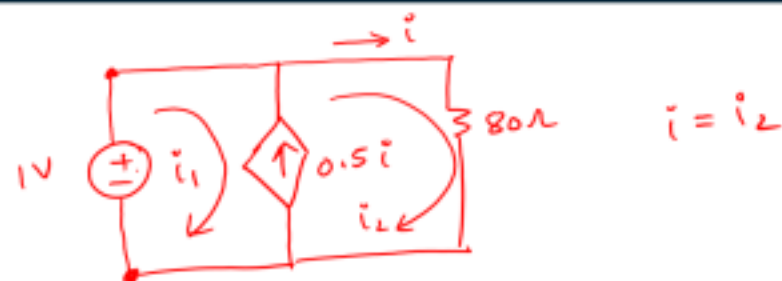
Find $i(t)$ for $t < 0$ and $t > 0$.



$$i(\infty) = 0$$

$$i(t) = 0.8 e^{-t/480} \text{ A}$$

$t > 0$



$$i_2 - i_1 = 0.5 i$$

$$\Rightarrow 0.5 i_2 = i_1 \Rightarrow i_1 = \frac{1}{160}$$

$$-1 + 80 i_2 = 0$$

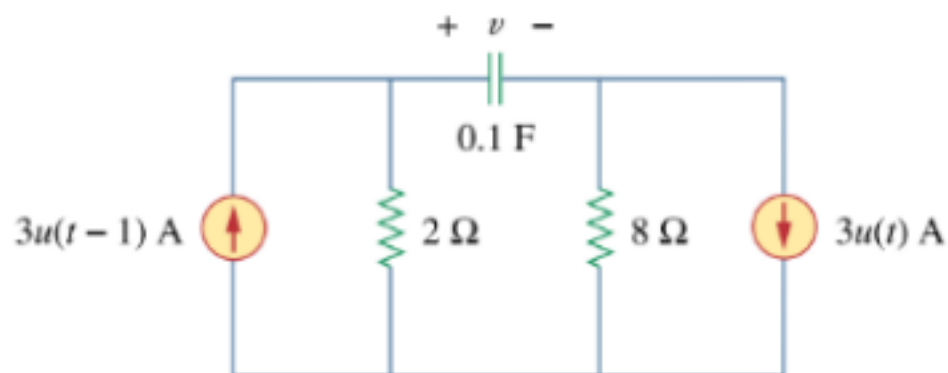
$$\Rightarrow i_2 = \frac{1}{80}$$

$$R_{th} = \frac{1}{i_1} = 160 \Omega$$

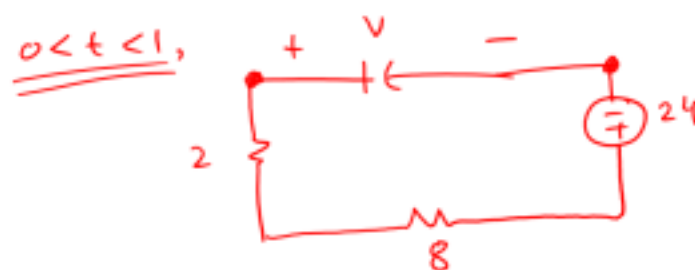
$$\tau = 160 \times 3 = 480$$

DC Circuit

Determine $v(t)$ for $t > 0$ in the circuit of Fig. 7.114 if $v(0) = 0$



$$v(t) = 24 + (0 - 24)e^{-t}$$
$$= \boxed{24(1 - e^{-t})}$$



$$v(\infty) = 24 \text{ V}$$

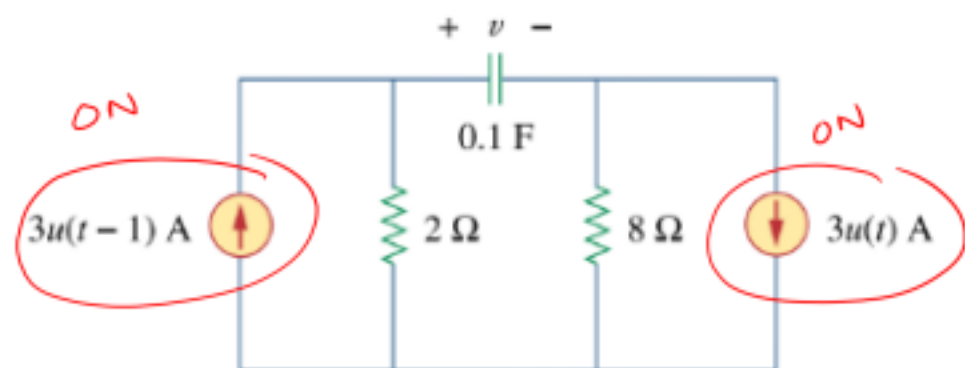
$$R_{Th} = 10 \Omega$$

$$\tau = 10 \times 0.1 = 1$$

DC Circuit

Determine $v(t)$ for $t > 0$ in the circuit of Fig. 7.114 if $v(0) = 0$

$$v(1) = 24(1 - e^{-1}) = \boxed{15.2}$$

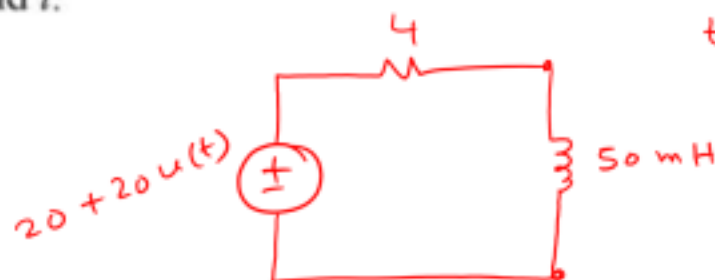
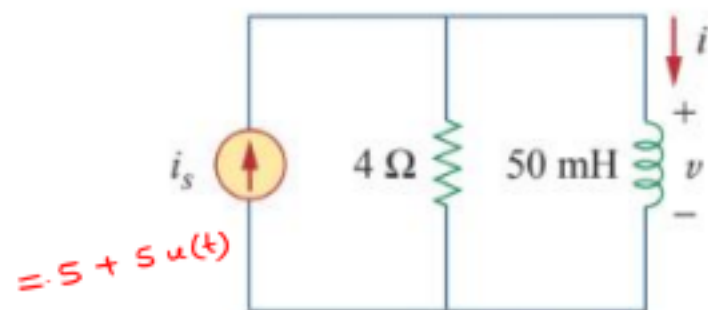


$$v - 24 - 6 = 0$$
$$\Rightarrow v(0) = 30 \text{ V}$$

$$v(t) = 30 + (15.2 - 30)e^{-(t-1)}$$
$$= \boxed{30 + 14.8 e^{-(t-1)} \text{ V}}$$

DC Circuit

In the circuit of Fig. 7.126, i_s changes from 5 A to 10 A at $t = 0$; that is, $i_s = (5 + 5u(t))$ A. Find v and i .



$$t < 0, \quad i(0^-) = i(0) = 5 \text{ A}$$

$$t > 0, \quad i(\infty) = 10 \text{ A}$$

$$R_{th} = 4 \Omega$$

$$\tau = \frac{50 \times 10^{-3}}{4}$$

$$= \frac{1}{80}$$

$$i(t) = 10 + (5 - 10)e^{-80t}$$
$$= \boxed{10 - 5e^{-80t} \text{ A}}$$

$$v = L \frac{di}{dt} = 50 \times 10^{-3} (400 e^{-80t})$$
$$= \boxed{20 e^{-80t} \text{ V}}$$

DC Circuit

For the op amp circuit in Fig. 7.138, let $R_1 = 10 \text{ k}\Omega$, $R_f = 20 \text{ k}\Omega$, $C = 20 \mu\text{F}$, and $v(0) = 1 \text{ V}$. Find v_o .

$$\textcircled{a} \Rightarrow \frac{v_1 - v}{R_1} = C \frac{dv}{dt}$$

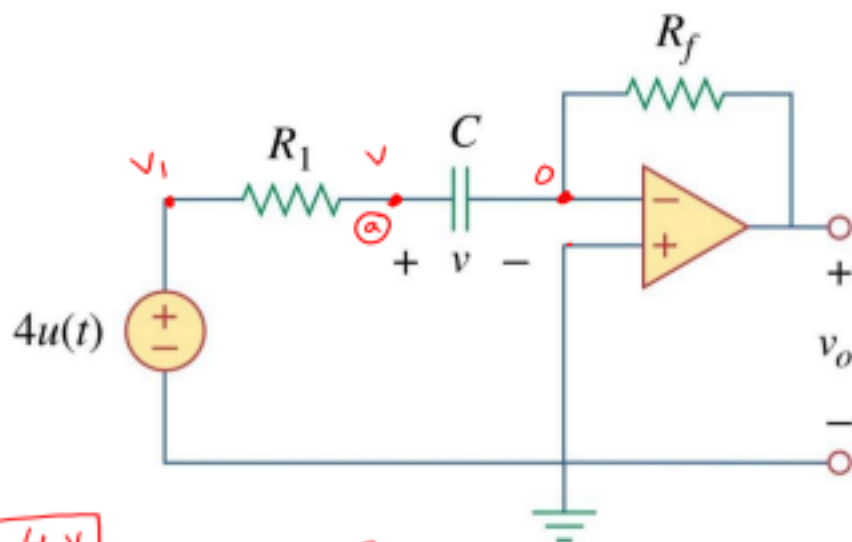
$$\Rightarrow \frac{v_1}{R_1 C} = \frac{v}{R_1 C} + \frac{dv}{dt}$$

$$\textcircled{b} \Rightarrow C \frac{dv}{dt} = -\frac{v_o}{R_f}$$

$$\Rightarrow v(t) = \begin{cases} 1 & ; t < 0 \\ 4 - 3e^{-5t} & ; t > 0 \end{cases}$$

$$\begin{aligned} -4 + v(\infty) &= 0 \\ \Rightarrow v(\infty) &= 4 \end{aligned} \quad \left| \quad v(t) = 4 + (1-4)e^{-5t} \right.$$

$$v_1 = 4 \text{ V}$$

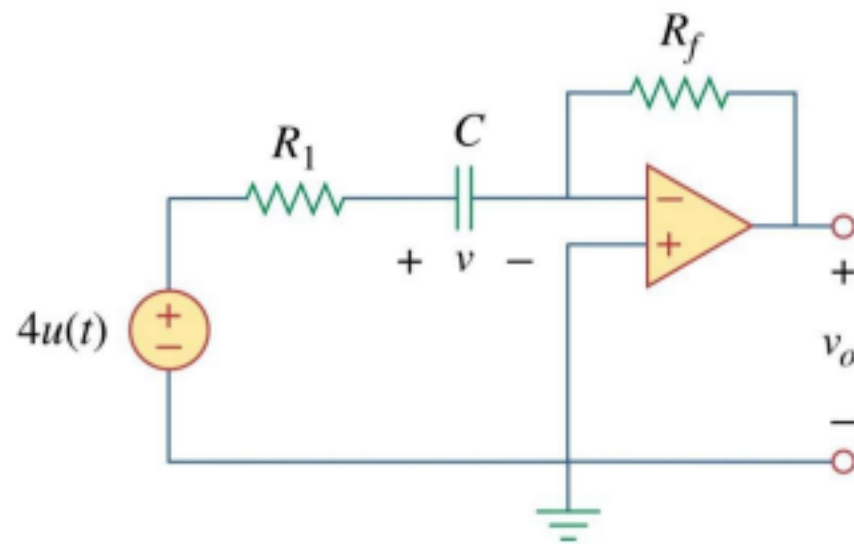


$$\begin{aligned} \tau &= R_1 C \\ &= 10 \times 10^3 \times 20 \times 10^{-6} \\ &= 1/5 \end{aligned}$$

DC Circuit

For the op amp circuit in Fig. 7.138, let $R_1 = 10 \text{ k}\Omega$, $R_f = 20 \text{ k}\Omega$, $C = 20 \mu\text{F}$, and $v(0) = 1 \text{ V}$. Find v_o .

$$\begin{aligned} \text{for } t > 0, \\ v_o &= -R_f C \frac{dv}{dt} \\ &= -20 \times 10^3 \times 20 \times 10^{-6} \times 15 e^{-5t} \\ &= \boxed{-6 e^{-5t} u(t)} \end{aligned}$$



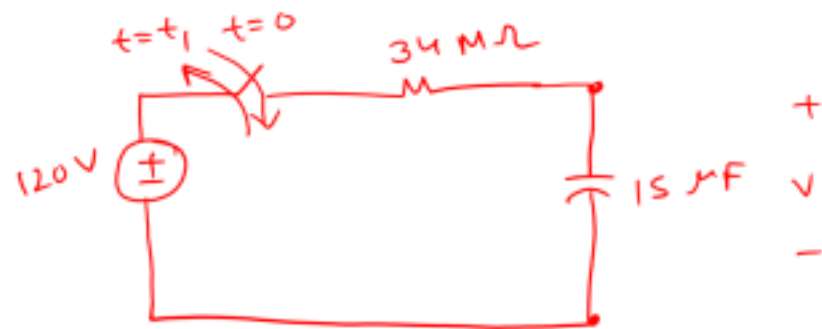
DC Circuit

An RC circuit consists of a series connection of a 120-V source, a switch, a 34-M Ω resistor, and a 15- μ F capacitor. The circuit is used in estimating the speed of a horse running a 4-km racetrack. The switch closes when the horse begins and opens when the horse crosses the finish line. Assuming that the capacitor charges to 85.6 V, calculate the speed of the horse.

$$v(t) = 120 (1 - e^{-t/510})$$

$$\Rightarrow 85.6 = 120 (1 - e^{-t_1/510})$$

$$\Rightarrow t_1 = ?$$



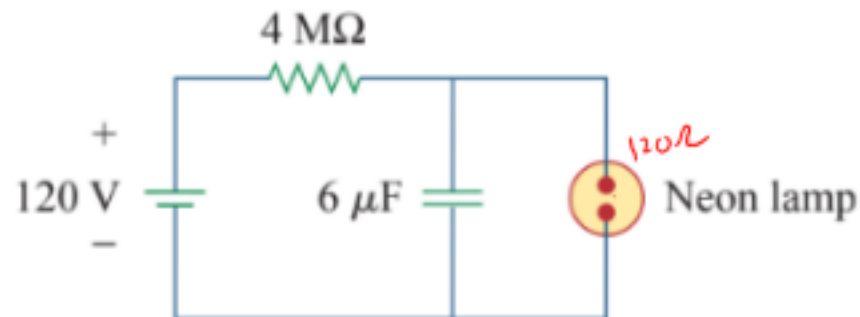
$$\begin{aligned} v(0) &= 0 \\ v(\infty) &= 120\text{ V} \end{aligned} \quad \left| \quad \begin{aligned} \tau &= 34 \times 10^6 \times 15 \times 10^{-6} \\ &= 510 \end{aligned} \right.$$

$$\text{speed} = \frac{4000\text{ m}}{t_1} = \boxed{v\text{ m/s}}$$

DC Circuit

A simple relaxation oscillator circuit is shown in Fig. 7.145. The neon lamp fires when its voltage reaches 75 V and turns off when its voltage drops to 30 V. Its resistance is 120 Ω when on and infinitely high when off.

- (a) For how long is the lamp on each time the capacitor discharges?
- (b) What is the time interval between light flashes?



a) Light ON $\rightarrow (75\text{V} \rightarrow 30\text{V})$

$$\tau = 120 \times 6 \times 10^{-6} = 7.2 \times 10^{-4}$$

$$V(0) = 75\text{ V}$$

$$V(\infty) = 0$$

$$V(t) = 75 e^{-t/7.2 \times 10^{-4}}$$

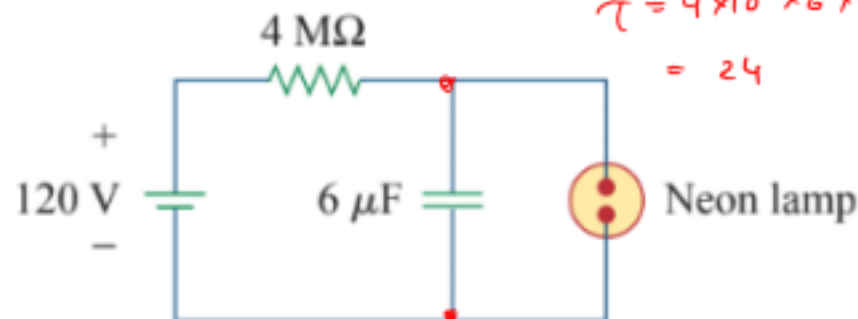
$$\Rightarrow 30 = 75 e^{-t/7.2 \times 10^{-4}}$$

$$\Rightarrow \boxed{t = ?}$$

DC Circuit

A simple relaxation oscillator circuit is shown in Fig. 7.145. The neon lamp fires when its voltage reaches 75 V and turns off when its voltage drops to 30 V . Its resistance is $120\ \Omega$ when on and infinitely high when off.

- (a) For how long is the lamp on each time the capacitor discharges? $t_{\text{int}} = t_1 \sim t_2$
- (b) What is the time interval between light flashes?



$$\tau = 4 \times 10^6 \times 6 \times 10^{-6} = 24$$

$$b) \begin{cases} v(t) - v(\infty) = [v(0) - v(\infty)] e^{-t/\tau} \\ v(t_1) - v(\infty) = [v(0) - v(\infty)] e^{-t_1/\tau} \\ v(t_2) - v(\infty) = [v(0) - v(\infty)] e^{-t_2/\tau} \end{cases}$$

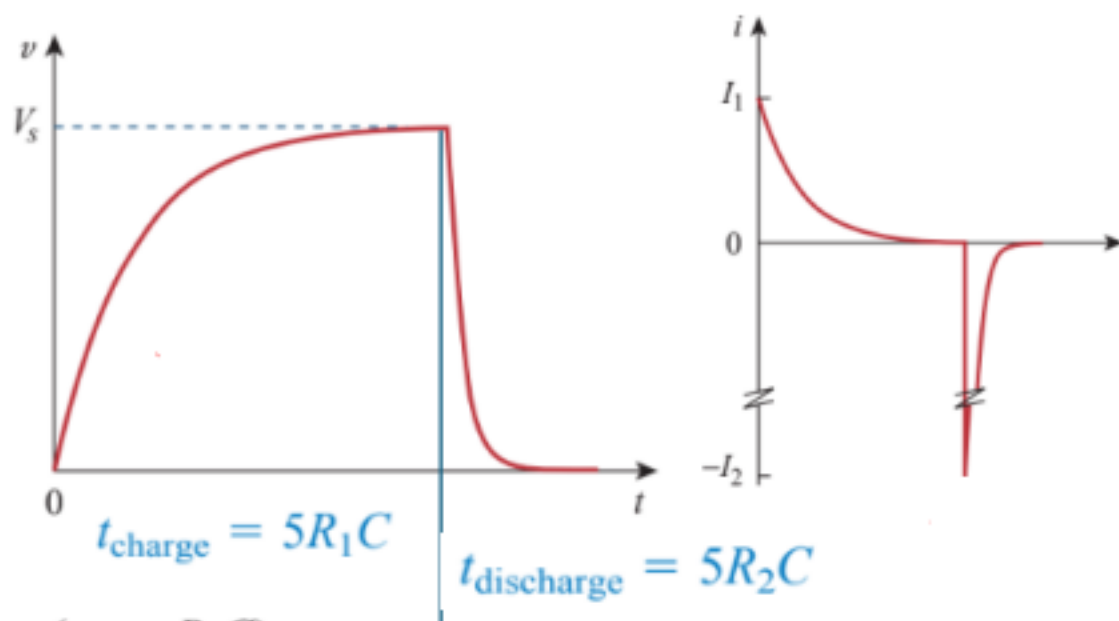
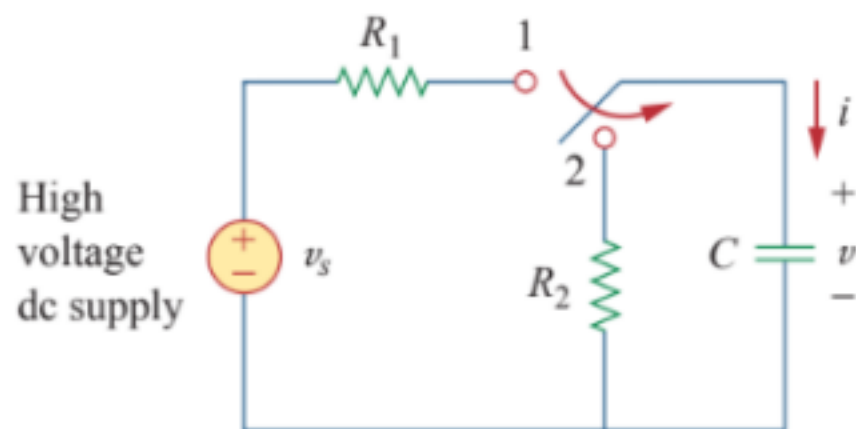
$$\frac{v(t_1) - v(\infty)}{v(t_2) - v(\infty)} = \frac{e^{-t_1/\tau}}{e^{-t_2/\tau}}$$

$$\Rightarrow \frac{75 - 120}{30 - 120} = e^{-(t_1 - t_2)/\tau}$$

$$\Rightarrow t_2 - t_1 = t_{\text{int}} = 16.64 \text{ sec}$$

DC Circuit

Photoflash Unit

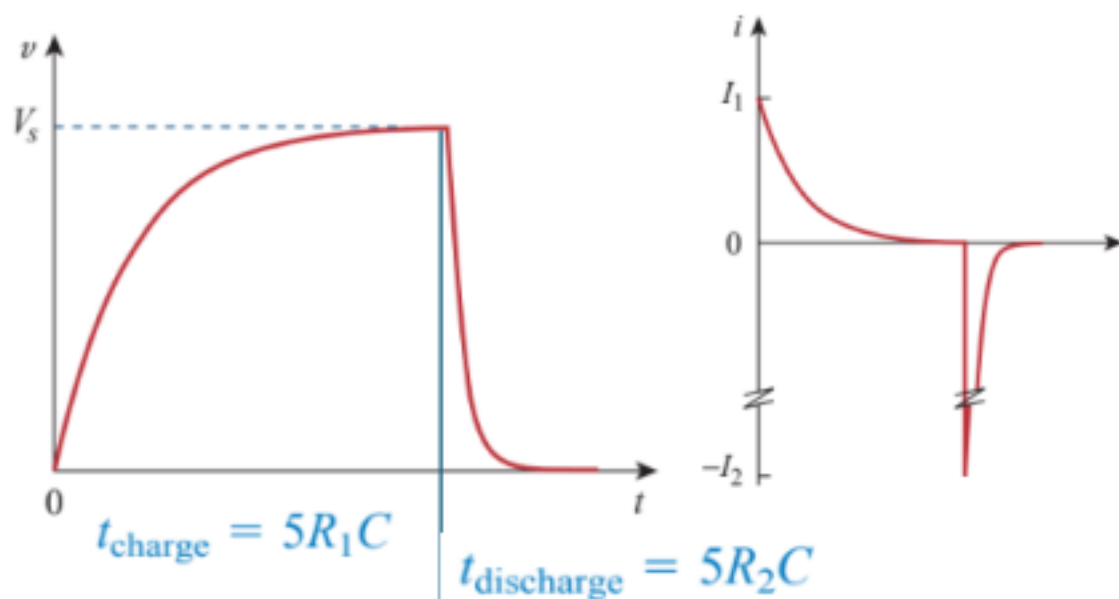
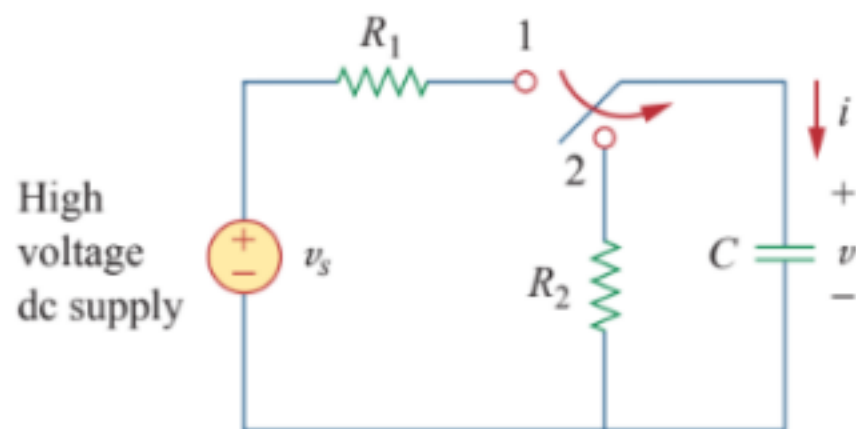


capacitor charges slowly due to the large time constant ($\tau_1 = R_1C$).

capacitor voltage rises gradually from zero to V_s , current decreases gradually from $I_1 = V_s/R_1$ to zero.

DC Circuit

Photoflash Unit



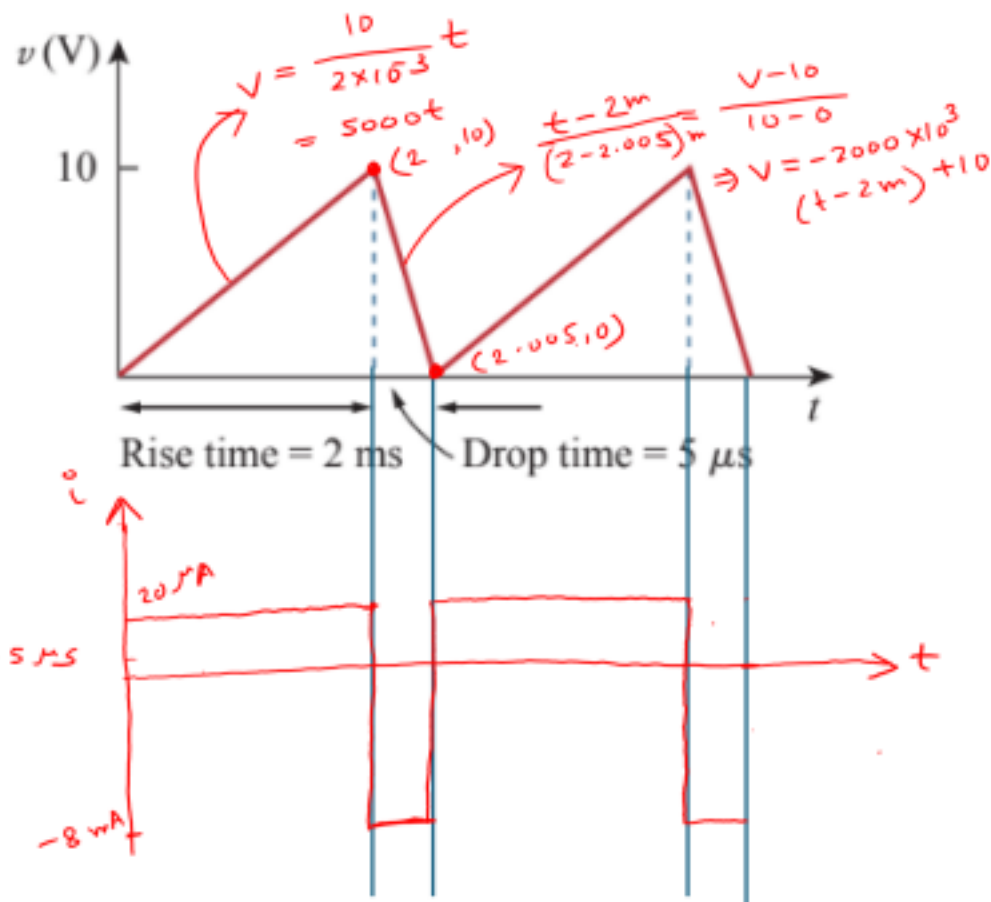
With the switch in position 2, the capacitor voltage is discharged. The low resistance R_2 of the photolamp permits a high discharge current with peak $I_2 = V_s/R_2$ in a short duration, as depicted in

DC Circuit

To move a spot of a cathode-ray tube across the screen requires a linear increase in the voltage across the deflection plates, as shown in Fig. 7.151. Given that the capacitance of the plates is 4 nF, sketch the current flowing through the plates.

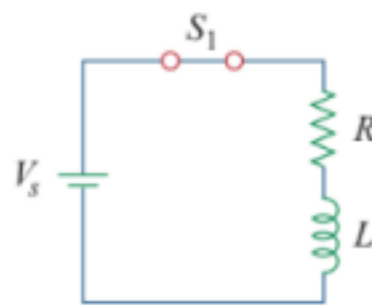
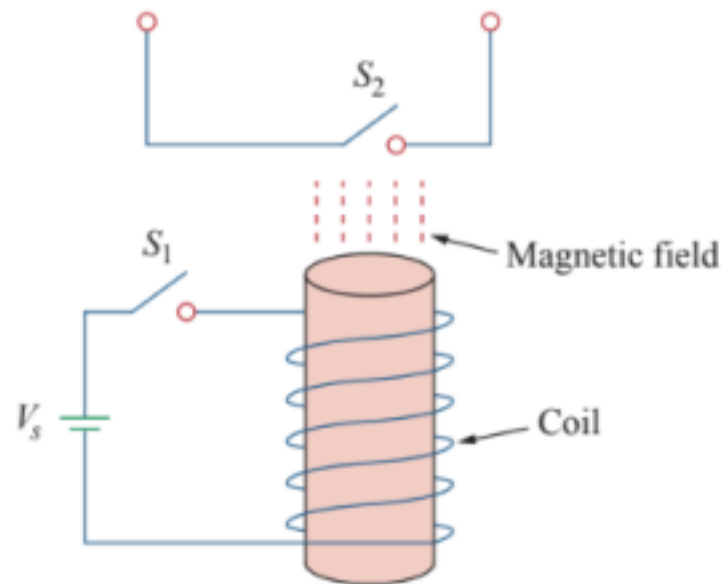
$$i = C \frac{dv}{dt}$$

$$= \begin{cases} 4 \times 10^{-9} \times 5000 & ; 0 < t < 2 \text{ ms} \\ = \boxed{20 \mu\text{A}} \\ -4 \times 10^{-9} \times 2000 \times 10^3 & ; 2 \text{ ms} < t < 2 \text{ ms} + 5 \mu\text{s} \\ = \boxed{-8 \text{ mA}} \end{cases}$$



DC Circuit

Relay Circuits



When switch S_1 is closed, the coil circuit is energized. The coil current gradually increases and produces a magnetic field. Eventually, the magnetic field is sufficiently strong to pull the movable contact in the other circuit and close switch S_2 . At this point, the relay is said to be pulled in. The time interval between the closure of switches S_1 and S_2 is called the relay delay time.

DC Circuit

A relay has a resistance of $200\ \Omega$ and an inductance of 500 mH . The relay contacts close when the current through the coil reaches 350 mA . What time elapses between the application of 110 V to the coil and contact closure?

$$\tau = \frac{L}{R} = \frac{500 \times 10^{-3}}{200} = 2.5 \times 10^{-3}$$

$$i(t) = i(\infty) + [i(0) - i(\infty)] e^{-t/\tau}$$

$$\Rightarrow 350\text{ m} = \frac{11}{20} + \left(0 - \frac{11}{20}\right) e^{-t_d/\tau}$$

$$t < 0, \quad S_1 \rightarrow \text{open}$$

$$i(0) = 0$$

$$i(\infty) = \frac{110}{200} = \frac{11}{20}$$

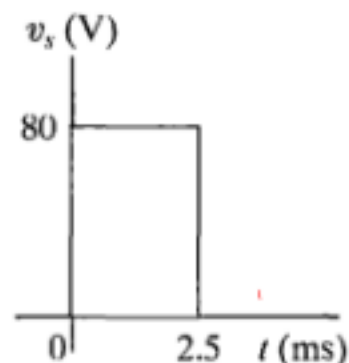
$$\Rightarrow \boxed{t_d = ?}$$

DC Circuit

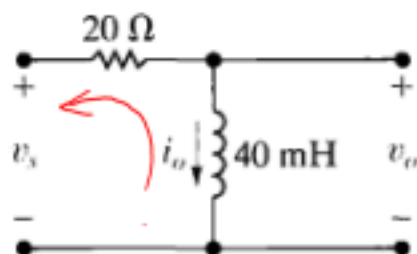
The voltage waveform shown in Fig. P7.79(a) is applied to the circuit of Fig. P7.79(b). The initial current in the inductor is zero.

- Calculate $v_o(t)$.
- Make a sketch of $v_o(t)$ versus t .
- Find i_o at $t = 5$ ms.

Figure P7.79



(a)



(b)

$$\begin{aligned} & a) \left\{ \begin{array}{l} 0 < t < 2.5\text{ ms}, \\ i(0^+) = 0 \\ v_o(0^+) = 80\text{ V} \\ v_o(\infty) = 0 \end{array} \right. \quad \tau = \frac{40 \times 10^{-3}}{20} = 2 \times 10^{-3} \\ & v_o(t) = 80 e^{-500t} \quad 0^+ < t < 2.5\text{ ms} \end{aligned}$$

$$\begin{aligned} & \left\{ \begin{array}{l} v_o(2.5\text{ ms}) = 23\text{ V} \\ i_o(2.5\text{ ms}) = \frac{80 - 23}{20} = 2.85\text{ A} \end{array} \right. \\ & v_o(2.5\text{ ms}^+) = (-2.85) \times 20 = -57.1\text{ A} \\ & v_o(\infty) = 0 \\ & v_o = -57.1 e^{-500(t - 2.5 \times 10^{-3})} \quad t > 2.5\text{ ms}^+ \end{aligned}$$